CHAPTER

73

ENGINE FUEL AND CONTROL

(LEAP-1B ENGINES)

BOEING

737-7/8/8200/9/10
AIRCRAFT MAINTENANCE MANUAL

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A 498.425	Sep 15/2026	
A 498.426	Sep 15/2026	
A 498.427	Sep 15/2026	
A 498.428	Sep 15/2026	
A 498.429	Sep 15/2026	
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A 498.432	Sep 15/2026	
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A 498.434	Sep 15/2026	
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A 498.437	Sep 15/2026	
A 498.438	Sep 15/2026	
A 498.439	Sep 15/2026	
A 498.440	Sep 15/2026	
A 498.441	Sep 15/2026	
A 498.442	Sep 15/2026	
A 498.443	Sep 15/2026	
A 498.444	Sep 15/2026	
A 498.445	Sep 15/2026	
A 498.446	Sep 15/2026	
A 498.447	Sep 15/2026	
A 498.448	Sep 15/2026	
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A 498.450	Sep 15/2026	
A 498.451	Sep 15/2026	
A 498.452	Sep 15/2026	
A 498.453	Sep 15/2026	
A 498.454	Sep 15/2026	
A 498.455	Sep 15/2026	
A 498.456	Sep 15/2026	

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73-11-05 (cont)		
A 498.457	Sep 15/2026	
A 498.458	Sep 15/2026	
A 498.459	Sep 15/2026	
A 498.460	Sep 15/2026	
A 498.461	Sep 15/2026	
A 498.462	Sep 15/2026	
A 498.463	Sep 15/2026	
A 498.464	Sep 15/2026	
A 498.465	Sep 15/2026	
A 498.466	Sep 15/2026	
A 498.467	Sep 15/2026	
A 498.468	Sep 15/2026	
73-11-05		
R 601	Sep 15/2026	
R 602	Sep 15/2026	
R 603	Sep 15/2026	
R 604	Sep 15/2026	
R 605	Sep 15/2026	
R 606	Sep 15/2026	
R 607	Sep 15/2026	
R 608	Sep 15/2026	
R 609	Sep 15/2026	
O 610	Sep 15/2026	
O 611	Sep 15/2026	
R 612	Sep 15/2026	
R 613	Sep 15/2026	
R 614	Sep 15/2026	
A 615	Sep 15/2026	
A 616	Sep 15/2026	
A 617	Sep 15/2026	
A 618	Sep 15/2026	
A 619	Sep 15/2026	
A 620	BLANK	
73-11-06		
401	May 15/2026	
402	May 15/2024	
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405	May 15/2026	
406	May 15/2026	
407	May 15/2026	
R 408	Sep 15/2026	
R 409	Sep 15/2026	
R 410	Sep 15/2026	
R 411	Sep 15/2026	
R 412	Sep 15/2026	
R 413	Sep 15/2026	
R 414	Sep 15/2026	
R 415	Sep 15/2026	
R 416	Sep 15/2026	
R 417	Sep 15/2026	
R 418	Sep 15/2026	
R 419	Sep 15/2026	
R 420	Sep 15/2026	
R 421	Sep 15/2026	
R 422	Sep 15/2026	
O 423	Sep 15/2026	
O 424	Sep 15/2026	
O 425	Sep 15/2026	
O 426	Sep 15/2026	
O 427	Sep 15/2026	
O 428	Sep 15/2026	
O 429	Sep 15/2026	
R 430	Sep 15/2026	
R 431	Sep 15/2026	
O 432	Sep 15/2026	
73-11-06		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	BLANK	
73-11-07		
401	May 15/2026	
402	May 15/2024	
403	May 15/2026	
404	May 15/2026	
405	May 15/2026	
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409	May 15/2024	
410	May 15/2024	
411	Jan 15/2025	
412	May 15/2024	
413	Jan 15/2025	
414	May 15/2024	
73-11-09		
401	May 15/2026	
402	May 15/2026	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	BLANK	
73-11-09		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
73-11-10		
401	May 15/2026	
402	May 15/2024	
403	May 15/2026	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	

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73-11-10 (cont)		
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	BLANK	
73-11-10		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
73-11-11		
401	May 15/2026	
402	May 15/2024	
403	May 15/2026	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	Jan 15/2026	
411	Jan 15/2026	
412	Jan 15/2026	
413	Jan 15/2026	
414	BLANK	
73-11-12		
401	May 15/2026	
402	May 15/2026	
403	May 15/2026	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	May 15/2024	
412	May 15/2024	

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73-11-12		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
73-21-00		
201	Jan 15/2025	
202	Jan 15/2025	
203	Jan 15/2025	
R 204	Sep 15/2026	
R 205	Sep 15/2026	
O 206	Sep 15/2026	
O 207	Sep 15/2026	
O 208	Sep 15/2026	
O 209	Sep 15/2026	
O 210	Sep 15/2026	
O 211	Sep 15/2026	
O 212	Sep 15/2026	
O 213	Sep 15/2026	
O 214	Sep 15/2026	
O 215	Sep 15/2026	
O 216	Sep 15/2026	
O 217	Sep 15/2026	
O 218	Sep 15/2026	
O 219	Sep 15/2026	
O 220	Sep 15/2026	
O 221	Sep 15/2026	
O 222	Sep 15/2026	
O 223	Sep 15/2026	
O 224	Sep 15/2026	
O 225	Sep 15/2026	
O 226	Sep 15/2026	
O 227	Sep 15/2026	
R 228	Sep 15/2026	
O 229	Sep 15/2026	
R 230	Sep 15/2026	
O 231	Sep 15/2026	

Subject/Page	Date	COC
73-21-00 (cont)		
O 232	Sep 15/2026	
R 233	Sep 15/2026	
O 234	Sep 15/2026	
R 235	Sep 15/2026	
R 236	Sep 15/2026	
R 237	Sep 15/2026	
O 238	Sep 15/2026	
O 239	Sep 15/2026	
O 240	Sep 15/2026	
O 241	Sep 15/2026	
O 242	Sep 15/2026	
O 243	Sep 15/2026	
O 244	Sep 15/2026	
O 245	Sep 15/2026	
O 246	Sep 15/2026	
O 247	Sep 15/2026	
O 248	Sep 15/2026	
O 249	Sep 15/2026	
O 250	Sep 15/2026	
O 251	Sep 15/2026	
O 252	Sep 15/2026	
O 253	Sep 15/2026	
O 254	Sep 15/2026	
O 255	Sep 15/2026	
O 256	Sep 15/2026	
O 257	Sep 15/2026	
O 258	Sep 15/2026	
O 259	Sep 15/2026	
R 260	Sep 15/2026	
O 261	Sep 15/2026	
O 262	Sep 15/2026	
O 263	Sep 15/2026	
O 264	Sep 15/2026	
A 265	Sep 15/2026	
A 266	Sep 15/2026	

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73-21-00		
601	May 15/2024	
602	May 15/2024	
73-21-04		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	Jan 15/2026	
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408	Jan 15/2026	
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410	Jan 15/2026	
411	Jan 15/2026	
412	Jan 15/2026	
413	Jan 15/2026	
414	Jan 15/2026	
415	Jan 15/2026	
416	BLANK	
73-21-04		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
73-21-05		
201	Jan 15/2025	
R 202	Sep 15/2026	
O 203	Sep 15/2026	
O 204	Sep 15/2026	
O 205	Sep 15/2026	
O 206	Sep 15/2026	
O 207	Sep 15/2026	
O 208	Sep 15/2026	
O 209	Sep 15/2026	
O 210	Sep 15/2026	
O 211	Sep 15/2026	
O 212	Sep 15/2026	
73-21-05 (cont)		
O 213	Sep 15/2026	
O 214	Sep 15/2026	
O 215	Sep 15/2026	
O 216	Sep 15/2026	
O 217	Sep 15/2026	
O 218	Sep 15/2026	
O 219	Sep 15/2026	
R 220	Sep 15/2026	
O 221	Sep 15/2026	
O 222	Sep 15/2026	
D 223	Sep 15/2026	
D 224	BLANK	
73-21-05		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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410	May 15/2024	
R 411	Sep 15/2026	
O 412	Sep 15/2026	
O 413	Sep 15/2026	
O 414	Sep 15/2026	
R 415	Sep 15/2026	
O 416	Sep 15/2026	
417	May 15/2024	
418	May 15/2024	
419	May 15/2024	
420	May 15/2024	
421	May 15/2024	
422	May 15/2024	
423	May 15/2024	
73-21-05 (cont)		
424	May 15/2024	
425	May 15/2024	
R 426	Sep 15/2026	
O 427	Sep 15/2026	
O 428	Sep 15/2026	
O 429	Sep 15/2026	
R 430	Sep 15/2026	
73-21-05		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
73-21-06		
401	May 15/2024	
402	May 15/2025	
403	May 15/2024	
R 404	Sep 15/2026	
405	May 15/2024	
406	Sep 15/2024	
407	May 15/2024	
408	May 15/2024	
409	Sep 15/2024	
410	May 15/2026	
R 411	Sep 15/2026	
R 412	Sep 15/2026	
413	May 15/2024	
414	BLANK	
73-21-06		
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602	Sep 15/2024	
603	Sep 15/2024	
604	Sep 15/2024	
605	May 15/2024	
606	BLANK	
73-21-08		
401	May 15/2024	
402	May 15/2024	

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73-21-08 (cont)		
403	Jan 15/2025	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
408	May 15/2024	
409	May 15/2024	
410	May 15/2024	
411	Jan 15/2025	
412	May 15/2024	
73-21-08		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
73-21-09		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
73-21-09		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
R 604	Sep 15/2026	
R 605	Sep 15/2026	
O 606	Sep 15/2026	
607	May 15/2024	
608	May 15/2024	
609	May 15/2024	
610	May 15/2024	
611	May 15/2024	
612	May 15/2024	
613	May 15/2024	
614	May 15/2024	
73-21-11		
201	May 15/2024	
202	May 15/2024	
203	May 15/2024	
204	May 15/2024	
205	May 15/2024	
206	May 15/2024	
R 207	Sep 15/2026	
208	May 15/2024	
209	May 15/2026	
210	May 15/2024	
211	May 15/2024	
212	May 15/2024	
73-21-11		
401	May 15/2024	
402	May 15/2024	
403	May 15/2026	
404	May 15/2024	
405	May 15/2026	
406	May 15/2024	
407	Sep 15/2024	
408	Sep 15/2024	
409	May 15/2024	
410	May 15/2024	
73-21-11		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
73-21-12		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
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73-21-63		
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402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	
406	May 15/2024	
407	May 15/2024	
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73-21-63		
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602	May 15/2024	
603	May 15/2024	
604	May 15/2024	
605	May 15/2024	
606	May 15/2024	
73-21-64		
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403	Sep 15/2024	
404	May 15/2024	
405	Sep 15/2024	
406	May 15/2026	
407	Sep 15/2024	
408	BLANK	
73-21-64		
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602	May 15/2024	
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73-21-67		
401	May 15/2024	
402	May 15/2024	
403	May 15/2024	
404	May 15/2024	
405	May 15/2024	

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73-21-67 (cont)		
406	May 15/2024	
407	May 15/2024	
408	BLANK	
73-21-67		
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602	May 15/2024	
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73-22-01		
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205	May 15/2024	
206	May 15/2024	
73-22-01		
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404	May 15/2024	
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413	May 15/2024	
414	May 15/2024	
415	May 15/2024	
416	May 15/2024	
417	May 15/2024	
418	May 15/2024	
419	May 15/2024	

Subject/Page	Date	COC
73-22-01 (cont)		
420	May 15/2024	
421	May 15/2024	
422	May 15/2024	
423	May 15/2024	
424	May 15/2026	
425	May 15/2026	
426	May 15/2026	
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436	May 15/2026	
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438	May 15/2026	
439	May 15/2026	
R 440	Sep 15/2026	
441	May 15/2026	
442	May 15/2026	
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446	May 15/2026	
447	May 15/2026	
448	May 15/2026	
R 449	Sep 15/2026	
450	May 15/2026	
451	May 15/2026	
452	May 15/2026	
453	May 15/2026	
454	May 15/2026	
455	May 15/2026	

Subject/Page	Date	COC
73-22-01 (cont)		
456	May 15/2026	
457	May 15/2026	
458	May 15/2026	
459	May 15/2026	
460	May 15/2026	
461	May 15/2026	
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482	May 15/2026	
483	May 15/2026	
484	May 15/2026	
485	May 15/2026	
R 486	Sep 15/2026	
487	May 15/2026	
488	May 15/2026	
489	May 15/2026	
490	May 15/2026	
491	May 15/2026	

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73-22-01 (cont)		
492	May 15/2026	
493	May 15/2026	
494	May 15/2026	
R 495	Sep 15/2026	
496	May 15/2026	
497	May 15/2026	
498	May 15/2026	
498.1	May 15/2026	
498.2	May 15/2026	
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498.4	May 15/2026	
498.5	May 15/2026	
498.6	May 15/2026	
498.7	May 15/2026	
498.8	May 15/2026	
498.9	May 15/2026	
498.10	May 15/2026	
498.11	May 15/2026	
498.12	May 15/2026	
498.13	May 15/2026	
498.14	May 15/2026	
498.15	May 15/2026	
498.16	May 15/2026	
498.17	May 15/2026	
498.18	May 15/2026	
498.19	May 15/2026	
498.20	May 15/2026	
498.21	May 15/2026	
498.22	May 15/2026	
498.23	May 15/2026	
498.24	May 15/2026	
498.25	May 15/2026	
498.26	May 15/2026	
498.27	May 15/2026	
498.28	May 15/2026	
498.29	May 15/2026	
73-22-01 (cont)		
498.30	May 15/2026	
498.31	May 15/2026	
498.32	May 15/2026	
498.33	May 15/2026	
498.34	May 15/2026	
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498.37	May 15/2026	
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498.40	May 15/2026	
498.41	May 15/2026	
498.42	May 15/2026	
498.43	May 15/2026	
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498.45	May 15/2026	
498.46	May 15/2026	
498.47	May 15/2026	
498.48	May 15/2026	
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498.50	May 15/2026	
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498.55	May 15/2026	
498.56	May 15/2026	
498.57	May 15/2026	
498.58	May 15/2026	
R 498.59	Sep 15/2026	
O 498.60	Sep 15/2026	
R 498.61	Sep 15/2026	
R 498.62	Sep 15/2026	
R 498.63	Sep 15/2026	
R 498.64	Sep 15/2026	
R 498.65	Sep 15/2026	
73-22-01 (cont)		
R 498.66	Sep 15/2026	
R 498.67	Sep 15/2026	
R 498.68	Sep 15/2026	
R 498.69	Sep 15/2026	
R 498.70	Sep 15/2026	
R 498.71	Sep 15/2026	
R 498.72	Sep 15/2026	
R 498.73	Sep 15/2026	
R 498.74	Sep 15/2026	
R 498.75	Sep 15/2026	
R 498.76	Sep 15/2026	
R 498.77	Sep 15/2026	
R 498.78	Sep 15/2026	
R 498.79	Sep 15/2026	
R 498.80	Sep 15/2026	
O 498.81	Sep 15/2026	
O 498.82	Sep 15/2026	
O 498.83	Sep 15/2026	
O 498.84	Sep 15/2026	
O 498.85	Sep 15/2026	
O 498.86	Sep 15/2026	
O 498.87	Sep 15/2026	
O 498.88	Sep 15/2026	
R 498.89	Sep 15/2026	
O 498.90	Sep 15/2026	
O 498.91	Sep 15/2026	
O 498.92	Sep 15/2026	
O 498.93	Sep 15/2026	
O 498.94	Sep 15/2026	
O 498.95	Sep 15/2026	
O 498.96	Sep 15/2026	
O 498.97	Sep 15/2026	
O 498.98	Sep 15/2026	
O 498.99	Sep 15/2026	
R 498.100	Sep 15/2026	
R 498.101	Sep 15/2026	

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73-22-01 (cont)		
R 498.102	Sep 15/2026	
R 498.103	Sep 15/2026	
R 498.104	Sep 15/2026	
R 498.105	Sep 15/2026	
R 498.106	Sep 15/2026	
R 498.107	Sep 15/2026	
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R 498.120	Sep 15/2026	
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R 498.122	Sep 15/2026	
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R 498.147	Sep 15/2026	
R 498.148	Sep 15/2026	
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R 498.151	Sep 15/2026	
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O 498.167	Sep 15/2026	
O 498.168	Sep 15/2026	
R 498.169	Sep 15/2026	
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R 498.172	Sep 15/2026	
R 498.173	Sep 15/2026	
73-22-01 (cont)		
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613	May 15/2024	
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73-22-01 (cont)		
615	May 15/2024	
616	May 15/2024	
617	May 15/2024	
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619	May 15/2024	
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73-22-01 (cont)		
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73-22-01		
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853	Jan 15/2025	

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855	Jan 15/2025	
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857	Jan 15/2025	
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73-31-00		
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602	May 15/2024	
73-31-01		
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408	May 15/2024	
73-31-01		
601	May 15/2024	
602	May 15/2024	
603	May 15/2024	
73-31-01 (cont)		
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737-7/8/8200/9/10
AIRCRAFT MAINTENANCE MANUAL

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SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ENGINE FUEL AND CONTROL	73-00-00		501	PTF ALL
Scheduled Maintenance Task for Short-Time Fault Interrogation TASK 73-00-00-700-801-G00			501	PTF ALL
Scheduled Maintenance Task for Long-Time Fault Interrogation TASK 73-00-00-700-802-G00			503	PTF ALL
Maintenance Messages Check for EEC Faults TASK 73-00-00-700-803-G00			505	PTF ALL
ENGINE FUEL AND CONTROL - DDG MAINTENANCE PROCEDURES	73-00-00		901	PTF ALL
DDG 73-30-01 Fuel Control ENG VALVE CLOSED Indicating System Inoperative - Preparation TASK 73-00-00-040-801-G00			901	PTF ALL
DDG 73-30-01 Fuel Control ENG VALVE CLOSED Indicating System Inoperative - Restoration TASK 73-00-00-440-801-G00			902	PTF ALL
FUEL DISTRIBUTION SYSTEM - INSPECTION/CHECK	73-11-00		601	PTF ALL
Engine Fuel Distribution System General Visual Inspection TASK 73-11-00-210-801-G00			601	PTF ALL
Fuel Filter Contamination Detailed Visual Inspection TASK 73-11-00-211-801-G00			607	PTF ALL
MAIN FUEL PUMP - REMOVAL/INSTALLATION	73-11-01		401	PTF ALL
Main Fuel Pump Removal TASK 73-11-01-000-801-G00			401	PTF ALL
Main Fuel Pump Installation TASK 73-11-01-400-801-G00			411	PTF ALL
FUEL FILTER CARTRIDGE - REMOVAL/INSTALLATION	73-11-02		401	PTF ALL
Fuel Filter Cartridge Removal TASK 73-11-02-000-801-G00			401	PTF ALL

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Fuel Filter Cartridge Installation TASK 73-11-02-400-801-G00			408	PTF ALL
MAIN FUEL FILTER HOUSING -REMOVAL/INSTALLATION	73-11-03		401	PTF ALL
Main Fuel Filter Housing Removal TASK 73-11-03-000-801-G00			401	PTF ALL
Main Fuel Filter Housing Installation TASK 73-11-03-400-801-G00			407	PTF ALL
FUEL METERING UNIT STRAINER - REMOVAL/INSTALLATION	73-11-04		401	PTF ALL
Fuel Metering Unit Strainer Removal TASK 73-11-04-000-801-G00			401	PTF ALL
Fuel Metering Unit Strainer Installation TASK 73-11-04-400-801-G00			405	PTF ALL
FUEL METERING UNIT STRAINER - INSPECTION/CHECK	73-11-04		601	PTF ALL
Fuel Metering Unit Strainer Inspection TASK 73-11-04-200-801-G00			601	PTF ALL
FUEL METERING UNIT STRAINER - CLEANING/PAINTING	73-11-04		701	PTF ALL
Fuel Metering Unit Strainer Cleaning TASK 73-11-04-100-801-G00			701	PTF ALL
FUEL NOZZLE - REMOVAL/INSTALLATION	73-11-05		401	PTF ALL
Fuel Nozzle Removal TASK 73-11-05-000-801-G00			401	PTF ALL
Fuel Nozzle Installation TASK 73-11-05-400-801-G00			498.193	PTF ALL
Fuel Nozzle Set Removal TASK 73-11-05-000-802-G00			498.326	PTF ALL
Fuel Nozzle Set Installation TASK 73-11-05-400-802-G00			498.444	PTF ALL
FUEL NOZZLE AND FUEL MANIFOLD - INSPECTION/CHECK	73-11-05		601	PTF ALL
Fuel Nozzle and Fuel Manifold Leak Check TASK 73-11-05-200-801-G00			601	PTF ALL

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SPLIT CONTROL UNIT - REMOVAL/INSTALLATION	73-11-06		401	PTF ALL
Split Control Unit Removal TASK 73-11-06-000-801-G00			401	PTF ALL
Split Control Unit Installation TASK 73-11-06-400-801-G00			424	PTF ALL
SPLIT CONTROL UNIT - INSPECTION/CHECK	73-11-06		601	PTF ALL
Split Control Unit Inspection TASK 73-11-06-210-801-G00			601	PTF ALL
MAIN FUEL SUPPLY HOSE - REMOVAL/INSTALLATION	73-11-07		401	PTF ALL
Main Fuel Supply Hose Removal TASK 73-11-07-000-801-G00			401	PTF ALL
Main Fuel Supply Hose Installation TASK 73-11-07-400-801-G00			411	PTF ALL
FMU FUEL STRAINER DIFFERENTIAL PRESSURE TRANSDUCER - REMOVAL/INSTALLATION	73-11-09		401	PTF ALL
FMU Fuel Strainer Differential Pressure Transducer Removal TASK 73-11-09-000-801-G00			401	PTF ALL
FMU Fuel Strainer Differential Pressure Transducer Installation TASK 73-11-09-400-801-G00			407	PTF ALL
FMU FUEL STRAINER DIFFERENTIAL PRESSURE TRANSDUCER - INSPECTION/CHECK	73-11-09		601	PTF ALL
FMU Fuel Strainer Differential Pressure Transducer Inspection TASK 73-11-09-210-801-G00			601	PTF ALL
FUEL FILTER DIFFERENTIAL PRESSURE TRANSDUCER - REMOVAL/INSTALLATION	73-11-10		401	PTF ALL
Fuel Filter Differential Pressure Transducer Removal TASK 73-11-10-000-801-G00			401	PTF ALL
Fuel Filter Differential Pressure Transducer Installation TASK 73-11-10-400-801-G00			407	PTF ALL

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FUEL FILTER DIFFERENTIAL PRESSURE TRANSDUCER - INSPECTION/CHECK	73-11-10		601	PTF ALL
Fuel Filter Differential Pressure Transducer Inspection TASK 73-11-10-210-801-G00			601	PTF ALL
SERVO FUEL HEATER - REMOVAL/INSTALLATION	73-11-11		401	PTF ALL
Servo Fuel Heater Removal TASK 73-11-11-000-801-G00			401	PTF ALL
Servo Fuel Heater Installation TASK 73-11-11-400-801-G00			410	PTF ALL
MAIN FUEL/OIL HEAT EXCHANGER DIFFERENTIAL PRESSURE SENSOR - REMOVAL/INSTALLATION	73-11-12		401	PTF ALL
Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Removal TASK 73-11-12-000-801-G00			401	PTF ALL
Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Installation TASK 73-11-12-400-801-G00			410	PTF ALL
MAIN FUEL/OIL HEAT EXCHANGER DIFFERENTIAL PRESSURE SENSOR - INSPECTION/CHECK	73-11-12		601	PTF ALL
Main Fuel/Oil Heat Exchanger Differential Pressure Sensor Inspection TASK 73-11-12-210-801-G00			601	PTF ALL
FUEL CONTROL SYSTEM - MAINTENANCE PRACTICES	73-21-00		201	PTF ALL
EEC System Configuration Data Functions Check TASK 73-21-00-470-801-G00			201	PTF ALL
EEC Shop Faults Function Check TASK 73-21-00-470-802-G00			205	PTF ALL
EEC Onboard Software Installation TASK 73-21-00-470-803-G00			235	PTF ALL
EEC Software Installation Using a PMAT TASK 73-21-00-470-805-G00			240	PTF ALL
EEC Onboard Software Installation Verification TASK 73-21-00-470-804-G00			260	PTF ALL

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EEC Maintenance Power Selection TASK 73-21-00-800-801-G00			261	PTF ALL
FUEL CONTROL SYSTEM - INSPECTION/CHECK	73-21-00		601	PTF ALL
Fuel Control System Inspection TASK 73-21-00-200-801-G00			601	PTF ALL
FUEL METERING UNIT - REMOVAL/INSTALLATION	73-21-04		401	PTF ALL
Fuel Metering Unit Removal TASK 73-21-04-000-801-G00			401	PTF ALL
Fuel Metering Unit Installation TASK 73-21-04-400-801-G00			410	PTF ALL
FUEL METERING UNIT - INSPECTION/CHECK	73-21-04		601	PTF ALL
Fuel Metering Unit (FMU) Inspection TASK 73-21-04-200-801-G00			601	PTF ALL
ELECTRONIC ENGINE CONTROL (EEC) - MAINTENANCE PRACTICES	73-21-05		201	PTF ALL
Electronic Engine Control (EEC) Data Download (On-Wing)(Selection) TASK 73-21-05-970-803-G00			201	PTF ALL
Electronic Engine Control (EEC) Data Download (On Wing) Using the ONS TASK 73-21-05-970-802-G00			201	PTF ALL
Electronic Engine Control (EEC) Data Download (On-Wing) Using a PMAT TASK 73-21-05-970-801-G00			206	PTF ALL
Electronic Engine Control (EEC) Data Download (On Wing) Using the Engine Health Management (EHM) Function TASK 73-21-05-970-804-G00			220	PTF ALL
ELECTRONIC ENGINE CONTROL (EEC) - REMOVAL/INSTALLATION	73-21-05		401	PTF ALL
Electronic Engine Control (EEC) Channel A Removal TASK 73-21-05-000-801-G00			401	PTF ALL
Electronic Engine Control (EEC) Channel A Installation TASK 73-21-05-400-801-G00			411	PTF ALL

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Electronic Engine Control (EEC) Channel B Removal TASK 73-21-05-000-802-G00			415	PTF ALL
Electronic Engine Control (EEC) Channel B Installation TASK 73-21-05-400-802-G00			426	PTF ALL
ELECTRONIC ENGINE CONTROL - INSPECTION/CHECK	73-21-05		601	PTF ALL
Electronic Engine Control Inspection TASK 73-21-05-210-801-G00			601	PTF ALL
T12 SENSOR - REMOVAL/INSTALLATION	73-21-06		401	PTF ALL
T12 Sensor Removal TASK 73-21-06-000-801-G00			401	PTF ALL
T12 Sensor Installation TASK 73-21-06-400-801-G00			410	PTF ALL
T12 SENSOR - INSPECTION/CHECK	73-21-06		601	PTF ALL
T12 Sensor Inspection TASK 73-21-06-210-801-G00			601	PTF ALL
PERMANENT MAGNET ALTERNATOR - REMOVAL/INSTALLATION	73-21-08		401	PTF ALL
Permanent Magnet Alternator Removal TASK 73-21-08-000-801-G00			401	PTF ALL
Permanent Magnet Alternator Installation TASK 73-21-08-400-801-G00			409	PTF ALL
PERMANENT MAGNET ALTERNATOR - INSPECTION/CHECK	73-21-08		601	PTF ALL
Permanent Magnet Alternator (PMA) Inspection TASK 73-21-08-210-801-G00			601	PTF ALL
DATA ENTRY PLUG - REMOVAL/INSTALLATION	73-21-09		401	PTF ALL
Data Entry Plug Removal TASK 73-21-09-000-801-G00			401	PTF ALL
Data Entry Plug Installation TASK 73-21-09-400-801-G00			405	PTF ALL

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Data Entry Plug Configuration Check TASK 73-21-09-700-801-G00			601	PTF ALL
Data Entry Plug Inspection TASK 73-21-09-211-801-G00			611	PTF ALL
PRESSURE SUB-SYSTEM - MAINTENANCE PRACTICES	73-21-11		201	PTF ALL
Pressure Sub-System and Sense Line Dry Out TASK 73-21-11-160-801-G00			201	PTF ALL
PRESSURE SUB-SYSTEM - REMOVAL/INSTALLATION	73-21-11		401	PTF ALL
Pressure Sub-System Removal TASK 73-21-11-000-801-G00			401	PTF ALL
Pressure Sub-System Installation TASK 73-21-11-400-801-G00			407	PTF ALL
PRESSURE SUB-SYSTEM - INSPECTION/CHECK	73-21-11		601	PTF ALL
Pressure Sub-System Inspection TASK 73-21-11-210-801-G00			601	PTF ALL
ENGINE CONTROL MODULE - REMOVAL/INSTALLATION	73-21-12		401	PTF ALL
Engine Control Module Removal TASK 73-21-12-000-801-G00			401	PTF ALL
Engine Control Module Installation TASK 73-21-12-400-801-G00			404	PTF ALL
T25 SENSOR - REMOVAL/INSTALLATION	73-21-63		401	PTF ALL
T25 Sensor Removal TASK 73-21-63-000-801-G00			401	PTF ALL
T25 Sensor Installation TASK 73-21-63-400-801-G00			405	PTF ALL
T25 SENSOR - INSPECTION/CHECK	73-21-63		601	PTF ALL
T25 Sensor Inspection TASK 73-21-63-210-801-G00			601	PTF ALL

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T3 SENSOR - REMOVAL/INSTALLATION	73-21-64		401	PTF ALL
T3 Sensor Removal TASK 73-21-64-000-801-G00			401	PTF ALL
T3 Sensor Installation TASK 73-21-64-400-801-G00			405	PTF ALL
T3 SENSOR - INSPECTION/CHECK	73-21-64		601	PTF ALL
T3 Sensor Inspection TASK 73-21-64-210-801-G00			601	PTF ALL
FUEL MANIFOLD PRESSURE TRANSDUCER - REMOVAL/INSTALLATION	73-21-67		401	PTF ALL
Fuel Manifold Pressure Transducer Removal TASK 73-21-67-000-801-G00			401	PTF ALL
Fuel Manifold Pressure Transducer Installation TASK 73-21-67-400-801-G00			405	PTF ALL
FUEL MANIFOLD PRESSURE TRANSDUCER - INSPECTION/CHECK	73-21-67		601	PTF ALL
Fuel Manifold Pressure Transducer Inspection TASK 73-21-67-210-801-G00			601	PTF ALL
ELECTRONIC ENGINE CONTROL (EEC) ELECTRICAL HARNESSES - MAINTENANCE PRACTICES	73-22-01		201	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ8A) Tape Wrap TASK 73-22-01-390-802-G00			201	PTF ALL
ELECTRONIC ENGINE CONTROL (EEC) ELECTRICAL HARNESSES - REMOVAL/INSTALLATION	73-22-01		401	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ3A) Removal TASK 73-22-01-000-801-G00			401	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ3A) Installation TASK 73-22-01-400-801-G00			408	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ3B) Removal TASK 73-22-01-000-802-G00			411	PTF ALL

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Electronic Engine Control (EEC) Electrical Harness (HJ3B) Installation TASK 73-22-01-400-802-G00			418	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ5A) Removal TASK 73-22-01-000-803-G00			421	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ5A) Installation TASK 73-22-01-400-803-G00			461	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ5B) Removal TASK 73-22-01-000-804-G00			471	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ5B) Installation TASK 73-22-01-400-804-G00			498.8	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ6A) Removal TASK 73-22-01-000-805-G00			498.17	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ6A) Installation TASK 73-22-01-400-805-G00			498.31	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ6B) Removal TASK 73-22-01-000-806-G00			498.35	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ6B) Installation TASK 73-22-01-400-806-G00			498.46	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ7) Removal TASK 73-22-01-000-807-G00			498.50	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ7) Installation TASK 73-22-01-400-807-G00			498.56	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ8A) Removal TASK 73-22-01-000-808-G00			498.59	PTF ALL

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CHAPTER 73
ENGINE FUEL AND CONTROL

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
Electronic Engine Control (EEC) Electrical Harness (HJ8A) Installation TASK 73-22-01-400-808-G00			498.100	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ8B) Removal TASK 73-22-01-000-809-G00			498.111	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (HJ8B) Installation TASK 73-22-01-400-809-G00			498.144	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (W4A) Removal TASK 73-22-01-000-810-G00			498.153	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (W4A) Installation TASK 73-22-01-400-810-G00			498.158	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (W4B) Removal TASK 73-22-01-000-811-G00			498.160	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (W4B) Installation TASK 73-22-01-400-811-G00			498.166	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (W2) Removal TASK 73-22-01-000-812-G00			498.169	PTF ALL
Electronic Engine Control (EEC) Electrical Harness (W2) Installation TASK 73-22-01-400-812-G00			498.189	PTF ALL
ENGINE CONTROL ELECTRICAL HARNESSES - INSPECTION/CHECK	73-22-01		601	PTF ALL
Engine Control Electrical Harnesses Inspection TASK 73-22-01-210-801-G00			601	PTF ALL
Engine Control Electrical Harness Connectors Inspection TASK 73-22-01-210-802-G00			654	PTF ALL
ENGINE CONTROL ELECTRICAL HARNESSES - CLEANING/PAINTING	73-22-01		701	PTF ALL
Engine Control Electrical Harness Cleaning TASK 73-22-01-100-801-G00			701	PTF ALL

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CHAPTER 73
ENGINE FUEL AND CONTROL

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
ENGINE WIRING HARNESS - REPAIRS	73-22-01		801	PTF ALL
Engine Wiring Harness Repair (Application of Silicone Tape Patch) TASK 73-22-01-900-801-G00			801	PTF ALL
Engine Wiring Harness Repair (Application of Stainless Steel Braid) TASK 73-22-01-900-802-G00			808	PTF ALL
Engine Wiring Harness Heat-Shrinkable Boot Repair TASK 73-22-01-900-803-G00			818	PTF ALL
Engine Wiring Harness - Heat-Shrinkable Tubing Repair (Application of Silicone Tape Patch) TASK 73-22-01-900-804-G00			830	PTF ALL
Engine Wiring Harnesses W200 and W300 - BMS 1367 Cable at the Terminal Lugs Repair TASK 73-22-01-300-801-G00			839	PTF ALL
Engine Wiring Harnesses W200 and W300 - Bobbin Replacement TASK 73-22-01-300-803-G00			848	PTF ALL
Engine Wiring Harnesses W200 and W300 - Anti-Rotation Bobbin Repair TASK 73-22-01-900-805-G00			856	PTF ALL
Engine Wiring Harness - Harness Firewall Area Repair TASK 73-22-01-300-802-G00			865	PTF ALL
FUEL FLOW INDICATING SYSTEM - INSPECTION/CHECK	73-31-00		601	PTF ALL
Fuel Flow Indicating System Inspection TASK 73-31-00-200-801-G00			601	PTF ALL
FUEL FLOW TRANSMITTER-REMOVAL/INSTALLATION	73-31-01		401	PTF ALL
Fuel Flow Transmitter Removal TASK 73-31-01-000-801-G00			401	PTF ALL
Fuel Flow Transmitter Installation TASK 73-31-01-400-801-G00			406	PTF ALL

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CHAPTER 73
ENGINE FUEL AND CONTROL

SUBJECT	CHAPTER SECTION SUBJECT	CONF	PAGE	EFFECT
FUEL FLOW TRANSMITTER - INSPECTION/CHECK	73-31-01		601	PTF ALL
Fuel Flow Transmitter Inspection TASK 73-31-01-210-801-G00			601	PTF ALL

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ENGINE FUEL AND CONTROL

1. General

A. This procedure has these tasks:

(1) Scheduled Maintenance Task for Short-Time Fault Interrogation

(2) Scheduled Maintenance Task for Long-Time Fault Interrogation

(3) Maintenance Messages Check.

TASK 73-00-00-700-801-G00

2. Scheduled Maintenance Task for Short-Time Fault Interrogation

A. General

(1) This task gives the instructions to interrogate the Scheduled Maintenance Task (SMT) message ENG X EEC C1 for Short Term Dispatch (STD) faults.

(2) For other SMT messages, do this task: Scheduled Maintenance Task (SMT) - Repair Time Limits, TASK 05-00-00-750-801.

(3) Deferral period begins at the point when Electronic Engine Control (EEC) fault is observed/deferred.

(4) The data that follows provide the allowable deferral period of the ENG X EEC C1 fault:

(a) STD Fault Remaining Deferral Period: R = 150 – Q/2, where R is the remaining flight hours and Q is SMT interval.

(b) SMT interval (Q) of 150 hours allows deferral period (R) of 75 hours.

(c) SMT interval (Q) of 70 hours allows deferral period (R) of 115 hours.

B. References

Reference	Title
05-00-00-750-801	Scheduled Maintenance Task (SMT) - Repair Time Limits (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Fault Interrogation

SUBTASK 73-00-00-860-003-G00

(1) Set the GND TESTS switch, on the P61-4 maintenance BITE panel, to the NORM position.

SUBTASK 73-00-00-860-004-G00

(2) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

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(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-00-00-860-005-G00

(3) Make sure that the thrust levers are at the idle and down position.

SUBTASK 73-00-00-860-002-G00

(4) Select the left or right Multi Function Display (MFD).

(a) Push the SYS button, on the MFD P9 control panel.

(b) Push the SEL button on SELECTOR knob, on the MFD P9 control panel.

SUBTASK 73-00-00-860-058-G00

(5) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the TEST position (TASK 73-21-00-800-801-G00).

F. Scheduled Maintenance Task for Short-Time Fault Interrogation

SUBTASK 73-00-00-860-008-G00

(1) If the SMT exists, use the MFD screen to find it.

(a) Select MAINT CTRL PGS on the MFD.

(b) Select LATCHED MESSAGE ERASE.

SUBTASK 73-00-00-860-061-G00

(2) If ENG X EEC C1 for EEC STD exists, do one of the steps that follow:

(a) Refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

(b) Continue operation with the fault provided the fault is corrected within remaining deferral period described in the general statement of this procedure.

NOTE: To help with the future maintenance plan to correct the fault, you may want to use the IFIM to determine and record the maintenance message(s) correlated to the ENG X EEC C1 SMT message.

NOTE: All economic faults have no time limits and can be corrected at operator's discretion.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-00-00-860-009-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-00-00-860-056-G00

(2) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the NORM position (TASK 73-21-00-800-801-G00).

END OF TASK

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TASK 73-00-00-700-802-G00

3. Scheduled Maintenance Task for Long-Time Fault Interrogation

A. General

(1) This task gives the instructions to interrogate the Scheduled Maintenance Task (SMT) message ENG X EEC C2 for Long Term Dispatch (LTD) faults.

(2) For other SMT messages, do this task: Scheduled Maintenance Task (SMT) - Repair Time Limits, TASK 05-00-00-750-801.

(3) Deferral period begins at the point when Electronic Engine Control (EEC) fault is observed/deferred.

(4) The data that follows provide the allowable deferral period of the ENG X EEC C2 fault:

(a) LTD Fault Remaining Deferral Period: R = 250 – Q/2, where R is the remaining flight hours and Q is the SMT interval.

(b) SMT interval (Q) of 300 hours allows deferral period (R) of 100 hours.

(c) SMT interval (Q) of 150 hours allows deferral period (R) of 175 hours.

(d) SMT interval (Q) of 70 hours allows deferral period (R) of 215 hours.

(5) It is recommended to check for the 73-4218X FUEL FILTER APPROACHING BYPASS and 79-4288X OIL FILTER APPROACHING BYPASS maintenance messages at the same time as the SMT check for long term dispatch faults.

NOTE: Where X refers to engine position (1-Left or 5-Right).

(a) This check is for maintenance awareness as these maintenance messages are economic and are not indicated by any flight deck effect.

(b) It is not necessary to correct economic faults in the SMT deferral period. Use your airline's best practice for when to take corrective action for the approaching bypass messages.

B. References

Reference	Title
05-00-00-750-801	Scheduled Maintenance Task (SMT) - Repair Time Limits (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Fault Interrogation

SUBTASK 73-00-00-860-012-G00

(1) Set the GRD TESTS switch, on the P61-4 maintenance BITE panel, to the NORM position.

SUBTASK 73-00-00-860-013-G00

(2) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

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(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-00-00-860-014-G00

(3) Make sure that the thrust levers are at the idle and down position.

SUBTASK 73-00-00-860-011-G00

(4) Select the left or right Multi Function Display (MFD).

(a) Push the SYS button, on the MFD P9 control panel.

(b) Push the SEL button on SELECTOR knob, on the MFD P9 control panel.

SUBTASK 73-00-00-860-059-G00

(5) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the TEST position (TASK 73-21-00-800-801-G00).

F. Scheduled Maintenance Task for Long-Time Fault Interrogation

SUBTASK 73-00-00-860-017-G00

(1) If the SMT exists, use the MFD screen to find it.

(a) Select MAINT CTRL PGS on the MFD.

(b) Select LATCHED MESSAGE ERASE.

SUBTASK 73-00-00-860-062-G00

(2) If ENG X EEC C2 for EEC LTD exists, do one of the steps that follow:

(a) Refer to the Interactive Fault Isolation Manual (IFIM) and do the applicable procedure(s).

(b) Continue operation with the fault provided the fault is corrected within remaining deferral period described in the general statement of this procedure.

NOTE: To help with the future maintenance plan to correct the fault, you may want to use the IFIM to determine and record the maintenance message(s) correlated to the ENG X EEC C2 SMT message.

SUBTASK 73-00-00-200-003-G00

(3) It is recommended to check for 73-4218X FUEL FILTER APPROACHING BYPASS and 79-4288X OIL FILTER APPROACHING BYPASS maintenance messages at this time.

NOTE: Maintenance messages 73-4218X and 79-4288X are economic faults that do not required corrective action in the SMT deferral period. This check is for maintenance awareness as these maintenance messages are not indicated by any flight deck effect or SMT message. Use your airline's best practice for when to take corrective action for these messages.

NOTE: All economic faults have no time limits and can be corrected at operator's discretion.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-00-00-860-018-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

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(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-00-00-860-057-G00

(2) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the NORM position (TASK 73-21-00-800-801-G00).

——— END OF TASK ———

TASK 73-00-00-700-803-G00

4. Maintenance Messages Check for EEC Faults

NOTE: This procedure is a scheduled maintenance task.

A. General

(1) This task gives instructions to use the Multi Function Display (MFD) to find and read maintenance messages for faults detected by the Electronic Engine Control (EEC). These maintenance messages can be found on the Line Maintenance or the Extended Maintenance pages.

(2) This task can be used to find economic dispatch faults detected by the EEC (EEC Economic faults).

(a) EEC Economic faults are ATA 71-80 maintenance messages which are not correlated to Flight Deck Effect (FDE), STATUS message, or Scheduled Maintenance Task (SMT) message.

NOTE: Maintenance messages which are not correlated to a FDE, STATUS message, or SMT message may also be called uncorrelated or non-correlated maintenance messages.

(b) EEC Economic faults have no time limits and may be corrected at operator's discretion.

(3) During this check you may find ATA 71-80 maintenance messages correlated to a FDE, STATUS message, or SMT message.

NOTE: ATA 71-80 maintenance messages correlated to a FDE, STATUS message, or SMT message are not EEC Economic faults and have their own dispatch limitations. Refer to TASK 73-21-00-470-802-G00 for the dispatch group for these maintenance messages.

(a) For ATA 71-80 maintenance messages correlated to the FDE or STATUS message, refer to Fault Interrogation for a Maintenance Message, TASK 31-65-00-750-801.

(b) For ATA 71-80 maintenance messages correlated to the SMT message, refer to the TASK 73-00-00-700-801-G00 for ENG X EEC C1 SMT message and TASK 73-00-00-700-802-G00 for ENG X EEC C2 SMT message.

B. References

Reference	Title
31-65-00-750-801	Fault Interrogation for a Maintenance Message (P/B 201)
46-13-02-700-804	Present Leg Faults Check (P/B 201)
46-13-02-700-805	Existing Faults Check (P/B 201)
46-13-02-700-806	Fault History Check (P/B 201)
73-21-00-470-802-G00	EEC Shop Faults Function Check (P/B 201)
73-21-00-800-801-G00	EEC Maintenance Power Selection (P/B 201)

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C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
211	Flight Compartment - Left
212	Flight Compartment - Right

E. Prepare for the Check

SUBTASK 73-00-00-860-021-G00

(1) Set the GND TEST switch on the maintenance BITE panel, P61-4, to the NORM position.

SUBTASK 73-00-00-860-022-G00

(2) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-00-00-860-023-G00

(3) Make sure that the thrust levers are at the idle and down position.

SUBTASK 73-00-00-860-020-G00

(4) Select the left or right MFD.

(a) Push the SYS button on the MFD control panel, P9.

(b) Push the SEL button on SELECTOR knob on the MFD control panel, P9.

SUBTASK 73-00-00-860-060-G00

(5) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the TEST position (TASK 73-21-00-800-801-G00).

F. Maintenance Messages Check for EEC Faults

SUBTASK 73-00-00-200-001-G00

(1) Do this task: Present Leg Faults Check, TASK 46-13-02-700-804.

(a) To check for EEC Economic faults, find any non-correlated (uncorrelated) maintenance messages in ATA 71-80 and make a record of them, if it is necessary.

NOTE: Economic faults detected the EECs are maintenance messages which do not have a correlated FDE, STATUS message, or SMT message.

SUBTASK 73-00-00-860-054-G00

(2) Make sure that the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD is set to the TEST position (TASK 73-21-00-800-801-G00).

NOTE: The EEC will do a self-test when you set the EEC MAINT POWER switch to the TEST position. Wait for a minimum of 30 seconds to let the EEC complete the test.

SUBTASK 73-00-00-200-002-G00

(3) Do this task: Existing Faults Check, TASK 46-13-02-700-805.

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(a) To check for EEC Economic faults, find any non-correlated (uncorrelated) maintenance messages in ATA 71-80 and make a record of them, if it is necessary.

NOTE: Economic faults detected the EECs are maintenance messages which do not have a correlated FDE, STATUS message, or SMT message.

NOTE: To check for maintenance messages from previous flight legs, refer to TASK 46-13-02-700-806.

G. Put the Airplane Back to Its Usual Condition

SUBTASK 73-00-00-860-027-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-00-00-860-055-G00

(2) Set the applicable EEC MAINT POWER switch on the MISC SYSTEM CTRLS page on the MFD to the NORM position (TASK 73-21-00-800-801-G00).

——— END OF TASK ———

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ENGINE FUEL AND CONTROL - DDG MAINTENANCE PROCEDURES

1. General

A. This procedure has maintenance tasks for the Master Minimum Equipment List (MMEL) maintenance requirements as shown in the Dispatch Deviations Guide (DDG). These tasks are used to prepare the airplane for flight with certain systems/components inoperative.

B. This procedure also has the tasks that put the airplane back to its usual condition.

C. This procedure has these tasks:

(1) DDG 73-30-01 Fuel Control ENG VALVE CLOSED Indicating System Inoperative - Preparation

(2) DDG 73-30-01 Fuel Control ENG VALVE CLOSED Indicating System Inoperative - Restoration.

TASK 73-00-00-040-801-G00

2. DDG 73-30-01 Fuel Control ENG VALVE CLOSED Indicating System Inoperative - Preparation

A. General

(1) This task gives the maintenance steps which prepare the airplane for flight with the Engine Valve Closed Indicating System inoperative.

(2) The two ENG VALVE CLOSED lights may be inoperative provided that the High Pressure ShutOff Valves (HPSOV) will operate correctly.

B. References

Reference	Title
71-00-00-910-802-G00	Start the Engine (Selection) (P/B 201)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Fuel Control ENG VALVE CLOSED Indicating System Deactivation

SUBTASK 73-00-00-860-050-G00

(1) For the left engine, open this circuit breaker and install a safety lock:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

For the right engine, open this circuit breaker and install a safety lock:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 73-00-00-720-001-G00

(2) Make sure that the applicable engine fuel shutoff valve operates normally.

(a) Start the applicable engine and let it stabilize at ground idle speed (TASK 71-00-00-910-802-G00).

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SUBTASK 73-00-00-860-041-G00

(3) For the left engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-00-00-860-042-G00

(4) For the right engine, open this circuit breaker and install safety tag:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2

SUBTASK 73-00-00-860-029-G00

(5) Put the applicable the start lever to the CUTOFF position.

(a) Make sure that the engine immediately begins to decelerate, and continues to decelerate at the usual rate to a complete stop.

NOTE: If the engine does not immediately begin to decelerate, there is a problem in the HPSOV system.

SUBTASK 73-00-00-860-044-G00

(6) For the left engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-00-00-860-045-G00

(7) For the right engine, remove the safety tag and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2

——— END OF TASK ———

TASK 73-00-00-440-801-G00

3. DDG 73-30-01 Fuel Control ENG VALVE CLOSED Indicating System Inoperative - Restoration

A. General

(1) This task puts the airplane back to its usual condition after operation with the engine ENG VALVE CLOSED Indicating System inoperative.

B. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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C. Fuel Control ENG VALVE CLOSED Indicating System Inoperative Activation

SUBTASK 73-00-00-860-051-G00

(1) For the left engine, remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

For the right engine, remove the safety lock and close this circuit breaker:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

——— END OF TASK ———

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FUEL DISTRIBUTION SYSTEM - INSPECTION/CHECK

1. General

A. This procedure has these tasks:

(1) Engine Fuel Distribution System General Visual Inspection

(2) Fuel filter contamination detailed visual inspection.

TASK 73-11-00-210-801-G00

2. Engine Fuel Distribution System General Visual Inspection

A. General

(1) This task provides the instructions on how to examine the engine fuel distribution system.

(2) This task is a scheduled maintenance task.

B. References

Reference	Title
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
70-00-05-200-801-G00	Tube and Hose Inspection Criteria and Damage Limits (P/B 201)
70-43-42-300-802-G00	Blending Procedures - Hand Blending (P/B 201)
71-00-00-700-807-G00	Test No. 31 - Wet Motor Ground Test (P/B 501)
71-00-00-910-808-G00	Dry Motor Procedure (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-01-000-801-G00	Main Fuel Pump Removal (P/B 401)
73-11-01-400-801-G00	Main Fuel Pump Installation (P/B 401)
73-11-02-000-801-G00	Fuel Filter Cartridge Removal (P/B 401)
73-11-02-400-801-G00	Fuel Filter Cartridge Installation (P/B 401)
73-11-03-000-801-G00	Main Fuel Filter Housing Removal (P/B 401)
73-11-03-400-801-G00	Main Fuel Filter Housing Installation (P/B 401)
73-11-05 P/B 401	FUEL NOZZLE - REMOVAL/INSTALLATION
73-11-05-400-801-G00	Fuel Nozzle Installation (P/B 401)
73-11-05-400-802-G00	Fuel Nozzle Set Installation (P/B 401)
73-11-06-000-801-G00	Split Control Unit Removal (P/B 401)
73-11-06-400-801-G00	Split Control Unit Installation (P/B 401)
73-21-04-000-801-G00	Fuel Metering Unit Removal (P/B 401)
73-21-04-400-801-G00	Fuel Metering Unit Installation (P/B 401)
73-31-01-000-801-G00	Fuel Flow Transmitter Removal (P/B 401)
73-31-01-400-801-G00	Fuel Flow Transmitter Installation (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

F. Prepare for the Inspection

SUBTASK 73-11-00-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-00-010-001-G00

(2) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

ECCN 9E991 BOEING PROPRIETARY - See title page for details

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Fuel Distribution System Inspection

SUBTASK 73-11-00-212-001-G00

(1) Examine the main fuel pump:

(a) Look for signs of fuel leaks.

1) If you see fuel leaks, replace the seals (TASK 73-11-01-000-801-G00 and TASK 73-11-01-400-801-G00).

(b) Make sure that the main fuel pump is not loose.

1) If the main fuel pump is loose, tighten the bolts that attach the pump to the AGB (TASK 73-11-01-400-801-G00).

(c) Look for nicks, dents, or scratches on the main fuel pump as follows:

1) Nicks, dents, or scratches are permitted if they are less than 0.010 in. (0.25 mm) in depth after you remove the high metal (TASK 70-43-42-300-802-G00).

2) Damage is not permitted on an edge, small radius, or corners of the pump.

3) The damage is not permitted across mating joints, sealing surfaces or seal grooves.

(d) Look for cracks in the main fuel pump.

1) If you see cracks, replace the main fuel pump (TASK 73-11-01-000-801-G00 and TASK 73-11-01-400-801-G00).

(e) Make sure that the fuel tubes are not loose.

1) If a fuel tube is loose, tighten the bolts that attach it to the pump (TASK 73-11-01-400-801-G00).

2) If the fuel tubes are not loose but there is a fuel leak, replace the fuel tube gasket seal (TASK 73-11-01-000-801-G00 and TASK 73-11-01-400-801-G00).

(f) Make sure that the electrical connectors are not loose (TASK 70-00-01-910-803-G00).

SUBTASK 73-11-00-210-001-G00

(2) Examine the main fuel filter housing for leaks, cracks, dents and loose electrical connectors as follows:

(a) If there is a leak on the fuel filter bowl, tighten the filter bowl (TASK 73-11-02-400-801-G00).

(b) If the leak continues, replace the preformed packings on the fuel filter cartridge (TASK 73-11-02-000-801-G00 and TASK 73-11-02-400-801-G00).

(c) If there is a leak on the fuel filter bowl drain plug, tighten the drain plug to 140.0 in-lb (15.82 N·m) to 150.0 in-lb (16.95 N·m).

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(d) If the leak continues, replace the preformed packing on the drain plug (TASK 73-11-02-000-801-G00 and TASK 73-11-02-400-801-G00).

(e) Cracks or dents are not permitted on the main fuel filter housing.

(f) Tighten the nuts on the main fuel filter housing if the main fuel filter housing is loose (TASK 73-11-03-400-801-G00).

(g) If there is a fuel leak between the main fuel filter housing and the fuel tube, do this step:

1) Replace the gasket seal between the main fuel filter housing assembly and the fuel tube (TASK 73-11-03-000-801-G00 and TASK 73-11-03-400-801-G00).

(h) Tighten the electrical connectors that are loose (TASK 70-00-01-910-803-G00).

SUBTASK 73-11-00-212-002-G00

(3) Examine the fuel nozzles as follows:

(a) Look for signs of fuel leaks from the fuel nozzle.

1) If you see fuel leaks, replace the fuel nozzle (PAGEBLOCK 73-11-05/401).

(b) If you cannot identify the location of the leak in the fuel manifold and fuel nozzle circuit, do this task: Test No. 31 - Wet Motor Ground Test, TASK 71-00-00-700-807-G00.

(c) Look for cracks in the mounting flange of the fuel nozzle.

1) If you see cracks, replace the fuel nozzle (PAGEBLOCK 73-11-05/401).

(d) Look for signs of air leaks from the mounting flange.

1) If you see signs of air leaks, replace the fuel nozzle seal ring (PAGEBLOCK 73-11-05/401).

(e) Look for nicks, dents, or scratches.

1) Nicks, dents, or scratches are permitted if they are less than 0.010 in. (0.25 mm) in depth after removal of the high metal (TASK 70-43-42-300-802-G00).

2) Damage across mating joints, sealing surfaces or seal grooves is not permitted.

(f) Make sure that the fuel nozzles are not loose.

1) If you find a loose fuel nozzle, tighten the bolts (TASK 73-11-05-400-801-G00 or TASK 73-11-05-400-802-G00).

2) If the fuel nozzle cannot be tightened due to damage of the threaded insert, do this step:

a) Replace the fuel nozzle (PAGEBLOCK 73-11-05/401).

SUBTASK 73-11-00-212-003-G00

(4) Examine the fuel metering unit as follows:

(a) Look for signs of fuel leaks.

1) If you see a fuel leaks, replace the seals (TASK 73-21-04-000-801-G00 and TASK 73-21-04-400-801-G00).

(b) Make sure that the fuel metering unit is not loose.

1) If the fuel metering unit is loose, tighten the bolts that attach the unit to engine (TASK 73-21-04-400-801-G00).

(c) Look for nicks, dents, or scratches on the fuel metering unit.

1) Nicks, dents or scratches are permitted if they are less than 0.010 in. (0.25 mm) in depth after removal of high metal.

2) Damage is not permitted on an edge, small radius, or corners of the unit.

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3) The damage is not permitted across mating joint, sealing surfaces or seal grooves.

(d) Look for cracks in the fuel metering unit.

1) If you see cracks, replace the fuel metering unit (TASK 73-21-04-000-801-G00 and TASK 73-21-04-400-801-G00).

(e) Look for loose electrical connectors and tighten as necessary (TASK 70-00-01-910-803-G00).

SUBTASK 73-11-00-220-001-G00

(5) Examine the fuel flow transmitter as follows:

(a) Look for signs of fuel leaks.

1) If you see fuel leaks, replace the gasket seals (TASK 73-31-01-000-801-G00 and TASK 73-31-01-400-801-G00).

(b) Make sure that the fuel flow transmitter is not loose.

1) If the fuel flow transmitter is loose, tighten the bolts (TASK 73-31-01-400-801-G00).

(c) Look for nicks, dents, or scratches on the fuel flow transmitter as follows:

1) Nicks, dents, or scratches are permitted if they are less than 0.025 in. (0.64 mm) in depth or 0.080 in. (2.03 mm) in width after removal of high metal.

(d) Look for damage on an edge, small radius, or corners of the fuel flow transmitter.

1) Damage is not permitted.

(e) Look for damage across mating joints, sealing surface or seal grooves.

1) Damage is not permitted.

(f) Look for cracks in the fuel flow transmitter.

1) If you see cracks, replace the fuel flow transmitter (TASK 73-31-01-000-801-G00 and TASK 73-31-01-400-801-G00).

(g) Make sure that the electrical connectors are not loose (TASK 70-00-01-910-803-G00).

SUBTASK 73-11-00-868-001-G00

(6) Examine the Split Control Unit (SCU) as follows:

(a) Look for signs of fuel leaks.

1) If you see fuel leaks, replace the gasket seals (TASK 73-11-06-000-801-G00 and TASK 73-11-06-400-801-G00).

(b) Make sure that the SCU is not loose.

1) If the SCU is loose, tighten the bolts that attach the unit to the engine (TASK 73-11-03-400-801-G00).

(c) Look for nicks, dents or scratches on the SCU.

1) Nicks, dents or scratches are permitted if they are less than 0.010 in. (0.25 mm) in depth after you remove the high metal.

(d) Look for damage on an edge, on small radius or corners of the assembly.

1) Damage is not permitted.

(e) Look for damage across mating joints, sealing surfaces or seal grooves.

1) Damage is not permitted.

(f) Look for cracks in the SCU.

1) If you see cracks, replace the SCU (TASK 73-11-06-000-801-G00 and TASK 73-11-06-400-801-G00).

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(g) Make sure that the electrical connectors are not loose (TASK 70-00-01-910-803-G00).

SUBTASK 73-11-00-210-002-G00

(7) Examine all the fuel tubes, hoses, tube-hoses and the manifolds (TASK 70-00-05-200-801-G00).

SUBTASK 73-11-00-360-001-G00

(8) If you find a leak at a location, do these steps:

(a) Make sure that the applicable START LEVER and ENGINE START switch circuit breakers are closed.

(b) Close the applicable thrust reversers (TASK 78-31-00-010-802-G00).

(c) Do this task: Test No. 31 - Wet Motor Ground Test, TASK 71-00-00-700-807-G00.

(d) Do this task: Dry Motor Procedure, TASK 71-00-00-910-808-G00.

(e) Open the applicable thrust reversers (TASK 78-31-00-010-801-G00).

(f) Do the repair of the leaks you find.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-00-410-001-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-11-00-860-002-G00

(2) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

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(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

——— END OF TASK ———

TASK 73-11-00-211-801-G00

3. Fuel Filter Contamination Detailed Visual Inspection

A. General

(1) This task gives the instructions on how to do a detailed examination of the fuel filter for contamination.

(2) You use one of the following procedures to complete this task:

(a) The preferred procedure is to look for fuel filter bypass maintenance messages on the Onboard Maintenance Function (OMF).

NOTE: If you find the fuel filter approaching bypass maintenance messages, you must replace the fuel filter element. You must also examine and clean (if necessary) the fuel strainer.

(b) The alternate procedure is to replace the fuel filter element and examine and clean (if necessary) the fuel strainer. This procedure is used when the OMF is not available to view the fault history.

B. References

Reference	Title
28-10-00-200-801	Detection Test for Microbial Growth (P/B 201)
28-10-00-600-801	Treatment of Fuel Tanks Contaminated with Microbial Growth (P/B 201)
46-13-02-700-806	Fault History Check (P/B 201)
73-11-01-000-801-G00	Main Fuel Pump Removal (P/B 401)
73-11-01-400-801-G00	Main Fuel Pump Installation (P/B 401)
73-11-02-000-801-G00	Fuel Filter Cartridge Removal (P/B 401)
73-11-02-400-801-G00	Fuel Filter Cartridge Installation (P/B 401)
73-11-04-100-801-G00	Fuel Metering Unit Strainer Cleaning (P/B 701)
73-11-04-200-801-G00	Fuel Metering Unit Strainer Inspection (P/B 601)
73-11-05 P/B 401	FUEL NOZZLE - REMOVAL/INSTALLATION
73-11-06-000-801-G00	Split Control Unit Removal (P/B 401)
73-11-06-400-801-G00	Split Control Unit Installation (P/B 401)
73-11-11-000-801-G00	Servo Fuel Heater Removal (P/B 401)
73-11-11-400-801-G00	Servo Fuel Heater Installation (P/B 401)
73-21-04-000-801-G00	Fuel Metering Unit Removal (P/B 401)
73-21-04-400-801-G00	Fuel Metering Unit Installation (P/B 401)
73-31-01-000-801-G00	Fuel Flow Transmitter Removal (P/B 401)
73-31-01-400-801-G00	Fuel Flow Transmitter Installation (P/B 401)
79-21-01-000-801-G00	Main Fuel/Oil Heat Exchanger Removal (P/B 401)
79-21-01-400-801-G00	Main Fuel/Oil Heat Exchanger Installation (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

D. Fuel Filter Cartridge Inspection (Preferred Procedure)

SUBTASK 73-11-00-212-004-G00

(1) Find maintenance messages for the fuel filter bypass as follows:

(a) Do this task: Fault History Check, TASK 46-13-02-700-806.

(b) Look for these maintenance messages in the last three flight legs:

1) 73–42170 FUEL FILTER BYPASS [ENGINE 1]

2) 73–42175 FUEL FILTER BYPASS [ENGINE 2]

3) 73–42180 FUEL FILTER APPROACHING BYPASS [ENGINE 1]

4) 73–42185 FUEL FILTER APPROACHING BYPASS [ENGINE 2]

5) 73–42190 FUEL FILTER IMPENDING BYPASS [ENGINE 1]

6) 73–42195 FUEL FILTER IMPENDING BYPASS [ENGINE 2].

(c) Do the troubleshooting in the Fault Isolation Manual (FIM) for the maintenance message that you find.

NOTE: If you do not find a maintenance message or the messages above, then this inspection task is complete.

(d) If you find one or more of the above maintenance messages, do these steps:

1) Remove the applicable fuel filter cartridge (TASK 73-11-02-000-801-G00).

2) Examine the inner wall of the fuel filter housing, the bottom of the fuel filter bowl, and the fuel filter cartridge for contaminants.

3) Install a new fuel filter cartridge (TASK 73-11-02-400-801-G00).

4) Examine the Fuel Metering Unit (FMU) strainer (TASK 73-11-04-200-801-G00).

5) If it is necessary, clean the FMU strainer (TASK 73-11-04-100-801-G00).

E. Fuel Filter Cartridge Inspection (Alternate Procedure)

SUBTASK 73-11-00-020-001-G00

(1) Remove the applicable fuel filter cartridge (TASK 73-11-02-000-801-G00).

SUBTASK 73-11-00-210-003-G00

(2) Examine the inner wall of the fuel filter housing, the bottom of the fuel filter bowl, and the fuel filter cartridge for contaminants.

SUBTASK 73-11-00-900-001-G00

(3) If it is necessary, install a new fuel filter cartridge (TASK 73-11-02-400-801-G00).

SUBTASK 73-11-00-210-004-G00

(4) Examine the Fuel Metering Unit (FMU) strainer (TASK 73-11-04-200-801-G00).

SUBTASK 73-11-00-170-001-G00

(5) If it is necessary, clean the FMU strainer (TASK 73-11-04-100-801-G00).

SUBTASK 73-11-00-100-001-G00

(6) For contamination by solids in small quantities (dirt, sand, metal particles or sealant particles), do this step:

NOTE: Contamination in small quantities is not a sign of a possible malfunction.

(a) Replace the fuel filter cartridge (TASK 73-11-02-000-801-G00 and TASK 73-11-02-400-801-G00).

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SUBTASK 73-11-00-100-002-G00

(7) For contamination by solids in large quantities (dirt, sand, metals or sealant particles but not bronze particles), do these steps:

NOTE: The steps for bronze particles contamination is in another step below.

NOTE: Possible causes are contaminated fuel, contaminated aircraft fuel tanks, damaged aircraft boost pump, damaged engine fuel pump and foreign material.

(a) Replace the fuel filter cartridge (TASK 73-11-02-000-801-G00 and TASK 73-11-02-400-801-G00).

(b) Do this task: Fuel Metering Unit Strainer Inspection, TASK 73-11-04-200-801-G00

(c) If it is necessary, clean the FMU strainer (TASK 73-11-04-100-801-G00)

(d) If the fuel filter bypass alert indication happens again, replace the fuel filter cartridge on the opposite engine (TASK 73-11-02-000-801-G00 and TASK 73-11-02-400-801-G00)

SUBTASK 73-11-00-100-003-G00

(8) For contamination by black sludge or slimy materials (biological contamination, fungus growth):

(a) Do a check the fuel filter cartridge on the opposite engine.

(b) If you find contamination on the filter cartridge, do these steps:

1) Do this task: Detection Test for Microbial Growth, TASK 28-10-00-200-801.

2) If it is necessary, do this task: Treatment of Fuel Tanks Contaminated with Microbial Growth, TASK 28-10-00-600-801.

SUBTASK 73-11-00-100-004-G00

(9) For contamination by bronze particles, do these steps:

NOTE: Bronze particles in the main fuel filter cartridge is a sign of a main fuel pump bearing failure.

(a) If the “FUEL FILTER APPROACHING BYPASS” or “FUEL FILTER IMPENDING BYPASS” message is on, do these steps:

NOTE: When you disconnect the fuel system components, make sure that you fully drain the associated tubes and blow with dry air to remove all bronze particles inside.

1) Replace the main fuel pump (TASK 73-11-01-000-801-G00 and TASK 73-11-01-400-801-G00)

2) Do this task: Fuel Metering Unit Strainer Inspection, TASK 73-11-04-200-801-G00.

3) Do this task: Fuel Metering Unit Strainer Cleaning, TASK 73-11-04-100-801-G00.

(b) If the “FUEL FILTER BYPASS” message is on, do these steps:

NOTE: When you disconnect the fuel system components, make sure that you fully drain the associated tubes and blow with dry air to remove all bronze particles inside.

1) Replace the main fuel pump (TASK 73-11-01-000-801-G00 and TASK 73-11-01-400-801-G00)

2) Replace the FMU (TASK 73-21-04-000-801-G00 and TASK 73-21-04-400-801-G00).

3) Replace the main fuel/oil heat exchanger (TASK 79-21-01-000-801-G00 and TASK 79-21-01-400-801-G00).

4) Replace the servo fuel heater (TASK 73-11-11-000-801-G00 and TASK 73-11-11-400-801-G00).

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5) Replace the Split Control Unit (SCU) (TASK 73-11-06-000-801-G00 and TASK 73-11-06-400-801-G00).

6) Replace the fuel flow transmitter (TASK 73-31-01-000-801-G00 and TASK 73-31-01-400-801-G00).

7) Replace the fuel nozzles (PAGEBLOCK 73-11-05/401).

——— END OF TASK ———

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MAIN FUEL PUMP - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Main Fuel Pump Removal

(2) Main Fuel Pump Installation.

TASK 73-11-01-000-801-G00

2. Main Fuel Pump Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the Main Fuel Pump (MFP).

(2) A minimum of two persons are required to handle the MFP if the jack stand and the MFP adapter support equipment are not used.

B. References

Reference	Title
70-41-00-400-802-G00	Safety Cable Installation (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14500	Stand - Jack Part #: 9471M17P01 Supplier: 58828
SPL-14572	Adapter - Main Fuel Pump Part #: 956A3608G01 Supplier: 58828
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
4	Preformed packing	73-11-01-01-070	PTF ALL
7	Preformed packing	73-11-01-01-070	PTF ALL
10	Preformed packing	73-11-01-01-050	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

ECCN 9E991 BOEING PROPRIETARY - See title page for details

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Prepare for the Removal

SUBTASK 73-11-01-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-01-860-002-G00

(2) Do these steps to make sure that the fuel control valve and spar valve stay in the closed position:

(a) Make sure that the L (R) START LEVER switch is in the CUTOFF position.

1) Install a DO NOT OPERATE tag, STD-858, on the L (R) START LEVER switch.

(b) Make sure that the applicable SPAR VALVE CLOSED and ENG VALVE CLOSED lights, on the P5 forward overhead panel, are on dimly.

SUBTASK 73-11-01-860-003-G00

(3) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-01-860-004-G00

(4) For the left engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 73-11-01-860-005-G00

(5) For the right engine, open these applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 73-11-01-010-001-G00

(6) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

I. Main Fuel Pump Removal

SUBTASK 73-11-01-680-001-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

CAUTION A LARGE VOLUME OF FUEL WILL DRAIN FROM THE FUEL SUPPLY SYSTEM. TO PREVENT A FUEL SPILL, YOU MUST USE A MINIMUM OF TWO CONTAINERS TO DRAIN THE FUEL SYSTEM.

(1) Drain the MFP [1] as follows:

(a) Put a fuel resistant container (5 gal), STD-201, below the MFP [1].

(b) Remove the safety cables [5] from the drain plug [3], drain plug [9], and drain plug [6].

(c) Remove the drain plug [3] from the MFP [1] as follows:

1) Remove the drain plug [3] from the MFP [1].

2) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

3) Remove and discard the preformed packing [4] from the drain plug [3].

(d) Remove the drain plug [6] from the MFP [1] as follows:

1) Remove the drain plug [6] from the MFP [1].

2) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

3) Remove and discard the preformed packing [7] from the drain plug [6].

(e) Remove the drain plug [9] from the MFP [1] as follows:

1) Remove the drain plug [9] from the MFP [1].

2) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

3) Remove and discard the preformed packing [10] from the drain plug [9].

(f) Install the drain plug [9] on the MFP [1] as follows:

1) Make sure that the mating faces are clean and clear of unwanted materials.

2) Lubricate the threads of the drain plug [9] with pure mineral vaseline, D00672 [CP5070].

3) Lubricate the new preformed packing [10] with the pure mineral vaseline, D00672 [CP5070].

4) Install the preformed packing [10] on the drain plug [9].

5) Install the drain plug [9] on its position on the MFP [1].

6) Tighten the drain plug [9] to 13.0 in-lb (1.5 N·m) - 16.0 in-lb (1.8 N·m).

(g) Install the drain plug [3] on the MFP [1] as follows:

1) Make sure that the mating faces are clean and clear of unwanted materials.

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2) Lubricate the threads of the drain plug [3] with pure mineral vaseline, D00672 [CP5070].

3) Lubricate the new preformed packing [4] with pure mineral vaseline, D00672 [CP5070].

4) Install the preformed packing [4] on the drain plug [3].

5) Install the drain plug [3] on its position on the MFP [1].

6) Tighten the drain plug [3] to 190.3 in-lb (21.5 N·m) - 210.0 in-lb (23.7 N·m).

(h) Install the drain plug [6] on the MFP [1] as follows:

1) Make sure that the mating faces are clean and clear of unwanted materials.

2) Lubricate the threads of the drain plug [6] with pure mineral vaseline, D00672 [CP5070].

3) Lubricate the new preformed packing [7] with pure mineral vaseline, D00672 [CP5070].

4) Install the preformed packing [7] on the drain plug [6].

5) Install the drain plug [6] on its position on the MFP [1].

6) Tighten the drain plug [6] to 190.3 in-lb (21.5 N·m) - 210.0 in-lb (23.7 N·m).

SUBTASK 73-11-01-420-002-G00

(2) Install a safety safety cable, G50065 [CP8006], on the drain plug [3], drain plug [6], and drain plug [9] (TASK 70-41-00-400-802-G00).

SUBTASK 73-11-01-020-001-G00

(3) Remove the bracket [12] from the AGB [2] as follows:

(a) Remove the two bolts [13] that attach the bracket [12] to the AGB [2].

(b) Remove the bracket [12] from the AGB [2].

SUBTASK 73-11-01-020-002-G00

(4) Disconnect the fuel tube [14] from the MFP [1] as follows:

(a) Remove the four bolts [15] from the fuel tube [14].

(b) Disconnect the fuel tube [14] from the MFP [1].

(c) Remove and discard the gasket seal [16].

(d) Install the protective covers on the openings.

SUBTASK 73-11-01-010-002-G00

(5) Disconnect the fuel tube [17] from the MFP [1] as follows:

(a) Remove the four bolts [18] from the fuel tube [17].

(b) Disconnect the fuel tube [17] from the MFP [1].

(c) Remove and discard the gasket seal [19].

(d) Install the protective covers on the openings.

SUBTASK 73-11-01-010-003-G00

(6) Disconnect the fuel tube [20] from the MFP [1] as follows:

(a) Remove the four bolts [21] from the fuel tube [20].

(b) Disconnect the fuel tube [20] from the MFP [1].

(c) Remove and discard the gasket seal [22].

(d) Install the protective covers on the openings.

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SUBTASK 73-11-01-010-004-G00

(7) Disconnect the fuel tube [23] from the MFP [1] as follows:

(a) Remove the four bolts [24] from the fuel tube [23].

(b) Disconnect the fuel tube [23] from the MFP [1].

(c) Remove and discard the gasket seal [25].

(d) Install the protective covers on the openings.

SUBTASK 73-11-01-490-001-G00

WARNING BE CAREFUL WHEN YOU LIFT THE MAIN FUEL PUMP WITHOUT AID. THE MAIN FUEL PUMP IS HEAVY. IF YOU DO NOT OBEY, INJURIES TO PERSONNEL OR DAMAGE TO THE MAIN FUEL PUMP CAN OCCUR.

CAUTION DO NOT PUT THE WEIGHT OF THE PUMP ON THE DRIVESHAFT SPLINES. THIS WILL PREVENT DAMAGE TO THE SPLINES.

(8) Remove the MFP [1] from the AGB input pad [30] as follows:

NOTE: The MFP weights 34.4 lb (15.6 kg).

(a) Loosen the five bolts [26] that attach the MFP [1] to the AGB input pad [30].

(b) If necessary, attach the adapter, SPL-14572, on the MFP [1].

NOTE: If the two-person alternative method is used, you do not have to install the MFP to the adapter, SPL-14572, and jack stand, SPL-14500.

(c) If necessary, move the jack stand, SPL-14500, to a position that will allow installation of the adapter, SPL-14572.

(d) If necessary, install the adapter, SPL-14572, on the jack stand, SPL-14500.

(e) Remove the five bolts [26] that attach the MFP [1] to the AGB input pad [30].

(f) Move the MFP [1] slowly to prevent damage to the shaft of the MFP [1] and AGB [2].

CAUTION MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE MFP. DAMAGE CAN OCCUR.

(g) Carefully pull the MFP [1] rearward until the splines of the drive shaft [27] disengage from the AGB input pad [30].

(h) Remove and discard the preformed packing [28] from the groove on the MFP [1].

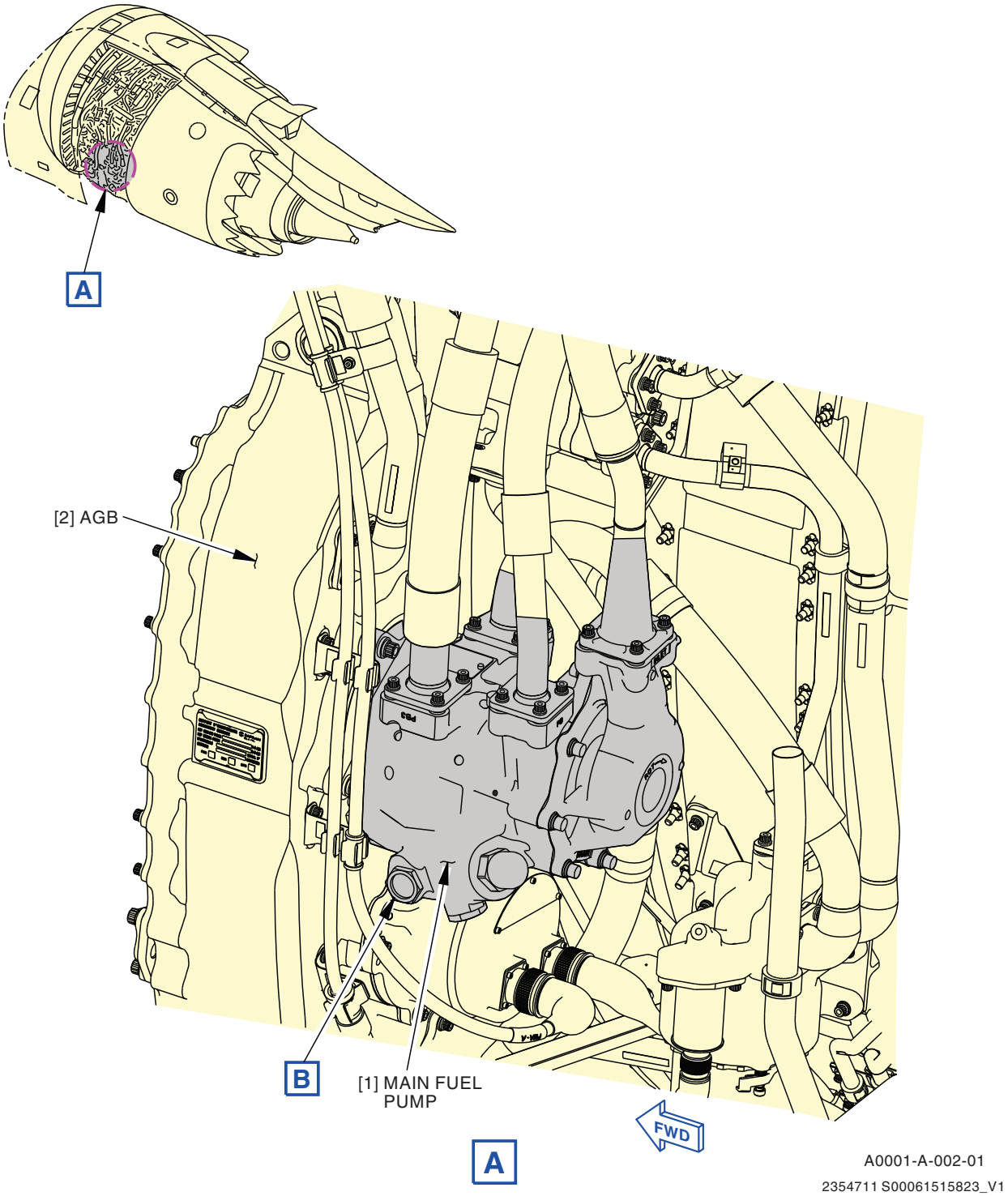
(i) Remove and discard the preformed packing [29] from the groove on the drive shaft [27].

(j) Install the protective covers on the openings.

(k) If necessary, remove the MFP [1] from the adapter, SPL-14572.

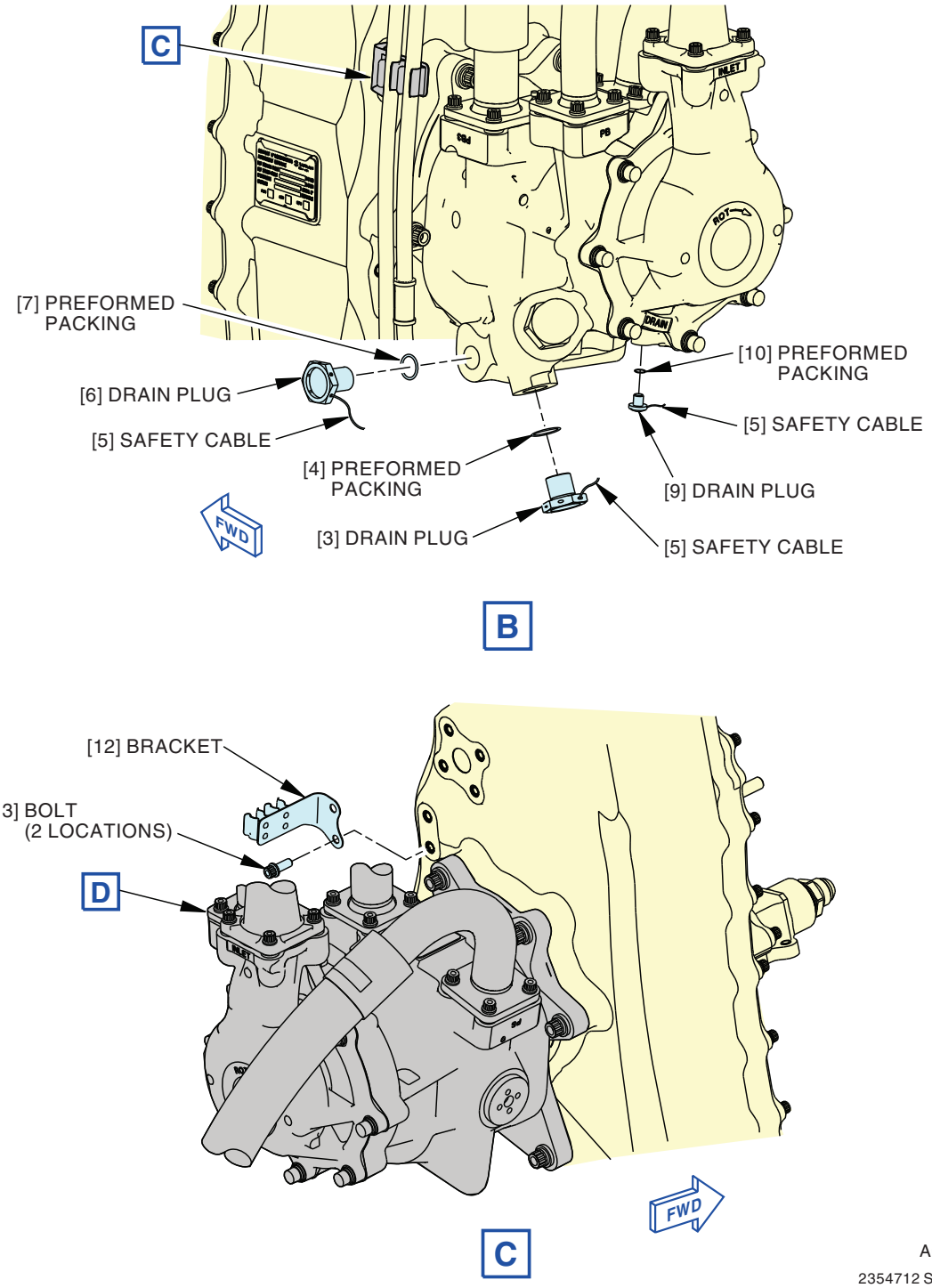
END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Fuel Pump Installation
Figure 401/73-11-01-990-801-G00 (Sheet 1 of 5)

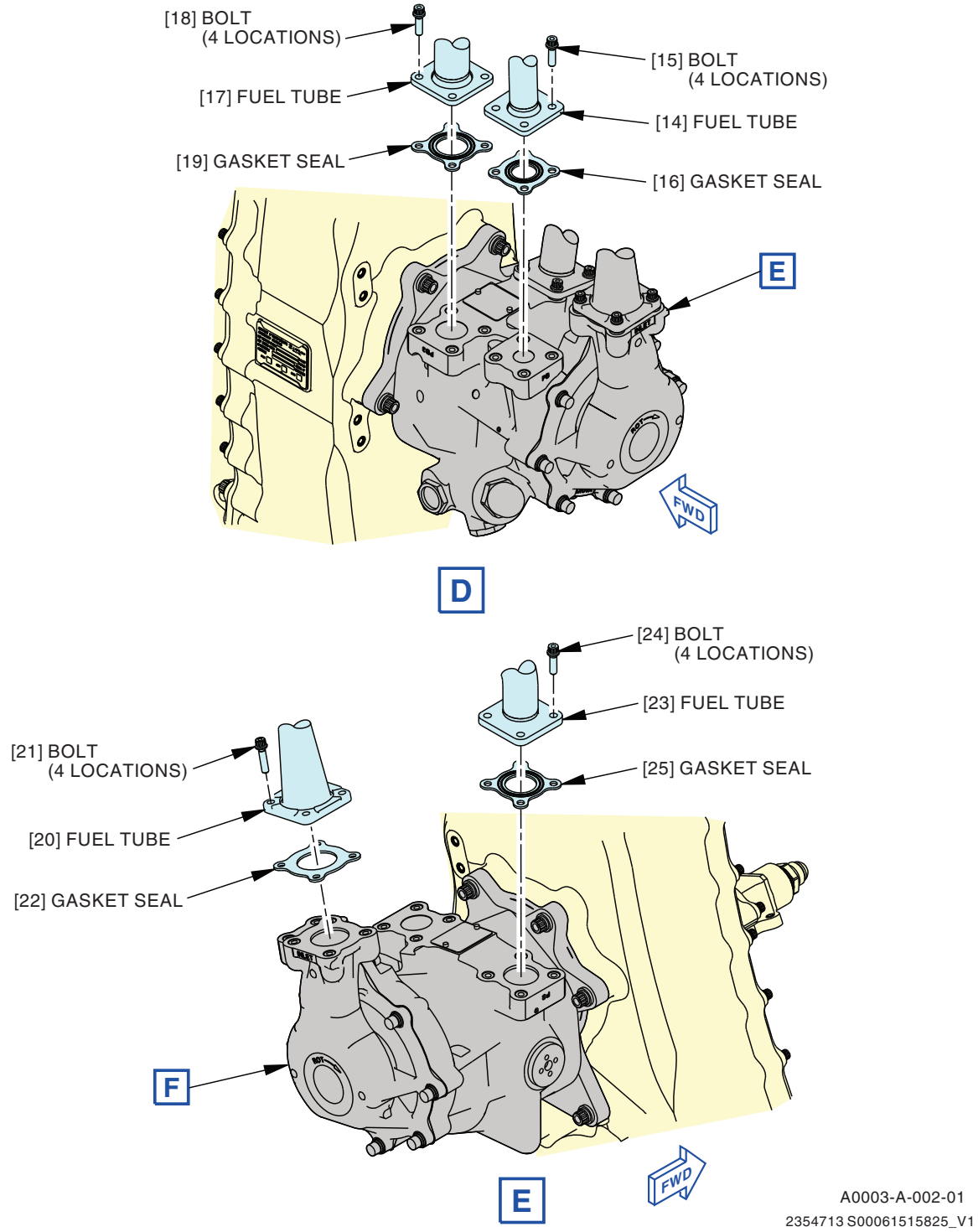
ECCN 9E991 BOEING PROPRIETARY - See title page for details



A0002-A-002-01
2354712 S00061515824_V3

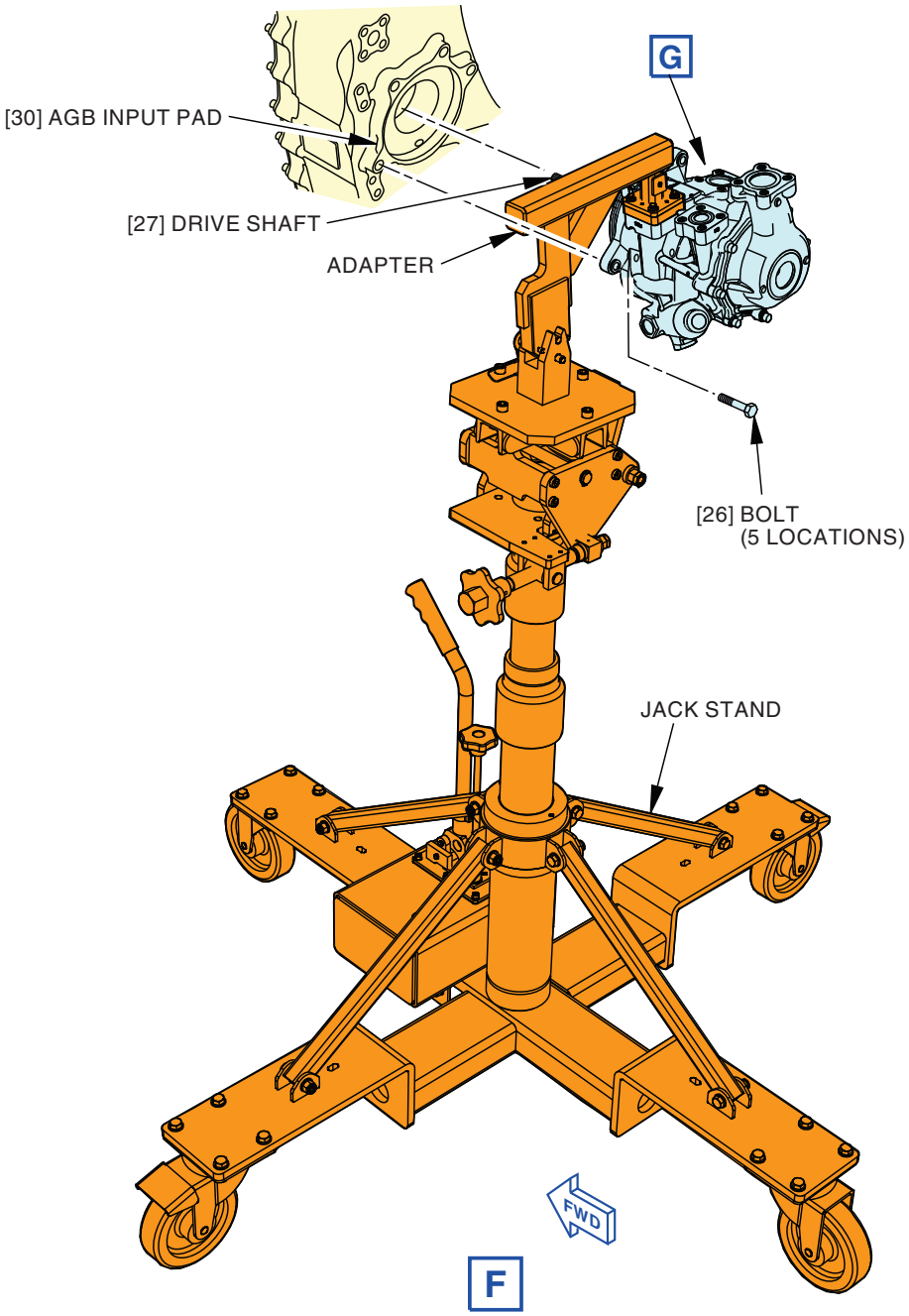
Main Fuel Pump Installation
Figure 401/73-11-01-990-801-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Fuel Pump Installation
Figure 401/73-11-01-990-801-G00 (Sheet 3 of 5)

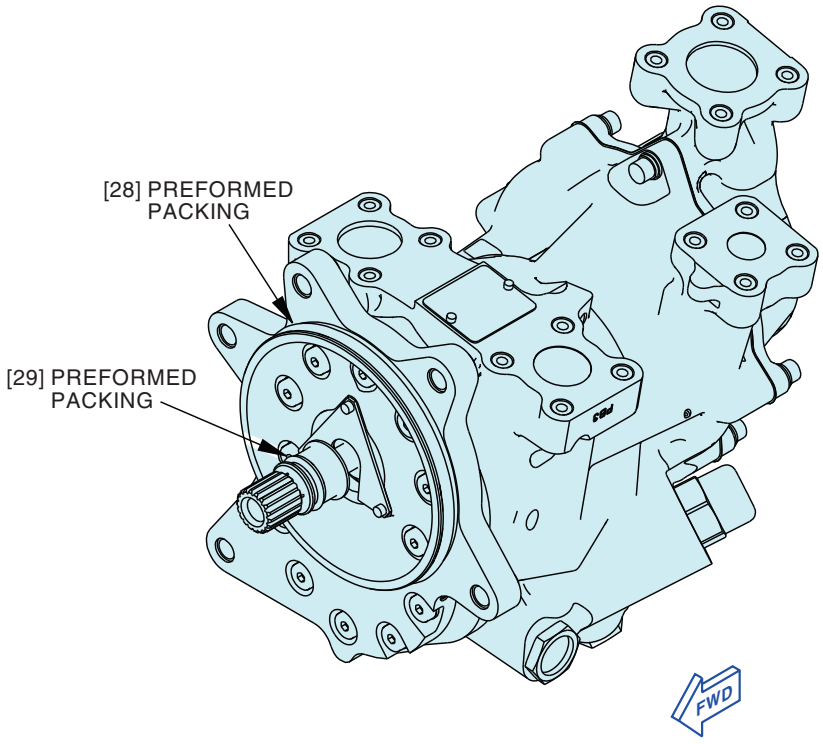
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5050363-00
2354714 S00061515826_V2

Main Fuel Pump Installation
Figure 401/73-11-01-990-801-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



G

A0009-A-001-01
2403644 S0000556185_V1

Main Fuel Pump Installation
Figure 401/73-11-01-990-801-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-11-01-400-801-G00

3. Main Fuel Pump Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the Main Fuel Pump (MFP).

(2) A minimum of two persons are required to handle the MFP if the jack stand and the MFP adapter support equipment are not used.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
72-00-00-910-801-G00	High Pressure Rotor (N2) Turning for Borescope (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-14500	Stand - Jack Part #: 9471M17P01 Supplier: 58828
SPL-14572	Adapter - Main Fuel Pump Part #: 956A3608G01 Supplier: 58828
SPL-14847	Kit, Installation - O-Ring, AGB Part #: 956A8610G01 Supplier: 58828
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G51335 [CP2674]	Lint-Free Cloth	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	MFP	73-11-01-01-040	PTF ALL
16	Gasket seal	73-11-01-01-160	PTF ALL
19	Gasket seal	73-11-01-01-140	PTF ALL
22	Gasket seal	73-11-51-01-025	PTF ALL
25	Gasket seal	73-11-01-01-120	PTF ALL
28	Preformed packing	73-11-01-01-020	PTF ALL
29	Preformed packing	73-11-01-01-030	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Main Fuel Pump Installation

SUBTASK 73-11-01-420-001-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

WARNING

BE CAREFUL WHEN YOU LIFT THE MAIN FUEL PUMP WITHOUT AID. THE MAIN FUEL PUMP IS HEAVY. IF YOU DO NOT OBEY, INJURIES TO PERSONNEL OR DAMAGE TO THE MAIN FUEL PUMP CAN OCCUR.

CAUTION

DO NOT CAUSE DAMAGE TO THE PACKING WHEN YOU INSTALL THE MFP. A DAMAGED PACKING CAN CAUSE ENGINE OIL LEAKS. OIL LEAKS CAN CAUSE AN IN-FLIGHT SHUTDOWN

CAUTION

DO NOT PUT THE WEIGHT OF THE PUMP ON THE DRIVESHAFT SPLINES. THIS WILL PREVENT DAMAGE TO THE SPLINES.

(1) Install the MFP [1] as follows:

NOTE: The MFP weighs 34.4 lb (15.6 kg).

(a) If necessary, install the MFP [1] on the adapter, SPL-14572.

NOTE: If the two-person alternative method is used, you do not have to install the MFP to the adapter, SPL-14572, and jack stand, SPL-14500.

(b) If necessary, attach the adapter, SPL-14572, to the jack stand, SPL-14500.

(c) Make sure that the mating faces are clean and clear of unwanted materials.

(d) Lubricate the threads of the five bolts [26] and the splines of the drive shaft [27] with engine oil, D00599 [CP2442].

(e) Lubricate the new preformed packing [28] and new preformed packing [29] with engine oil, D00599 [CP2442].

(f) Install the new preformed packing [29] on the groove of the drive shaft [27].

NOTE: If it is necessary, use the O-ring installation kit, SPL-14847.

(g) Install the new preformed packing [28] onto the groove of the MFP [1].

(h) If necessary, move the jack stand, SPL-14500 to align the drive shaft [27] of the MFP [1] with the splines on the AGB [2].

1) If it is necessary, turn the N2 rotor to do the alignment (TASK 72-00-00-910-801-G00).

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CAUTION

MAKE SURE THAT YOU DO NOT HIT THE TUBES AND BRACKETS WHEN YOU MOVE THE MFP. DAMAGE CAN OCCUR.

(i) Carefully push the drive shaft [27] of the MFP [1] into the AGB [2] as follows:

(j) Make sure that the mating flange of the MFP [1] fully engages with the AGB input pad [30].

(k) Install the five bolts [26] on the MFP [1].

(l) Tighten the bolts [26] in a criss-cross pattern as follows:

1) Tighten the bolts to 95.6 in-lb (10.8 N·m) - 104.4 in-lb (11.8 N·m).

2) Tighten the bolts to 191 in-lb (21.6 N·m) - 210.0 in-lb (23.7 N·m).

3) Tighten the bolts to 380.6 in-lb (43.0 N·m) - 420.0 in-lb (47.5 N·m).

(m) If necessary, remove the adapter, SPL-14572, from the MFP [1].

(n) If necessary, move the jack stand, SPL-14500, away from the engine.

(o) Connect the fuel tube [23] to the MFP [1] as follows:

1) Remove the protective covers from the openings.

2) Make sure that the mating faces are clean and clear of unwanted materials.

3) Lubricate the threads and pressure faces of the four bolts [24] with graphite mineral vaseline, D00601 [CP2101].

4) Lubricate each side of the elastomer part of a new gasket seal [25], with vaseline, D00672 [CP5070], or oil, D00599 [CP2442].

5) Align the new gasket seal [25] and the fuel tube [23] with the applicable port on the MFP [1].

6) Install the four bolts [24] on the fuel tube [23].

7) Tighten the bolts [24] in a criss cross pattern to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).

8) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

(p) Connect the fuel tube [20] to the MFP [1] as follows:

1) Remove the protective covers from the openings.

2) Make sure that the mating faces are clean and clear of unwanted materials.

3) Lubricate the threads and pressure faces of the four bolts [21] with graphite mineral vaseline, D00601 [CP2101].

4) Lubricate each side of the elastomer part of a new gasket seal [22] with vaseline, D00672 [CP5070], or oil, D00599 [CP2442].

5) Align the new gasket seal [22] and the fuel tube [20] with the applicable port on the MFP [1].

6) Install the four bolts [21] on the fuel tube [20].

7) Tighten the bolts [21] in a criss cross pattern to 51.0 in-lb (5.8 N·m) - 55.0 in-lb (6.2 N·m).

8) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

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(q) Connect the fuel tube [17] to the MFP [1] as follows:

1) Remove the protective covers from the openings.
2) Make sure that the mating faces are clean and clear of unwanted materials.
3) Lubricate the threads and pressure faces of the four bolts [18] with graphite mineral vaseline, D00601 [CP2101].
4) Lubricate each side of the elastomer part of a new gasket seal [19] with vaseline, D00672 [CP5070], or oil, D00599 [CP2442].
5) Align the new gasket seal [19] and the fuel tube [17] with the applicable port on the MFP [1].
6) Install the four bolts [18] on the fuel tube [17].
7) Tighten the bolts [18] in a criss cross pattern to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).
8) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

(r) Connect the fuel tube [14] to the MFP [1] as follows:

1) Remove the protective covers from the openings.
2) Make sure that the mating faces are clean and clear of unwanted materials.
3) Lubricate the threads and pressure faces of the four bolts [15] with graphite mineral vaseline, D00601 [CP2101].
4) Lubricate each side of the elastomer part of a new gasket seal [16] with vaseline, D00672 [CP5070], or oil, D00599 [CP2442].
5) Align the new gasket seal [16] and the fuel tube [14] with the applicable port on the MFP [1].
6) Install the four bolts [15] on the fuel tube [14].
7) Tighten the bolts [15] in a criss cross pattern to 109.31 in-lb (12.35 N·m) - 120.81 in-lb (13.65 N·m).
8) Remove the unwanted vaseline, D00672 [CP5070], and the graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

(s) Install the bracket [12] on the AGB [2] as follows:

1) Make sure that the mating faces are clean and clear of unwanted materials.
2) Align the bracket [12] with the applicable position on the AGB [2].
3) Install the two bolts [13] to attach the bracket [12] to the AGB [2].
4) Tighten the bolts [13] to 84 in-lb (9.5 N·m) - 93 in-lb (10.5 N·m).

(t) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-01-860-006-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

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SUBTASK 73-11-01-860-007-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-01-860-008-G00

(3) For the left engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	5	C01320	ENGINE FUEL ENGINE 1 HPSOV CONT
E	6	C01395	ENGINE FUEL ENGINE 1 HPSOV IND

SUBTASK 73-11-01-860-009-G00

(4) For the right engine, remove the safety tags and close these applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
E	3	C01321	ENGINE FUEL ENGINE 2 HPSOV CONT
E	4	C01396	ENGINE FUEL ENGINE 2 HPSOV IND

SUBTASK 73-11-01-410-001-G00

(5) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

J. Main Fuel Pump Installation Test

SUBTASK 73-11-01-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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FUEL FILTER CARTRIDGE - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Fuel filter cartridge removal

(2) Fuel filter cartridge installation.

TASK 73-11-02-000-801-G00

2. Fuel Filter Cartridge Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the fuel filter cartridge.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
5	Preformed packing	73-11-03-01-030	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Prepare for the Removal

SUBTASK 73-11-02-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

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(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-02-860-002-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-02-010-001-G00

(3) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

I. Fuel Filter Cartridge Removal

SUBTASK 73-11-02-680-001-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Drain the fuel from the main fuel filter housing [2]:

(a) Put the fuel resistant container (5 gal), STD-201, below the fuel filter bowl [3].

(b) Remove the drain plug [4] from the fuel filter bowl [3]:

1) Cut and remove the safety cable [6] or safety wire [6] from the drain plug [4].

2) Remove the drain plug [4] from the fuel filter bowl [3].

3) Remove and discard the preformed packing [5] from the groove of the drain plug [4].

4) Let the fuel drain into the fuel resistant container (5 gal), STD-201.

(c) Install the drain plug [4] on the fuel filter bowl [3]:

1) Make sure that the mating faces are clean and clear of unwanted materials.

2) Lubricate the new preformed packing [5] with vaseline, D00672 [CP5070].

3) Install the preformed packing [5] on the groove of the drain plug [4].

CAUTION MAKE SURE THAT YOU INSTALL THE PREFORMED PACKING AND DRAIN PLUG. IF YOU DO NOT INSTALL THE PACKING AND DRAIN PLUG, FUEL CAN LEAK FROM THE BOWL AND CAN CAUSE A SHUTDOWN OF THE ENGINE.

4) Install the drain plug [4] on the fuel filter bowl [3].

5) Tighten the drain plug [4] to 140.0 in-lb (15.82 N·m) - 150.0 in-lb (16.95 N·m).

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6) Install the safety cable [6] (safety cable, G50065 [CP8006]) or safety wire [6] (safety wire, G02345 [CP8001]) on the drain plug [4].

NOTE: It is acceptable to wait to install lockwire until after the filter bowl has been removed and installed again so that a socket can be used on the filter bowl boss to make removal and installation easier.

(d) Remove the fuel filter cartridge [1]:

1) Make sure that you drain all the fuel from the fuel filter bowl [3].

CAUTION

DO NOT LET THE FUEL FILTER BOWL FALL. DAMAGE TO THE FUEL FILTER BOWL CAN OCCUR.

2) Carefully push the latch [7] to remove the fuel filter bowl [3] from the main fuel filter housing [2].

NOTE: The latch will remain in the open position.

NOTE: Be careful, the latch is very easy to break.

3) Make sure that you hold the fuel filter bowl [3] vertically.

4) Turn the fuel filter bowl [3] to remove it from the main fuel filter housing [2].

NOTE: It is acceptable to use a wrench or socket on the filter bowl boss to remove the filter bowl if it is difficult to remove by hand.

5) Remove and discard the fuel filter cartridge [1].

6) Remove and discard the preformed packing [9] and two back-up rings [8] from the groove of the main fuel filter housing [2].

(e) Install the protective covers on the main fuel filter housing [2] and fuel filter bowl [3].

END OF TASK

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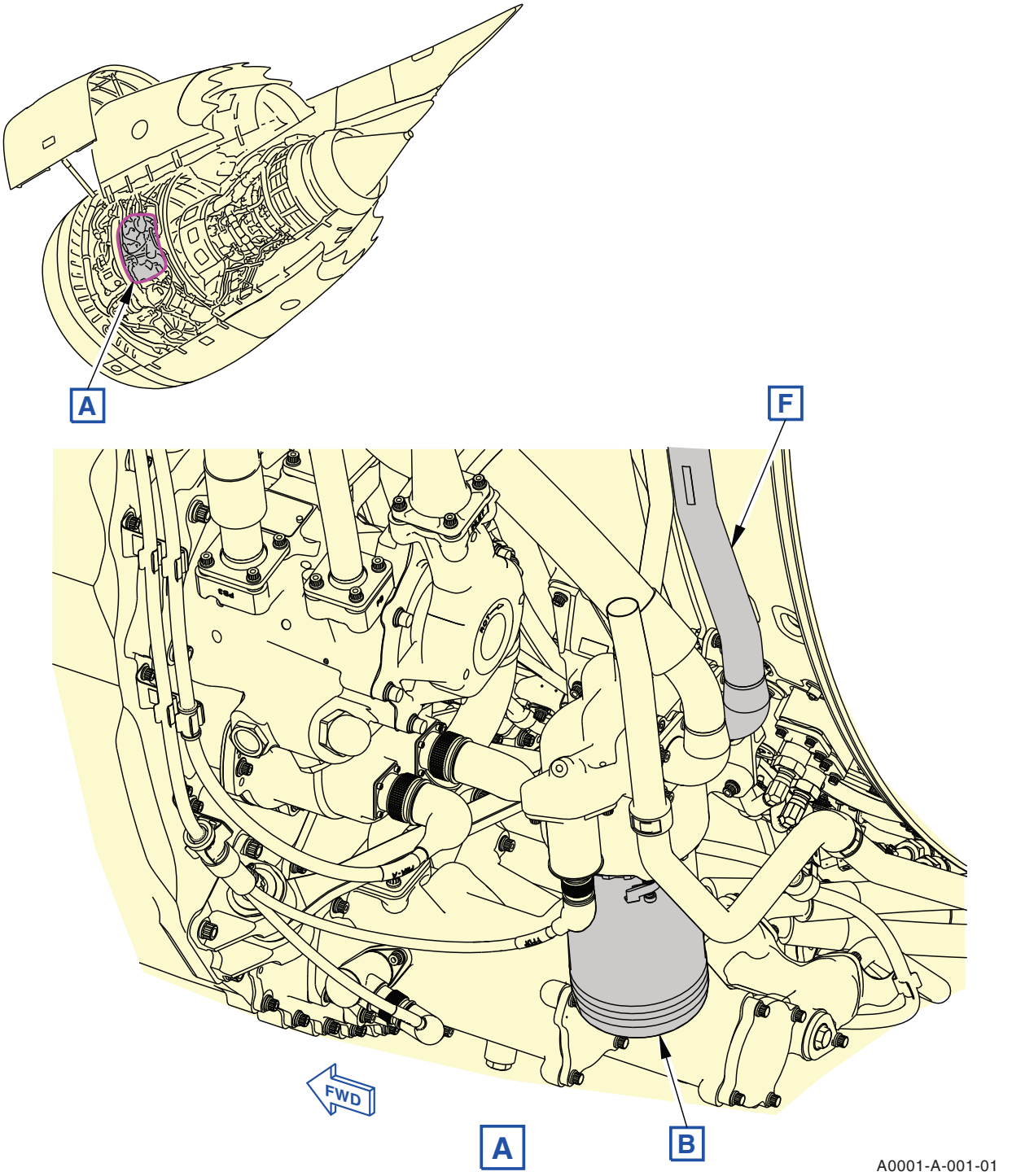
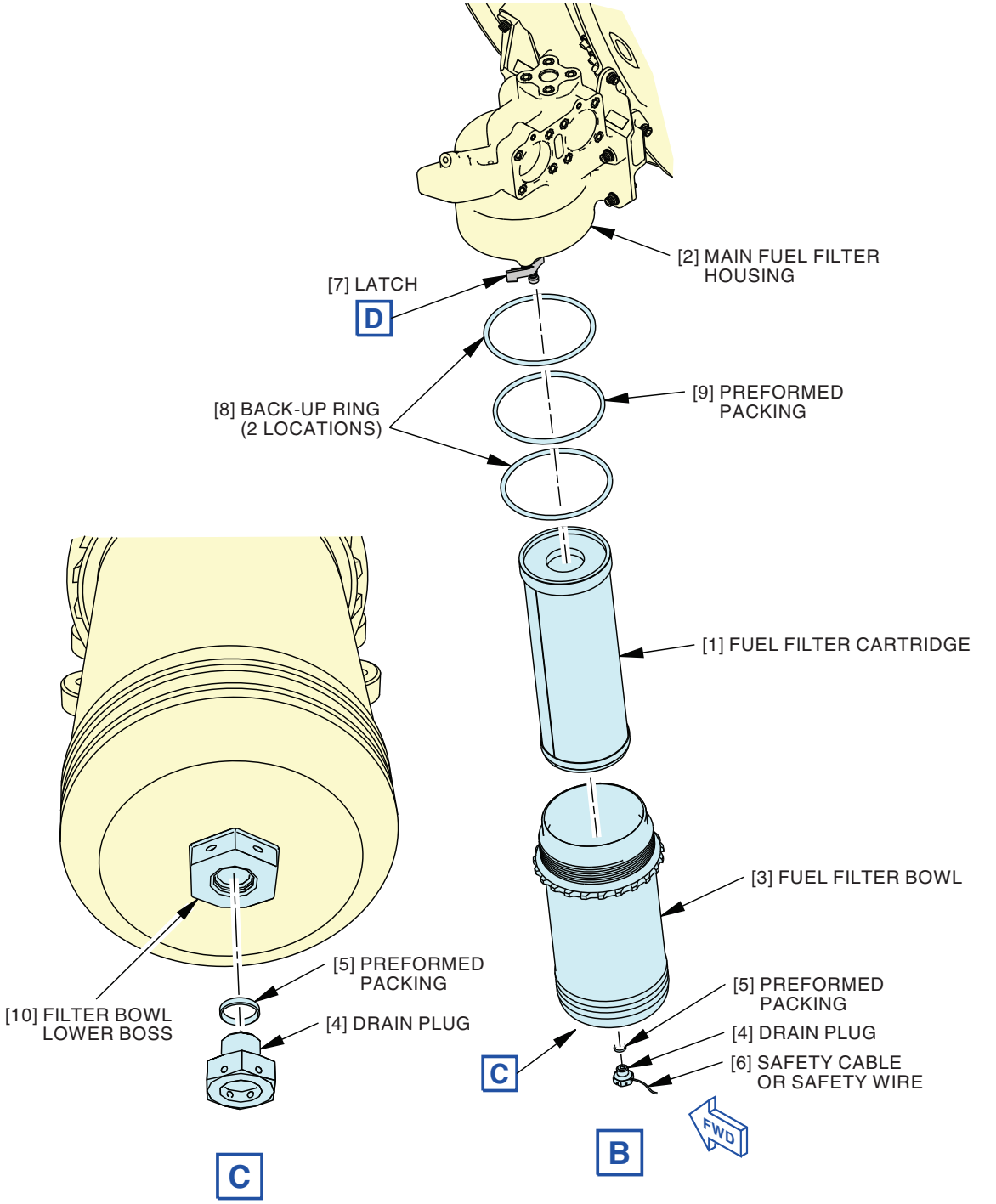


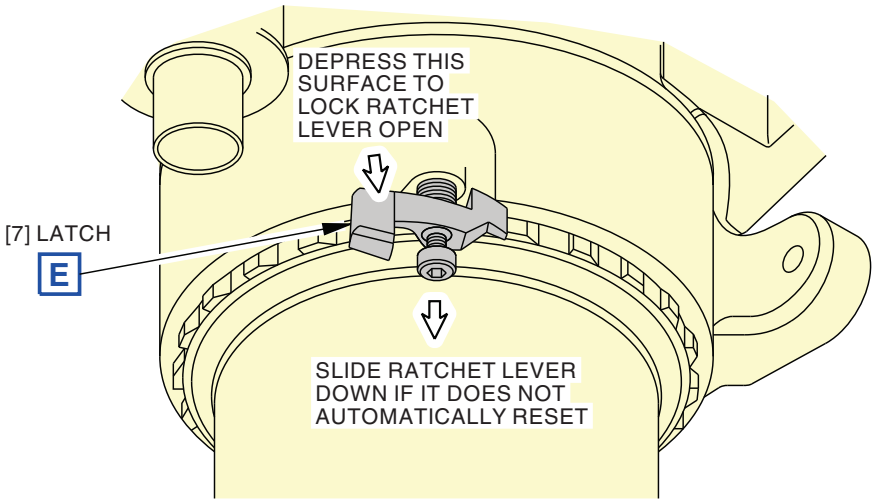
Fuel Filter Cartridge Installation
Figure 401/73-11-02-990-801-G00 (Sheet 1 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

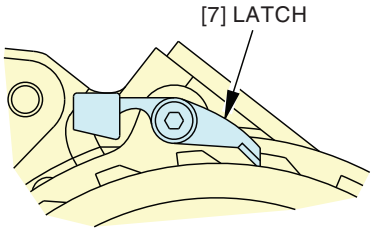


Fuel Filter Cartridge Installation
Figure 401/73-11-02-990-801-G00 (Sheet 2 of 4)

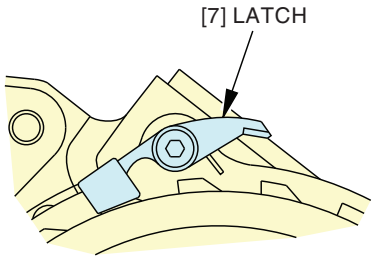
ECCN 9E991 BOEING PROPRIETARY - See title page for details



LATCH
D



RATCHET LEVER LOCKED CLOSED



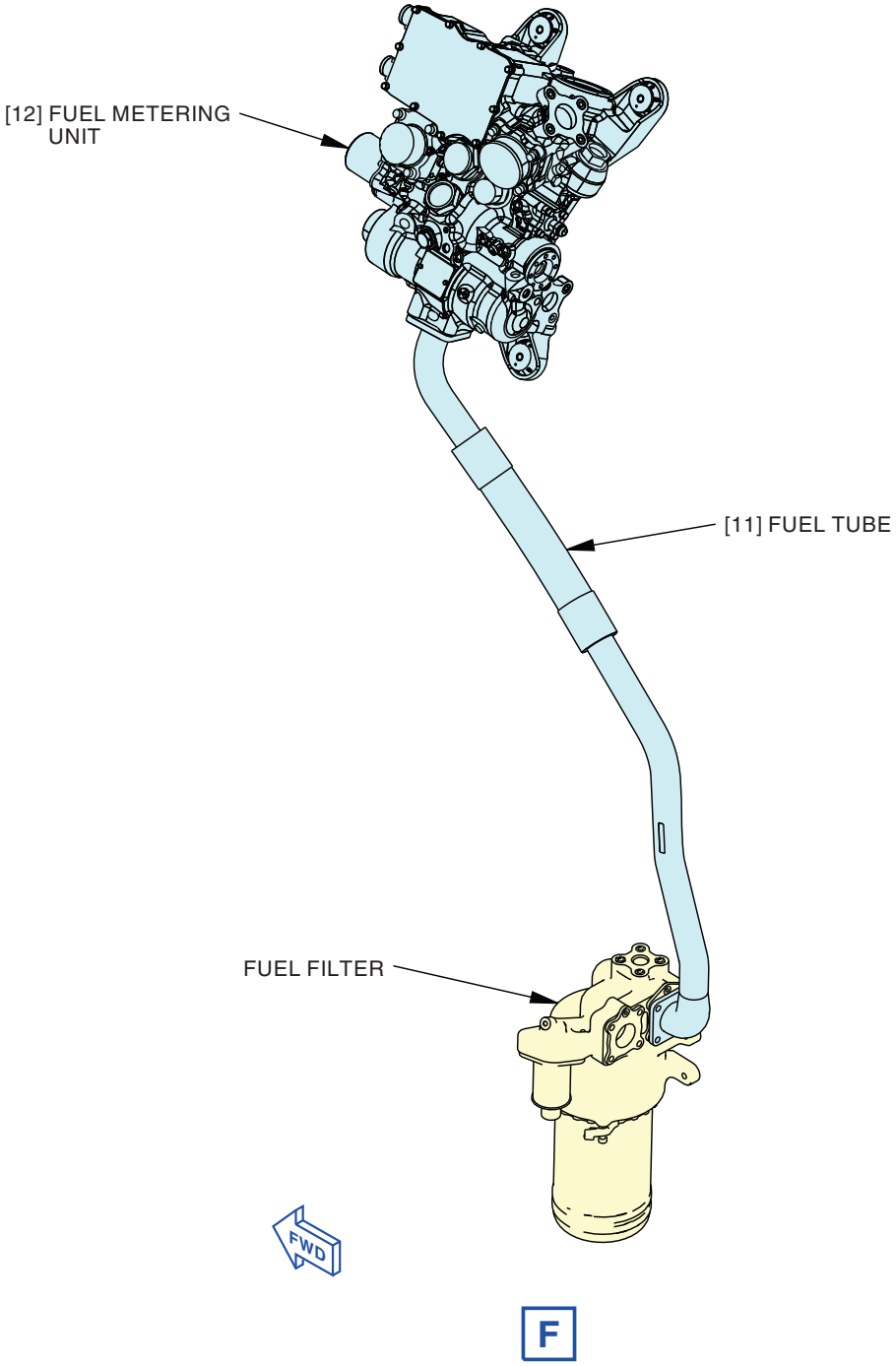
RATCHET LEVER LOCKED OPEN

LATCH
(VIEW IN THE UP DIRECTION)
E

5039036-00
2585462 S0000623960_V1

Fuel Filter Cartridge Installation
Figure 401/73-11-02-990-801-G00 (Sheet 3 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Filter Cartridge Installation
Figure 401/73-11-02-990-801-G00 (Sheet 4 of 4)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-11-02-400-801-G00

3. Fuel Filter Cartridge Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the fuel filter cartridge.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-21-04-000-801-G00	Fuel Metering Unit Removal (P/B 401)
73-21-04-400-801-G00	Fuel Metering Unit Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G02345 [CP8001]	Wire - Safety, 0.032 Inch (0.8 mm) Diameter	AMS 5687
G50065 [CP8006]	Cable, Safety, Stainless Steel, 0.032 inch (0.8 mm) Diameter	M50 TF 9 CL-A

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Fuel filter cartridge	73-11-03-01-050	PTF ALL
8	Back-up ring	73-11-03-01-060	PTF ALL
9	Preformed packing	73-11-03-01-070	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Fuel Filter Cartridge Installation

SUBTASK 73-11-02-420-001-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Install the fuel filter cartridge [1]:

(a) Remove the protective covers from the main fuel filter housing [2] and fuel filter bowl [3].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the two new back-up rings [8] and new preformed packing [9] with vaseline, D00672 [CP5070].

(d) Install a new preformed packing [9] and two back-up rings [8] onto the groove of the main fuel filter housing [2].

1) Make sure that the preformed packing [9] and two back-up rings [8] are correctly installed and fully seated in the main fuel filter housing [2] groove.

NOTE: Use of an inspection mirror is recommended.

(e) Lubricate the extremities of the new fuel filter cartridge [1] with vaseline, D00672 [CP5070].

(f) Make sure that the preformed packing [9] installed in the fuel filter cartridge [1] is sufficiently lubricated.

NOTE: Preformed packing may be removed and reinstalled to help with lubrication.

(g) Put the new fuel filter cartridge [1] into position on the main fuel filter housing [2].

NOTE: It is acceptable to install the filter cartridge into either the main fuel filter housing or fuel filter bowl.

(h) Lubricate the upper section of the fuel filter bowl [3] with vaseline, D00672 [CP5070], but do not lubricate the threads.

NOTE: It is optional, but not necessary, to lubricate the threads with oil, D00599 [CP2442], to aid installation.

(i) Make sure that the latch [7] is open.

(j) Align the fuel filter bowl [3] in a correct position on the main fuel filter housing [2].

(k) Turn the fuel filter bowl [3] in a clockwise direction with your hands to engage the fuel filter bowl [3] with the threads in the port in the main fuel filter housing [2].

(l) Turn the fuel filter bowl [3] until it is not easy to turn with your hands.

NOTE: It will become harder to turn the fuel filter bowl as the preformed packings engage the main fuel filter housing.

(m) Install a wrench or socket over the filter bowl boss [10] on the bottom of the bowl.

CAUTION

DO NOT USE MORE TORQUE THAN 600 INCH-POUNDS (68 NEWTON-METERS) WHEN YOU INSTALL THE FUEL FILTER BOWL. IF YOU DO, DAMAGE CAN OCCUR.

CAUTION

MAKE SURE TO APPLY TORQUE TO THE FUEL FILTER BOWL CAREFULLY. IF THERE IS DAMAGE TO THE PACKING'S O-RINGS OR INSTALLATION NOT DONE CORRECTLY, FUEL LEAKAGE FROM THE FUEL FILTER BOWL OR CONTAMINATION IN THE FUEL SYSTEM CAN OCCUR. IF YOU DO NOT OBEY, DAMAGE TO ENGINE COMPONENTS CAN OCCUR.

(n) If a sudden decrease in torque occurs when the fuel filter bowl [3] is installed, do these steps:

NOTE: When a sudden decrease in torque is noticed, it can be accompanied by snapping noise. It is possible that the preformed packing moved. The packing must be examined for damage.

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1) Remove the fuel filter bowl [3].

2) Examine the preformed packing [9] and back-up rings [8] for damage.

a) Remove all unwanted material from the fuel filter bowl [3] and main fuel filter housing [2].

b) Lubricate the new back-up rings [8] and new preformed packing [9] with vaseline, D00672 [CP5070].

c) Install the preformed packing [9] and back-up rings [8] onto the groove of the main fuel filter housing [2].

3) Do an inspection of the fuel tube [11] between the fuel filter and fuel metering unit [12].

a) Remove all unwanted material from the fuel tube [11].

4) Replace the fuel metering unit [12] (TASK 73-21-04-000-801-G00 and TASK 73-21-04-400-801-G00).

(o) Turn the fuel filter bowl [3] in a clockwise direction to fully install the fuel filter bowl [3] into the main fuel filter housing [2].

1) Pull the latch [7] down in the direction of the filter bowl drain plug and release it.

NOTE: The latch must return to the closed position and make contact with the fuel filter bowl. The latch can sit at any position along the lock ring either on top of the teeth or in the grooves.

2) Do not tighten or loosen the fuel filter bowl [3] to change the alignment.

(p) Install the safety cable [6] (safety cable, G50065 [CP8006]) or safety wire [6] (safety wire, G02345 [CP8001]) on the drain plug [4] if not already installed.

(q) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-02-410-001-G00

(1) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

SUBTASK 73-11-02-860-006-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-02-860-008-G00

(3) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

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J. Fuel Filter Cartridge Installation Test

SUBTASK 73-11-02-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

MAIN FUEL FILTER HOUSING -REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Main Fuel Filter Housing Removal

(2) Main Fuel Filter Housing Installation.

TASK 73-11-03-000-801-G00

2. Main Fuel Filter Housing Removal

(Figure 401)

A. General

(1) This task gives the instructions to remove the main fuel filter housing.

B. References

Reference	Title
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
73-11-02-000-801-G00	Fuel Filter Cartridge Removal (P/B 401)
73-11-10-000-801-G00	Fuel Filter Differential Pressure Transducer Removal (P/B 401)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-858	Tag - DO NOT OPERATE

D. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

E. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

F. Prepare for the Removal

SUBTASK 73-11-03-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

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SUBTASK 73-11-03-860-002-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-03-860-005-G00

(3) On the applicable engine, open the left fan cowl panel (TASK 71-11-04-010-801-G00).

(a) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

G. Main Fuel Filter Housing Removal

SUBTASK 73-11-03-860-006-G00

(1) Put a fuel resistant container (5 gal), STD-201 below the main fuel filter housing.

SUBTASK 73-11-03-680-001-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Remove the main fuel filter housing [1]:

(a) Remove the fuel filter cartridge (TASK 73-11-02-000-801-G00).

(b) Remove the main fuel filter pump differential pressure transducer sensor [2] (TASK 73-11-10-000-801-G00).

(c) Disconnect the fuel tube [7] from the main fuel filter housing [1]:

1) Remove the four bolts [9] with integrated washers from the fuel tube [7].

2) Disconnect the fuel tube [7] from the main fuel filter housing [1].

3) Remove and discard the gasket seal [10].

(d) Disconnect the fuel tube [11] from the main fuel filter housing [1]:

1) Remove the four bolts [12] from the fuel tube [11].

2) Disconnect the fuel tube [11] from the main fuel filter housing [1].

3) Remove and discard the gasket seal [13].

(e) Disconnect the fuel tube [14] from the main fuel filter housing [1]:

1) Remove the four bolts [15] from the fuel tube [14].

2) Disconnect the fuel tube [14] from the main fuel filter housing [1].

3) Remove and discard the gasket seal [16].

(f) Remove the main fuel filter housing [1] from the mounting bracket [19]:

1) Remove the four bolts [17] with integrated washers that attach the main fuel filter housing [1] to the mounting bracket [19].

2) Remove the main fuel filter housing [1] from the mounting bracket [19].

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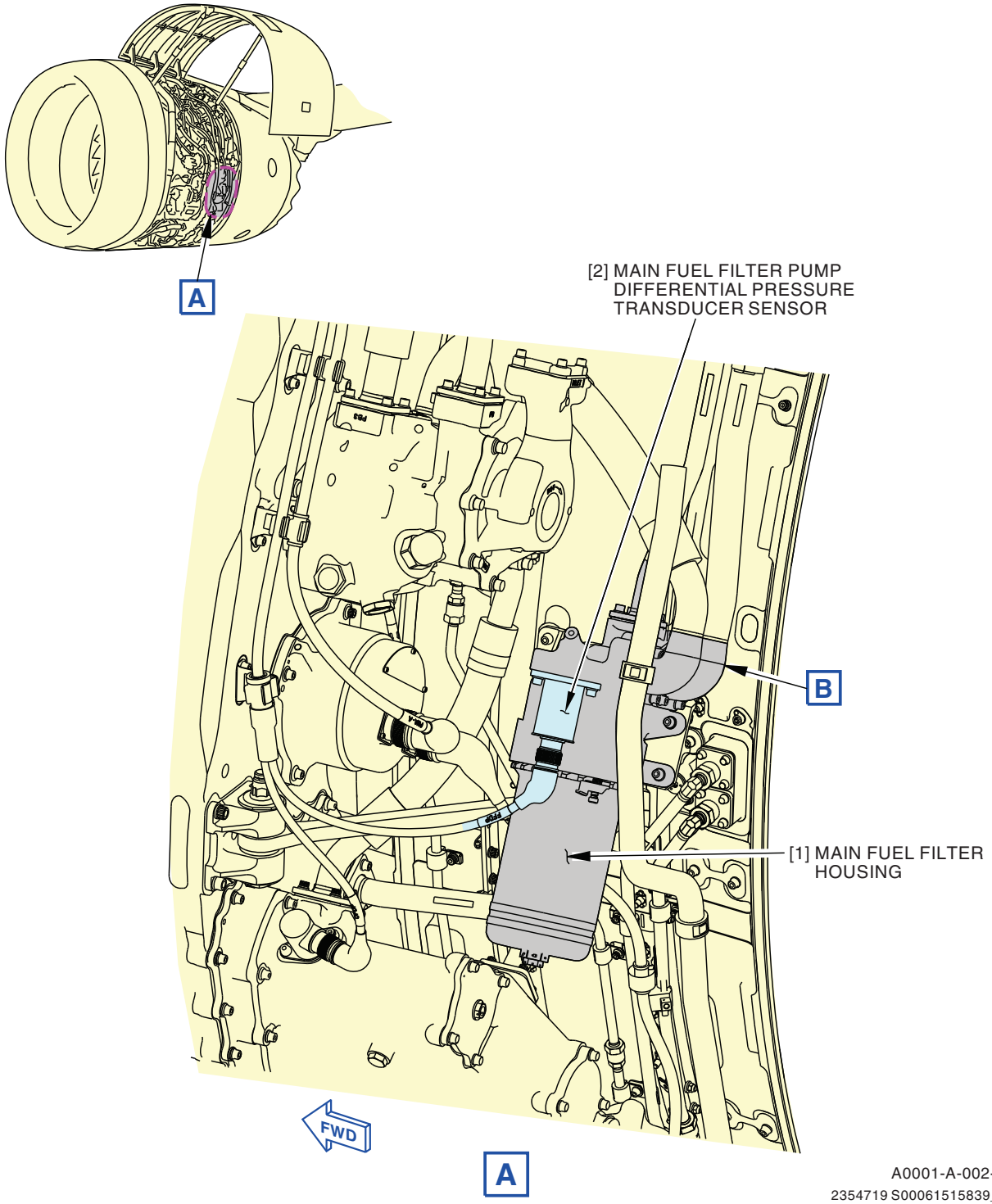
(g) Install the protective covers and caps on all openings.

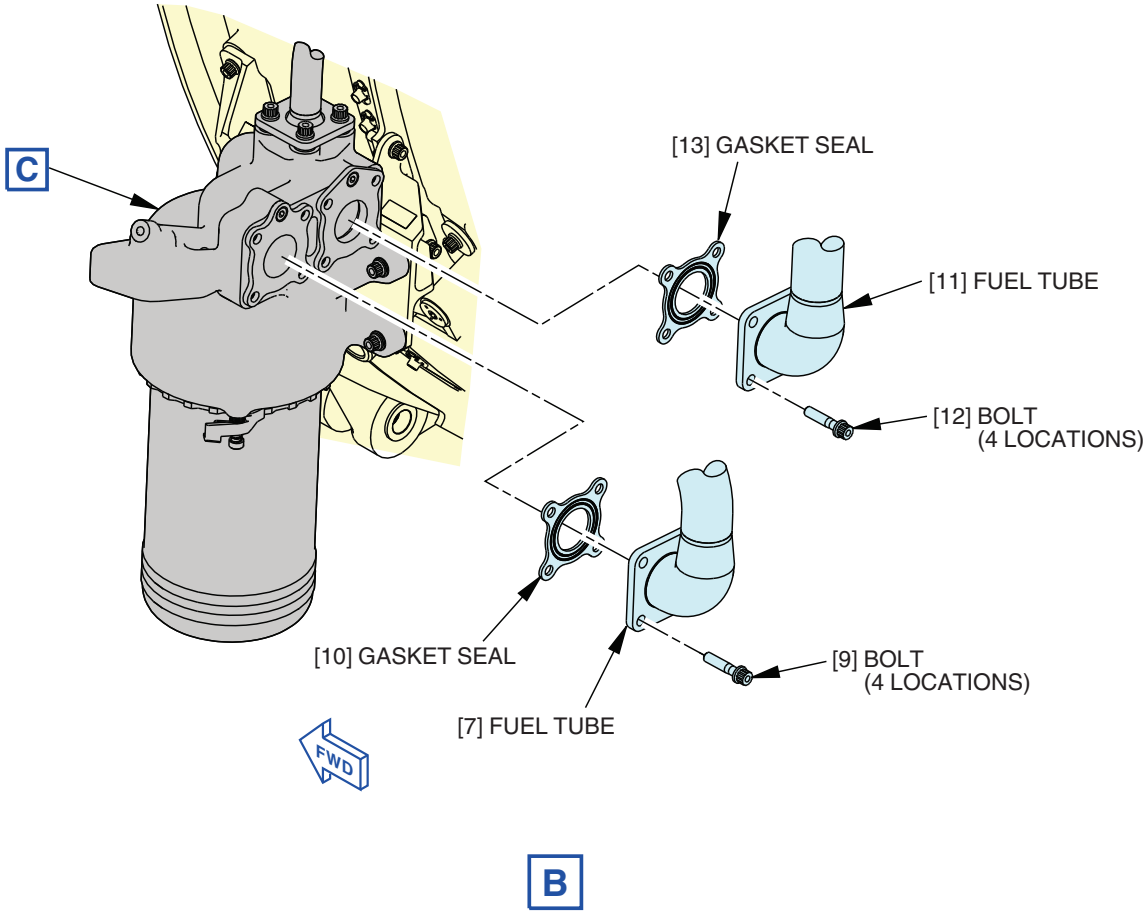
——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Fuel Filter Housing Installation
Figure 401/73-11-03-990-801-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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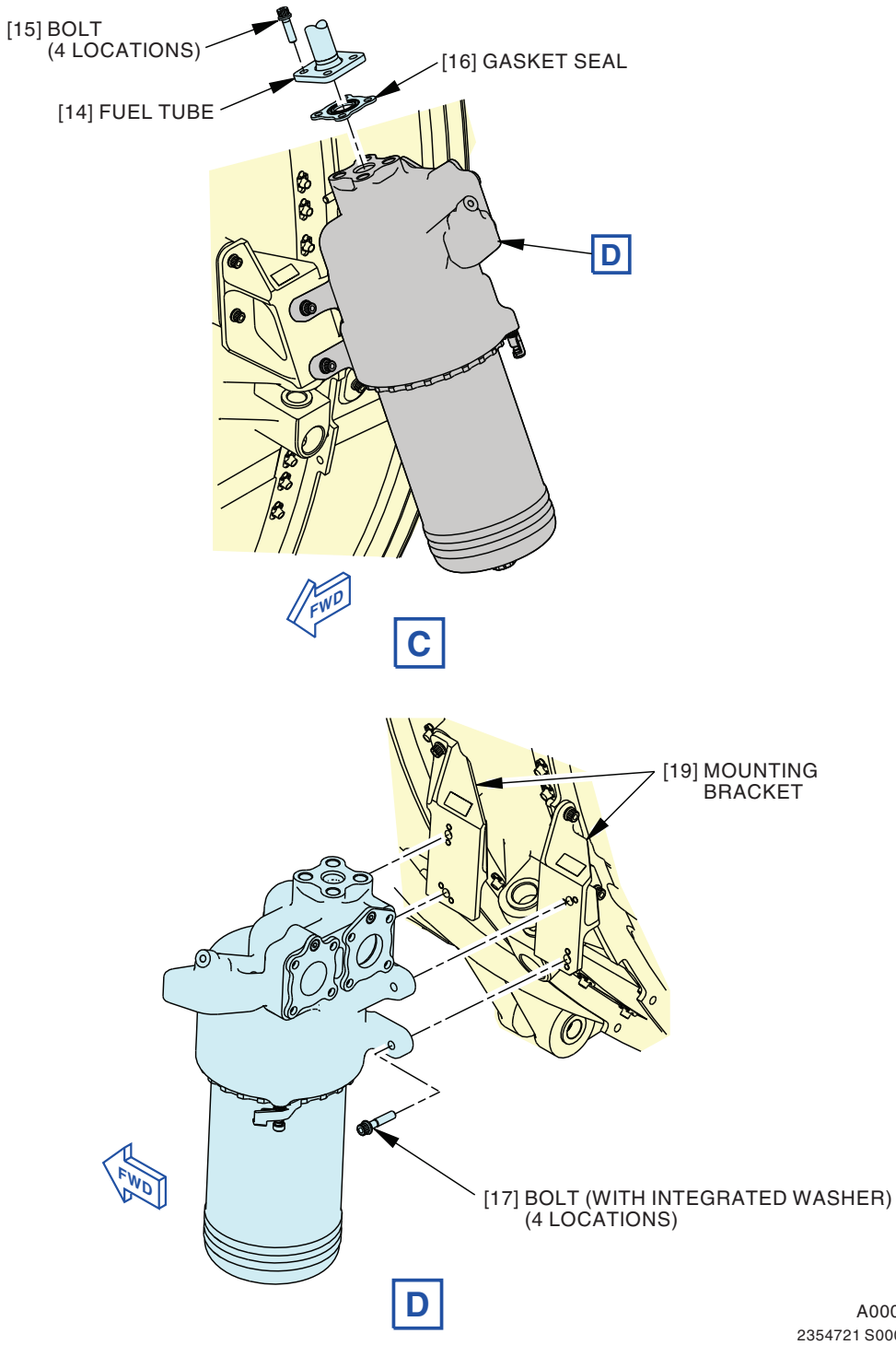
Main Fuel Filter Housing Installation
Figure 401/73-11-03-990-801-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Main Fuel Filter Housing Installation
Figure 401/73-11-03-990-801-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

TASK 73-11-03-400-801-G00

3. Main Fuel Filter Housing Installation

(Figure 401)

A. General

(1) This task gives the instructions to install the main fuel filter housing.

(2) You must do the tests that are listed in the power plant test reference table after you install the main fuel filter housing.

B. References

Reference	Title
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
73-11-02-400-801-G00	Fuel Filter Cartridge Installation (P/B 401)
73-11-10-400-801-G00	Fuel Filter Differential Pressure Transducer Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	
D00672 [CP5070]	Vaseline - Pure Mineral	V V-P-236
G51335 [CP2674]	Lint-Free Cloth	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
1	Main fuel filter housing	73-11-03-01-010	PTF ALL
10	Gasket seal	73-11-03-01-110	PTF ALL
13	Gasket seal	73-11-03-01-090	PTF ALL
16	Gasket seal	73-11-03-01-130	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

H. Main Fuel Filter Housing Installation

SUBTASK 73-11-03-420-001-G00

(1) Install the main fuel filter housing [1]:

(a) Remove the protective covers and caps from the openings.

(b) Make sure that the mating faces are clean and clear of unwanted materials.

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(c) Install the main fuel filter housing [1] on the mounting bracket [19]:

1) Lubricate the threads and pressure faces of the four bolts [17] with oil, D00599 [CP2442].
2) Install the main fuel filter housing [1] on the mounting bracket [19].
3) Install the four bolts [17] with integrated washers on the mounting bracket [19].
4) Tighten the bolts [17] in a criss cross pattern to 67.45 in-lb (7.6 N·m) - 74.55 in-lb (8.4 N·m).

(d) Connect the fuel tube [14] to the main fuel filter housing [1]:

1) Lubricate the threads and pressure faces of the four bolts [15] and with graphite mineral vaseline, D00601 [CP2101].
2) Lubricate each side of the elastomer part of a new gasket seal [16] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].
3) Install the new gasket seal [16] and fuel tube [14] on the main fuel filter housing [1].
4) Install the four bolts [15] on the fuel tube [14].
5) Tighten the bolts [15] in a criss cross pattern to 109.75 in-lb (12.4 N·m) - 120.27 in-lb (13.6 N·m).
6) Remove the unwanted vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], or engine oil, D00599 [CP2442], with a lint-free cloth, G51335 [CP2674].

(e) Connect the fuel tube [11] to the main fuel filter housing [1]:

1) Lubricate the threads and pressure faces of the four bolts [12] with graphite mineral vaseline, D00601 [CP2101].
2) Lubricate each side of the elastomer part of a new gasket seal [13] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].
3) Install the new gasket seal [13] and fuel tube [11] on the main fuel filter housing [1].
4) Install the four bolts [12] on the fuel tube [11].
5) Tighten the bolts [12] in a criss cross pattern to 109.75 in-lb (12.4 N·m) - 120.27 in-lb (13.6 N·m).
6) Remove the unwanted vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0021

(f) Connect the fuel tube [7] to the main fuel filter housing [1]:

1) Lubricate the threads and pressure faces of the four bolts [9] with graphite mineral vaseline, D00601 [CP2101].
2) Lubricate each side of the elastomer part of a new gasket seal [10] with vaseline, D00672 [CP5070], or engine oil, D00599 [CP2442].
3) Install the new gasket seal [10] and fuel tube [7] on the main fuel filter housing [1].
4) Install the four bolts [9] that attach the fuel tube [7] to the main fuel filter housing [1].
5) Tighten the bolts [9] in a criss cross pattern to 109.75 in-lb (12.4 N·m) - 120.27 in-lb (13.6 N·m).
6) Remove the unwanted vaseline, D00672 [CP5070], and graphite mineral vaseline, D00601 [CP2101], with a lint-free cloth, G51335 [CP2674].

PTF ALL

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(g) If not already done, install the fuel filter cartridge (TASK 73-11-02-400-801-G00).

(h) Install the main fuel filter pump differential pressure transducer sensor [2] (TASK 73-11-10-400-801-G00).

(i) Clean the work area and clear the tools and other items.

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-03-860-007-G00

(1) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

SUBTASK 73-11-03-860-008-G00

(2) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-03-410-001-G00

(3) On the applicable engine, close the left fan cowl panel (TASK 71-11-04-410-801-G00).

(a) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2

J. Main Fuel Filter Housing Installation Test

SUBTASK 73-11-03-730-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL METERING UNIT STRAINER - REMOVAL/INSTALLATION

1. General

A. This procedure has two tasks:

(1) Fuel metering unit strainer removal

(2) Fuel metering unit strainer installation.

TASK 73-11-04-000-801-G00

2. Fuel Metering Unit Strainer Removal

(Figure 401)

A. General

(1) This task provides the instructions on how to remove the fuel metering unit strainer.

(2) The fuel metering unit strainer is referred to as the FMU strainer.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
73-11-04-200-801-G00	Fuel Metering Unit Strainer Inspection (P/B 601)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-201	Container - Fuel Resistant, 5-Gallon (19-Liter)
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
3	Preformed packing	73-21-04-01-023	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1

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(Continued)

Number	Name/Location
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Prepare for the Removal

SUBTASK 73-11-04-860-001-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-04-040-001-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-04-860-002-G00

(3) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 73-11-04-860-003-G00

(4) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 73-11-04-860-004-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

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WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

I. Fuel Metering Unit Strainer Removal

SUBTASK 73-11-04-680-001-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. KEEP THE FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE, WHICH CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(1) Drain the fuel from the FMU [1] as follows:

(a) Put a fuel resistant container (5 gal), STD-201 below the FMU [1].

(b) Remove the drain plug [2] from the FMU [1].

1) Let the fuel drain from the FMU [1] into the fuel resistant container (5 gal), STD-201.

(c) Remove and discard the preformed packing [3] from the drain plug [2].

(d) Lubricate the new preformed packing [3] with engine oil, D00599 [CP2442].

(e) Install the preformed packing [3] on the drain plug [2].

(f) Put graphite mineral vaseline, D00601 [CP2101] on the threads and under the heads of the drain plug [2].

(g) Make sure that the mating faces are clean and clear of unwanted materials.

(h) Install the drain plug [2] on the FMU [1].

(i) Tighten the drain plug [2] to 32.0 - 35.0 inch-pounds (3.6 - 4.0 Newton-meters).

SUBTASK 73-11-04-020-001-G00

(2) Remove the FMU strainer [4] as follows:

(a) Remove the four bolts [6] and the four washers [5] that attach the FMU strainer [4] to the FMU [1].

(b) Remove the FMU strainer [4] from the FMU [1].

NOTE: Do not disassemble the FMU strainer [4]. It is not necessary to disassemble the FMU strainer [4] to clean it.

(c) Remove and discard the preformed packing [7] and the preformed packing [8].

(d) Install the protective cover on the FMU strainer [4] port on the FMU [1].

SUBTASK 73-11-04-200-001-G00

(3) Do an inspection of the FMU strainer [4] (TASK 73-11-04-200-801-G00).

END OF TASK

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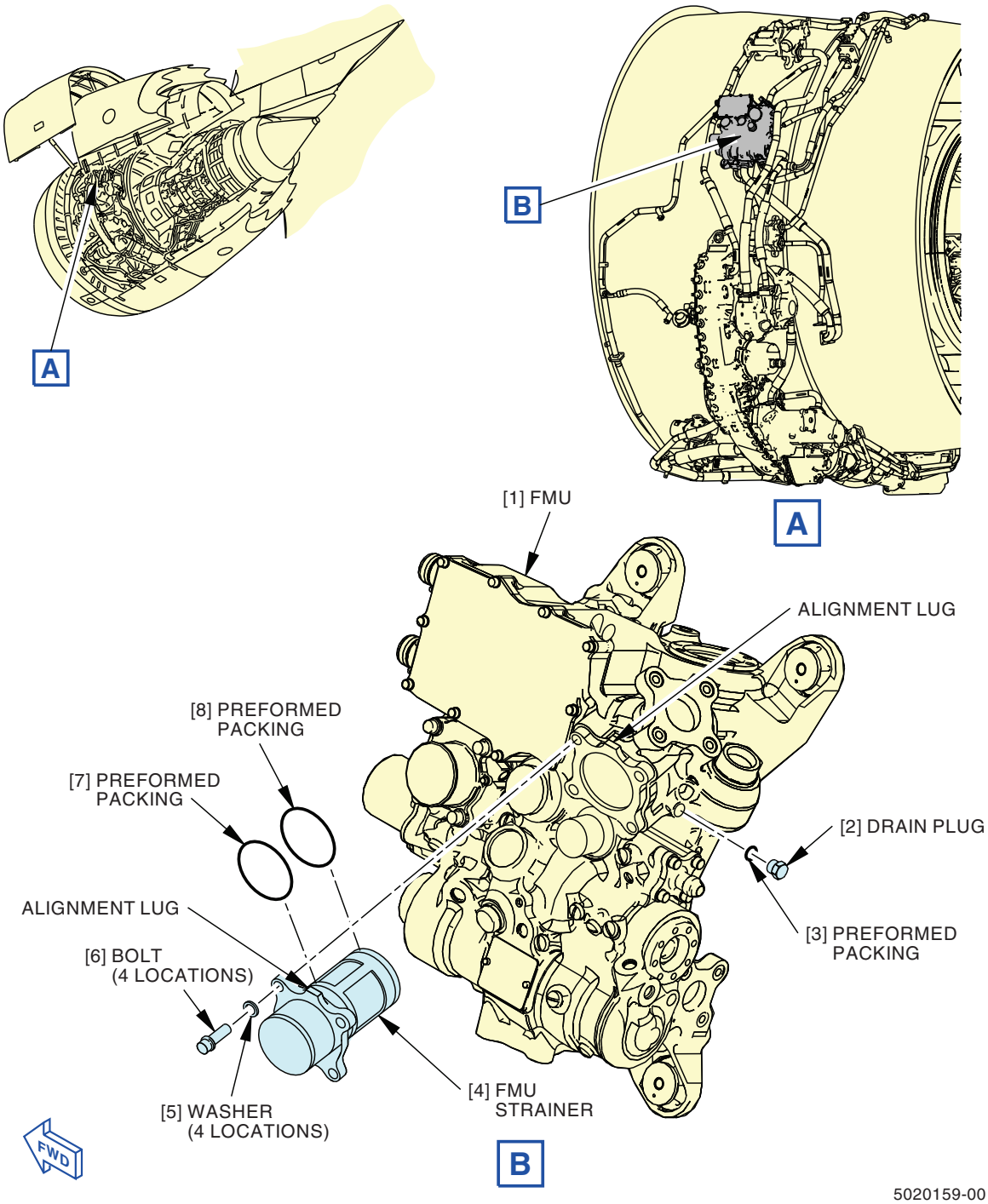

Fuel Metering Unit Strainer Installation
Figure 401/73-11-04-990-801-G00

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TASK 73-11-04-400-801-G00

3. Fuel Metering Unit Strainer Installation

(Figure 401)

A. General

(1) This task provides the instructions on how to install the fuel metering unit strainer.

(2) The fuel metering unit strainer is referred to as the FMU strainer.

(3) You must do the tests that are listed in the power plant test reference table after you install the FMU strainer.

B. References

Reference	Title
27-81-00-440-801	Leading Edge Flaps and Slats - Activation (P/B 201)
71-00-00-800-804-G00	Power Plant Test Reference Table (P/B 501)
71-11-04-410-801-G00	Close the Fan Cowl Panels (Selection) (P/B 201)
78-31-00-010-802-G00	Close the Thrust Reverser (Selection) (P/B 201)
78-31-00-440-801-G00	Thrust Reverser Activation After Ground Maintenance (P/B 201)

C. Tools/Equipment

Reference	Description
STD-858	Tag - DO NOT OPERATE

D. Consumable Materials

Reference	Description	Specification
D00599 [CP2442]	Oil - Engine	
D00601 [CP2101]	Vaseline - Graphite Mineral	

E. Expendables/Parts

AMM Item	Description	AIPC Reference	AIPC Effectivity
7	Preformed packing	73-21-04-01-030	PTF ALL
8	Preformed packing	73-21-04-01-050	PTF ALL

F. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

G. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

H. Fuel Metering Unit Strainer Installation

SUBTASK 73-11-04-420-001-G00

(1) Install the FMU strainer [4] as follows:

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(a) Remove the protective cover from the FMU strainer port on the FMU [1]

(b) Make sure that the mating faces are clean and clear of unwanted materials.

(c) Lubricate the new new preformed packing [7] and the new preformed packing [8] with the engine oil, D00599 [CP2442].

(d) Install the preformed packing [7] and the preformed packing [8] onto the packing grooves of the FMU strainer [4].

(e) Apply the graphite graphite mineral vaseline, D00601 [CP2101] on the threads and below the heads of the bolts [6].

(f) Align the alignment lug on the FMU strainer [4] to the alignment lug on the FMU [1] housing, and apply firm hand pressure to seat it.

NOTE: To avoid damage to the FMU strainer [4], it is important to make sure that it is correctly aligned and seated in the FMU housing before tightening the four bolts. The correct alignment and seating is indicated by 360 degree contact/engagement of the preformed packings with little or no visible gap between the strainer flange and the mating surface of the housing.

(g) Install the four bolts [6] and the four washers [5] that attach the FMU strainer [4] to the FMU [1].

(h) Tighten the bolts [6] in one quarter turn increments, one at a time, in a crisscross pattern to 73.0 - 87.0 pound-inches (8.2 - 9.8 Newton-meters).

I. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-04-860-005-G00

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO CLOSE THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(1) Do these tasks in sequence to safely close the applicable left or right thrust reversers:

(a) Do this task: Close the Thrust Reverser (Selection), TASK 78-31-00-010-802-G00.

1) Close these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

(b) Do this task: Close the Fan Cowl Panels (Selection), TASK 71-11-04-410-801-G00.

1) Close these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

(c) Do this task: Thrust Reverser Activation After Ground Maintenance, TASK 78-31-00-440-801-G00.

SUBTASK 73-11-04-860-006-G00

(2) Do this task: Leading Edge Flaps and Slats - Activation, TASK 27-81-00-440-801.

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SUBTASK 73-11-04-440-001-G00

(3) Remove the safety tags and close the applicable circuit breakers:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-04-860-007-G00

(4) Do these steps to remove the DO NOT OPERATE tags, STD-858, from the applicable ENGINE START switch and START LEVER switch:

(a) On the P5 overhead panel, remove the DO NOT OPERATE tag, STD-858, from the applicable ENGINE START switch.

(b) On the P10 control stand, remove the DO NOT OPERATE tag, STD-858, from the applicable START LEVER switch.

J. Fuel Metering Unit Strainer Installation Test

SUBTASK 73-11-04-710-001-G00

(1) Do the test(s) that are given in the Power Plant Test Reference Table (TASK 71-00-00-800-804-G00).

——— END OF TASK ———

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FUEL METERING UNIT STRAINER - INSPECTION/CHECK

1. General

A. This procedure has one task:

(1) Fuel metering unit strainer inspection.

TASK 73-11-04-200-801-G00

2. Fuel Metering Unit Strainer Inspection

A. General

(1) This task provides the instructions on how to do the inspection on the fuel metering unit strainer.

(2) The fuel metering unit strainer is referred to as the FMU strainer.

B. References

Reference	Title
73-11-04-000-801-G00	Fuel Metering Unit Strainer Removal (P/B 401)
73-11-04-100-801-G00	Fuel Metering Unit Strainer Cleaning (P/B 701)
73-11-04-400-801-G00	Fuel Metering Unit Strainer Installation (P/B 401)

C. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

D. Prepare for the Inspection

SUBTASK 73-11-04-010-001-G00

(1) Do this task: Fuel Metering Unit Strainer Removal, TASK 73-11-04-000-801-G00.

E. Fuel Metering Unit Strainer Inspection

SUBTASK 73-11-04-212-001-G00

(1) Do the inspection of the FMU strainer as follows:

(a) Do a visual check of the FMU strainer for the contamination material that is caught by the strainer.

1) If you find the material that is caught by the strainer, do these steps:

a) Record the type and quantity of the contamination material that you find on a record sheet.

b) Do this task: Fuel Metering Unit Strainer Cleaning, TASK 73-11-04-100-801-G00.

2) Do a visual check on the inlet screen.

a) If you see the damage, replace the FMU strainer (TASK 73-11-04-000-801-G00 and TASK 73-11-04-400-801-G00).

F. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-04-410-001-G00

(1) Do this task: Fuel Metering Unit Strainer Installation, TASK 73-11-04-400-801-G00.

END OF TASK

ECCN 9E991 BOEING PROPRIETARY - See title page for details

FUEL METERING UNIT STRAINER - CLEANING/PAINTING

1. General

A. This procedure has one task:

(1) Fuel metering unit strainer cleaning.

TASK 73-11-04-100-801-G00

2. Fuel Metering Unit Strainer Cleaning

A. General

(1) This task provides the instructions on how to clean the Fuel Metering Unit strainer.

(2) The Fuel Metering Unit strainer is referred to as the FMU strainer.

B. References

Reference	Title
73-11-04-000-801-G00	Fuel Metering Unit Strainer Removal (P/B 401)
73-11-04-400-801-G00	Fuel Metering Unit Strainer Installation (P/B 401)

C. Tools/Equipment

Reference	Description
STD-128	Brush - Paint

D. Consumable Materials

Reference	Description	Specification
B00074	Solvent - Degreasing	MIL-PRF-680 (Supersedes P-D-680)
B00130	Alcohol - Isopropyl	TT-I-735

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

F. Prepare for the Cleaning

SUBTASK 73-11-04-010-002-G00

(1) Do this task: Fuel Metering Unit Strainer Removal, TASK 73-11-04-000-801-G00.

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G. Fuel Metering Unit Strainer Cleaning

SUBTASK 73-11-04-110-001-G00

WARNING

DO NOT GET SOLVENTS IN YOUR MOUTH OR EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM SOLVENTS. PUT ON A RESPIRATOR, PROTECTIVE SPLASH GOGGLES, AND GLOVES WHEN YOU USE SOLVENTS. KEEP THE SOLVENTS AWAY FROM SPARKS, FLAME, AND HEAT. SOLVENTS ARE POISONOUS AND FLAMMABLE. SOLVENTS CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(1) Clean the fuel metering unit (FMU) strainer in solvent, B00074 or alcohol, B00130 with a paint brush, STD-128.

NOTE: Keep the material that you remove from the FMU strainer. This material is necessary to troubleshoot the fuel system.

NOTE: Do not disassemble the Fuel Metering Unit strainer assembly when you clean it. If you disassemble the Fuel Metering Unit (FMU) strainer assembly, you can cause damage to it.

(a) Let the FMU strainer assembly dry fully.

SUBTASK 73-11-04-212-002-G00

(2) Examine the FMU strainer assembly for damage.

(a) Replace the FMU strainer assembly if you find damage.

H. Put the Airplane Back to Its Usual Condition

SUBTASK 73-11-04-410-002-G00

(1) Do this task: Fuel Metering Unit Strainer Installation, TASK 73-11-04-400-801-G00.

——— END OF TASK ———

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FUEL NOZZLE - REMOVAL/INSTALLATION

1. General

A. This procedure has these tasks:

(1) Fuel Nozzle Removal

(2) Fuel Nozzle Installation

(3) Fuel Nozzle Set Removal

(4) Fuel Nozzle Set Installation.

TASK 73-11-05-000-801-G00

2. Fuel Nozzle Removal

(Figure 401, Figure 402, Figure 403, Figure 404, Figure 405, Figure 406, Figure 407, Figure 408, Figure 409, Figure 410, Figure 411, Figure 412, Figure 413, Figure 414, Figure 415, Figure 416, Figure 417, Figure 418, Figure 419, Figure 420, Figure 421, Figure 422, Figure 423, and Figure 424)

A. General

(1) This task gives the instructions to remove one or more fuel nozzles.

(2) There are eighteen fuel nozzles installed on the combustion case.

(3) The fuel nozzle position numbers are in a CW (Clockwise) sequence Aft Looking Forward (ALF).

NOTE: The offset hole on the combustion case aft flange is the TVCL (Top Vertical Centerline). The first fuel nozzle clockwise, aft looking forward, from the TVCL is No. 1.

(4) The task is divided into eighteen removal topics.

(5) You can remove a fuel nozzle individually per given instructions in the individual topic.

(6) It is recommended to dry motor the engine up to 5 minutes to help cool down the engine (Dry Motor Procedure, TASK 71-00-00-910-808-G00).

(a) Make sure that the Exhaust Gas Temperature (EGT) is below 150°F (66°C).

NOTE: The dry motoring will help to cool down the engine to start the work of the fuel nozzle [1] replacement.

B. References

Reference	Title
27-81-00-040-801	Leading Edge Flaps and Slats - Deactivation (P/B 201)
27-81-00-860-804	Leading Edge Flaps and Slats Retraction (P/B 201)
36-11-01-000-801	Engine Pneumatic Duct Removal (P/B 401)
36-11-04-000-801	PRSOV Removal (P/B 401)
36-11-07-000-801	HPSOV Removal (P/B 401)
36-12-01-000-802	Bleed Air Precooler Removal with Engine Installed (P/B 401)
70-00-01-910-803-G00	Electrical Connector Disconnection and Connection (P/B 201)
71-00-00-910-808-G00	Dry Motor Procedure (P/B 201)
71-11-04-010-801-G00	Open the Fan Cowl Panels (Selection) (P/B 201)
73-21-64-000-801-G00	T3 Sensor Removal (P/B 401)
74-21-02-000-801-G00	Left Igniter Plug Removal (P/B 401)
74-21-02-000-802-G00	Right Igniter Plug Removal (P/B 401)
75-23-02-000-801-G00	Left Modulated Turbine Cooling (MTC) Valve Removal (P/B 401)

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(Continued)

Reference	Title
75-23-02-000-802-G00	Right Modulated Turbine Cooling (MTC) Valve Removal (P/B 401)
75-24-01-000-801-G00	Low Pressure Turbine Active Clearance Control (LPTACC) Valve Removal (P/B 401)
75-24-04-000-801-G00	High Pressure Turbine Active Clearance Control (HPTACC) Valve Removal (P/B 401)
75-34-01-000-801-G00	Reverse Bleed Check Valve Assembly Removal (P/B 401)
77-21-04-000-801-G00	Exhaust Gas Temperature Sensor Harness 1 Removal (P/B 401)
77-21-05-000-801-G00	Exhaust Gas Temperature Sensor Harness 2 Removal (P/B 401)
77-31-06-000-801-G00	Turbine Center Frame Accelerometer Removal (P/B 401)
78-31-00-010-801-G00	Open the Thrust Reverser (Selection) (P/B 201)
78-31-00-040-801-G00	Thrust Reverser Deactivation For Ground Maintenance (P/B 201)

C. Tools/Equipment

NOTE: When more than one tool part number is listed under the same "Reference" number, the tools shown are alternates to each other within the same airplane series. Tool part numbers that are replaced or non-procurable are preceded by "Opt:", which stands for Optional.

Reference	Description
SPL-25114	Device - Ultrasonic Part #: 9447M49P01 Supplier: 58828
SPL-25115	Kit - Adapters, Ultrasonic Part #: 956A1742P01 Supplier: 58828
SPL-26374	Removal Restraint Bolt - Fuel Nozzle Part #: 956A3605G01 Supplier: 58828
STD-200	Container - Fuel Resistant, 10-gallon (38-Liter)
STD-562	Lens - Magnifying, 10x
STD-858	Tag - DO NOT OPERATE
STD-1242	Hammer - Standard
STD-1285	Container - Fuel Resistant, 10 Gallon (38 Liter)
STD-3842	Punch - Drift
STD-6847	Ratchet

D. Consumable Materials

Reference	Description	Specification
D50186 [CP2691]	Fluid - Penetrating - Aerokroil	
G00834	Cloth - Lint-free Cotton	
G51403 [CP2166]	Tape - Polyester-Backed, Adhesive	

E. Location Zones

Zone	Area
411	Engine 1 - Engine
421	Engine 2 - Engine

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F. Access Panels

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

G. Prepare for the Removal

SUBTASK 73-11-05-860-019-G00

(1) Do these steps to make sure that the ENGINE START switch and the START LEVER switch are not operated:

(a) Make sure that the applicable ENGINE START switch, on the P5 overhead panel, is in the AUTO or OFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable ENGINE START switch.

(b) Make sure that the applicable START LEVER switch, on the P10 control stand, is in the CUTOFF position.

1) Put a DO NOT OPERATE tag, STD-858, on the applicable START LEVER switch.

SUBTASK 73-11-05-860-020-G00

(2) Open the applicable circuit breakers and install safety tags:

F/O Electrical System Panel, P6-3

Row	Col	Number	Name
B	3	C00360	FUEL SPAR VALVE ENG 2
B	4	C00359	FUEL SPAR VALVE ENG 1

SUBTASK 73-11-05-860-023-G00

(3) Do this task: Leading Edge Flaps and Slats Retraction, TASK 27-81-00-860-804.

SUBTASK 73-11-05-860-024-G00

(4) Do this task: Leading Edge Flaps and Slats - Deactivation, TASK 27-81-00-040-801.

SUBTASK 73-11-05-860-021-G00

(5) Do these tasks in sequence to safely open the left and right thrust reversers on the applicable engine:

WARNING DO THE DEACTIVATION PROCEDURE FOR THE THRUST REVERSER TO PREVENT THE OPERATION OF THE THRUST REVERSER. ACCIDENTAL OPERATION OF THE THRUST REVERSER CAN CAUSE INJURIES TO PERSONS OR DAMAGE TO EQUIPMENT.

(a) Do this task: Thrust Reverser Deactivation For Ground Maintenance, TASK 78-31-00-040-801-G00.

(b) Open the applicable left or right fan cowl panels (TASK 71-11-04-010-801-G00).

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1) Open these access panels:

Number	Name/Location
413	Left Fan Cowl, Engine 1
414	Right Fan Cowl, Engine 1
423	Left Fan Cowl, Engine 2
424	Right Fan Cowl, Engine 2

WARNING OBEY THE INSTRUCTIONS IN THE PROCEDURE TO OPEN THE THRUST REVERSERS. IF YOU DO NOT OBEY THE INSTRUCTIONS, INJURIES TO PERSONS AND DAMAGE TO EQUIPMENT CAN OCCUR.

(c) Open the applicable left or right thrust reversers (TASK 78-31-00-010-801-G00).

1) Open these access panels:

Number	Name/Location
415	Left Thrust Reverser, Engine 1
416	Right Thrust Reverser, Engine 1
425	Left Thrust Reverser, Engine 2
426	Right Thrust Reverser, Engine 2

SUBTASK 73-11-05-710-001-G00

(6) Read and obey the recommendation, while you do this procedure:

(a) Use good lighting and a 10x magnifying lens, STD-562, to help you see the left and right engine core areas.

(b) Catch the fuel that will drain from the combustor area in a 10 gallon (38 liter) fuel resistant container, STD-1285.

(c) Use lint-free cloth, G00834, or rags around the combustion case [5].

SUBTASK 73-11-05-000-034-G00

CAUTION BE PREPARED TO REPLACE DAMAGED INSERTS WHEN REMOVING FUEL NOZZLES AND CASE MOUNTED COMPONENTS. IF THE DAMAGED INSERTS ARE NOT REPAIRED OR REPLACED, DAMAGE TO THE ENGINE CAN OCCUR.

(7) To prevent broken bolts and damaged inserts during the removal process, do the following steps:

NOTE: The steps below apply to the 72 fuel nozzle pad bolts, and any bolts that can be removed from the engine cases to get access to the fuel nozzle pads (i.e., ECS pad, TBV pad, and/or 7th stage air manifolds). This is done to keep the possible risk of broken bolts and damaged inserts to a minimum.

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WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

(a) Identify all the fuel nozzles that you will remove and do an initial soak of the fuel nozzle bolts with penetrating fluid, D50186 [CP2691], at the start of the job.

NOTE: It is recommended to let the lubricant soak into the threads for a one hour.

NOTE: It is recommended to let the capillary action of penetrating fluid, D50186 [CP2691], penetrate while you remove the external hardware at the same time before you break the torque of the fuel nozzle bolts.

1) If it is necessary, use the removal restraint bolt, SPL-26374, to remove the fuel nozzle bolts.

NOTE: To help with the removal of the bolts, the use of the removal restraint bolt, SPL-26374, is recommended to hold the fuel nozzle in the position.

CAUTION IF THE FUEL NOZZLE BOLT SEIZES TO THE INSERT, IT CAN MOVE THE FUEL NOZZLE PAD. THIS CONDITION CAN CAUSE DAMAGE TO THE MIXER.

a) Remove the first engine bolt by application of the break-away torque.

b) If the bolt does not come free, the bolt is seized, try to remove the next bolt, until all engine bolts will be removed or identified as seized.

<1> Make sure that one bolt is removed at a time.

c) If the bolt does come free, completely remove the bolt.

<1> If the location where the engine bolt was removed has the bracket, install the seating bolt at the location.

<a> Hand tight the seating bolt.

<b> Do not tighten the seating bolt to more than 70 in-lb (7.9 N·m).

<2> If there is no bracket at this location, install the seating bolt and spacer.

<a> Hand tight the seating bolt.

<b> Do not tighten the seating bolt to more than 70 in-lb (7.9 N·m).

2) When the seating bolts are installed, remove the seized engine bolts.

3) When the seized engine bolts are removed, remove the seating bolts and spacers.

(b) If it is necessary, identify all the case pad bolts that you will remove and do an initial soak of all the case pad bolts listed below at the very start of the job.

1) At the fuel nozzle [1] position 2.

a) The four bolts [22] that attach the transient bleed air tube [19] to the case (View D, Figure 403).

2) At the fuel nozzle [1] position 3.

a) The four bolts [66] that attach the 4th-stage cooling air manifold [58] to the compressor case [65] (View E, Figure 404).

3) At the fuel nozzle [1] position 4.

a) The four bolts [91] that attach the 7th-stage air tube [90] to the HPT case [94] (View C, Figure 405).

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b) The four bolts [99] that attach the 7th-stage air tube [90] to the HPC extension case [101] (View E, Figure 405).

4) At the fuel nozzle [1] position 7.

a) The four bolts [225] that attach the start bleed tube [224] to the HPC extension case [101] (View F, Figure 408).

b) The two bolts [220] that attach the start bleed tube [224] to the combustion case [5] (View E, Figure 408).

5) At the fuel nozzle [1] position 8.

a) The four bolts [225] that attach the start bleed tube [224] to the HPC extension case [101] (View F, Figure 408).

b) The two bolts [220] that attach the start bleed bracket [219] to the combustion case [5] (View E, Figure 408).

c) The four bolts [260] that attach the ECS middle flange [262] on the combustion case [5].

6) At the fuel nozzle [1] position 9.

a) The four bolts [414] that attach the 4th stage air manifold [419] to the compressor case [65].

7) At the fuel nozzle [1] position 10.

a) The four bolts [311] that attach the 7-th stage air tube [310] to the compressor case [65].

b) The four bolts [316] that attach the 7-th stage air tube [310] to the HPT port [318].

c) The four bolts [414] that attach the 4th stage air manifold [419] to the compressor case [65].

8) At the fuel nozzle [1] position 12.

a) The four bolts [414] that attach the 4th stage air manifold [419] to the compressor case [65].

9) At the fuel nozzle [1] position 13.

a) The four bolts [453] that attach the 7th-stage air manifold [452] to the compressor case [65].

b) The four bolts [457] that attach the 7th-stage air manifold [452] to the compressor case [65].

10) At the fuel nozzle [1] position 17.

a) The four bolts [561] that attach the 4th-stage cooling air manifold [560] to the compressor case [65].

b) The four bolts [575] that attach the 7th-stage air tube [574] to the HPC extension case [101].

c) The four bolts [579] that attach the 7th-stage air tube [574] to the HPT port [581].

(c) Let the application soak for one hour.

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SUBTASK 73-11-05-000-035-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(8) Loosen the fuel manifolds B-nuts as follows and do these steps to drain the three fuel manifold sections as follows:

NOTE: There are no fuel manifold drain plugs on the engine. The three fuel manifold sections are drained by decreasing the fuel pressure and draining each fuel manifold tube. Each fuel manifold tube is independent and must be drained at its lowest connection point on each manifold section. Each manifold section has two separate manifold tubes.

NOTE: Both fuel manifold section tubes that supply the fuel nozzle to be removed must be drained before the fuel nozzle removal. The fuel manifolds can have a small amount of pressure remaining after the fuel manifold tube is drained.

(a) Remove the thrust reverser compression rod No. 2 (Figure 424).

(b) Put the fuel resistant container (10 gal), STD-200, below the fuel manifold drain point of the fuel nozzle [1] you will remove.

(c) Remove all the safety wire from the fuel nozzle B-nuts [4] where the fuel manifolds are drained.

(d) Loosen the fuel manifold B-nuts [4] at the lower 6 o'clock position.

1) Fully remove the B-nuts [4] from the fuel nozzle [1].

(e) Loosen the fuel manifold B-nuts [4] at the upper 12 o'clock position.

1) Fully remove the B-nuts [4] from the fuel nozzle [1].

(f) Drain all the fuel into fuel resistant container (10 gal), STD-200.

(g) Loosen the remaining B-nuts [4] from the engine.

SUBTASK 73-11-05-020-196-G00

CAUTION BE PREPARED TO REPLACE DAMAGED INSERTS WHEN REMOVING FUEL NOZZLES AND CASE MOUNTED COMPONENTS. IF THE DAMAGED INSERTS ARE NOT REPAIRED OR REPLACED, DAMAGE TO THE ENGINE CAN OCCUR.

(9) If you will remove multiple or the full set of fuel nozzles at the same time, it is preferential to remove the nozzles in sets:

NOTE: It may be necessary to loosen and partially remove an adjacent fuel nozzle in order to provide access to rotate the fuel nozzle out of the combustion case during fuel nozzle removal. Replace the fuel nozzle seal assembly if you loosen three or more fuel nozzle bolts.

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0124

NOTE: The left side Modulated Turbine Cooling (MTC) valve and actuator (ALF) are removed.

PTF ALL

NOTE: The right side MTC valve is installed between Fuel nozzles 3 & 4 and left side MTC valve is installed between Fuel nozzles 12 & 13. Nozzle removal in sets may avoid MTC valve removal.

NOTE: The fuel nozzle removal sequence is for reference only and may be changed as necessary.

(a) Fuel Nozzle (FN) at positions 1, 2, 3, 18, and 17 (removed at the 1:30 o'clock position)

NOTE: FN 2 and FN 3 need to be turned right to get FN 1 removed.

1) The FN 1 turn to the RIGHT and out of the case.
2) The FN 2 turn to the LEFT and out of the case.
3) The FN 3 turn to the LEFT and out of the case.
4) The FN 18 turn to the RIGHT and out of the case.
5) The FN 17 turn to the RIGHT and out of the case.

(b) FN 4, 5, 6, 7, 8, and 9 (removed at the 3 to 5 o'clock position)

NOTE: FN 5 and FN 6 need to be turned right to get FN 4 removed.

1) The FN 4 turn DOWN and out of the case.
2) The FN 5 turn UP and out of the case.
3) The FN 6 turn UP and out of the case.
4) The FN 7 turn UP and out of the case.
5) The FN 8 turn to the RIGHT and out of the case.
6) The FN 9 turn to the RIGHT and out of the case.

(c) The FN 11, 10, and 12 (removed at the 7 to 8 o'clock position)

NOTE: FN 12 needs to be turned up to get FN 11 removed.

1) The FN 11 turn UP and out of the case.
2) The FN 10 turn LEFT and out of the case.
3) The FN 12 turn DOWN and out of the case.

(d) FN 16, 15, 14, and 13 (removed at the 9 o'clock position).

NOTE: FN 15 and FN 14 need to be turned down to get FN 16 removed.

1) The FN 16 turn DOWN and out of the case.
2) The FN 15 turn UP and out of the case.
3) The FN 13 turn UP and out of the case.

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H. Fuel Nozzle Removal (Position 1)

SUBTASK 73-11-05-010-003-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) If necessary to gain access, disconnect the bleed air precooler (TASK 36-12-01-000-802).

NOTE: It is possible to remove fuel nozzles in positions 1 and 18 without disconnecting the bleed air precooler but access is limited. If additional access is desired, disconnect the bleed air precooler.

SUBTASK 73-11-05-000-036-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Remove the fuel nozzle [1] (position 1) as follows (View C, Figure 402):

(a) Disconnect the pilot primary main (PPM) and pilot secondary (PS) B-nuts [4] from the fuel nozzle [1] (position 1).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 1).

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(c) Remove the four fuel nozzle bolts [2] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [2].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [2] again.

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a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts.

4) Start the removal of the fuel nozzle bolts [2].

a) Use a ratchet, STD-6847, to turn the bolts [2] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [2] slowly.

NOTE: If you turn the fuel nozzle bolts [2] too fast, they can get hot and seize.

b) Turn the machine bolts [2] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [2] again.

e) If the bolts [2] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [2].

7) Discard the fuel nozzle bolts [2].

(d) Remove the fuel nozzle [1] (position 1) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 1) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 1) and move the mounting flange aft and down.

3) Turn the fuel nozzle [1] (position 1) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(e) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(f) Remove and discard the seal [3] from the fuel nozzle [1] (position 1) or the combustion case [5].

(g) Install protective covers on the fuel nozzle [1], fuel nozzle ports, PS and PPM fuel manifold connections.

(h) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(i) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(j) Put caps on the fuel nozzle inlet fittings and heat shield.

(k) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(l) Make a record of the engine TSN and CSN.

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I. Fuel Nozzle Removal (Position 2)

SUBTASK 73-11-05-010-004-G00

WARNING THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the engine crossover duct assembly between the bleed air precooler at the 12:00 o'clock position and Pressure Regulating and Shutoff Valve (PRSOV) at the 2:30 o'clock position (TASK 36-11-01-000-801).

SUBTASK 73-11-05-000-037-G00

(2) Remove the transient bleed air tube [19] as follows (View C and D, Figure 403):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0229

(a) Remove the three bolts [18] that attach the transient bleed air tube [19] to support bracket [20].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0229

(b) Remove the two bolts [18] that attach the transient bleed air tube [19] to support bracket [20].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(c) Remove the coupling clamp [21] from the transient bleed valve [47].

(d) Remove and discard the seal [21A] from the transient bleed air tube [19].

(e) Remove the four bolts [22] that attach transient bleed air tube [19] to the combustion case [5].

(f) Remove the transient bleed air tube [19] and discard the metal seal ring [23] from the combustion case [5].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

NOTE: The transient bleed air tube [19] is referenced as the reverse bleed check valve assembly in Post LEAP-1B Service Bulletin 72-0378.

(g) Remove the reverse bleed check valve assembly (TASK 75-34-01-000-801-G00).

PTF ALL

SUBTASK 73-11-05-000-038-G00

(3) Remove the booster anti-ice tube [37A] as follows (View E, Figure 403):

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(a) Remove the four bolts [42A] and two strap clamps [43A] that attach the booster anti-ice tube [37A] to the support brackets.

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(b) Remove the two bolts [42A] and strap clamp [43A] that attach the booster anti-ice tube [37A] to the support bracket.

PTF ALL

(c) Remove the coupling clamp [46A] and seal [45A] from the booster anti-ice tube [38A].

(d) Remove the coupling clamp [46A] and seal [45A] from the booster anti-ice tube [39A].

(e) Discard the seals [45A].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0144

(f) Remove the bolt [47A] that attaches the loop clamp [48A] to the support bracket.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0144

(g) Remove the two bolts [47A] that attach the strap clamp [43A] to the support bracket.

PTF ALL

(h) Remove the coupling clamp [40A] and seal [41A] from the booster anti-ice tube [44A].

1) Discard the seal [41A].

(i) Remove the booster anti-ice tube [37A].

SUBTASK 73-11-05-000-124-G00

(4) Remove the booster anti-ice tube [44A] as follows (View G, Figure 403):

(a) Remove the two bolts [50A] and clamp [49A] that attach the booster anti-ice tube [44A] to the support bracket.

(b) Remove the two bolts [53A] and clamp [49A] that attach the booster anti-ice tube [44A] to the support bracket.

(c) Remove the coupling clamp [51A] and seal [52A] that attach the booster anti-ice tube [44A] to the SBV valve.

1) Discard the seal [52A].

(d) Remove the booster anti-ice tube [44A] from the engine.

SUBTASK 73-11-05-000-039-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Remove the fuel nozzle [1] (position 2) as follows (View L, Figure 403):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 2).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 2).

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(c) Remove the four fuel nozzle bolts [40] as follows:

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1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [40].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [40] again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [40].

4) Start the removal of the fuel nozzle bolts [40].

a) Use a ratchet, STD-6847, to turn the bolts [40] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [40] slowly.

NOTE: If you turn the fuel nozzle bolts [40] too fast, they can get hot and seize.

b) Turn the machine bolts [40] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [40] again.

e) If the bolts [40] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [40].

7) Discard the fuel nozzle bolts [40].

(d) Remove the booster anti-ice tube bracket [45] as follows:

NOTE: The booster anti-ice bracket [45] is two brackets assembled together that span across the fuel nozzle [1] (position 2) and fuel nozzle [1] (position 3).

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CAUTION

BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

1) Remove the bolt [46] that attaches the bracket [45] to the fuel nozzle [1] (position 3).

a) Carefully shock the bolt [46] head with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolt shank, threads, or bolt point.

b) Break torque on the bolt [46].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING

REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION

DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

c) Apply penetrating fluid, D50186 [CP2691], to the bolt [46] again.

<1> Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolt.

d) Start the removal of the bolt [46].

<1> Use a ratchet, STD-6847, to turn the bolt [46] slowly.

e) If bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

<1> If it is necessary, use a ratchet, STD-6847, and turn the bolt [46].

NOTE: If you turn the bolt too fast, it can get hot and seize.

<2> Turn the machine bolt [46] counterclockwise.

<3> Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

<4> Loosen the bolt [46] again.

<5> If the bolt [46] cannot be removed, repeat the steps above.

f) Remove the bolt [46].

g) Discard the bolt [46].

(e) Remove the fuel nozzle [1] (position 2) from the combustion case [5] as follows:

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1) Hold the fuel nozzle [1] (position 2) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 2) and move the mounting flange aft and down.

3) Turn the fuel nozzle [1] (position 2) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(f) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(g) Remove and discard the seal [41] from the fuel nozzle [1] (position 2) or the combustion case [5].

(h) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(i) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(j) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(k) Put caps on the fuel nozzle inlet fittings and heat shield.

(l) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(m) Make a record of the engine TSN and CSN.

J. Fuel Nozzle Removal (Position 3)

SUBTASK 73-11-05-020-197-G00

WARNING THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the engine crossover duct assembly between the bleed air precooler at the 12:00 o'clock position and PRSOV at the 2:30 o'clock position (TASK 36-11-01-000-801).

SUBTASK 73-11-05-000-040-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(2) Remove the booster anti-ice tube [37A] as follows (View E, Figure 403):

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(a) Remove the four bolts [42A] and two strap clamps [43A] that attach the booster anti-ice tube [37A] to the support brackets.

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(b) Remove the two bolts [42A] and strap clamp [43A] that attach the booster anti-ice tube [37A] to the support bracket.

PTF ALL

(c) Remove the coupling clamp [46A] and seal [45A] from the booster anti-ice tube [38A].

(d) Remove the coupling clamp [46A] and seal [45A] from the booster anti-ice tube [39A].

(e) Discard the seals [45A].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0144

(f) Remove the bolt [47A] that attaches the loop clamp [48A] to the support bracket.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0144

(g) Remove the two bolts [47A] that attach the strap clamp [43A] to the support bracket.

PTF ALL

(h) Remove the coupling clamp [40A] and seal [41A] from the booster anti-ice tube [44A].

1) Discard the seal [41A].

(i) Remove the booster anti-ice tube [37A].

SUBTASK 73-11-05-000-125-G00

(3) Remove the booster anti-ice tube [44A] as follows (View G, Figure 403):

(a) Remove the two bolts [50A] and clamp [49A] that attach the booster anti-ice tube [44A] to the support bracket.

(b) Remove the two bolts [53A] and clamp [49A] that attach the booster anti-ice tube [44A] to the support bracket.

(c) Remove the coupling clamp [51A] and seal [52A] that attach the booster anti-ice tube [44A] to the SBV valve.

1) Discard the seal [52A].

(d) Remove the booster anti-ice tube [44A] from the engine.

SUBTASK 73-11-05-020-198-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

CAUTION BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

(4) Remove the 4th-stage cooling air manifold [58] as follows (View C, D, and E, Figure 404):

(a) Disconnect the LPT air supply tube [59] from the 4th-stage cooling air manifold [58].

(b) Remove the coupling clamp [61] that attaches the 4th-stage cooling air manifold [58] to the air supply adapter tube [62].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0004

(c) Remove the two bolts [64] that attach the 4th-stage cooling air manifold [58] to the air tube bracket [63].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0004

(d) Remove the two bolts [634] and strap clamp [635] that attach the 4th-stage cooling air manifold [58] to the air tube bracket [63].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0004

(e) Remove the bolt [80] that attaches the harness support bracket [78] to the compressor case [65].

PTF ALL

(f) Remove the two bolts [66] that attach the harness support bracket [78] and the 4th-stage cooling air manifold [58] to the compressor case [65].

(g) Remove the bolt [66] that attaches the support bracket [79] and the 4th-stage cooling air manifold [58] to the compressor case [65].

(h) Remove the bolt [66] that attaches the bolt [58] to the compressor case [65].

(i) Remove the 4th-stage cooling air manifold [58] and discard the seal ring [67] from the 4th-stage cooling air manifold [58] or the compressor case [65].

(j) Discard the seal [60] from the 4th-stage cooling air manifold [58] or the air supply adapter tube [62].

SUBTASK 73-11-05-860-022-G00

(5) If it is necessary, remove the right MTC valve (TASK 75-23-02-000-802-G00).

SUBTASK 73-11-05-000-041-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(6) Remove the fuel nozzle [1] (position 3) as follows (View G, Figure 404):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 3).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle position 3).

CAUTION

BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(c) Remove the four fuel nozzle bolts [69] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [69].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

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WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [69] again.
   a) Let the lubrication soak into the threads for another hour.
      NOTE: Do not use power tools when you remove the bolts [69].
4) Start the removal of the fuel nozzle bolts [69].
   a) Use a ratchet, STD-6847, to turn the bolts [69] slowly.
5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.
   NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.
   a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [69] slowly.
      NOTE: If you turn the fuel nozzle bolts [69] too fast, they can get hot and seize.
   b) Turn the machine bolts counterclockwise.
   c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.
   d) Loosen the bolts [69] again.
   e) If the bolts [69] cannot be removed, repeat the steps above.
6) Remove the four fuel nozzle bolts [69].
7) Discard the fuel nozzle bolts [69].

(d) Remove the booster anti-ice tube bracket [74] as follows:

NOTE: The booster anti-ice bracket [74] is two brackets assembled together that span across the fuel nozzle [1] (position 2) and fuel nozzle [1] (position 3).

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

1) Remove the two bolts [68] that attach the bracket [74] to the fuel nozzle [1] (position 2).
   a) Carefully shock the bolts [68] heads with a standard hammer, STD-1242, and drift punch, STD-3842.
      NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

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b) Break torque on the bolts [68].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING

REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION

DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts [68] again.

<1> Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [68].

d) Start the removal of the bolts [68].

<1> Use a ratchet, STD-6847, to turn the bolts [68] slowly.

e) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

<1> If it is necessary, use a ratchet, STD-6847, and turn the bolts [68] slowly.

NOTE: If you turn the bolts [68] too fast, they can get hot and seize.

<2> Turn the machine bolts counterclockwise.

<3> Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

<4> Loosen the bolts [68] again.

<5> If the bolts [68] cannot be removed, repeat the steps above.

f) Remove the bolts [68].

g) Discard the bolts [68].

(e) Remove the bolts [76] that attach the loop clamps [77], fuel manifold [71], and fuel manifold [73] to the bracket [75].

(f) Remove the bracket [75] from the engine core.

(g) Remove the fuel nozzle [1] (position 3) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 3) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 3) and move the mounting flange aft and down.

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3) Turn the fuel nozzle [1] (position 3) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(h) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(i) Remove and discard the seal [70] from the fuel nozzle [1] (position 3) or the combustion case [5].

(j) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and PS and PPM fuel manifold connections.

(k) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(l) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(m) Put caps on the fuel nozzle inlet fittings and heat shield.

(n) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(o) Make a record of the engine TSN and CSN.

K. Fuel Nozzle Removal (Position 4)

SUBTASK 73-11-05-000-042-G00

WARNING THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the engine crossover duct assembly between the bleed air precooler at the 12:00 o'clock position and PRSOV at the 2:30 o'clock position (TASK 36-11-01-000-801).

SUBTASK 73-11-05-000-043-G00

(2) Remove the PRSOV at the 2:30 o'clock position (TASK 36-11-04-000-801).

SUBTASK 73-11-05-020-199-G00

CAUTION BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(3) Remove the 7th-stage air tube [90] as follows (View C, D, and E, Figure 405):

(a) Remove the four bolts [91] that attach the bracket [93] and 7th-stage air tube [90] to the HPT case [94].

(b) Remove the bolt [87] and loop clamp [88] that attaches the air tube [89] to the bracket [93].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(c) Remove the two bolts [97] that attach the 7th-stage air tube [90] to the bracket [98].

(d) Remove the two bolts [95] that attach the 7th-stage air tube [90] flange to the 7th-stage air tube bracket [96].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(e) Remove the two bolts [95] that attach the 7th-stage air tube [90] flange to the 7th-stage air tube bracket [96].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0005

(f) Remove the two bolts [632] and strap clamp [633] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

PTF ALL

(g) Remove the bolts [99] that attach the 7th-stage air tube [90] to the HPC extension case [101].

(h) Remove the 7th-stage air tube [90] from the HPC extension case [101] and discard the seal ring [100].

(i) Remove the 7th-stage air tube [90] from the engine and discard the seal ring [92] and seal ring [100].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-020-200-G00

(4) Remove the bracket [98] as follows (Figure 406):

(a) Remove the two bolts [126] that attach the SBV TBV manifold [125] to the bracket [98] (View J, Figure 406).

(b) Remove the bolt [152] that attaches the CCC air manifold [120] to the bracket [98] (View S, Figure 406).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-044-G00

(5) Remove the bracket [98] as follows (Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] and SBV TBV manifold [125] to the bracket [98]

(b) Remove the bolt [152] that attaches the CCC air manifold [120] to the bracket [98].

(c) Remove the two bolts [191] that attach the strap clamp [191A] and BAI air tube [27] to the bracket [98].

(d) Remove the bolt [197] that attaches loop clamp [205] and drain tube [208] to the bracket [98].

(e) Remove the two bolts [206] that attach the bracket [98] to the support bracket [207].

PTF ALL

SUBTASK 73-11-05-010-005-G00

(6) If it is necessary, remove the right MTC valve (TASK 75-23-02-000-802-G00).

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SUBTASK 73-11-05-000-045-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) Remove the fuel nozzle [1] (position 4) as follows (View G, Figure 405):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 4).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 4).

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(c) Remove the four fuel nozzle bolts [102] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [102].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [102] again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [102].

4) Start the removal of the fuel nozzle bolts [102].

a) Use a ratchet, STD-6847, to turn the bolts [102] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

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a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [102] slowly.

NOTE: If you turn the fuel nozzle bolts [102] too fast, they can get hot and seize.

b) Turn the machine bolts [102] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts [102] shaft and in the bolt holes.

d) Loosen the bolts [102] again.

e) If the bolts [102] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [102].

7) Discard the fuel nozzle bolts [102].

(d) Remove the fuel nozzle [1] (position 4) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 4) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 4) and move the mounting flange aft and down.

3) Turn the fuel nozzle [1] (position 4) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(e) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(f) Remove and discard the seal [103] from the fuel nozzle [1] (position 4) or the combustion case [5].

(g) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and PS and PPM fuel manifold connections.

(h) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

I (i) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(j) Put caps on the fuel nozzle inlet fittings and heat shield.

(k) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(l) Make a record of the engine TSN and CSN.

L. Fuel Nozzle Removal (Position 5)

SUBTASK 73-11-05-010-006-G00

WARNING

THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(WARNING PRECEDES)

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the Environmental Control System (ECS) valves and ducts as necessary to get access:

(a) Remove the PRSOV (TASK 36-11-04-000-801).

(b) Remove the engine mix manifold duct (TASK 36-11-01-000-801).

SUBTASK 73-11-05-020-201-G00

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO LOOSEN THE ELECTRICAL CONNECTOR. DO NOT USE PLIERS THAT HAVE METAL JAWS. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(2) Disconnect the HJ8A harness [118] as follows (View E, F, and G, Figure 406):

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(a) Remove the bolt [119] that attaches the HJ8A harness [118] wye to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(b) Remove the bolt [119] that attaches the HJ8A harness [118] wye to the bracket [202].

PTF ALL

(c) Remove the bolt [121] that attaches loop clamp [122] to the CCC air manifold [120].

(d) Remove the HJ8A harness [118] from the push clamp [178] on the CCC air manifold [120].

(e) Disconnect the electrical connector [116] and electrical connector [117] of the HJ8A harness [118] from the TBV valve [123] and SBV valve [124] (TASK 70-00-01-910-803-G00).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-046-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(3) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

(c) Remove the two bolts [131] that attach the two loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] that attach the two loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the tube bundle bracket [138] to the support bracket [139].

(h) Remove the three bolts [140] that attach the three hinged loop clamps [141] to the fuel manifold brackets [142] and fuel manifold bracket [143].

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube brackets [148].

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-047-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(4) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

(c) Remove the two bolts [131] that attach the loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] and two bolts [134A] that attach the four loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the clamp assembly [136A] to the support bracket [139].

(h) Remove the two bolts [140] and bolt [140A] that attach the three hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube bracket [148].

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 73-11-05-000-048-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(5) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(c) Remove the bolts [129] that attach the two loop clamps [130] to the bracket [202].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(d) Remove the two bolts [131] that attach the loop clamps [132] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(e) Remove the two bolts [131] that attach the two loop clamps [132] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(f) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(g) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valves [124] (View CC).

(h) Remove the two bolts [134] and two bolts [131A] that attach the four loop clamps [135] to the manifold bracket [133].

(i) Remove the three bolts [137] and bolt [137A] that attach the four loop clamps [136B] to the support bracket [139].

(j) Remove the two bolts [140] that attach the two hinged loop clamps [141] to the fuel manifold brackets [142] and fuel manifold brackets [144].

(k) Remove the bolt [140] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142].

(l) Remove the bolt [179] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142] to the fuel manifold bracket [143].

(m) Remove the two bolts [145] that attach the fuel hydraulic support block [146] to the tube bracket [148].

(n) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(o) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-049-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(6) Remove the CCC air manifold [120] as follows (View R, S, T, U, V, W, and X, Figure 406):

(a) Remove the bolt [152] that attaches the CCC air manifold [120] to the bracket [98].

(b) Remove the two bolts [153] that attach the strap clamp [154] to the bracket [155].

(c) Remove the bolt [156] that attaches the loop clamp [157] to the support bracket [158].

(d) Remove the bolt [159] that attaches the CCC air manifold [120] to the support bracket [160].

(e) Loosed the hose clamp [163] that attaches the upper end of rubber bellows [161] to the CCC air manifold [120].

(f) Remove the two bolts [209] that attach the CCC air manifold [120] to the CCC duct support bracket [210].

(g) Loosen the hose clamp [180] that attaches the aft end of the flexible duct [162] to the CCC air manifold [120].

(h) Remove the CCC air manifold [120] from the rubber bellows [161] and flexible duct [162].

(i) Remove the bracket [98].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0135 OR POST SB-CFM-LEAP-1B-72-0050 OR POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-000-050-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(7) Remove the CCC air manifold [120] as follows (View R, S, T, U, V, W, and X, Figure 406):

(a) Remove the bolt [152] that attaches the CCC air manifold [120] to the bracket [98].

(b) Remove the two bolts [153] that attach the strap clamp [154] to the bracket [155].

(c) Remove the bolt [156] that attaches the loop clamp [157] to the support bracket [158].

(d) Remove the bolt [159] that attaches the CCC air manifold [120] to the support bracket [160].

(e) Loosen the hose clamp [163] that attaches the upper end of rubber bellows [161] to the CCC air manifold [120].

(f) Loosen the hose clamp [180] that attaches the aft end of flexible duct [162] to the CCC air manifold [120].

PTF ALL; ALL 737 MAX, PRE LEAP-1B SERVICE BULLETIN 72-0050 OR POST LEAP-1B SERVICE BULLETIN 72-0050

(g) Remove the two bolts [209] that attaches the CCC air manifold [120] to the CCC duct support bracket [210].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0135 OR POST SB-CFM-LEAP-1B-72-0050 OR POST SB-CFM-LEAP-1B-72-0378

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0135 OR POST SB-CFM-LEAP-1B-72-0050 OR POST SB-CFM-LEAP-1B-72-0378 (Continued)

(h) Remove the bolt [189] that attaches the bracket [155] to the fuel nozzle [1] (position 6) (View C, Figure 407).

(i) Remove the CCC air manifold [120] from the rubber bellows [161] and flexible duct [162].

(j) Remove the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

NOTE: Engines having POST 72-0217 or POST 72-0378 hardware do not require removal of the CCC air manifold [120].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

SUBTASK 73-11-05-010-007-G00

(8) Do these steps to remove the bracket [202] (View C, Figure 406):

(a) Remove two bolts [131A] and bolt [189] from nozzle position 6.

(b) Remove the bracket [202] from the engine core.

PTF ALL

SUBTASK 73-11-05-000-051-G00

CAUTION BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(9) Remove the 7th-stage air tube [90] as follows (View C, D, and E, Figure 405):

(a) Remove the four bolts [91] that attach the bracket [93] and 7th-stage air tube [90] to the HPT case [94].

(b) Remove the bolt [87] and loop clamp [88] that attaches the air tube [89] to the bracket [93].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(c) Remove the two bolts [97] that attach the 7th-stage air tube [90] to the bracket [98].

(d) Remove the two bolts [95] that attach the 7th-stage air tube [90] flange to the 7th-stage air tube bracket [96].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(e) Remove the two bolts [95] that attach the 7th-stage air tube [90] flange to the 7th-stage air tube bracket [96].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0005

(f) Remove the two bolts [632] and strap clamp [633] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(g) Remove the bolts [99] that attach the 7th-stage air tube [90] to the HPC extension case [101].

(h) Remove the 7th-stage air tube [90] from the HPC extension case [101] and discard the seal ring [100].

(i) Remove the 7th-stage air tube [90] from the engine and discard the seal ring [92] and seal ring [100].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-052-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(10) Remove the PS3 tube [164] as follows (View Y, Z, AA, and BB, Figure 406):

(a) Remove the bolt [167] that attaches the loop clamp [168] to the support bracket [166].

(b) Remove the bolt [169] that attaches the loop clamp [170] to the tube bracket [171].

(c) Disconnect the PS3 tube [164] from the PS3 port fitting [165] on the combustion case [5].

(d) Disconnect the PS3 tube [164] from the PS3 sense line [172].

(e) Remove the PS3 tube [164].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135 AND POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-053-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(11) Remove the PS3 tube [164] as follows (View Y, Z, AA, BB, and DD, Figure 406):

(a) Remove the bolt [167] that attaches the loop clamp [168] to the support bracket [166].

(b) Remove the bolt [169] that attaches the loop clamp [170] to the tube bracket [171].

(c) Disconnect the PS3 tube [164] from the PS3 port fitting [165] on the combustion case [5].

(d) Disconnect the PS3 tube [164] from the PS3 sense line [172].

(e) Remove the bolt [185] that attaches the loop clamp [186] and PS3 tube [164] to the support bracket [181].

(f) Remove the PS3 tube [164].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0135 AND POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-000-054-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(12) Remove the PS3 tube [164] as follows (View Y, Z, AA, and BB, Figure 406):

(a) Remove the bolt [636] that attaches the loop clamp [637] to the support bracket [638].

(b) Remove the bolt [639] that attaches the loop clamp [640] to the support bracket [641].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(c) Remove the bolt [642] that attaches the loop clamp [643] to the tube bracket [171].

(d) Cut and remove the safety wire or safety cable from the PS3 tube [164] that connects to the PS3 port fitting [165] on the combustion case [5].

(e) Disconnect the PS3 tube [164] from the PS3 port fitting [165] on the combustion case [5].

(f) Cut and remove the safety wire or safety cable from the PS3 tube [164] that connects to the sense line [172].

(g) Disconnect the PS3 tube [164] from the PS3 sense line [172].

(h) Remove the PS3 tube [164].

PTF ALL

SUBTASK 73-11-05-000-055-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(13) Remove the fuel nozzle [1] (position 5) as follows (View C, Figure 406):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 5).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 5).

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(c) Remove the four fuel nozzle bolts [173] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [173].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [173] again.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [173].

4) Start the removal of the fuel nozzle bolts [173].

a) Use a ratchet, STD-6847, to turn the bolts [173] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [173] slowly.

NOTE: If you turn the fuel nozzle bolts [173] too fast, they can get hot and seize.

b) Turn the machine bolts [173] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [173] again.

e) If the bolts [173] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [173].

7) Discard the fuel nozzle bolts [173].

(d) Remove the bracket [202A] from the engine core (View C, Figure 406).

(e) Remove the fuel nozzle [1] (position 5) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 5) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 5) and move the mounting flange aft and down

3) Turn the fuel nozzle [1] (position 5) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(f) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(g) Remove and discard the seal [175] from the fuel nozzles [1] (position 5) or the combustion case [5].

(h) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(i) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(j) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzles [1].

(k) Put caps on the fuel nozzle inlet fittings and heat shield.

(l) Make a record of the serial number and related cup position of the removed fuel nozzles [1].

(m) Make a record of the engine TSN and CSN.

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M. Fuel Nozzle Removal (Position 6)

SUBTASK 73-11-05-010-008-G00

WARNING THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the ECS valves and ducts as necessary to get access:

(a) Remove the High Pressure Shutoff Valve (HPSOV) (TASK 36-11-07-000-801).

(b) Remove the engine mix manifold duct (TASK 36-11-01-000-801).

SUBTASK 73-11-05-020-202-G00

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO LOOSEN THE ELECTRICAL CONNECTOR. DO NOT USE PLIERS THAT HAVE METAL JAWS. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(2) Disconnect the HJ8A harness [118] as follows (View E, F, and G, Figure 406):

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0050 OR POST SB-CFM-LEAP-1B-72-0135 OR PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(a) Remove the bolt [119] that attaches the HJ8A harness [118] wye to the CCC air manifold [120].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(b) Remove the bolt [119] that attaches the HJ8A harness [118] wye to the bracket [202].

PTF ALL

(c) Remove the bolt [121] that attaches loop clamp [122] to the CCC air manifold [120].

(d) Remove the HJ8A harness [118] from the push clamp [178] on the CCC air manifold [120].

(e) Disconnect the electrical connector [116] and electrical connector [117] of the HJ8A harness [118] from the TBV valve [123] and SBV valve [124] (TASK 70-00-01-910-803-G00).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-056-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(3) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(c) Remove the two bolts [131] that attach the two loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] that attach the two loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the tube bundle bracket [138] to the support bracket [139].

(h) Remove the three bolts [140] that attach the three hinged loop clamps [141] to the fuel manifold brackets [142] and fuel manifold bracket [143].

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube brackets [148].

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-057-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(4) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

(c) Remove the two bolts [131] that attach the loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] and two bolts [134A] that attach the four loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the clamp assembly [136A] to the support bracket [139].

(h) Remove the two bolts [140] and bolt [140A] that attach the three hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube bracket [148].

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 73-11-05-000-058-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(5) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0217

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(c) Remove the bolts [129] that attach the two loop clamps [130] to the bracket [202].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0217

(d) Remove the two bolts [131] that attach the loop clamps [132] to the CCC air manifold [120].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(e) Remove the two bolts [131] that attach the two loop clamps [132] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(f) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(g) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valves [124] (View CC).

(h) Remove the two bolts [134] and two bolts [131A] that attach the four loop clamps [135] to the manifold bracket [133].

(i) Remove the three bolts [137] and bolt [137A] that attach the four loop clamps [136B] to the support bracket [139].

(j) Remove the two bolts [140] that attach the two hinged loop clamps [141] to the fuel manifold brackets [142] and fuel manifold brackets [144].

(k) Remove the bolt [140] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(l) Remove the bolt [179] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142] to the fuel manifold bracket [143].

(m) Remove the two bolts [145] that attach the fuel hydraulic support block [146] to the tube bracket [148].

(n) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(o) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-059-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(6) Remove the CCC air manifold [120] as follows (View R, S, T, U, V, W, and X, Figure 406):

(a) Remove the bolt [152] that attaches the CCC air manifold [120] to the bracket [98].

(b) Remove the two bolts [153] that attach the strap clamp [154] to the bracket [155].

(c) Remove the bolt [156] that attaches the loop clamp [157] to the support bracket [158].

(d) Remove the bolt [159] that attaches the CCC air manifold [120] to the support bracket [160].

(e) Loosed the hose clamp [163] that attaches the upper end of rubber bellows [161] to the CCC air manifold [120].

(f) Remove the two bolts [209] that attaches the CCC air manifold [120] to the CCC duct support bracket [210].

(g) Loosen the hose clamp [180] that attaches the aft end of the flexible duct [162] to the CCC air manifold [120].

(h) Remove the CCC air manifold [120] from the rubber bellows [161] and flexible duct [162].

(i) Remove the bracket [98].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 OR POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-060-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(7) Remove the CCC air manifold [120] as follows (View R, S, T, U, V, W, and X, Figure 406):

(a) Remove the bolt [152] that attaches the CCC air manifold [120] to the bracket [98].

(b) Remove the two bolts [153] that attach the strap clamp [154] to the bracket [155].

(c) Remove the bolt [156] that attaches the loop clamp [157] to the support bracket [158].

(d) Remove the bolt [159] that attaches the CCC air manifold [120] to the support bracket [160].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 OR POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(e) Loosen the hose clamp [163] that attaches the upper end of rubber bellows [161] to the CCC air manifold [120].

(f) Loosen the hose clamp [180] that attaches the aft end of flexible duct [162] to the CCC air manifold [120].

PTF ALL; ALL 737 MAX, PRE LEAP-1B SERVICE BULLETIN 72-0050 OR POST LEAP-1B SERVICE BULLETIN 72-0050

(g) Remove the two bolts [209] that attaches the CCC air manifold [120] to the CCC duct support bracket [210].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 OR POST LEAP-1B SERVICE BULLETIN 72-0050

(h) Remove the bolt [189] that attaches the bracket [155] to the fuel nozzle [1] (position 6) (View C, Figure 407).

(i) Remove the CCC air manifold [120] from the rubber bellows [161] and flexible duct [162].

(j) Remove the CCC air manifold [120].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

NOTE: Engines not having POST 72-0217 hardware do not require removal of the CCC air manifold [120].

SUBTASK 73-11-05-000-061-G00

(8) Do these steps to remove the bracket [202] (View C, Figure 406):

(a) Remove two bolts [131A] and bolt [189] from nozzle position 6.

(b) Remove the bracket [202] from the engine core.

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-062-G00

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(9) Remove the CCC air manifold [120] as follows (View R, S, T, U, V, W, DD, and X, Figure 406):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0217

(a) Remove the bolt [152] that attaches the CCC air manifold [120] to the bracket [98].

(b) Remove the two bolts [153] that attach the strap clamp [154] to the bracket [155].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(c) Remove the bolt [156] that attaches the loop clamp [157] to the support bracket [158].

(d) Remove the bolt [159] that attaches the CCC air manifold [120] to the support brackets [160].

(e) Loosen the hose clamp [163] that attaches the upper end of rubber bellows [161] to the CCC air manifold [120].

(f) Loosen the hose clamp [180] that attaches the aft end of flexible duct [162] to the CCC air manifold [120].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(g) Remove the two bolts [209] that attach the CCC air manifold [120] to the CCC duct support bracket [210].

(h) Remove the bolt [189] that attach the bracket [155] to the fuel nozzle [1] (position 6) (View C, Figure 407).

(i) Remove the CCC air manifold [120] from the rubber bellows [161] and flexible duct [162].

(j) Remove the CCC air manifold [120].

PTF ALL

SUBTASK 73-11-05-000-063-G00

CAUTION

BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(10) Remove the 7th-stage air tube [90] as follows (View C, D, and E, Figure 405):

(a) Remove the four bolts [91] that attach the bracket [93] and 7th-stage air tube [90] to the HPT case [94].

(b) Remove the bolt [87] and loop clamp [88] that attaches the air tube [89] to the bracket [93].

(c) Remove the two bolts [97] that attach the 7th-stage air tube [90] to the bracket [98].

(d) Remove the two bolts [95] that attach the 7th-stage air tube [90] flange to the 7th-stage air tube bracket [96].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0005

(e) Remove the two bolts [632] and strap clamp [633] that attach the 7th-stage air tube [90] to the 7th-stage air tube bracket [96].

PTF ALL

(f) Remove the bolts [99] that attach the 7th-stage air tube [90] to the HPC extension case [101].

(g) Remove the 7th-stage air tube [90] from the HPC extension case [101] and discard the seal ring [100].

(h) Remove the 7th-stage air tube [90] from the engine and discard the seal ring [92] and seal ring [100].

SUBTASK 73-11-05-000-064-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(11) Remove the fuel nozzle [1] (position 6) as follows (View C and D, Figure 407):

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(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 6).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 6).

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(c) Remove the fuel nozzle bolt [189] and three bolts [193] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolt [189] and bolts [193].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts.

4) Start the removal of the fuel nozzle bolts.

a) Use a ratchet, STD-6847, to turn the bolts slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts slowly.

NOTE: If you turn the fuel nozzle bolts too fast, they can get hot and seize.

b) Turn the machine bolts counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts again.

e) If the bolts cannot be removed, repeat the steps above.

6) Remove the fuel nozzle bolt [189] and three bolts [193].

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7) Discard the fuel nozzle bolt [189] and bolts [193].

(d) Remove the two bolts [201] that attach the loop clamps [199] to the bracket [194].

(e) Remove the fuel nozzle [1] (position 6) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 6) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 6) and move the mounting flange aft and down

3) Turn the fuel nozzle [1] (position 6) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(f) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(g) Remove and discard the seal [195] from the fuel nozzle [1] (position 6) or the combustion case [5].

(h) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(i) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(j) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzles [1].

(k) Put caps on the fuel nozzle inlet fittings and heat shield.

(l) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(m) Make a record of the engine TSN and CSN.

N. Fuel Nozzle Removal (Position 7)

SUBTASK 73-11-05-010-009-G00

WARNING THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the ECS valves and ducts as necessary to get access:

(a) Remove the HPSOV (TASK 36-11-07-000-801).

(b) Remove the lower CDP duct assembly (TASK 36-11-01-000-801).

SUBTASK 73-11-05-020-203-G00

CAUTION BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(2) Remove the start bleed tube [224] as follows (View F, Figure 408):

(a) Remove the four bolts [225] that attaches the start bleed tube [224] to the HPC extension case [101].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0134

(b) Remove the two bolts [223] that attach the start bleed tube [224] to the start bleed bracket [219].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0134

(c) Remove the two bolts [223] and start bleed clamp [610] that attach the start bleed tube [224] to the start bleed bracket [219].

PTF ALL

(d) Remove the coupling clamp [227] and seal [222] that attaches the start bleed tube [224] to the SBV valve [124].

1) Discard the seal [222].

(e) Remove and discard the seal [226] from the start bleed tube [224] or the HPC extension case [101].

SUBTASK 73-11-05-020-204-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Remove the fuel nozzle [1] (position 7) as follows (View C, Figure 408):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 7).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 7).

(c) Remove the bolts [220] that attaches the start bleed bracket [219] to the combustion case [5].

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(d) Remove the two fuel nozzle bolts [213] and two bolts [383] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

2) Break torque on the fuel nozzle bolts [213] and bolts [383].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING

REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION

DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts.

4) Start the removal of the fuel nozzle bolts.

a) Use a ratchet, STD-6847, to turn the bolts slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts slowly.

NOTE: If you turn the fuel nozzle bolts too fast, they can get hot and seize.

b) Turn the machine bolts counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts again.

e) If the bolts cannot be removed, repeat the steps above.

6) Remove the two fuel nozzle bolts [213] and two bolts [383].

7) Discard the fuel nozzle bolts [213] and bolts [383].

(e) Remove the two bolts [384] that attach the brackets [385] to the support bracket [214].

(f) Remove the support bracket [214] for clearance.

(g) Remove the two bolts [217] that attach the loop clamps [216] to the support bracket [215].

(h) Remove the bolt [218] that support bracket [215] to the bracket [203].

(i) Remove the fuel nozzle [1] (position 7) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 7) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 7) and move the mounting flange aft and down

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3) Turn the fuel nozzle [1] (position 7) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(j) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(k) Remove and discard the seal [221] from the fuel nozzles [1] (position 7) or the combustion case [5].

(l) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(m) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(n) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(o) Put caps on the fuel nozzle inlet fittings and heat shield.

(p) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(q) Make a record of the engine TSN and CSN.

O. Fuel Nozzle Removal (Position 8)

SUBTASK 73-11-05-010-010-G00

WARNING THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the lower CDP duct assembly to get access (TASK 36-11-01-000-801).

SUBTASK 73-11-05-020-205-G00

CAUTION BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(2) Remove the start bleed tube [224] as follows (View F, Figure 408):

(a) Remove the four bolts [225] that attach the start bleed tube [224] to the HPC extension case [101].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0134

(b) Remove the two bolts [223] that attach the start bleed tube [224] to the start bleed bracket [219].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0134

(c) Remove the two bolts [223] and start bleed clamp [610] that attach the start bleed tube [224] to the start bleed bracket [219].

PTF ALL

(d) Remove the coupling clamp [227] and seal [222] that attaches the start bleed tube [224] to the SBV valve [124].

1) Discard the seal [222].

(e) Remove and discard the seal [226] from the start bleed tube [224] or the HPC extension case [101].

SUBTASK 73-11-05-020-206-G00

CAUTION BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

(3) Remove the start bleed bracket [219] as follows (View E, Figure 408):

(a) Remove the two bolts [220] that attach the start bleed bracket [219] to the combustion case [5].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-020-207-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(4) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

(c) Remove the two bolts [131] that attach the two loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] that attach the two loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the tube bundle bracket [138] to the support bracket [139].

(h) Remove the three bolts [140] that attach the three hinged loop clamps [141] to the fuel manifold brackets [142] and fuel manifold bracket [143].

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube brackets [148].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-020-208-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(5) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

(c) Remove the two bolts [131] that attach the loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] and two bolts [134A] that attach the four loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the clamp assembly [136A] to the support bracket [139].

(h) Remove the two bolts [140] and bolt [140A] that attach the three hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube bracket [148].

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 73-11-05-020-209-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(6) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0217

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(c) Remove the bolts [129] that attach the two loop clamps [130] to the bracket [202].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0217

(d) Remove the two bolts [131] that attach the loop clamps [132] to the CCC air manifold [120].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0217

(e) Remove the two bolts [131] that attach the two loop clamps [132] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(f) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(g) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valves [124] (View CC).

(h) Remove the two bolts [134] and two bolts [131A] that attach the four loop clamps [135] to the manifold bracket [133].

(i) Remove the three bolts [137] and bolt [137A] that attach the four loop clamps [136B] to the support bracket [139].

(j) Remove the two bolts [140] that attach the two hinged loop clamps [141] to the fuel manifold brackets [142] and fuel manifold brackets [144].

(k) Remove the bolt [140] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142].

(l) Remove the bolt [179] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142] to the fuel manifold bracket [143].

(m) Remove the two bolts [145] that attach the fuel hydraulic support block [146] to the tube bracket [148].

(n) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(o) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL

SUBTASK 73-11-05-020-210-G00

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(7) Remove the ECS middle flange [262] as follows (View E, Figure 409):

(a) Disconnect the sense lines manifold [277] B-nut from the ECS middle flange [262].

(b) Remove the four bolts [260] that attach the ECS middle flange [262] on the combustion case [5].

(c) Remove and discard the metal ring seal [261].

(d) Remove the ECS middle flange [262] from the engine.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-05-020-211-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(8) Remove the fuel nozzle [1] (position 8) as follows (View C, Figure 409):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 8).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 8).

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(c) Remove the four fuel nozzle bolts [2] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [2].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [2] again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [2].

4) Start the removal of the fuel nozzle bolts [2].

a) Use a ratchet, STD-6847, to turn the bolts [2] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [2] slowly.

NOTE: If you turn the fuel nozzle bolts [2] too fast, they can get hot and seize.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

b) Turn the machine bolts [2] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [2] again.

e) If the bolts [2] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [2].

7) Discard the fuel nozzle bolts [2].

(d) Remove the fuel nozzle [1] (position 8) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 8) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 8) and move the mounting flange aft and down

3) Turn the fuel nozzle [1] (position 8) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(e) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(f) Remove and discard the seal [3] from the fuel nozzle [1] (position 8) or the combustion case [5].

(g) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(h) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(i) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(j) Put caps on the fuel nozzle inlet fittings and heat shield.

(k) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(l) Make a record of the engine TSN and CSN.

P. Fuel Nozzle Removal (Position 9)

SUBTASK 73-11-05-010-011-G00

WARNING THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the lower CDP duct assembly to get access (TASK 36-11-01-000-801).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-05-020-212-G00

CAUTION

BE CAREFUL WHEN YOU ENGAGE AND DISENGAGE THE IGNITION LEAD CONNECTORS. YOU MUST USE AN AXIAL FORCE ONLY TO PUSH OR PULL THE LEAD. DO NOT SHAKE THE IGNITION LEAD CONNECTORS. IF YOU DO NOT FOLLOW THIS PROCEDURE, YOU CAN CAUSE DAMAGE TO THE EQUIPMENT.

(2) Disconnect the ignition leads [296] as follows (View F, Figure 410):

(a) Remove the two bolts [297] that attaches the two loop clamps [298] to the sense line bracket [397].

(b) Remove the igniter cooling shroud halves and left ignition lead from the igniter (TASK 74-21-02-000-801-G00).

(c) Remove the igniter cooling shroud halves and right ignition lead from the igniter (TASK 74-21-02-000-802-G00).

SUBTASK 73-11-05-000-065-G00

(3) Loosen the bifurcation drain manifold [265] as follows (View D, E, F, and R, Figure 410):

WARNING

THE OIL SUPPLY MANIFOLD IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL OR INSTALLATION TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(a) Disconnect the B-nut on anti-ice drain tube [269].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(b) Disconnect the B-nut [303] at the ACC valve drain tube [291].

(c) Remove the two bolts [266] that attach the strap clamp [267] and bifurcation drain manifold [265] to the support bracket [268].

(d) Disconnect the bifurcation drain manifold [265] at four locations in the TCF/HPT area [304] and B-nuts at the FWD end of the manifold (View D, Figure 410).

(e) Remove the bolt [309] that attaches the bifurcation drain manifold [265] to the bracket [308].

(f) Remove the bolt [481] that attaches the clamp [482] to the bifurcation drain manifold [265].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(g) Remove the two bolts [266] that attach the block clamp [651], drain tube [652], drain tube [653], drain tube [654], drain tube [655] to the support bracket [268].

PTF ALL

(h) Remove the two bolts [297] that attach the two loop clamps [298] and ignition lead [296] to the sense line bracket [397].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(i) Disconnect the drain tube [652], drain tube [653], drain tube [654], and drain tube [655] from the bifurcation drain manifold [265] and from the AFT end that connects the drain tubes to the auxiliary drain oil tube [305], C drain oli tube [306], and B drain oil tube [307].

PTF ALL

ECCN 9E991 BOEING PROPRIETARY - See title page for details

SUBTASK 73-11-05-000-066-G00

(4) Do these steps to remove the ACC valve drain tube [291] :

(a) Loosen the B-nut [303] on the ACC valve drain tube [291] at the bifurcation drain manifold [265].

(b) Remove the bolt [292] that attaches the loop clamp [293] to the fuel manifold bracket [142].

(c) Remove the bolt [294] that attaches the loop clamp [295] to the tube bracket [148].

(d) Disconnect the drain tube [435] B-nut from the ACC valve drain tube [291].

SUBTASK 73-11-05-000-067-G00

CAUTION

BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(5) Remove the 4th stage air manifold [419] as follows (View E, X, Y, and Z, Figure 411):

(a) Disconnect the Low Pressure Turbine (LPT) air supply tube B-nut [425] from the 4th stage air manifold [419].

(b) Remove the coupling clamp [422] that attaches the 4th stage air manifold [419] to the air adapter [426].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(c) Remove the two bolts [420] that attach the 4th stage air manifold [419] to the fuel tube bracket [421].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(d) Remove the two bolts [615A] and strap clamp [616A] that attach the 4th stage air manifold [419] to the bracket [617A].

PTF ALL

(e) Remove the four bolts [414] that attach the 4th stage air manifold [419] to the compressor case [65].

(f) Discard the seal [415] from the 4th stage air manifold [419] or the compressor case [65].

(g) Discard the seal [423] from the 4th stage air manifold [419] or the air adapter [426].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

SUBTASK 73-11-05-000-068-G00

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(6) Remove the oil tube [662] as follows (View A, S, T, U, V, W, X, and Y, Figure 410):

(a) Disconnect the oil tube [662] at its FWD end from the oil tube [664] and oil tube [663].

(b) Disconnect the oil tube [662] at the TCF area from the air adapter [426].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135 (Continued)

(c) Remove the bolt [668] from the clamp [669] that attaches the oil tube [662] to the support bracket [670].

(d) Remove the bolt [668] from the clamp [669] that attaches the oil tube [662] to the support bracket [671].

(e) Remove the bolts [668] from the clamp [669] that attach the oil tube [662] to the bracket [674].

(f) Remove the bolt [668] from the clamp [669] that attaches the oil tube [662] to the bracket [675] .

(g) Remove the bolt [668] from the clamp [669] that attaches the oil tube [662] to the bracket [676].

(h) Remove the bolt [668] from the clamp [669] that attaches the oil tube [662] to the bracket [677].

PTF ALL

SUBTASK 73-11-05-000-069-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(7) Remove the oil tube [661] as follows (View A, S, T, U, V, W, and X, Figure 410):

(a) Disconnect the oil tube [661] at the FWD end from the oil tube [666].

(b) Disconnect the oil tube [661] at the AFT end from the oil tube [678].

(c) Remove the bolt [672] from the clamp [673] that attaches the oil tube [661] to the support bracket [671].

(d) Remove the bolts [672] from the clamps [673] that attach the oil tube [661] to the bracket [674].

(e) Remove the bolt [672] from the clamp [673] that attaches the oil tube [661] to the bracket [675].

(f) Remove the bolt [672] from the clamp [673] that attaches the oil tube [661] to the bracket [676].

SUBTASK 73-11-05-000-070-G00

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(8) Remove the fuel manifold [376] and fuel manifold [377] from the fuel nozzle [1] (positions 7, 8, and 10) (View G, Figure 410).

(a) Disconnect the fuel manifold [376] from the fuel nozzle [1] (positions 7, 8, and 10).

(b) Disconnect the fuel manifold [377] from the fuel nozzle [1] (positions 7, 8, and 10).

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(c) Disconnect the B-nuts [395] that connect the fuel hose [376A] to fuel manifold [376] and fuel manifold [377] to fuel manifold [377A].

(d) Remove the two bolts [373] from the two loop clamps [374] that attach the fuel manifold [377] to the bracket [396].

(e) Remove the two bolts [373] from the two loop clamps [374] that attach the fuel manifold [376] to the bracket [396].

SUBTASK 73-11-05-000-071-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(9) Remove the fuel nozzle [1] (position 9) as follows (View C, Figure 410):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 9).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle position 9).

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(c) Remove the four fuel nozzle bolts [2] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [2].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [2] again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [2].

4) Start the removal of the fuel nozzle bolts [2].

a) Use a ratchet, STD-6847, to turn the bolts [2] slowly.

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5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [2] slowly.

NOTE: If you turn the fuel nozzle bolts [2] too fast, they can get hot and seize.

b) Turn the machine bolts [2] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [2] again.

e) If the bolts [2] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [2].

7) Discard the fuel nozzle bolts [2].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(d) Remove the bracket [650].

PTF ALL

(e) Remove the fuel nozzle [1] (position 9) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 9) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] .from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 9) and move the mounting flange aft and down

3) Turn the fuel nozzle [1] (position 9) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(f) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(g) Remove and discard the seal [3] from the fuel nozzle [1] (position 9) or the combustion case [5].

(h) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(i) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(j) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(k) Put caps on the fuel nozzle inlet fittings and heat shield.

(l) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(m) Make a record of the engine TSN and CSN.

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Q. Fuel Nozzle Removal (Position 10)

SUBTASK 73-11-05-010-012-G00

WARNING THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the upper CDP duct assembly to get access (TASK 36-11-01-000-801).

SUBTASK 73-11-05-000-072-G00

CAUTION BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(2) Remove the 7-th stage air tube [310] as follows (View E, F, G, and H, Figure 411):

(a) Remove the four bolts [311] that attach the 7-th stage air tube [310] to the compressor case [65].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0135

(b) Remove the four bolts [315] that attach the 7-th stage air tube [310] to the support bracket [313] and support bracket [314].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0135

(c) Remove the bolts [611] that attach the block clamp [612] to the bracket [613].

(d) Remove the two bolts [614] that attach the bracket [613] to the support bracket [313].

(e) Remove the bracket [613].

(f) Remove the two bolts [615] and strap clamps [616] that attach the 7-th stage air tube [310] to the brackets [617].

PTF ALL

(g) Remove the four bolts [316] that attach the 7-th stage air tube [310] to the HPT port [318].

(h) Remove and discard the seal [312] from the 7-th stage air tube [310] or the compressor case [65].

(i) Remove and discard the seal [317] from the 7-th stage air tube [310] or the HPT port [318].

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SUBTASK 73-11-05-000-073-G00

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(3) Remove the HPT ACC valve drain tube [352] as follows (View R, S, T, U, V, and W, Figure 411):

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(a) Put a fuel resistant container (10 gal), STD-200, below the HPT ACC valve drain tube [352] that you will remove (View W, Figure 411).

(b) Disconnect the HPT ACC valve drain tube [352] and B-nut [361] from the drain tube [436] and let the fuel drain into the fuel resistant container (10 gal), STD-200.

(c) Remove the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142] and fuel manifold bracket [144].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(d) Remove the bolt [362] that attaches the loop clamp [363] to the tube bracket [147] and tube bracket [148].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(e) Remove the bolt [362] that attaches the loop clamp [363] to the tube bracket [147].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(f) Disconnect the B-nut [365] that attaches the HPT ACC valve drain tube [352] to the LPT ACC valve drain tube [364].

(g) Remove the bolt [353] that attaches the loop clamp [354] to the support bracket [355].

(h) Remove the bolt [356] that attaches the loop clamp [357] to the ACC drain bracket [358].

(i) Disconnect the B-nut [380] that attaches the HPT ACC valve drain tube [352] to the ACC valve [379].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-020-217-G00

(4) Remove the fuel hydraulic drain tube [383] as follows (View R, U, V, and W, Figure 411):

(a) Put a fuel resistant container (10 gal), STD-200, below the fuel hydraulic drain tube [383] that you will remove (View W, Figure 411).

(b) Disconnect the fuel hydraulic drain tube [383] and B-nut [361] from the drain tube [436] and let the fuel drain into the fuel resistant container (10 gal), STD-200.

(c) Remove the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(d) Remove the bolt [362] that attaches the loop clamp [363] to the tube bracket [147].

(e) Remove the bolt [356] that attaches the loop clamp [357] to the ACC drain bracket [358].

NOTE: It is not necessary to remove the HPT ACC valve drain tube [352] from the engine.

(f) Disconnect the B-nut [384] that attaches the HPT ACC valve drain tube [352] to the fuel hydraulic drain tube [383].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-074-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(5) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

(c) Remove the two bolts [131] that attach the two loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] that attach the two loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the tube bundle bracket [138] to the support bracket [139].

(h) Remove the three bolts [140] that attach the three hinged loop clamps [141] to the fuel manifold brackets [142] and fuel manifold bracket [143].

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube brackets [148].

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-075-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(6) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

(c) Remove the two bolts [131] that attach the loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] and two bolts [134A] that attach the four loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the clamp assembly [136A] to the support bracket [139].

(h) Remove the two bolts [140] and bolt [140A] that attach the three hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube bracket [148].

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 73-11-05-000-076-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(7) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(c) Remove the bolts [129] that attach the two loop clamps [130] to the bracket [202].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(d) Remove the two bolts [131] that attach the loop clamps [132] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(e) Remove the two bolts [131] that attach the two loop clamps [132] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(f) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(g) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valves [124] (View CC).

(h) Remove the two bolts [134] and two bolts [131A] that attach the four loop clamps [135] to the manifold bracket [133].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(i) Remove the three bolts [137] and bolt [137A] that attach the four loop clamps [136B] to the support bracket [139].

(j) Remove the two bolts [140] that attach the two hinged loop clamps [141] to the fuel manifold brackets [142] and fuel manifold brackets [144].

(k) Remove the bolt [140] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142].

(l) Remove the bolt [179] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142] to the fuel manifold bracket [143].

(m) Remove the two bolts [145] that attach the fuel hydraulic support block [146] to the tube bracket [148].

(n) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(o) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL

SUBTASK 73-11-05-000-077-G00

(8) Remove the fuel manifold bracket [142] as follows (View N, Figure 406):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

(a) Remove the three bolts [140] that attach the SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [143] with the three hinged loop clamps [141].

(b) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

(c) Remove the two bolts [140] that attach the SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [143] with the two hinged loop clamps [141].

(d) Remove the bolt [140A] that attaches the SBV TBV manifold [125] to the fuel manifold bracket [142] with a hinged loop clamps [141].

(e) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(f) Remove the two bolts [140] that attach the SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [144] with the two hinged loop clamps [141].

(g) Remove the bolt [140] that attaches the SBV TBV manifold [125] to the fuel manifold bracket [142] and fuel manifold bracket [143] with a hinged loop clamps [141].

(h) Remove the bolt [179] that attaches the fuel manifold bracket [142] with a hinged loop clamps [141].

PTF ALL

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SUBTASK 73-11-05-000-078-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(9) Remove the fuel nozzle [1] (position 10) as follows (View C, Figure 411):

(a) If the fuel nozzles are difficult to remove, these steps are recommended for easy removal:

NOTE: Fuel nozzles installed in positions 10, 11, and 12 can be difficult to remove. The steps that follow are recommended to help in the removal of the fuel nozzles.

1) Turn the fuel nozzle [1] (position 10) towards the right side (aft looking forward) of the engine and partially remove from the combustion case [5] by moving it towards the fuel nozzle [1] (position 9).

NOTE: You will not be able to remove the fuel nozzle [1] (position 10) completely at this time, but it is necessary to move to get access to the fuel nozzle [1] (position 11).

2) Remove the fuel nozzle [1] (position 11) from the combustion case [5] by turning it in the direction of the right side (aft looking forward) of the engine and moving it in the direction of the fuel nozzle [1] (position 10).

3) Move the fuel nozzle [1] (position 10) to its installed position on the combustion case [5].

4) Turn the fuel nozzle [1] (position 10) in the direction of the left side (ALF) of the engine and remove it from the combustion case [5] completely.

5) Remove the fuel nozzle [1] (position 12) from the combustion case [5] by turning it in the direction of the fuel nozzle [1] (position 11) and fuel nozzle [1] (position 10).

(b) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 10).

(c) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 10).

(d) Remove the two bolts [373] that attach the loop clamps [374] to the support bracket [372].

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(e) Remove the four fuel nozzle bolts [366] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [366].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

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WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [366] again.
   a) Let the lubrication soak into the threads for another hour.
      NOTE: Do not use power tools when you remove the bolts [366].
4) Start the removal of the fuel nozzle bolts [366].
   a) Use a ratchet, STD-6847, to turn the bolts [366] slowly.
5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.
   NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.
   a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [366] slowly.
      NOTE: If you turn the fuel nozzle bolts [366] too fast, they can get hot and seize.
   b) Turn the machine bolts [366] counterclockwise.
   c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.
   d) Loosen the bolts [366] again.
   e) If the bolts [366] cannot be removed, repeat the steps above.
6) Remove the four fuel nozzle bolts [366].
7) Discard the fuel nozzle bolts [366].

(f) Remove the fuel nozzle [1] (position 10) from the combustion case [5] as follows:
1) Hold the fuel nozzle [1] (position 10) with your hand and move the mounting flange aft and down.
   NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).
2) Lift the fuel nozzle [1] (position 10) and move the mounting flange aft and down.
3) Turn the fuel nozzle [1] (position 10) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(g) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(h) Remove and discard the seal [378] from the fuel nozzles [1] (position 10) or the combustion case [5].

(i) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

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(j) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(k) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(l) Put caps on the fuel nozzle inlet fittings and heat shield.

(m) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(n) Make a record of the engine TSN and CSN.

R. Fuel Nozzle Removal (Position 11)

SUBTASK 73-11-05-010-013-G00

WARNING THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the upper CDP duct assembly to get access (TASK 36-11-01-000-801).

SUBTASK 73-11-05-000-079-G00

CAUTION USE PLIERS THAT HAVE TEFLON JAWS TO LOOSEN THE ELECTRICAL CONNECTOR. DO NOT USE PLIERS THAT HAVE METAL JAWS. DAMAGE TO THE ELECTRICAL CONNECTOR CAN OCCUR.

(2) Disconnect the electrical connector [438] from the fuel manifold pressure sensor [437] (TASK 70-00-01-910-803-G00) (View K, Figure 411).

SUBTASK 73-11-05-000-080-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Remove the fuel manifold hose [329] as follows (View K, Figure 411):

NOTE: The fuel manifold hose [329] can also be identified as pilot primary and main enriched (PPMe) fuel manifold.

(a) Disconnect the fuel manifold hose [329] and B-nut [331] from the SCU [151]

(b) Remove the bolt [328] that attaches the loop clamp [330] and fuel manifold hose [329] to the support bracket [326].

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SUBTASK 73-11-05-000-081-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Remove the fuel tube hose [319] as follows (View K, Figure 411):

NOTE: The fuel tube hose [319] can also be identified as pilot secondary (Psec) fuel manifold.

(a) Disconnect the fuel tube hose [319] and B-nut [322] from the SCU [151].

(b) Remove the bolt [320] that attaches the loop clamp [321] and fuel tube hose [319] to the support bracket [326].

SUBTASK 73-11-05-000-082-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Remove the fuel hose manifold [323] as follows (View J, K, M, and N, Figure 411):

NOTE: The fuel hose manifold [323] can also be identified as pilot primary and main non-enriched (PPMne) fuel manifold.

(a) Put a fuel resistant container (10 gal), STD-200, below fuel hose manifold [323] you will remove.

(b) Disconnect the fuel hose manifold [323] and B-nut [327] at the SCU [151] and let it drain into the fuel resistant container (10 gal), STD-200.

(c) Remove the bolt [324] that attaches the loop clamp [325] to the support bracket [326].

(d) Remove the bolt [338] that attaches the loop clamp [339] to the support bracket [340].

(e) Remove the bolt [341] that attaches the loop clamp [342] to the support bracket [343].

(f) Disconnect the fuel hose manifold [323] B-nut at the left primary fuel manifold [344] and let it drain into the fuel resistant container (10 gal), STD-200.

(g) Disconnect the fuel hose manifold [323] and B-nut [346] at the fuel tube [347] and let it drain into the fuel resistant container (10 gal), STD-200.

SUBTASK 73-11-05-000-083-G00

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(6) Remove the HPT ACC valve drain tube [352] as follows (View R, S, T, U, V, and W, Figure 411):

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(a) Put a fuel resistant container (10 gal), STD-200, below the HPT ACC valve drain tube [352] to be removed (View W).

(b) Disconnect the HPT ACC valve drain tube [352] and B-nut [361] from the drain tube [436] and let it drain into the fuel resistant container (10 gal), STD-200.

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PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378 (Continued)

(c) Remove the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142] and fuel manifold bracket [144].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(d) Remove the bolt [362] that attaches the loop clamp [363] to the tube bracket [147] and tube bracket [148].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(e) Remove the bolt [362] that attaches the loop clamp [363] to the tube bracket [147].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(f) Disconnect the B-nut [365] that attaches the HPT ACC valve drain tube [352] to the LPT ACC valve drain tube [364].

(g) Remove the bolt [353] that attaches the loop clamp [354] to the support bracket [355].

(h) Remove the bolt [356] that attaches the loop clamp [357] to the ACC drain bracket [358].

(i) Disconnect the B-nut [380] that attaches the HPT ACC valve drain tube [352] to the ACC valve [379].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-020-218-G00

(7) Remove the fuel hydraulic drain tube [383] as follows (View R, U, V, and W, Figure 411):

(a) Put a fuel resistant container (10 gal), STD-200, below the fuel hydraulic drain tube [383] that you will remove (View W, Figure 411).

(b) Disconnect the fuel hydraulic drain tube [383] and B-nut [361] from the drain tube [436] and let the fuel drain into the fuel resistant container (10 gal), STD-200.

(c) Remove the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(d) Remove the bolt [362] that attaches the loop clamp [363] to the tube bracket [147].

(e) Remove the bolt [356] that attaches the loop clamp [357] to the ACC drain bracket [358].

NOTE: It is not necessary to remove the HPT ACC valve drain tube [352] from the engine.

(f) Disconnect the B-nut [384] that attaches the HPT ACC valve drain tube [352] to the fuel hydraulic drain tube [383].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-084-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(8) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(c) Remove the two bolts [131] that attach the two loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] that attach the two loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the tube bundle bracket [138] to the support bracket [139].

(h) Remove the three bolts [140] that attach the three hinged loop clamps [141] to the fuel manifold brackets [142] and fuel manifold bracket [143].

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube brackets [148].

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-085-G00

(9) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

(c) Remove the two bolts [131] that attach the loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] and two bolts [134A] that attach the four loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the clamp assembly [136A] to the support bracket [139].

(h) Remove the two bolts [140] and bolt [140A] that attach the three hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube bracket [148].

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 73-11-05-000-086-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(10) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(c) Remove the bolts [129] that attach the two loop clamps [130] to the bracket [202].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(d) Remove the two bolts [131] that attach the loop clamps [132] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(e) Remove the two bolts [131] that attach the two loop clamps [132] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(f) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(g) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valves [124] (View CC).

(h) Remove the two bolts [134] and two bolts [131A] that attach the four loop clamps [135] to the manifold bracket [133].

(i) Remove the three bolts [137] and bolt [137A] that attach the four loop clamps [136B] to the support bracket [139].

(j) Remove the two bolts [140] that attach the two hinged loop clamps [141] to the fuel manifold brackets [142] and fuel manifold brackets [144].

(k) Remove the bolt [140] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142].

(l) Remove the bolt [179] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142] to the fuel manifold bracket [143].

(m) Remove the two bolts [145] that attach the fuel hydraulic support block [146] to the tube bracket [148].

(n) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(o) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL

SUBTASK 73-11-05-000-087-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(11) Remove the fuel nozzle [1] (position 11) as follows (View C, Figure 412):

(a) If the fuel nozzles are difficult to remove, these steps are recommended for easy removal:

NOTE: Fuel nozzles installed in positions 10, 11, and 12 can be difficult to remove. The steps that follow are recommended to help in the removal of the fuel nozzles.

1) Turn the fuel nozzle [1] (position 10) towards the right side (aft looking forward) of the engine and partially remove from the combustion case [5] by moving it towards the fuel nozzle [1] (position 9).

NOTE: You will not be able to remove the fuel nozzle [1] (position 10) completely at this time, but it is necessary to move to get access to the fuel nozzle [1] (position 11).

2) Remove the fuel nozzle [1] (position 11) from the combustion case [5] by turning it in the direction of the right side (aft looking forward) of the engine and moving it in the direction of the fuel nozzle [1] (position 10).

3) Move the fuel nozzle [1] (position 10) to its installed position on the combustion case [5].

4) Turn the fuel nozzle [1] (position 10) in the direction of the left side (aft looking forward) of the engine and remove it from the combustion case [5] completely.

5) Remove the fuel nozzle [1] (position 12) from the combustion case [5] by turning it in the direction of the fuel nozzle [1] (position 11) and fuel nozzle [1] (position 10).

(b) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 10).

(c) Remove the safety wire from the B-nuts [4] (at the fuel nozzle position 10).

(d) Remove the bolt [443] that attaches the loop clamps [442] and fuel manifold [440] to the fuel supply bracket [391].

(e) Remove the bolt [444] that attaches the loop clamp [441] and fuel manifold [439] to the fuel supply bracket [391].

(f) Remove the bolt [393] that attaches the fuel manifold bracket [394] to the fuel supply bracket [391].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(g) Remove the bolt [390] that attaches the tube bracket [147] to the fuel nozzle [1] (position 12).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(h) Remove the bolt [390] that attaches the tube bracket [147A] to the fuel nozzle [1] (position 12).

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 (Continued)

(i) Remove the bolt [390A] that attaches the tube bracket [147A] to the tube bracket [147].

PTF ALL

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(j) Remove the four fuel nozzle bolts [390] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [390].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [390] again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [390].

4) Start the removal of the fuel nozzle bolts [390].

a) Use a ratchet, STD-6847, to turn the bolts [390] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [390] slowly.

NOTE: If you turn the fuel nozzle bolts [390] too fast, they can get hot and seize.

b) Turn the machine bolts [390] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [390] again.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

e) If the bolts [390] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [390].

7) Discard the fuel nozzle bolts [390].

(k) Remove the fuel nozzle [1] (position 11) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 11) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 11) and move the mounting flange aft and down.

3) Turn the fuel nozzle [1] (position 11) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(l) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(m) Remove and discard the seal [392] from the fuel nozzles [1] (position 11) or the combustion case [5].

(n) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(o) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(p) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(q) Put caps on the fuel nozzle inlet fittings and heat shield.

(r) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(s) Make a record of the engine TSN and CSN.

S. Fuel Nozzle Removal (Position 12)

SUBTASK 73-11-05-010-014-G00

WARNING THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the upper CDP duct assembly to get access (TASK 36-11-01-000-801).

SUBTASK 73-11-05-000-088-G00

(2) Disconnect the electrical connector [438] from the fuel manifold pressure sensor [437] (TASK 70-00-01-910-803-G00), (View K, Figure 411).

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SUBTASK 73-11-05-000-089-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Remove the fuel tube hose [319] as follows (View K, N, and M, Figure 411):

NOTE: The fuel tube hose [319] can also be identified as pilot secondary (Psec) fuel manifold.

(a) Disconnect the fuel tube hose [319] and B-nut [322] at the SCU [151].

(b) Remove the bolt [320] that attaches loop clamp [321] and fuel tube hose [319] to the support bracket [326].

(c) Disconnect the fuel tube hose [319] at the fuel tube [382] and B-nut [381].

(d) Disconnect the fuel tube hose [319] and B-nut [348] from the fuel manifold [351].

(e) Remove the bolt [341] that attaches the loop clamp [342] and fuel tube hose [319] to the support bracket [343].

SUBTASK 73-11-05-000-090-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(4) Remove the fuel manifold hose [329] as follows (View K, Figure 411):

NOTE: The fuel manifold hose [329] can also be identified as pilot primary and main enriched (PPMe) fuel manifold.

(a) Disconnect the fuel manifold hose [329] and B-nut [331] from the SCU [151]

(b) Remove the bolt [328] that attaches the loop clamp [330] and fuel manifold hose [329] to the support bracket [326].

SUBTASK 73-11-05-000-091-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Remove the fuel hose manifold [323] as follows (View K, N, and M, Figure 411):

NOTE: The fuel hose manifold [323] can also be identified as pilot primary and main non-enriched (PPMne) fuel manifold.

(a) Disconnect the fuel hose manifold [323] and B-nut [327] from the SCU [151].

(b) Remove the bolt [324] that attaches the loop clamp [325] and fuel hose manifold [323] to the support bracket [326].

(c) Disconnect the left primary fuel manifold [344] and B-nut [375] from the fuel hose manifold [323].

(d) Disconnect the fuel hose manifold [323] and B-nut [346] from the fuel tube [347].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(e) Remove the bolt [341] that attaches the loop clamps [342] and fuel hose manifold [323] to the support bracket [343].

(f) Remove the bolt [338] that attaches the loop clamp [339] and fuel hose manifold [323] to the support bracket [340].

SUBTASK 73-11-05-000-092-G00

(6) Remove the LPT ACC valve drain tube [364] as follows (View D, Figure 413):

(a) Put a fuel resistant container (10 gal), STD-200, below the LPT ACC valve drain tube [364] that you will remove.

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(b) Disconnect the B-nut [430] from the HPT ACC valve drain tube [352] and let it drain into the fuel resistant container (10 gal), STD-200.

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(c) Disconnect the B-nut [430] from the fuel hydraulic drain tube [383] and let it drain into the fuel resistant container (10 gal), STD-200.

PTF ALL

(d) Disconnect the B-nut [410] from the LPT ACC fuel manifold [469] and let it drain into the fuel resistant container (10 gal), STD-200.

(e) Remove the bolt [411] that attaches the loop clamp [412] to the drain tube bracket [413].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-093-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(7) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

(c) Remove the two bolts [131] that attach the two loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] that attach the two loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the tube bundle bracket [138] to the support bracket [139].

(h) Remove the three bolts [140] that attach the three hinged loop clamps [141] to the fuel manifold brackets [142] and fuel manifold bracket [143].

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0050 (Continued)

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube brackets [148].

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0050

SUBTASK 73-11-05-000-094-G00

(8) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

(c) Remove the two bolts [131] that attach the loop clamps [132] to the CCC air manifold [120].

(d) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(e) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(f) Remove the two bolts [134] and two bolts [134A] that attach the four loop clamps [135] to the manifold bracket [133].

(g) Remove the two bolts [137] that attach the clamp assembly [136A] to the support bracket [139].

(h) Remove the two bolts [140] and bolt [140A] that attach the three hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [143].

(i) Remove the two bolts [179] that attach the fuel manifold bracket [144] to the fuel manifold bracket [142].

(j) Remove the four bolts [145] that attach the four hinged loop clamps [146] to the tube bracket [147] and tube bracket [148].

(k) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(l) Remove and discard the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

SUBTASK 73-11-05-000-095-G00

(9) Remove the SBV TBV manifold [125] as follows (View J, K, L, M, N, P, Q, and CC, Figure 406):

(a) Remove the two bolts [126] that attach the two loop clamps [127] to the bracket [98].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(b) Remove the two bolts [129] that attach the two loop clamps [130] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(c) Remove the two bolts [129] that attach the two loop clamps [130] to the bracket [202].

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378 (Continued)

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0217 OR PRE SB-CFM-LEAP-1B-72-0378

(d) Remove the two bolts [131] that attach the two loop clamps [132] to the CCC air manifold [120].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0217 OR POST SB-CFM-LEAP-1B-72-0378

(e) Remove the two bolts [131] that attach the two loop clamps [132] to the bracket [202].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(f) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the TBV valve [123] (View CC).

(g) Disconnect the two B-nuts that attach the SBV TBV manifold [125] to the SBV valve [124] (View CC).

(h) Remove the two bolts [134] and two bolts [134A] that attach the four loop clamps [135] to the manifold bracket [133].

(i) Remove the three bolts [137] and bolt [137A] that attach the four loop clamps [136B] to the support bracket [139].

(j) Remove the two bolts [140] that attach the two hinged loop clamps [141] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(k) Remove the bolt [140] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142].

(l) Remove the bolt [179] that attaches the hinged loop clamp [141] to the fuel manifold bracket [142] to the fuel manifold bracket [143].

(m) Remove the two bolts [145] that attach the fuel hydraulic support block [146] to the tube bracket [148].

(n) Remove the bolts [149] that attach the SBV TBV manifold [125] to the SCU [151].

(o) Remove the seal gasket [150] from the SBV TBV manifold [125] or the SCU [151].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-000-096-G00

(10) Remove the HPT ACC valve drain tube [352] as follows (View R, S, T, U, V, and W, Figure 411):

(a) Put a fuel resistant container (10 gal), STD-200, below the HPT ACC valve drain tube [352] to be removed (View W, Figure 411).

(b) Disconnect the HPT ACC valve drain tube [352] and B-nut [361] from the drain tube [436] and drain into the fuel resistant container (10 gal), STD-200.

(c) Remove the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142] and fuel manifold bracket [144].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(d) Remove the bolt [362] that attaches the loop clamp [363] to the tube bracket [147] and tube bracket [148].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(e) Remove the bolt [362] that attaches the loop clamp [363] to the tube bracket [147].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378 (Continued)

(f) Remove the bolt [353] that attaches the loop clamp [354] to the support bracket [355].

(g) Remove the bolt [356] that attaches the loop clamp [357] to the ACC drain bracket [358].

(h) Disconnect the B-nut [380] that attaches the HPT ACC valve drain tube [352] to the ACC valve [379].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-020-219-G00

(11) Remove the fuel hydraulic drain tube [383] as follows (View R, U, V, and W, Figure 411):

(a) Put a fuel resistant container (10 gal), STD-200, below the fuel hydraulic drain tube [383] that you will remove (View W, Figure 411).

(b) Disconnect the fuel hydraulic drain tube [383] and B-nut [361] from the drain tube [436] and let the fuel drain into the fuel resistant container (10 gal), STD-200.

(c) Remove the bolt [359] that attaches the loop clamp [360] to the fuel manifold bracket [142] and fuel manifold bracket [144].

(d) Remove the bolt [362] that attaches the loop clamp [363] to the tube bracket [147].

(e) Remove the bolt [356] that attaches the loop clamp [357] to the ACC drain bracket [358].

NOTE: It is not necessary to remove the HPT ACC valve drain tube [352] from the engine.

(f) Disconnect the B-nut [384] that attaches the HPT ACC valve drain tube [352] to the fuel hydraulic drain tube [383].

PTF ALL

SUBTASK 73-11-05-020-213-G00

(12) If it is necessary, remove the left MTC valve (TASK 75-23-02-000-801-G00).

SUBTASK 73-11-05-000-097-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(13) Remove the fuel nozzle [1] (position 12) as follows (View C, Figure 413):

NOTE: Fuel nozzles installed in positions 10, 11, and 12 can be difficult to remove. The steps that follow are recommended for easy removal:

(a) If the fuel nozzles are difficult to remove, these steps are recommended for easy removal:

NOTE: Fuel nozzles installed in positions 10, 11, and 12 can be difficult to remove. The steps that follow are recommended to help in the removal of the fuel nozzles.

1) Turn the fuel nozzle [1] (position 10) towards the right side (aft looking forward) of the engine and partially remove from the combustion case [5] by moving it towards the fuel nozzle [1] (position 9).

NOTE: You will not be able to remove the fuel nozzle [1] (position 10) completely at this time, but it is necessary to move to get access to the fuel nozzle [1] (position 11).

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2) Remove the fuel nozzle [1] (position 11) from the combustion case [5] by turning it in the direction of the right side (aft looking forward) of the engine and moving it in the direction of the fuel nozzle [1] (position 10).

3) Move the fuel nozzle [1] (position 10) to its installed position on the combustion case [5].

4) Turn the fuel nozzle [1] (position 10) in the direction of the left side (aft looking forward) of the engine and remove it from the combustion case [5] completely.

5) Remove the fuel nozzle [1] (position 12) from the combustion case [5] by turning it in the direction of the fuel nozzle [1] (position 11) and fuel nozzle [1] (position 10).

(b) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 12).

(c) Remove the safety wire from the B-nuts [4] (at the fuel nozzle position 12).

(d) Remove the bolt [427] that attaches the tube bracket [147] to the fuel nozzle [1] (position 11).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(e) Remove the bolt [427] that attaches the tube bracket [147] to the fuel nozzle [1] (position 12).

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(f) Remove the bolt [427] that attaches the tube bracket [147A] to the fuel nozzle [1] (position 12).

(g) Remove the bolt [390A] that attaches the tube bracket [147A] to the tube bracket [147].

PTF ALL

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(h) Remove the four fuel nozzle bolts [427] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [427].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [427] again.

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a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [427].

4) Start the removal of the fuel nozzle bolts [427].

a) Use a ratchet, STD-6847, to turn the bolts [427] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [427] slowly.

NOTE: If you turn the fuel nozzle bolts [427] too fast, they can get hot and seize.

b) Turn the machine bolts [427] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [427] again.

e) If the bolts [427] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [427].

7) Discard the fuel nozzle bolts [427].

(i) Remove the fuel nozzle [1] (position 12) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 12) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 12) and move the mounting flange aft and down.

3) Turn the fuel nozzle [1] (position 12) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(j) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(k) Remove and discard the seal [429] from the fuel nozzle [1] (position 12) or the combustion case [5].

(l) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(m) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(n) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(o) Put caps on the fuel nozzle inlet fittings and heat shield.

(p) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(q) Make a record of the engine TSN and CSN.

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T. Fuel Nozzle Removal (Position 13)

SUBTASK 73-11-05-020-214-G00

WARNING THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the upper CDP duct assembly to get access (TASK 36-11-01-000-801).

SUBTASK 73-11-05-000-098-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

CAUTION BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

(2) Remove the 4th stage air manifold [419] as follows (View E, F, G, H, and J, Figure 413):

(a) Disconnect the LPT air supply tube [424] and B-nut [425] from the 4th stage air manifold [419].

(b) Remove the coupling clamp [422] that attaches the 4th stage air manifold [419] to the air adapter [426].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0006

(c) Remove the two bolts [420] that attach the 4th stage air manifold [419] to the fuel tube bracket [421].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0006

(d) Remove the two bolts [628] and strap clamp [629] that attach the 4th stage air manifold [419] to the fuel tube bracket [421].

PTF ALL

(e) Remove the two bolts [416] that attach the strap clamp [417] to the support bracket [418].

(f) Remove the four bolts [414] that attach the 4th stage air manifold [419] to the compressor case [65].

(g) Discard the seal [415] from the 4th stage air manifold [419] or the compressor case [65].

(h) Discard the seal [423] from the 4th stage air manifold [419] or air adapter [426].

SUBTASK 73-11-05-000-099-G00

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

(CAUTION PRECEDES)

CAUTION

BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

(3) Remove the 7th-stage air manifold [452] as follows (View D, E, F, G, H, and J, Figure 414):

(a) Disconnect the air tube [460] and B-nut [461] from the 7th-stage air manifold [452].

(b) Remove the four bolts [453] that attach the 7th-stage air manifold [452] to the compressor case [65].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0007

(c) Remove the two bolts [450] that attach the 7th-stage air manifold [452] to the support bracket [451].

(d) Remove the two bolts [455] that attach the 7th-stage air manifold [452] to the support bracket [456].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0007

(e) Remove the two bolts [624] and strap clamp [625] that attach the 7th-stage air manifold [452] to the support bracket [451].

(f) Remove the two bolts [626] and strap clamp [627] that attach the 7th-stage air manifold [452] to the support bracket [456].

PTF ALL

(g) Remove the four bolts [457] that attach the 7th-stage air manifold [452] to the HPT port [459].

(h) Discard the seal [454] from the 7th-stage air manifold [452] or the compressor case [65].

(i) Discard the seal [458] from the 7th-stage air manifold [452] or the HPT port [459].

SUBTASK 73-11-05-000-100-G00

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(4) Remove the LPT ACC valve drain tube [364] as follows (View D, Figure 413):

(a) Remove the bolt [411] that attaches the loop clamp [412] to the drain tube bracket [413].

PTF ALL; AIRPLANES PRE SB-CFM-LEAP-1B-72-0378

(b) Disconnect the LPT ACC valve drain tube [364] and B-nut [430] that attach the LPT ACC valve drain tube [364] to the HPT ACC valve drain tube [352].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(c) Disconnect the LPT ACC valve drain tube [364] and B-nut [430] that attach the LPT ACC valve drain tube [364] to the fuel hydraulic drain tube [383].

PTF ALL

(d) Disconnect the LPT ACC valve drain tube [364] and B-nut [410] from the LPT ACC fuel manifold [469].

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SUBTASK 73-11-05-000-101-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(5) Remove the LPT ACC fuel manifold [469] as follows (View K, L, and M, Figure 414):

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(a) Remove the two bolts [466] that attach the loop clamps [467] to the support bracket [468].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0071 AND POST SB-CFM-LEAP-1B-72-0378

(b) Remove the two bolts [502A] that attach the two loop clamps [503A] to the support bracket [501A].

PTF ALL

(c) Disconnect the LPT ACC valve drain tube [364] and B-nut [465] from the LPT ACC fuel manifold [469].

(d) Disconnect the LPT ACC fuel manifold [469] and B-nut [463] from the hydraulic fuel manifold [462].

(e) Remove the four bolts [470] that attach the LPT ACC fuel manifold [469] to the ACC valve [472].

(f) Remove and discard the flange seal gasket [471].

SUBTASK 73-11-05-020-215-G00

(6) If it is necessary, remove the left MTC valve (TASK 75-23-02-000-801-G00).

SUBTASK 73-11-05-000-102-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(7) Remove the fuel nozzle [1] (position 13) as follows (View C, Figure 414):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 13).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 13).

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(c) Remove the four fuel nozzle bolts [474] as follows:

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1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [474].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [474] again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [474].

4) Start the removal of the fuel nozzle bolts [474].

a) Use a ratchet, STD-6847, to turn the bolts [474] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [474] slowly.

NOTE: If you turn the fuel nozzle bolts [474] too fast, they can get hot and seize.

b) Turn the machine bolts [474] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [474] again.

e) If the bolts [474] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [474].

7) Discard the fuel nozzle bolts [474].

(d) Remove the bolt [474] that attaches the drain tube bracket [428] to the fuel nozzle [1] (position 12).

(e) Remove the fuel nozzle [1] (position 13) from the combustion case [5] as follows:

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1) Hold the fuel nozzle [1] (position 13) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 13) and move the mounting flange aft and down

3) Turn the fuel nozzle [1] (position 13) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(f) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(g) Remove and discard the seal [475] from the fuel nozzle [1] (position 13) or the combustion case [5].

(h) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(i) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

I (j) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(k) Put caps on the fuel nozzle inlet fittings and heat shield.

(l) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(m) Make a record of the engine TSN and CSN.

U. Fuel Nozzle Removal (Position 14)

SUBTASK 73-11-05-020-216-G00

WARNING

THE ECS DUCT IS HEAVY AND AWKWARD AND MUST BE SUPPORTED AND SECURED PRIOR TO REMOVAL TO HELP PREVENT INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the upper CDP duct assembly to get access (TASK 36-11-01-000-801).

SUBTASK 73-11-05-000-103-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Remove the LPT ACC fuel manifold [469] as follows (View K, L, and M, Figure 414):

(a) Disconnect the LPT ACC valve drain tube [364] and B-nut [473] from the LPT ACC fuel manifold [469].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(b) Remove the two bolts [466] that attach the loop clamps [467] to the support bracket [468].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(c) Remove the two bolts [502A] that attach the two loop clamps [503A] to the support bracket [501A].

PTF ALL

(d) Disconnect the LPT ACC fuel tube [464] and B-nut [465] from the LPT ACC fuel manifold [469].

(e) Disconnect the LPT ACC fuel manifold [469] and B-nut [463] from the hydraulic fuel manifold [462].

(f) Remove the four bolts [470] that attach the LPT ACC fuel manifold [469] to the ACC valve [472].

(g) Remove and discard the flange seal gasket [471].

SUBTASK 73-11-05-000-104-G00

(3) Remove the Low Pressure Turbine Active Clearance Control (LPTACC) valve (TASK 75-24-01-000-801-G00).

SUBTASK 73-11-05-000-105-G00

(4) Remove the ACC wye duct (TASK 75-24-01-000-801-G00).

SUBTASK 73-11-05-000-106-G00

CAUTION

BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(5) Remove the 7th-stage air manifold [452] as follows (View D, E, F, G, H, and J, Figure 414):

(a) Disconnect the air tube [460] and B-nut [461] from the 7th-stage air manifold [452].

(b) Remove the four bolts [453] that attach the 7th-stage air manifold [452] to the compressor case [65].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0007

(c) Remove the two bolts [450] that attach the 7th-stage air manifold [452] to the support bracket [451].

(d) Remove the two bolts [455] that attach the 7th-stage air manifold [452] to the support bracket [456].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0007

(e) Remove the two bolts [624] and strap clamp [625] that attach the 7th-stage air manifold [452] to the support bracket [451].

(f) Remove the bolts [626] and strap clamp [627] that attach the 7th-stage air manifold [452] to the support bracket [456].

PTF ALL

(g) Remove the four bolts [457] that attach the 7th-stage air manifold [452] to the HPT port [459].

(h) Discard the seal [454] from the 7th-stage air manifold [452] or the compressor case [65].

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(i) Discard the seal [458] from the 7th-stage air manifold [452] or the HPT port [459].

SUBTASK 73-11-05-000-107-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(6) Remove the fuel nozzle [1] (position 14) as follows (View C, Figure 415):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 14).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 14).

(c) Remove the two bolts [502] that attach the two loop clamps [503] to the support bracket [501].

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(d) Remove the four fuel nozzle bolts [500] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [500].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [500] again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [500].

4) Start the removal of the fuel nozzle bolts [500].

a) Use a ratchet, STD-6847, to turn the bolts [500] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

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a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [500] slowly.

NOTE: If you turn the fuel nozzle bolts [500] too fast, they can get hot and seize.

b) Turn the machine bolts [500] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [500] again.

e) If the bolts [500] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [500].

7) Discard the fuel nozzle bolts [500].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071

(e) Remove the two bolts [502A] that attach the two loop clamps [503A] to the support bracket [501A].

PTF ALL

(f) Remove the fuel nozzle [1] (position 14) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 14) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 14) and move the mounting flange aft and down.

3) Turn the fuel nozzle [1] (position 14) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(g) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(h) Remove and discard the seal [506] from the fuel nozzle [1] (position 14) or the combustion case [5].

(i) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(j) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(k) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(l) Put caps on the fuel nozzle inlet fittings and heat shield.

(m) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(n) Make a record of the engine TSN and CSN.

V. Fuel Nozzle Removal (Position 15)

SUBTASK 73-11-05-000-108-G00

WARNING

YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

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(WARNING PRECEDES)

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the two bolts [525A] that attach the LPTACC valve [526B] to the support bracket [526] and support bracket [526A] (View E, Figure 416).

SUBTASK 73-11-05-000-109-G00

(2) Remove the High Pressure Turbine Active Clearance Control (HPTACC) valve (TASK 75-24-04-000-801-G00).

SUBTASK 73-11-05-000-110-G00

(3) Remove the ACC wye duct (TASK 75-24-01-000-801-G00).

SUBTASK 73-11-05-000-111-G00

(4) Remove the bolt [525] that attaches the support bracket [526] to the support bracket [527] (View E, Figure 416).

(a) Remove the support bracket [526] from the engine.

SUBTASK 73-11-05-000-112-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(5) Remove the fuel nozzle [1] (position 15) as follows (View C, Figure 416):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 15).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 15).

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(c) Remove the four fuel nozzle bolts [520] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [520].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

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WARNING: REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION: DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [520] again.
   a) Let the lubrication soak into the threads for another hour.
      NOTE: Do not use power tools when you remove the bolts [520].
4) Start the removal of the fuel nozzle bolts [520].
   a) Use a ratchet, STD-6847, to turn the bolts [520] slowly.
5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.
   NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.
   a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [520] slowly.
      NOTE: If you turn the fuel nozzle bolts [520] too fast, they can get hot and seize.
   b) Turn the machine bolts [520] counterclockwise.
   c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.
   d) Loosen the bolts [520] again.
   e) If the bolts [520] cannot be removed, repeat the steps above.
6) Remove the four fuel nozzle bolts [520].
7) Discard the fuel nozzle bolts [520].

(d) Remove the support bracket [468] with the support bracket [521] and support bracket [522] attached.

(e) Remove the fuel nozzle [1] (position 15) from the combustion case [5] as follows:
1) Hold the fuel nozzle [1] (position 15) with your hand and move the mounting flange aft and down.
   NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).
2) Lift the fuel nozzle [1] (position 15) and move the mounting flange aft and down.
3) Turn the fuel nozzle [1] (position 15) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(f) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(g) Remove and discard the seal [524] from the fuel nozzle [1] (position 15) or the combustion case [5].

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(h) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(i) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(j) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(k) Put caps on the fuel nozzle inlet fittings and heat shield.

(l) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(m) Make a record of the engine TSN and CSN.

W. Fuel Nozzle Removal (Position 16)

WARNING YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

SUBTASK 73-11-05-000-113-G00

(1) Remove the HPTACC valve (TASK 75-24-04-000-801-G00).

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

SUBTASK 73-11-05-000-114-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Disconnect the fuel hydraulic manifold [447] as follows (View D, E, F, G, and H, Figure 417):

(a) Remove the four bolts [448] that attach the fuel hydraulic manifold [447] to the SCU [151].

(b) Disconnect the fuel hydraulic manifold [447] B-nut from the ACC fuel tube [446].

(c) Disconnect the fuel hydraulic manifold [447] B-nut from the hydraulic fuel tube [529].

(d) Disconnect the hydraulic fuel tube [528]B-nut from the fuel hydraulic manifold [447].

(e) Disconnect the LPT ACC fuel manifold [469] B-nut from the fuel hydraulic manifold [447].

(f) Disconnect the fuel hydraulic manifold [447] B-nut from the fuel manifold [588].

(g) Disconnect the fuel manifold [588] B-nut from the fuel hydraulic manifold [447].

(h) Remove the three bolts [478] that attach the three loop clamps [477] and fuel hydraulic manifold [447] to the support bracket [445].

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PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124 (Continued)

(i) Remove the bolt [530] that attaches the loop clamp [531] and fuel hydraulic manifold [447] to the support bracket [532].

(j) Remove the bolt [549] that attaches the loop clamp [548] and fuel hydraulic manifold [447] to the support bracket [550].

(k) Remove the bolt [538] that attaches the loop clamp [539] and fuel hydraulic manifold [447] to the support bracket [550].

(l) Remove the bolt [536] that attaches the loop clamp [537] and fuel hydraulic manifold [447] to the support bracket [533].

(m) Remove the bolt [534] that attaches the loop clamp [535] and fuel hydraulic manifold [447] to the support bracket [533].

(n) Remove the bolt [552] that attaches the loop clamp [551] and fuel hydraulic manifold [447] to the support bracket [533].

(o) Remove the bolt [555] that attaches the loop clamp [554] and fuel hydraulic manifold [447] to the support bracket [533].

(p) Remove the bolt [557] that attaches the loop clamp [556] and fuel hydraulic manifold [447] to the support bracket [587].

(q) Remove the bolt [559] that attaches the loop clamp [558] and fuel hydraulic manifold [447] to the support bracket [587].

(r) Remove the bolt [592] that attaches the loop clamp [593] and fuel hydraulic manifold [447] to the support bracket [594].

(s) Remove the bolt [590] that attaches the loop clamp [591] and fuel hydraulic manifold [447] to the support bracket [594].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0071

(t) Remove the bolt [589] that attaches the fuel hydraulic manifold [447] to the compressor case [65].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0071 AND PRE LEAP-1B SERVICE BULLETIN 75-0004

(u) Remove the bolt [589] that attaches the loop clamp [597A] and fuel hydraulic manifold [447] to the support bracket [599].

PTF ALL; POST LEAP-1B SERVICE BULLETIN 75-0004

(v) Remove the bolt [589] that attaches the loop clamp [597A] and fuel hydraulic manifold [447] to the support bracket [599A].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 72-0124

(w) Remove the two bolts [598] that attach the two loop clamps [597] and fuel hydraulic manifold [447] to the support bracket [599].

(x) Remove and discard the gasket seal [476] from between the fuel hydraulic manifold [447] and SCU [151].

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PTF ALL; POST LEAP-1B SERVICE BULLETIN 72-0124

SUBTASK 73-11-05-000-115-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Disconnect the fuel hydraulic manifold [447] as follows (View D, E, F, and G, Figure 417):

(a) Remove the four bolts [448] that attach the fuel hydraulic manifold [447] to the SCU [151].

(b) Cut and remove the safety wire or safety cable from the fuel hydraulic manifold [447] B-nuts.

(c) Disconnect the fuel hydraulic manifold [447] B-nut from the ACC fuel tube [446].

(d) Disconnect the fuel hydraulic manifold [447] B-nut from the hydraulic fuel tube [528].

(e) Disconnect the fuel hydraulic manifold [447] B-nuts from the hydraulic fuel tube [529] at two locations.

(f) Remove the two bolts [478] that attach the fuel hydraulic support block [477], fuel hydraulic manifold [447], and ACC fuel tube [446] to the support bracket [445].

(g) Remove the bolt [538] that attaches the loop clamp [539] and fuel hydraulic manifold [447] to the support bracket [550].

(h) Remove the two bolts [534] that attach the two loop clamps [535] and fuel hydraulic manifold [447] to the support bracket [533].

(i) Remove and discard the gasket seal [476] from between the fuel hydraulic manifold [447] and SCU [151].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

SUBTASK 73-11-05-020-220-G00

(4) Disconnect the fuel hydraulic manifold [447] as follows (View D, E, F, G, XA, H, and XB, Figure 417):

(a) Remove the four bolts [448] that attach the fuel hydraulic manifold [447] to the SCU [151].

(b) Cut and remove the safety wire or safety cable from the fuel hydraulic manifold [447] B-nuts.

(c) Disconnect the fuel hydraulic manifold [447] B-nut from the ACC fuel tube [446].

(d) Disconnect the fuel hydraulic manifold [447] B-nut from the hydraulic fuel tube [528A].

(e) Disconnect the fuel hydraulic manifold [447] B-nuts from the hydraulic fuel tube [529A] at two locations.

(f) Remove the two bolts [478] that attach the fuel hydraulic support block [477], fuel hydraulic manifold [447], and ACC fuel tube [446] to the support bracket [445].

(g) Remove the bolt [538] that attaches the loop clamp [539] and fuel hydraulic manifold [447] to the support bracket [550].

(h) Remove the bolt [549] that attaches the loop clamp [548] and fuel hydraulic manifold [447] to the support bracket [550].

(i) Remove the two bolts [534] that attach the two loop clamps [535] and fuel hydraulic manifold [447] to the support bracket [533].

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PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378 (Continued)

(j) Remove the bolt [530] that attaches the loop clamp [531] and fuel hydraulic manifold [447] to the support bracket [532].

(k) Remove the two bolts [552] that attach the loop clamp [551] and fuel hydraulic manifold [447] to the support bracket [553].

(l) Remove the bolt [601] that attaches the loop clamp [600] and fuel hydraulic manifold [447] to the support bracket [602].

(m) Remove the bolt [557] that attaches the loop clamp [556] and fuel hydraulic manifold [447] to the support bracket [587].

(n) Remove the bolt [604] that attaches the loop clamp [605] and fuel hydraulic manifold [447] to the support bracket [603].

(o) Cut and remove the safety wire or safety cable from the fuel hydraulic manifold [447] B-nuts at the two locations.

(p) Disconnect the fuel hydraulic manifold [447] B-nuts from the hydraulic fuel tube [529] and hydraulic fuel tube [528A] at two locations.

(q) Remove the bolt [592] that attaches the loop clamp [593] and fuel hydraulic manifold [447] to the support bracket [594].

(r) Remove the bolt [606] that attaches the loop clamp [607] and fuel hydraulic manifold [447] to the support bracket [608].

(s) Remove the two bolts [598] that attach the two loop clamps [597] and fuel hydraulic manifold [447] to the support bracket [599].

(t) Cut and remove the safety wire or safety cable from the fuel hydraulic manifold [447] B-nuts at the two locations.

(u) Disconnect the fuel manifold [588] B-nut from the fuel hydraulic manifold [447].

(v) Remove and discard the gasket seal [476] from between the fuel hydraulic manifold [447] and SCU [151].

PTF ALL

SUBTASK 73-11-05-000-116-G00

(5) Remove the T3 sensor (TASK 73-21-64-000-801-G00).

SUBTASK 73-11-05-000-117-G00

(6) Disconnect the EGT sensor harness 1 and 2 as follows (TASK 77-21-04-000-801-G00 and TASK 77-21-05-000-801-G00):

(a) Disconnect the EGT sensor harness 1 and 2 at the connectors.

(b) Remove the EGT sensor harnesses from the mounting bracket.

SUBTASK 73-11-05-000-118-G00

(7) Disconnect the turbine center frame accelerometer at the connector and remove the turbine center frame accelerometer from the mounting bracket as follows:

(a) Do the applicable steps in this task: Turbine Center Frame Accelerometer Removal, TASK 77-31-06-000-801-G00.

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SUBTASK 73-11-05-000-119-G00

WARNING DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(8) Remove the fuel nozzle [1] (position 16) as follows (View C and J, Figure 417):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 16).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 16).

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(c) Remove the four fuel nozzle bolts [540] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [540].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [540] again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [540].

4) Start the removal of the fuel nozzle bolts [540].

a) Use a ratchet, STD-6847, to turn the bolts [540] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

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a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [540] slowly.

NOTE: If you turn the fuel nozzle bolts [540] too fast, they can get hot and seize.

b) Turn the machine bolts [540] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [540] again.

e) If the bolts [540] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [540].

7) Discard the fuel nozzle bolts [540].

(d) Remove the two bolts [543] that attach EGT harness [542] and two loop clamps [544] to the support bracket [541].

(e) Remove the bolt [545] that attaches the loop clamp [546] and T3 sensor [595] to the support bracket [541].

(f) Remove the bolt [545A] that attaches the loop clamp [546A] and TCF accelerometer cable [595A] to the support bracket [541].

(g) Remove the two bolts [596] that attach the support bracket [541] to the support bracket [584].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(h) Remove the two bolts [609] that attach the loop clamps [610], W200 electrical harness [611], and W300 electrical harness [612] to the support bracket [541] (View C and XC, Figure 417).

PTF ALL

(i) Remove the fuel nozzle [1] (position 16) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 16) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 16) and move the mounting flange aft and down.

3) Turn the fuel nozzle [1] (position 16) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(j) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(k) Remove and discard the seal [547] from the fuel nozzle [1] (position 14) or the combustion case [5].

(l) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(m) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(n) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(o) Put caps on the fuel nozzle inlet fittings and heat shield.

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(p) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(q) Make a record of the engine TSN and CSN.

X. Fuel Nozzle Removal (Position 17)

SUBTASK 73-11-05-000-120-G00

WARNING

YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

BE CAREFUL WHEN YOU BREAK THE TORQUE FROM THE BOLTS IN THE ENGINE CASES. IF YOU DO NOT OBEY, DAMAGE TO INSERTS CAN OCCUR.

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

(1) Remove the 4th-stage cooling air manifold [560] as follows (View D, E, F, G, H, and J, Figure 418):

(a) Disconnect the 4th-stage cooling air manifold [560] at the 4th-stage air tube [570] and B-nut [569].

(b) Remove the bolt [566] that attaches the loop clamp [567] to the support bracket [568].

(c) Remove the coupling clamp [571] that attaches the 4th-stage cooling air manifold [560] to the adapter [573].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0008

(d) Remove the two bolts [564] that attach the 4th-stage cooling air manifold [560] to the support bracket [565].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-75-0008 AND POST SB-CFM-LEAP-1B-72-0378

(e) Remove the two bolts [620] and strap clamp [621] that attach the 4th-stage cooling air manifold [560] to the support bracket [565].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(f) Remove the reverse bleed valve assembly (TASK 75-34-01-000-801-G00).

NOTE: The support bracket [562] is two brackets assembled together that provides attachment for the booster anti ice tube [67], HJ8A harness [72], and HJ8B harness [73].

(g) Remove the two bolts [68] and strap clamp [66] that attach the booster anti ice tube [67] to the support bracket [562] (View F, Figure 418).

(h) Remove the HJ8A harness [72] and HJ8B harness [73] from the push clamps [71] on the support bracket [562] (View F, Figure 418).

(i) Remove the HJ8A harness [72], W200 harness [74], and W300 harness [75] from the push clamps [71] on the harness support bracket [78] (View F, Figure 418).

ECCN 9E991 BOEING PROPRIETARY - See title page for details

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378 (Continued)

(j) Remove the two bolts [69] that attach the support bracket [76], support bracket [77], and harness support bracket [78] (View F, Figure 418).

1) Remove the harness support bracket [78] from the engine.

(k) Remove the two bolts [70] that attach the 4th-stage cooling air manifold [560] to the support bracket [79] (View FA, Figure 418).

PTF ALL

(l) Remove the two bolts [561] that attach the support bracket [562] and 4th-stage cooling air manifold [560] to the compressor case [65].

(m) Remove the remaining two bolts [561] that attach the 4th-stage cooling air manifold [560] to the compressor case [65].

(n) Remove the seal [563] from the 4th-stage cooling air manifold [560] or the compressor case [65].

(o) Remove the seal [572] from the 4th-stage cooling air manifold [560] or the adapter.

SUBTASK 73-11-05-000-121-G00

(2) Remove the 7th-stage air tube [574] as follows (View D, K, L, M, and N, Figure 418):

(a) Remove the four bolts [579] that attach the 7th-stage air tube [574] to the support bracket [578] and HPT port [581].

PTF ALL; PRE LEAP-1B SERVICE BULLETIN 75-0008

(b) Remove the two bolts [577] that attach the 7th-stage air tube [574] to the support bracket [565].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-75-0008 AND POST SB-CFM-LEAP-1B-72-0378

(c) Remove the two bolts [623] and strap clamp [622] that attach the 7th-stage air tube [574] to the support bracket [565].

PTF ALL

(d) Remove the four bolts [575] that attach the 7th-stage air tube [574] to the HPC extension case [101].

(e) Remove the seal [576] from the 7th-stage air tube [574] or the HPC extension case [101].

(f) Remove the seal [580] from the 7th-stage air tube [574] or the HPT port [581].

SUBTASK 73-11-05-000-122-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(3) Remove the fuel nozzle [1] (position 17) as follows (View C, Figure 418):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 17).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 17).

(c) Remove the two bolts [582] that attach the two loop clamps [583] to the support bracket [581A].

(d) Remove the two bolts [596] that attach the support bracket [541] to the support bracket [584].

ECCN 9E991 BOEING PROPRIETARY - See title page for details

CAUTION

BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(e) Remove the four fuel nozzle bolts [585] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [585].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING

REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION

DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [585] again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [585].

4) Start the removal of the fuel nozzle bolts [585].

a) Use a ratchet, STD-6847, to turn the bolts [585] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [585] slowly.

NOTE: If you turn the fuel nozzle bolts [585] too fast, they can get hot and seize.

b) Turn the machine bolts [585] counterclockwise.

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [585] again.

e) If the bolts [585] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [585].

7) Discard the fuel nozzle bolts [585].

(f) Remove the fuel nozzle [1] (position 17) from the combustion case [5] as follows:

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1) Hold the fuel nozzle [1] (position 17) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle [1] from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 17) and move the mounting flange aft and down.

3) Turn the fuel nozzle [1] (position 17) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(g) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(h) Remove and discard the seal [586] from the fuel nozzle [1] (position 14) or the combustion case [5].

(i) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(j) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(k) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

(l) Put caps on the fuel nozzle inlet fittings and heat shield.

(m) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(n) Make a record of the engine TSN and CSN.

Y. Fuel Nozzle Removal (Position 18)

WARNING

YOU MUST SUPPORT AND MAKE THE ECS DUCT SAFE BEFORE YOU INSTALL IT. THE ECS DUCT IS HEAVY AND HARD TO HANDLE. IF YOU DO NOT OBEY THESE INSTRUCTIONS, INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT CAN OCCUR.

CAUTION

MAKE SURE THAT YOU COVER ALL PORTS, OPENINGS, MATING SURFACES, AND CASES. IF YOU DO NOT OBEY, FOD CAN GO INTO THE COMPONENTS AND DAMAGE CAN OCCUR.

SUBTASK 73-11-05-010-015-G00

(1) If necessary to get access, disconnect the bleed air precooler (TASK 36-12-01-000-802).

NOTE: It is possible to remove fuel nozzles in position 1 and 18 without disconnecting the bleed air precooler but access is limited. If additional access is desired, disconnect the bleed air precooler.

SUBTASK 73-11-05-000-123-G00

WARNING

DO NOT GET FUEL IN YOUR MOUTH, EYES, OR ON YOUR SKIN. DO NOT BREATHE THE FUMES FROM THE FUEL. PUT ON GOGGLES, AND GLOVES WHEN YOU USE FUEL. KEEP FUEL AWAY FROM SPARKS, FLAME, AND HEAT. FUEL IS POISONOUS AND FLAMMABLE. FUEL CAN CAUSE INJURIES TO PERSONNEL AND DAMAGE TO EQUIPMENT.

(2) Remove the fuel nozzle [1] (position 18) as follows (View C, Figure 419):

(a) Disconnect the PPM and PS B-nuts [4] from the fuel nozzle [1] (position 18).

(b) Remove the safety wire from the B-nuts [4] (at the fuel nozzle [1] position 18).

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(c) Remove the two bolts [602] that attach the B-nuts [4] and two loop clamps [603] to the bracket [601].

PTF ALL; AIRPLANES POST SB-CFM-LEAP-1B-72-0378

(d) Remove the two bolts [614] that attach the support bracket [615] to the support bracket [616] (View E, Figure 419).

PTF ALL

CAUTION BE CAREFUL WHEN YOU SHOCK THE MACHINE BOLTS. IF YOU ARE NOT CAREFUL, YOU CAN CAUSE DAMAGE TO THE CASE OR SURROUNDING COMPONENTS.

(e) Remove the four fuel nozzle bolts [600] as follows:

1) Carefully shock the fuel nozzle bolt heads with a standard hammer, STD-1242, and drift punch, STD-3842.

NOTE: This shock can help to break loose any material buildup on bolts shank, threads, or bolt point.

2) Break torque on the fuel nozzle bolts [600].

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

WARNING REFER TO THE PRODUCT LABEL AND THE MANUFACTURER'S (MATERIAL) SAFETY DATA SHEET (SDS) FOR INSTRUCTIONS ON THE HAZARDS, STORAGE, SAFE HANDLING AND PROPER USE OF THESE PRODUCTS.

CAUTION DO NOT REMOVE THE BOLTS TOO QUICKLY. THE BOLTS CAN BECOME HOT AND SEIZE. IF YOU DO NOT OBEY, DAMAGE TO BOLTS AND THREADS CAN OCCUR.

3) Apply penetrating fluid, D50186 [CP2691], to the fuel nozzle bolts [600] again.

a) Let the lubrication soak into the threads for another hour.

NOTE: Do not use power tools when you remove the bolts [600].

4) Start the removal of the fuel nozzle bolts [600].

a) Use a ratchet, STD-6847, to turn the bolts [600] slowly.

5) If any bolt becomes too tight, apply penetrating fluid, D50186 [CP2691], onto the bolt shaft and in the bolt hole.

NOTE: Torque more than 35 ft-lb (47.5 N·m) can break the bolt. If the bolt is difficult to remove, use of the ultrasonic device, SPL-25114, with ultrasonic adapter kit, SPL-25115, is recommended.

a) If it is necessary, use a ratchet, STD-6847, and turn the fuel nozzle bolts [600] slowly.

NOTE: If you turn the fuel nozzle bolts [600] too fast, they can get hot and seize.

b) Turn the machine bolts [600] counterclockwise.

ECCN 9E991 BOEING PROPRIETARY - See title page for details

c) Apply penetrating fluid, D50186 [CP2691], to the bolts shaft and in the bolt holes.

d) Loosen the bolts [600] again.

e) If the bolts [600] cannot be removed, repeat the steps above.

6) Remove the four fuel nozzle bolts [600].

7) Discard the fuel nozzle bolts [600].

(f) Remove the fuel nozzle [1] (position 18) from the combustion case [5] as follows:

1) Hold the fuel nozzle [1] (position 18) with your hand and move the mounting flange aft and down.

NOTE: This will let you disengage the tip of the fuel nozzle from the combustor radial mixer (cup).

2) Lift the fuel nozzle [1] (position 18) and move the mounting flange aft and down

3) Turn the fuel nozzle [1] (position 18) 180 degrees to remove the fuel nozzle [1] from the combustion case port.

(g) Drain all residual fuel from the fuel nozzle [1] into the fuel resistant container (10 gal), STD-200.

(h) Remove and discard the seal [604] from the fuel nozzle [1] (position 14) or the combustion case [5].

(i) Install protective covers on the fuel nozzle [1], fuel nozzle ports, and fuel manifold connections.

(j) Cover the fuel nozzle pad with 3M 851 Tape, G51403 [CP2166], on the combustion case [5].

(k) Attach a DO NOT OPERATE tags, STD-858, to the fuel nozzle [1].

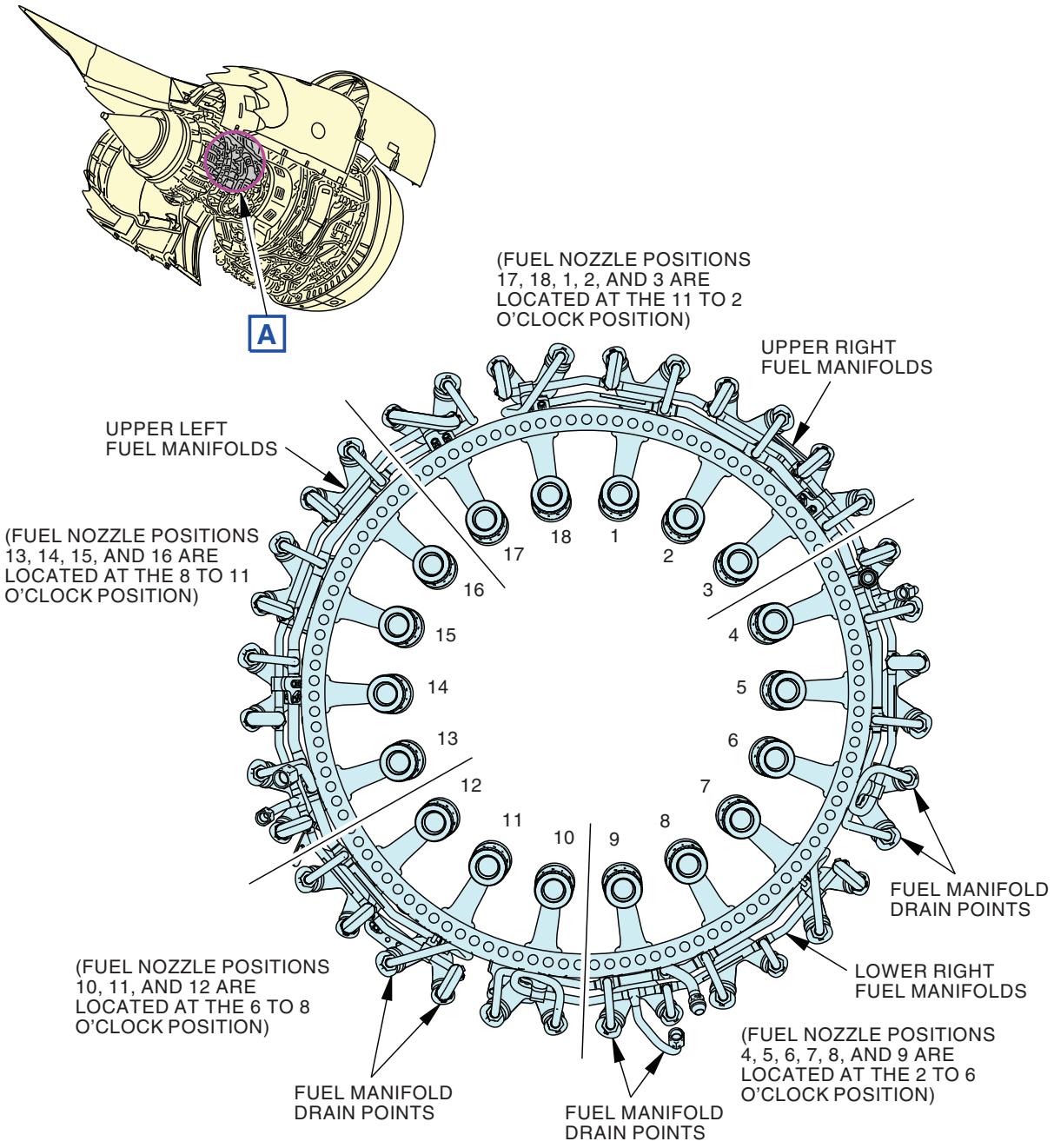
(l) Put caps on the fuel nozzle inlet fittings and aft heat shield.

(m) Make a record of the serial number and related cup position of the removed fuel nozzle [1].

(n) Make a record of the engine TSN and CSN.

——— END OF TASK ———

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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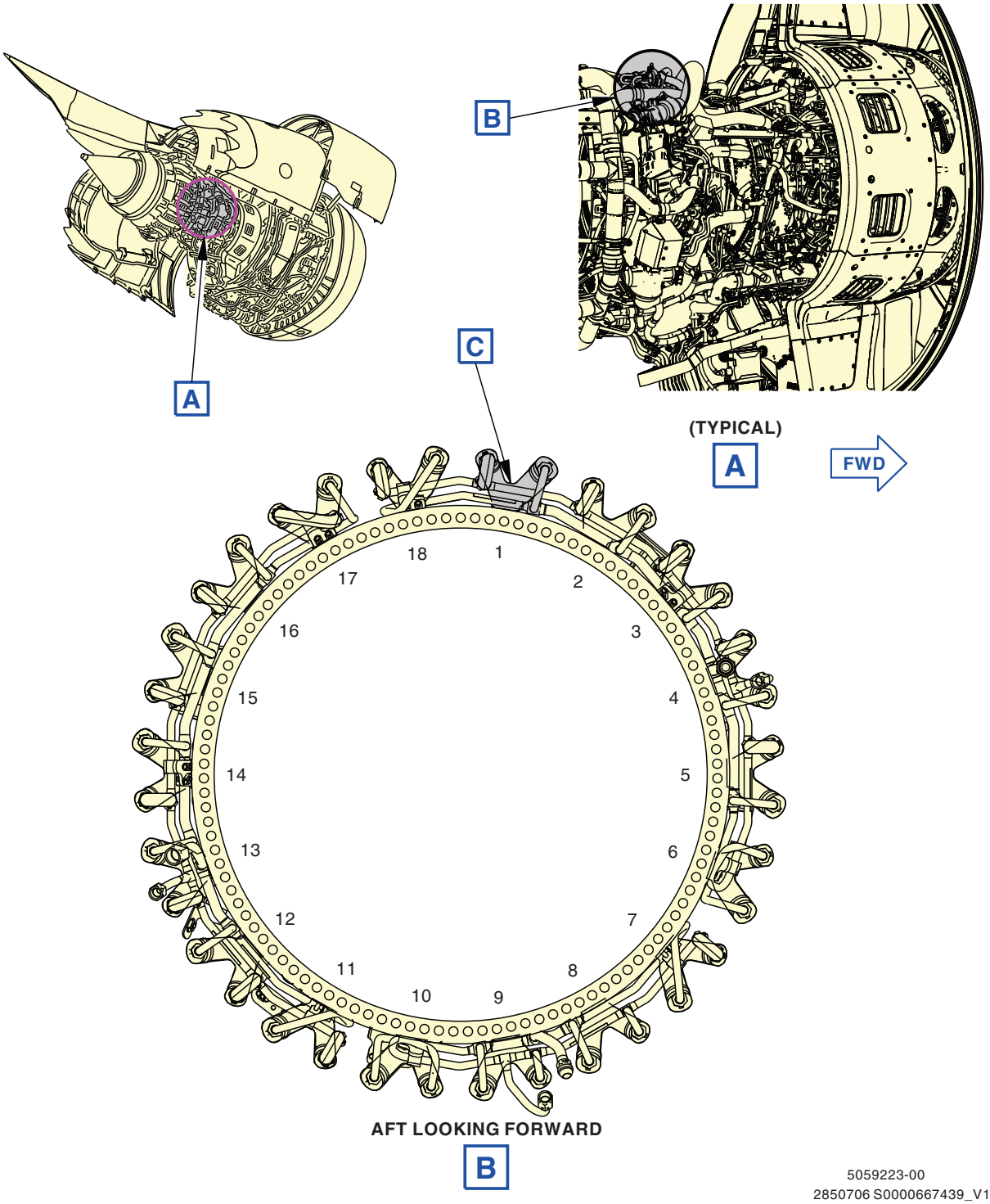
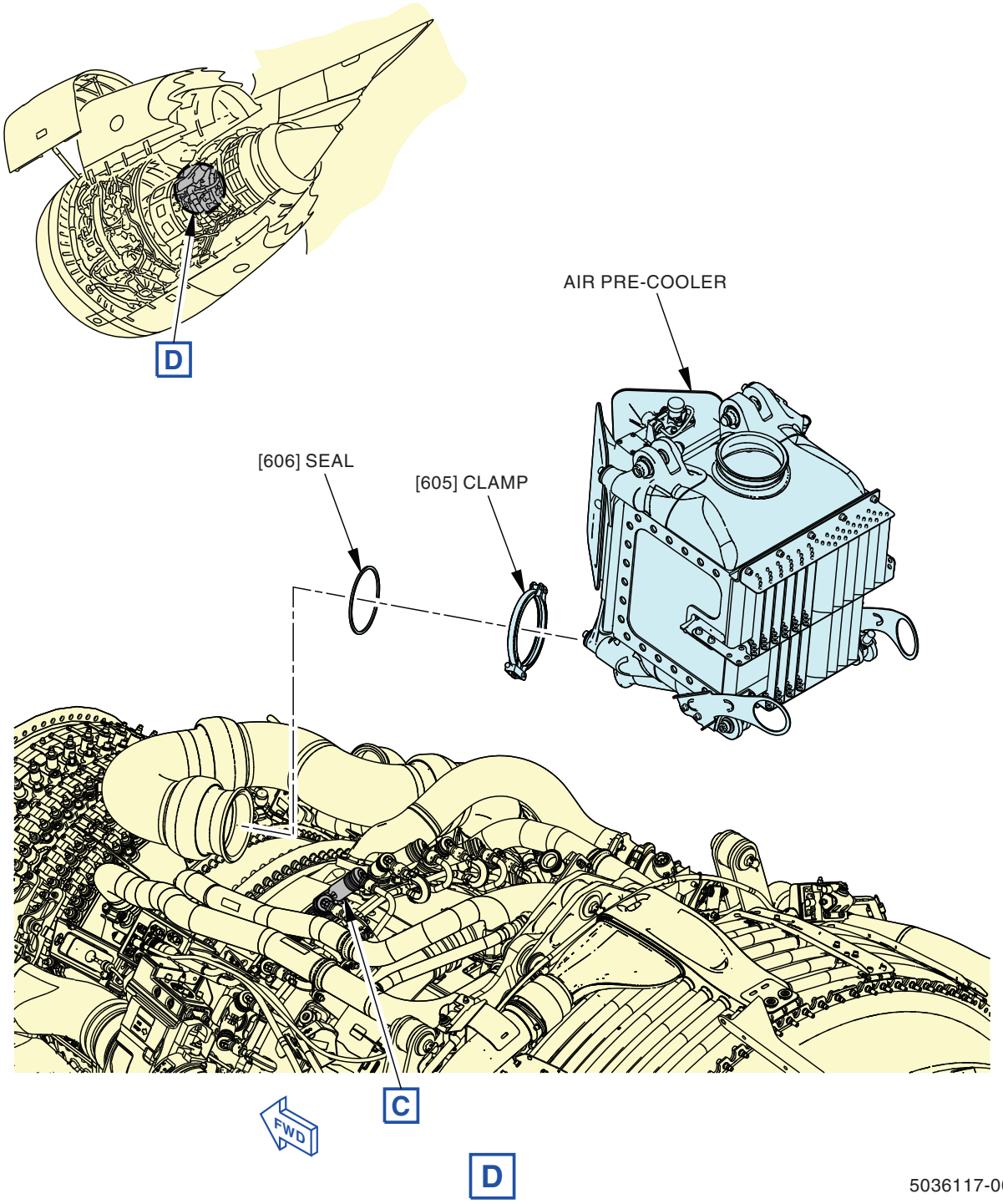
Fuel Nozzle Positions and Fuel Manifold Drain Points
Figure 401/73-11-05-990-887-G00

ECCN 9E991 BOEING PROPRIETARY - See title page for details



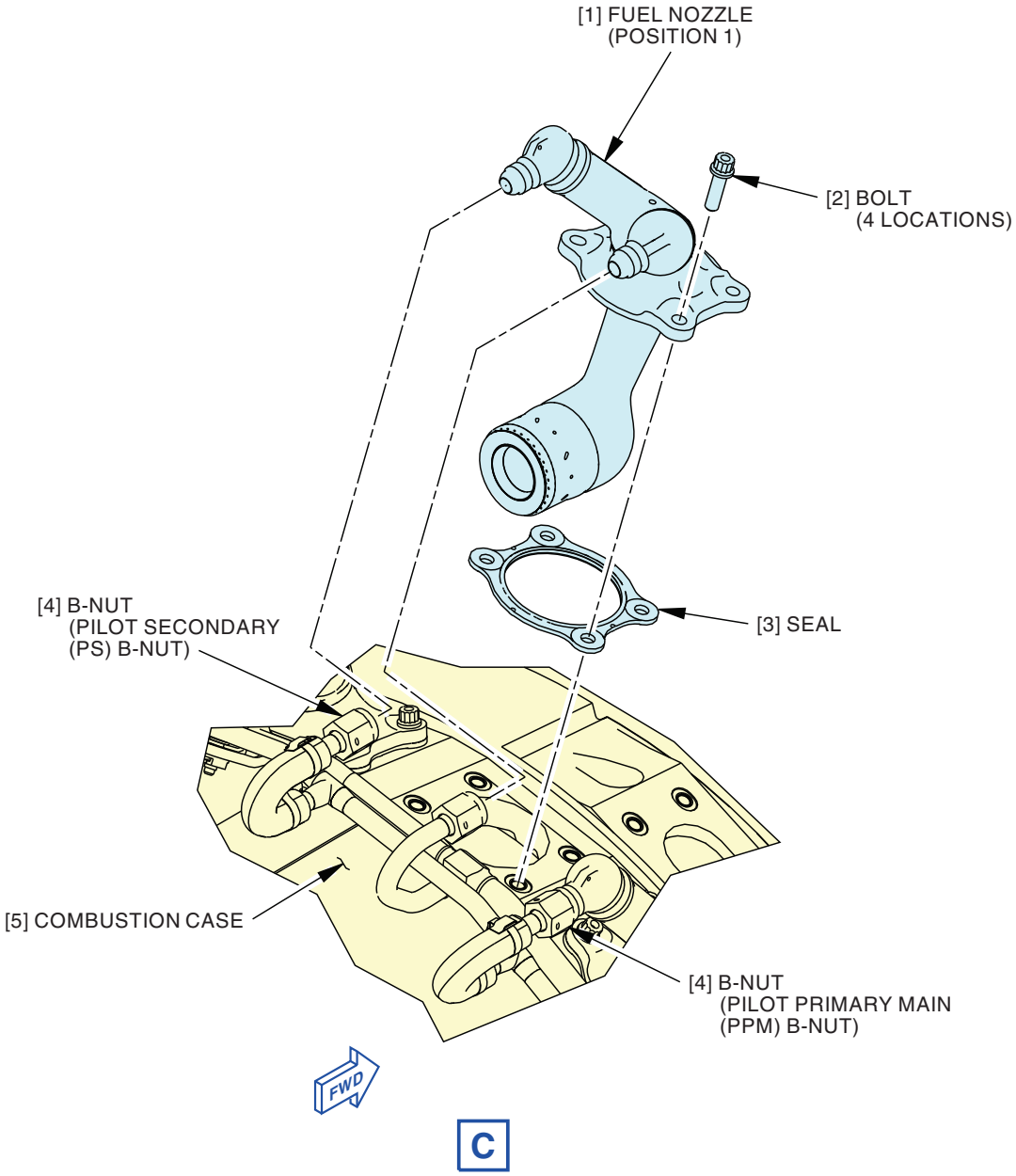
Fuel Nozzle (Position 1) Installation
Figure 402/73-11-05-990-888-G00 (Sheet 1 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



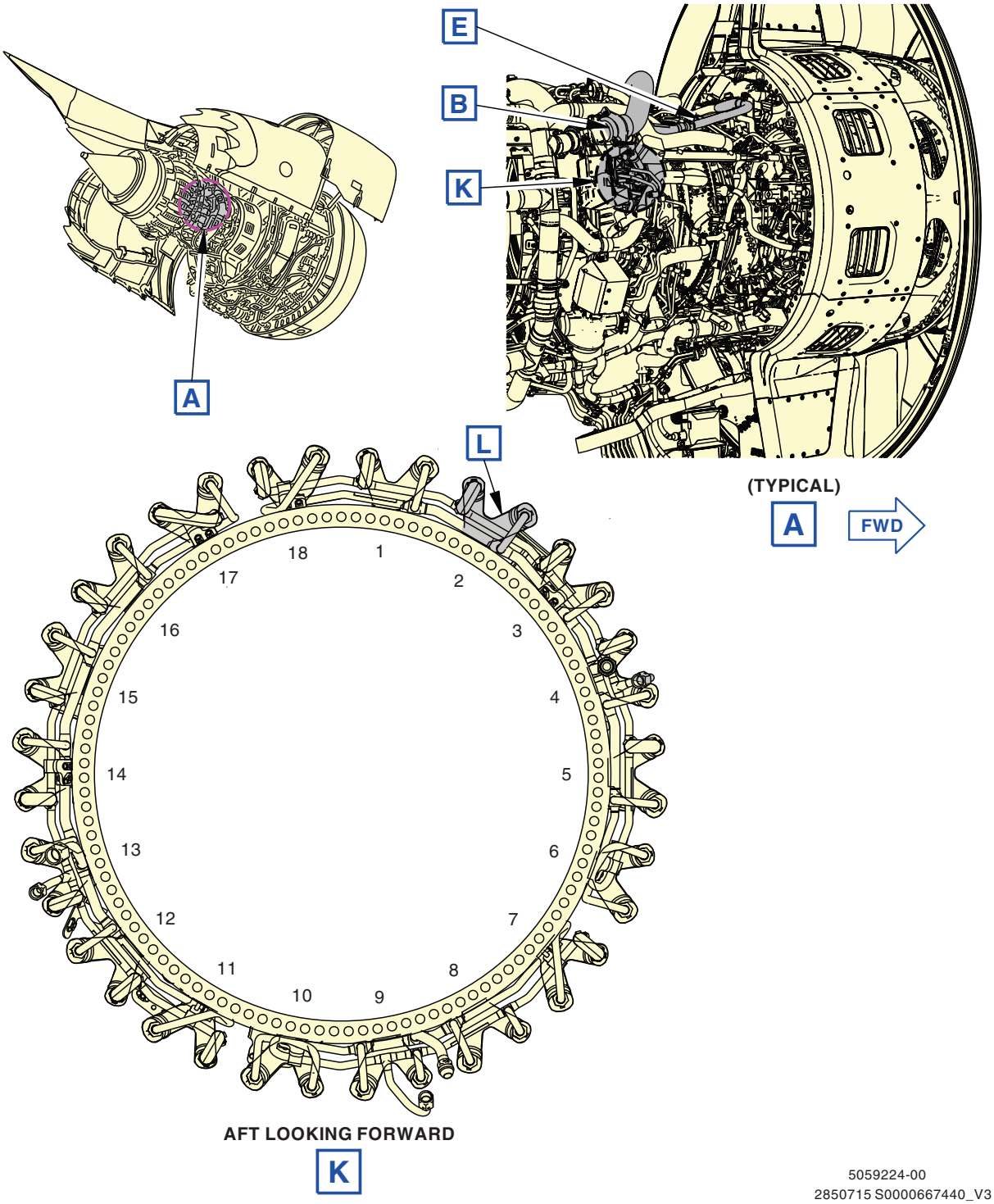
Fuel Nozzle (Position 1) Installation
Figure 402/73-11-05-990-888-G00 (Sheet 2 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



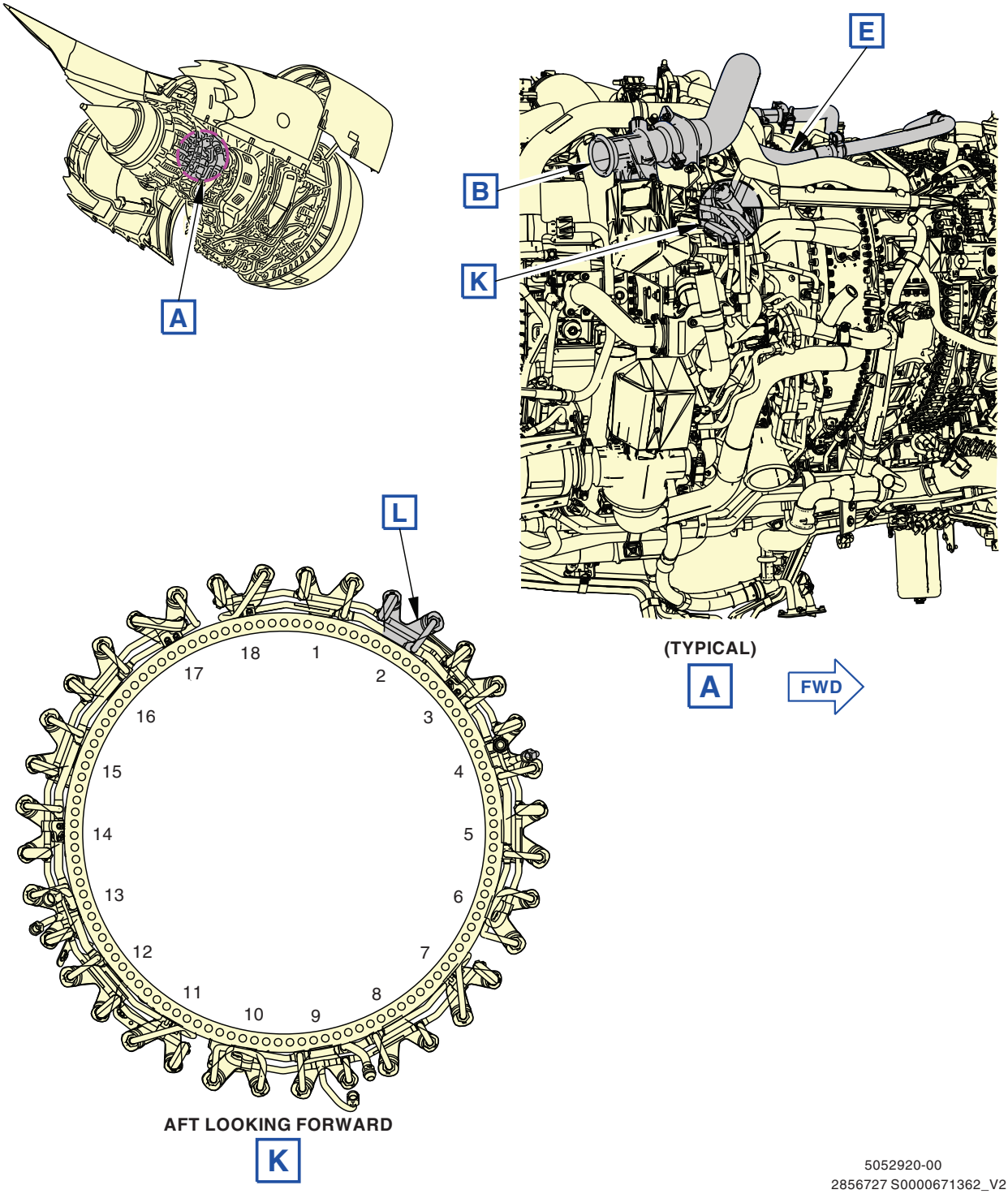
Fuel Nozzle (Position 1) Installation
Figure 402/73-11-05-990-888-G00 (Sheet 3 of 3)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



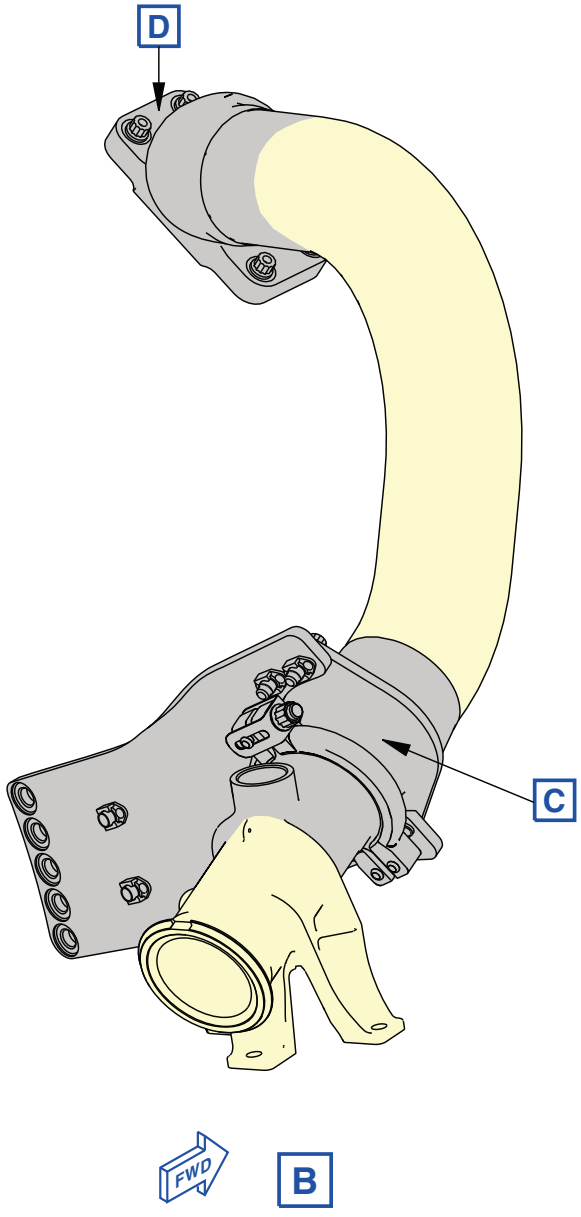
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 1 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 2 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021981-00

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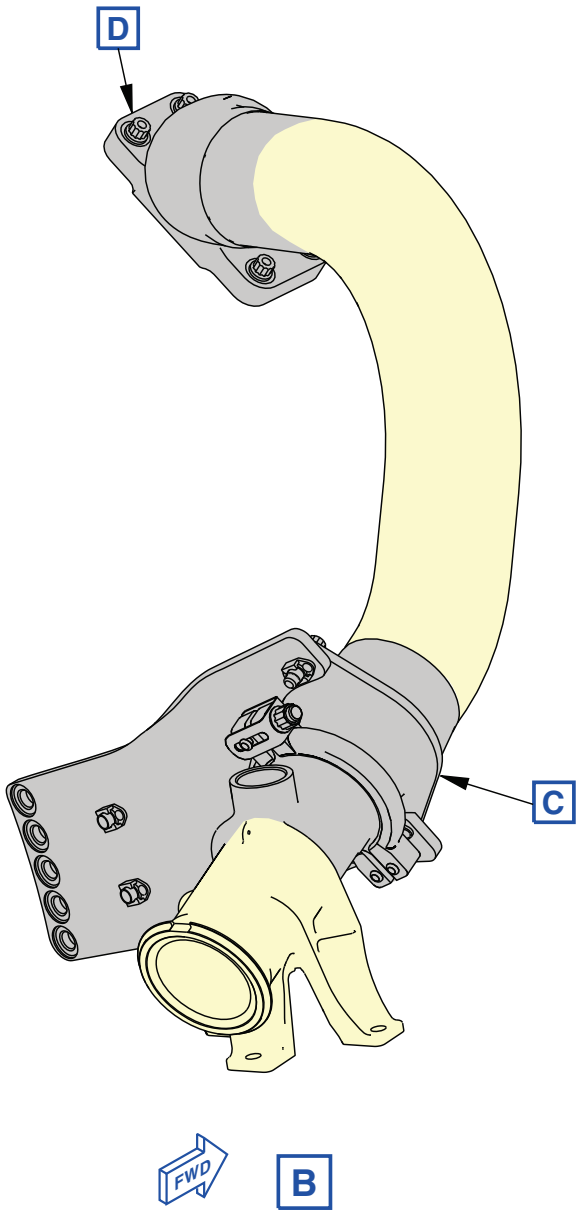
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 3 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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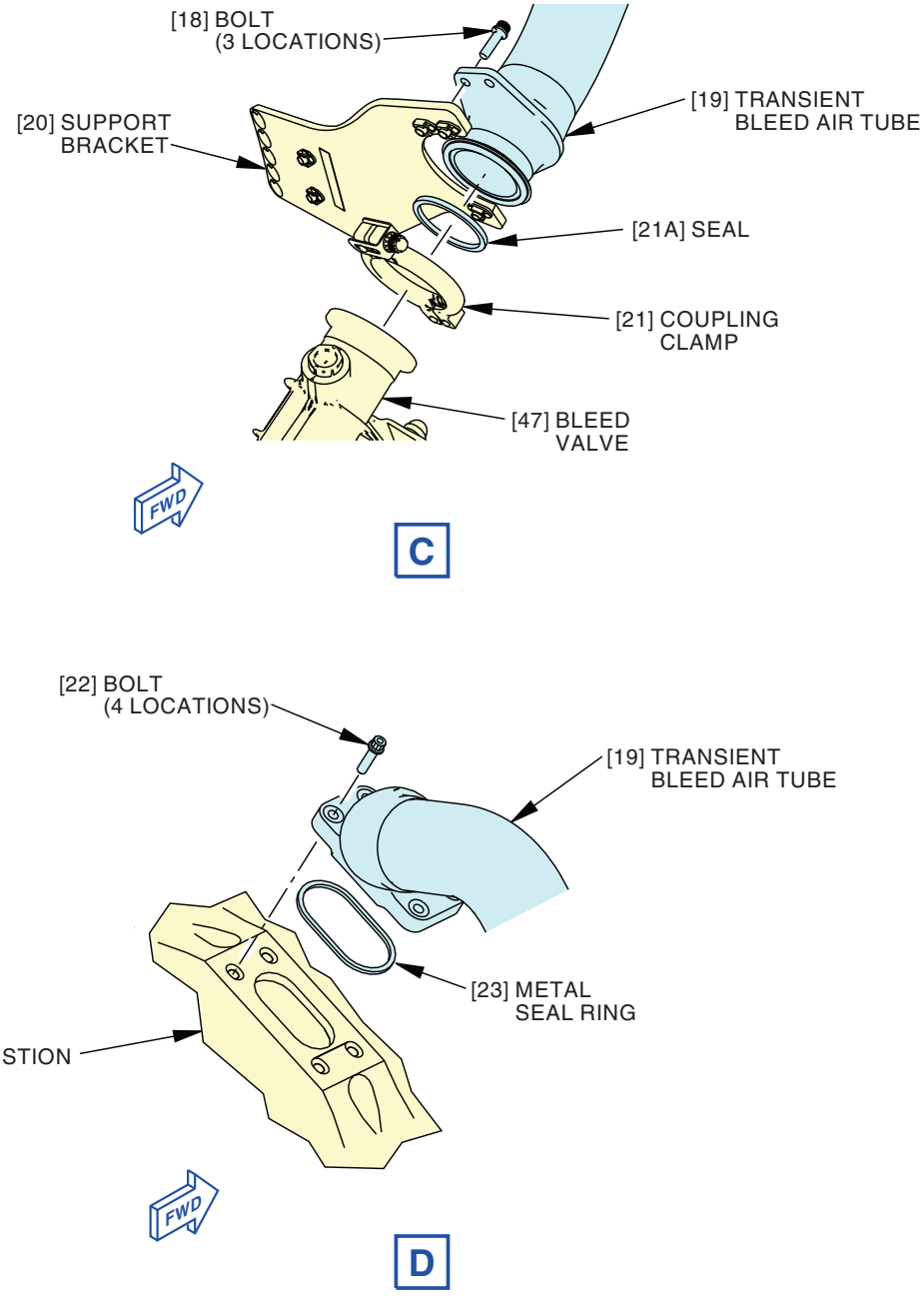
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 4 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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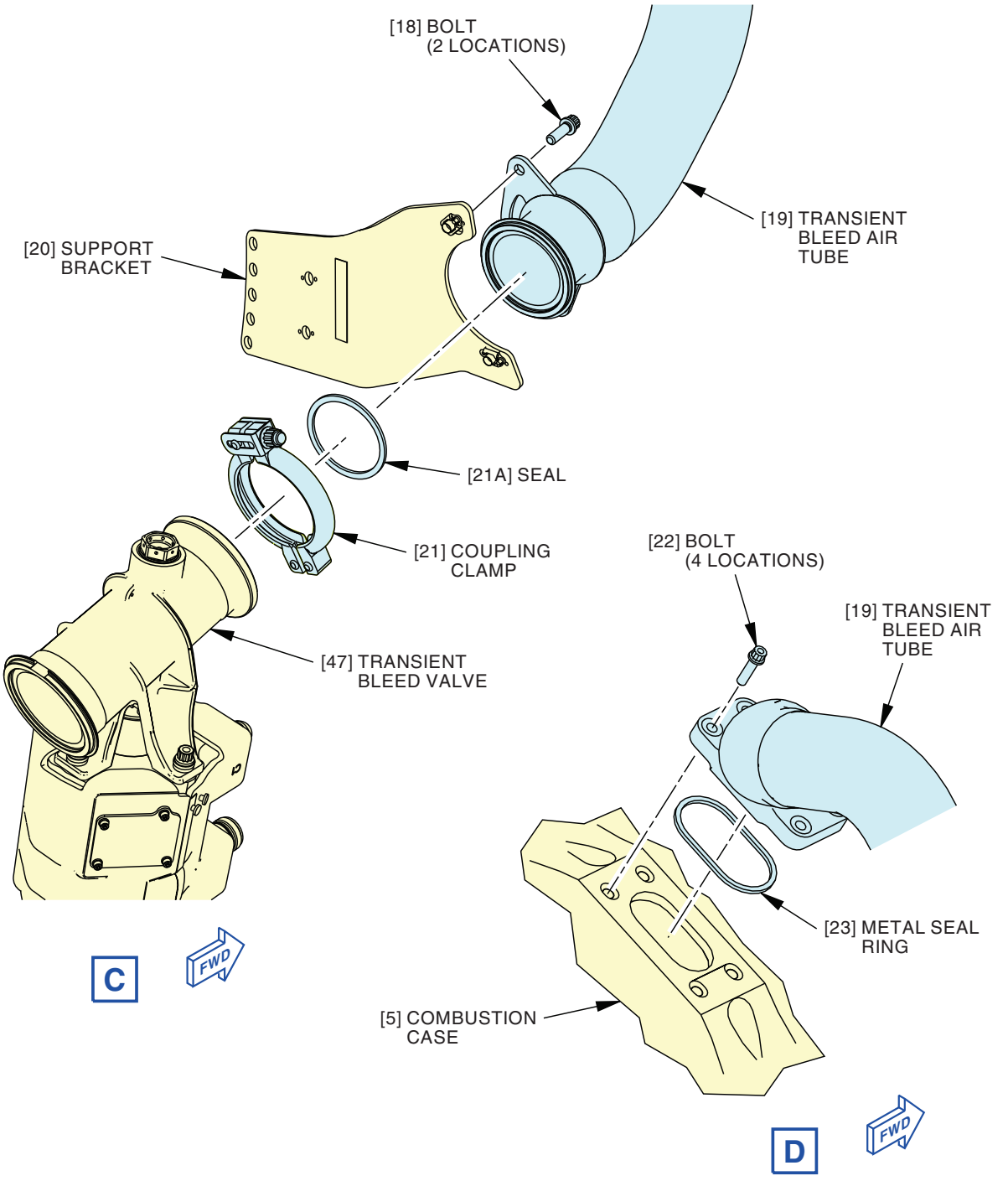
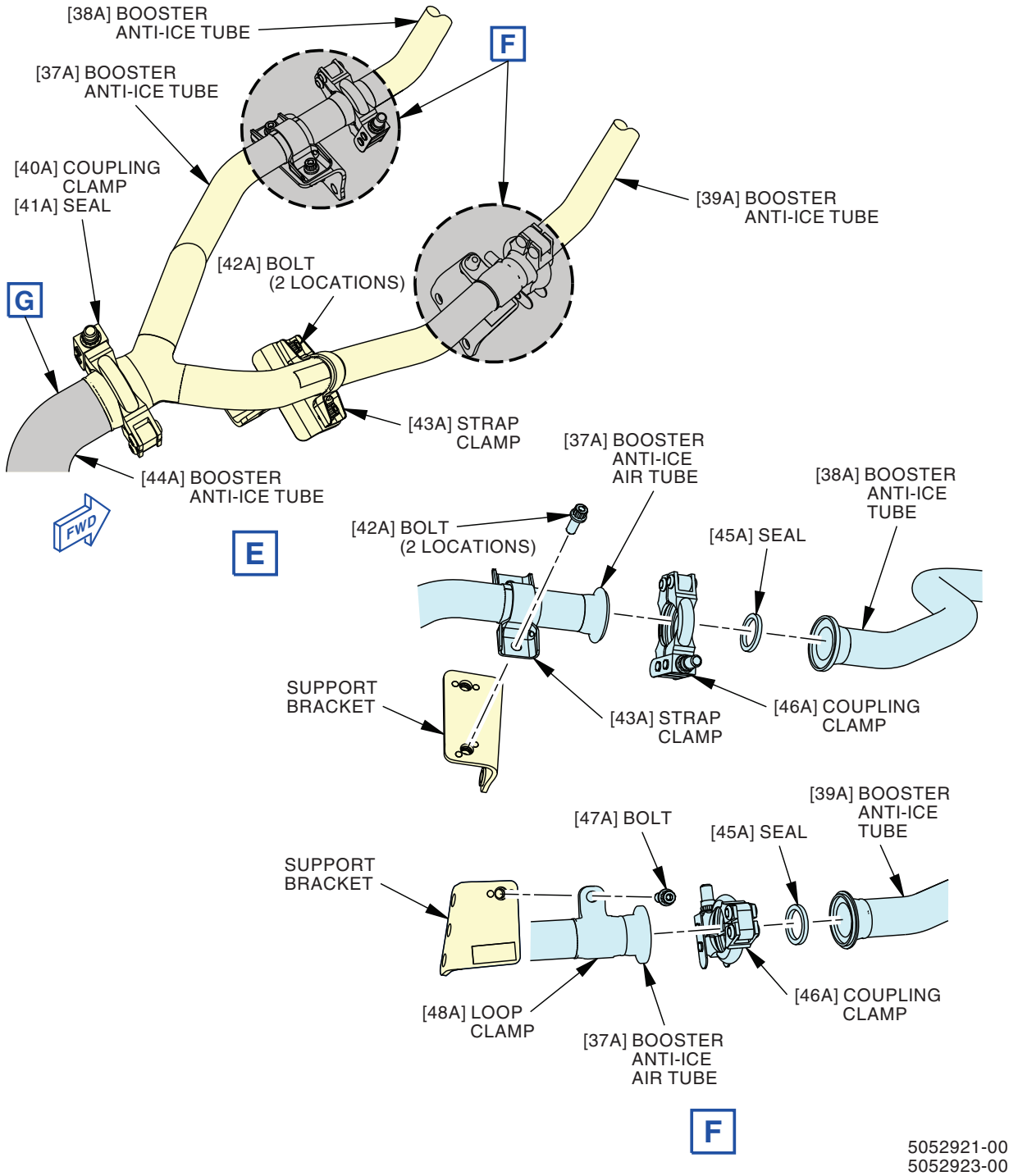
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 5 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 6 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5052921-00
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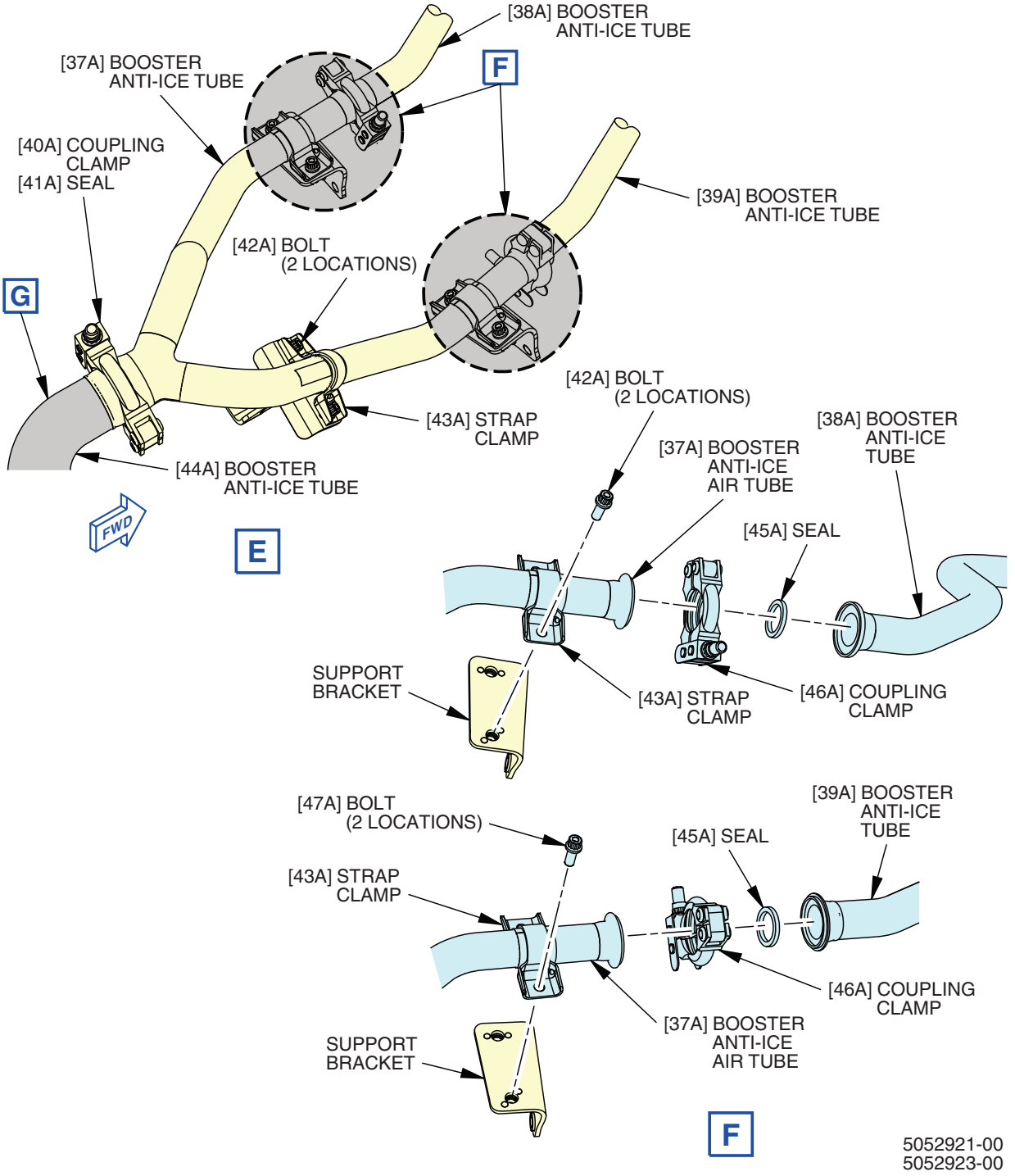
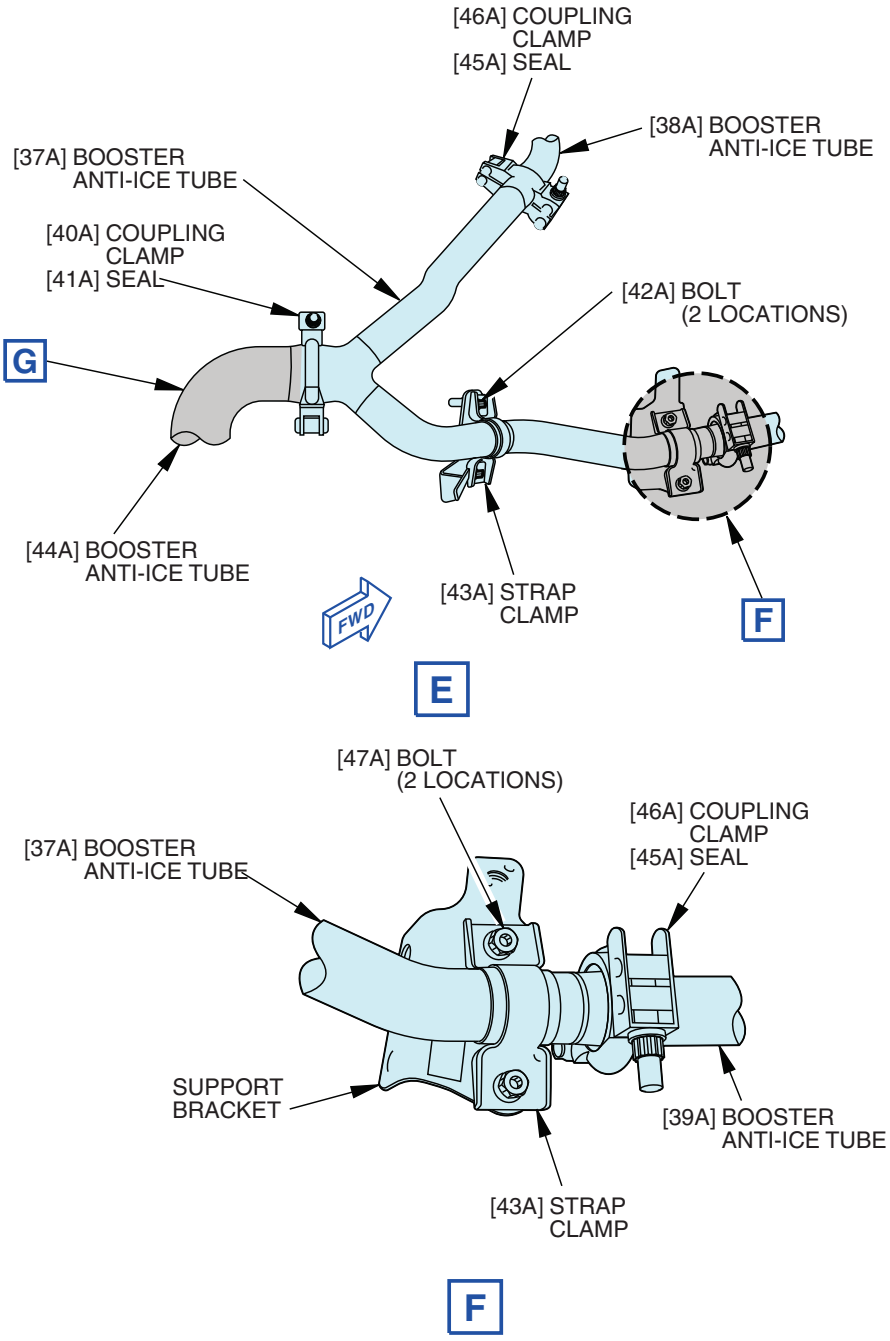
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 7 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 8 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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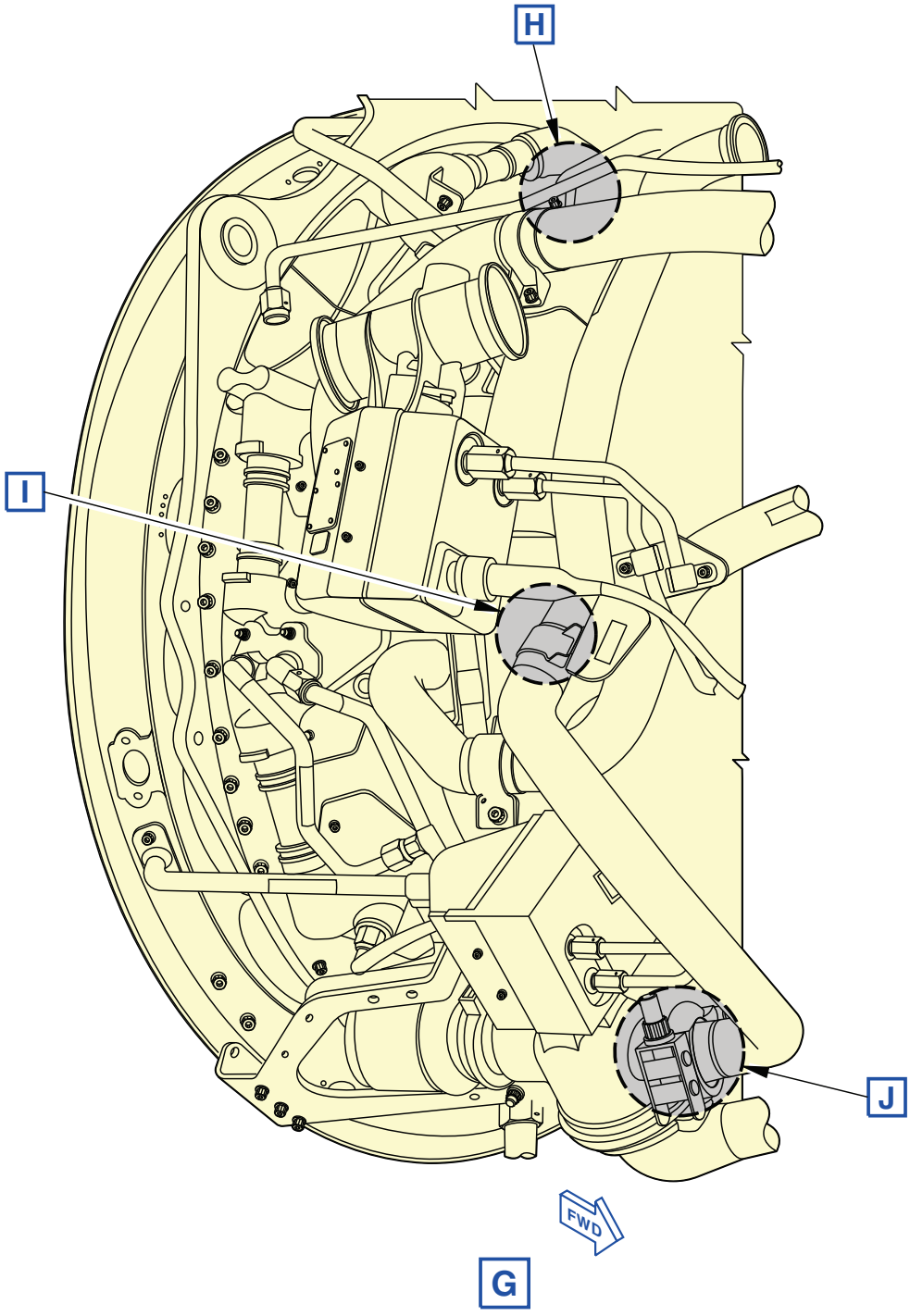
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 9 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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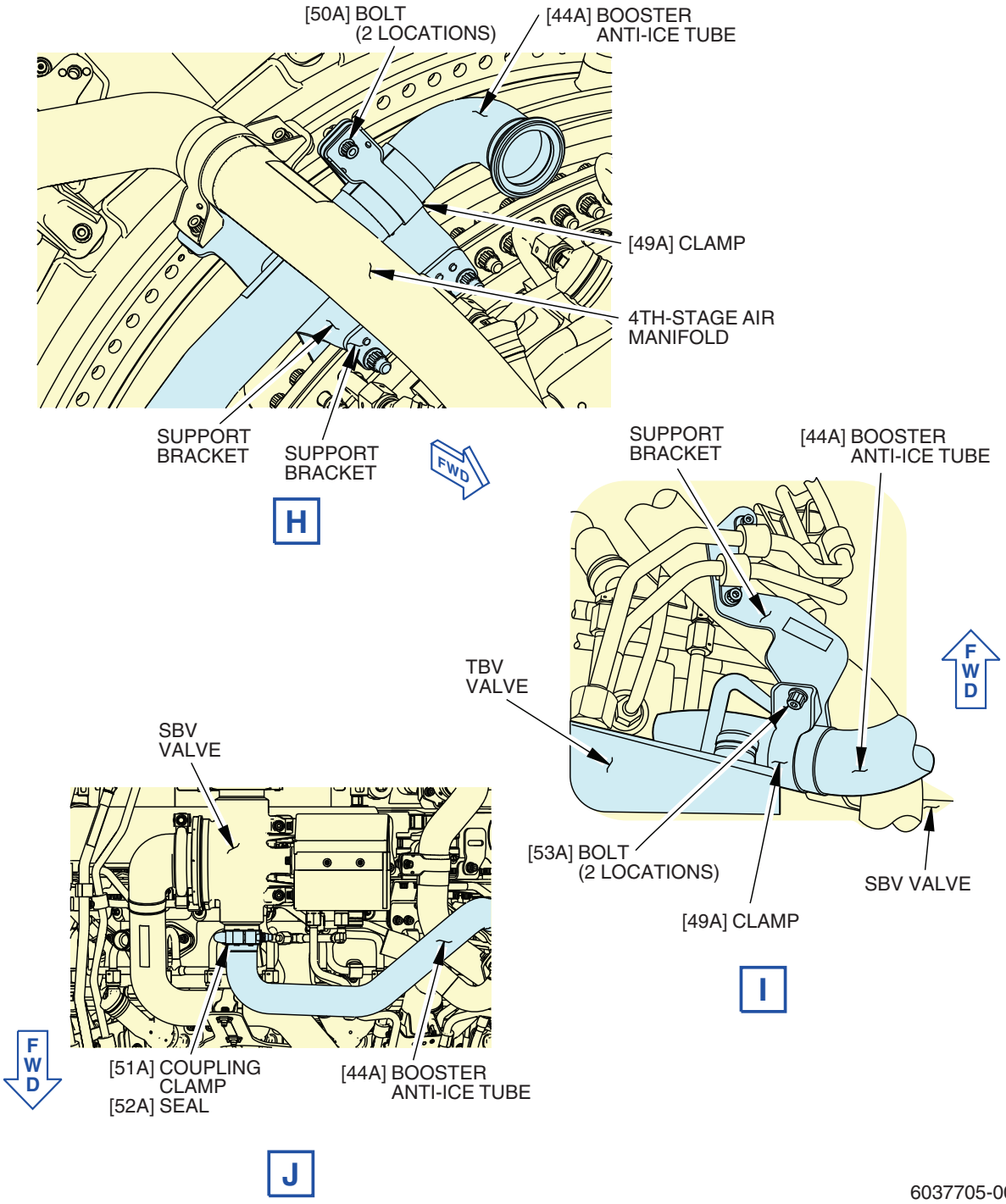
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 10 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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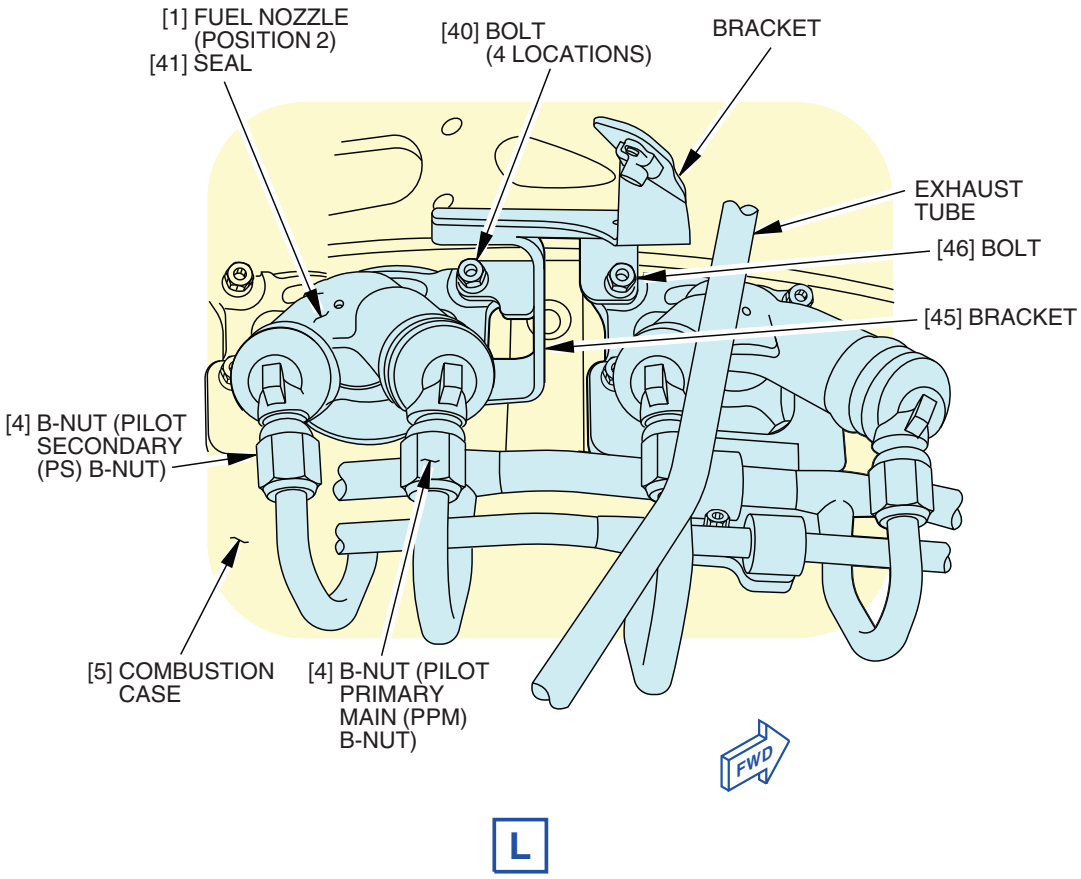
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 11 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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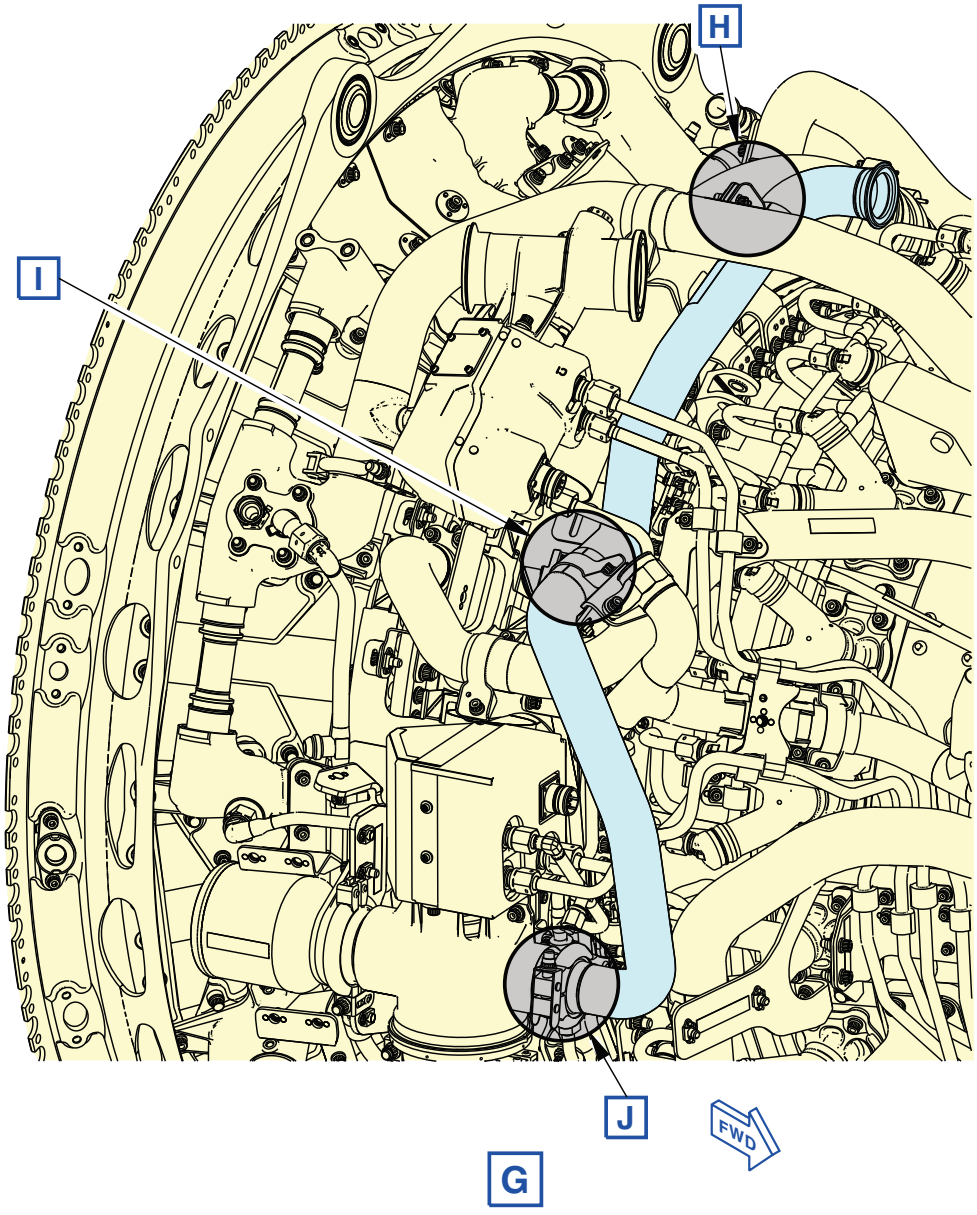
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 12 of 15)

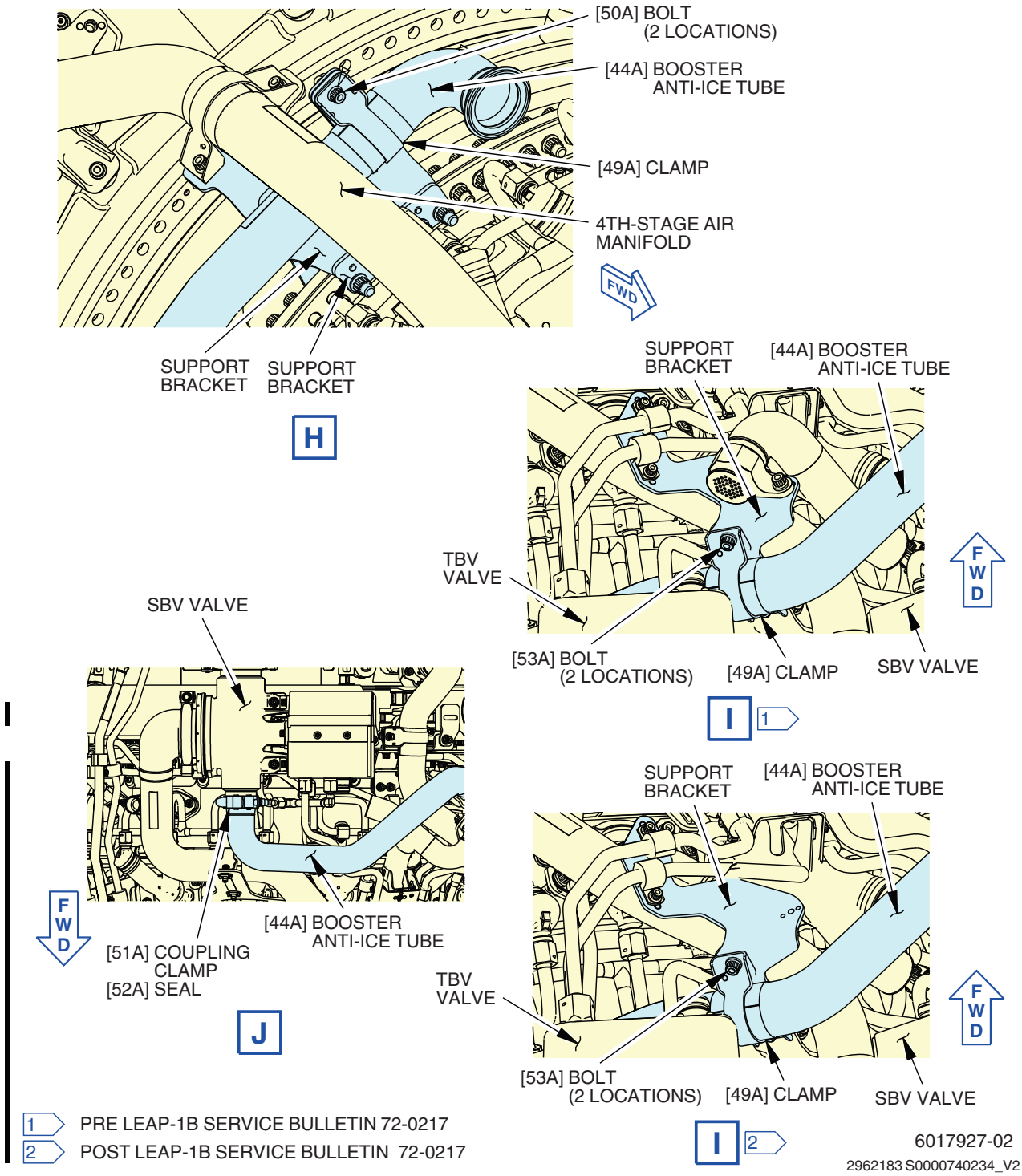
ECCN 9E991 BOEING PROPRIETARY - See title page for details



6017926-00
2962182 S0000740232_V1

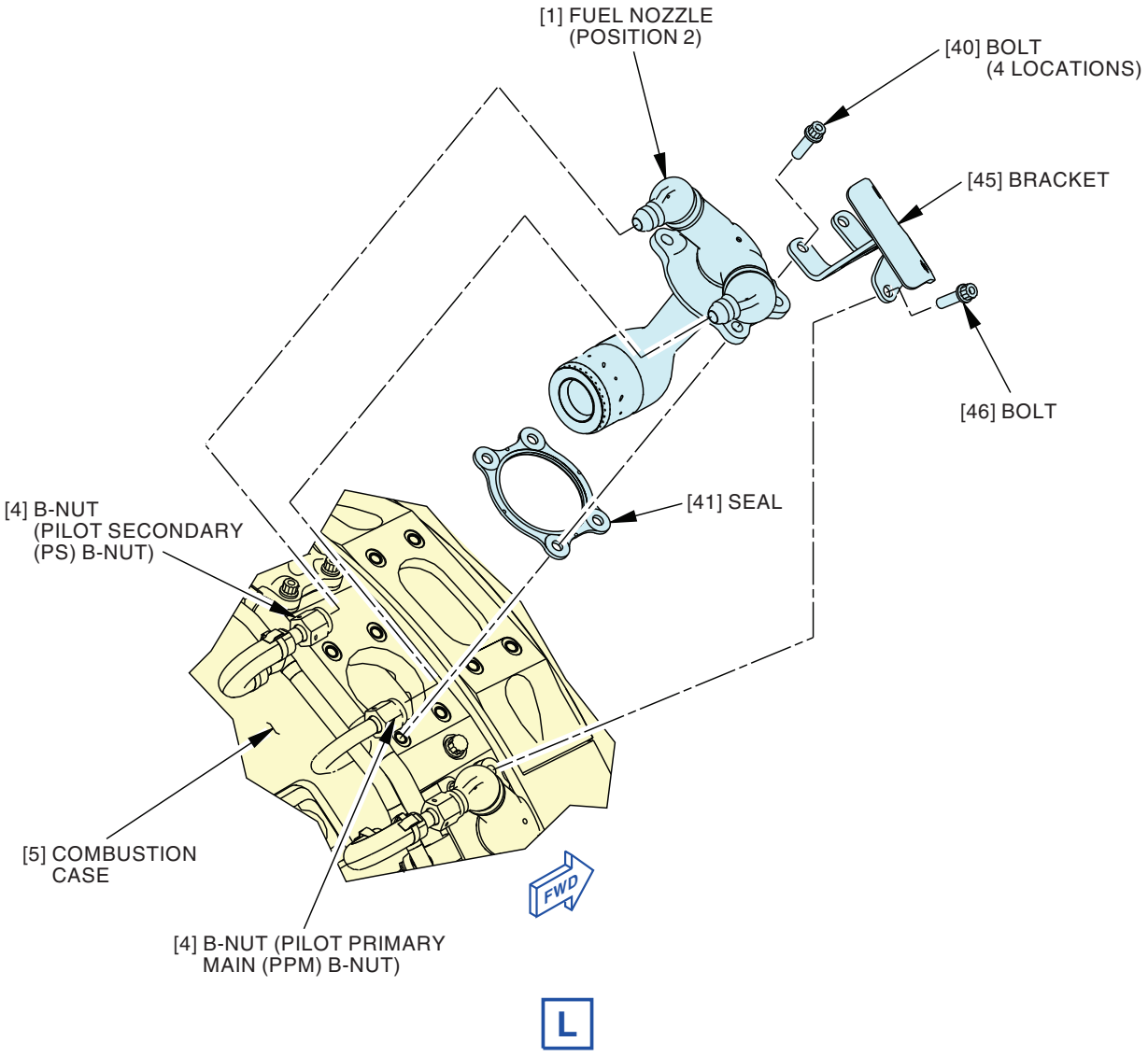
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 13 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



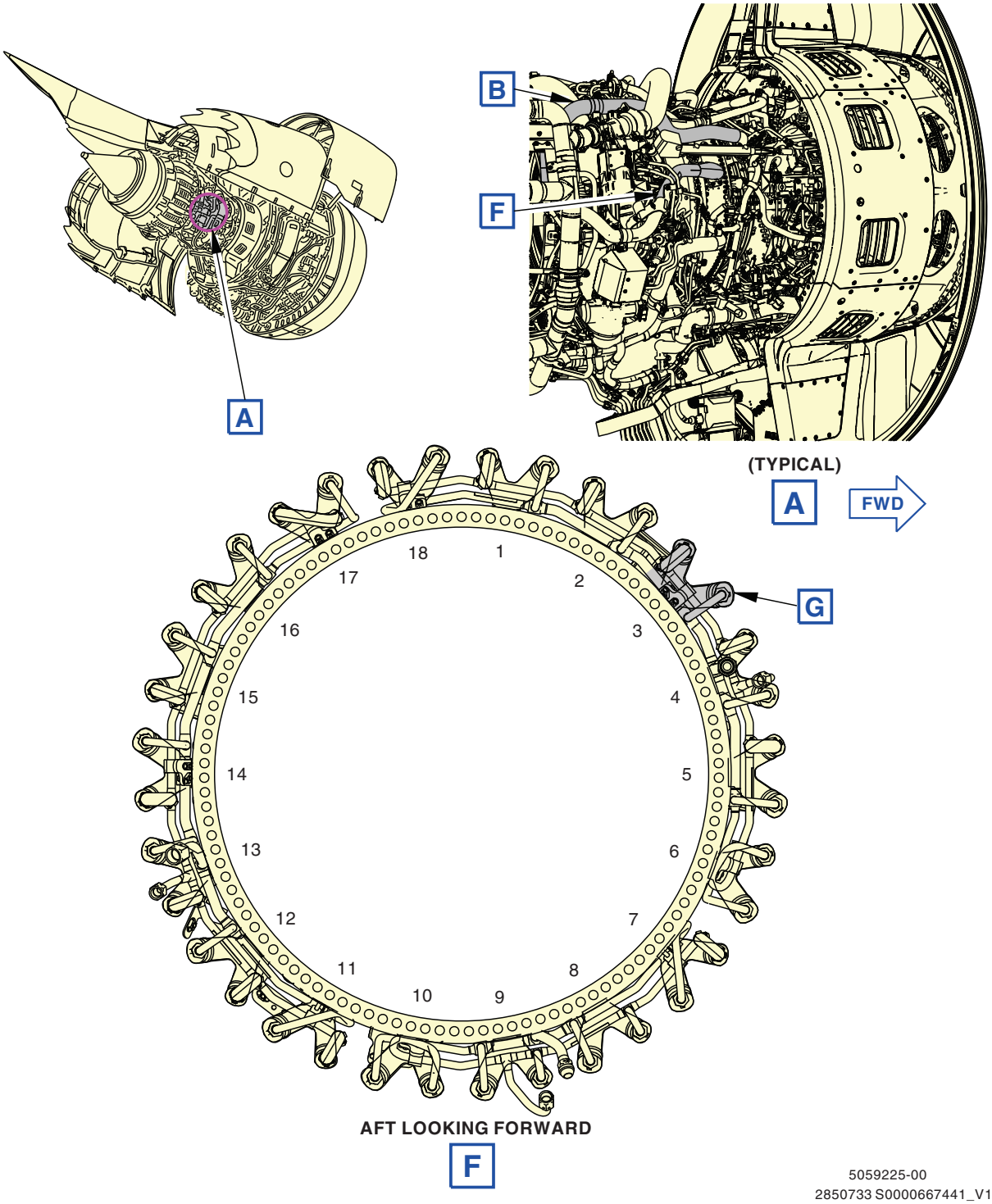
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 14 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



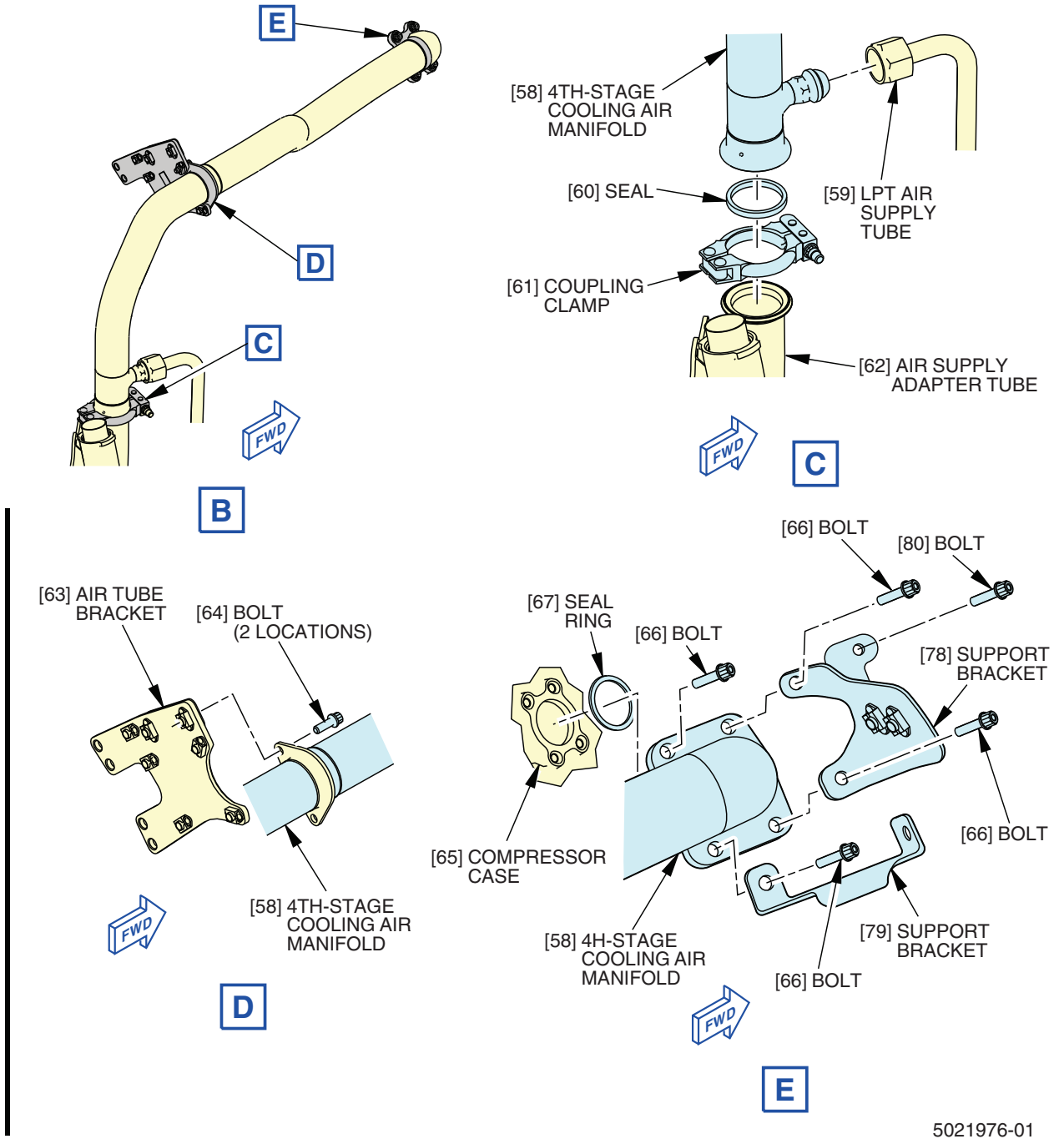
Fuel Nozzle (Position 2) Installation
Figure 403/73-11-05-990-889-G00 (Sheet 15 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



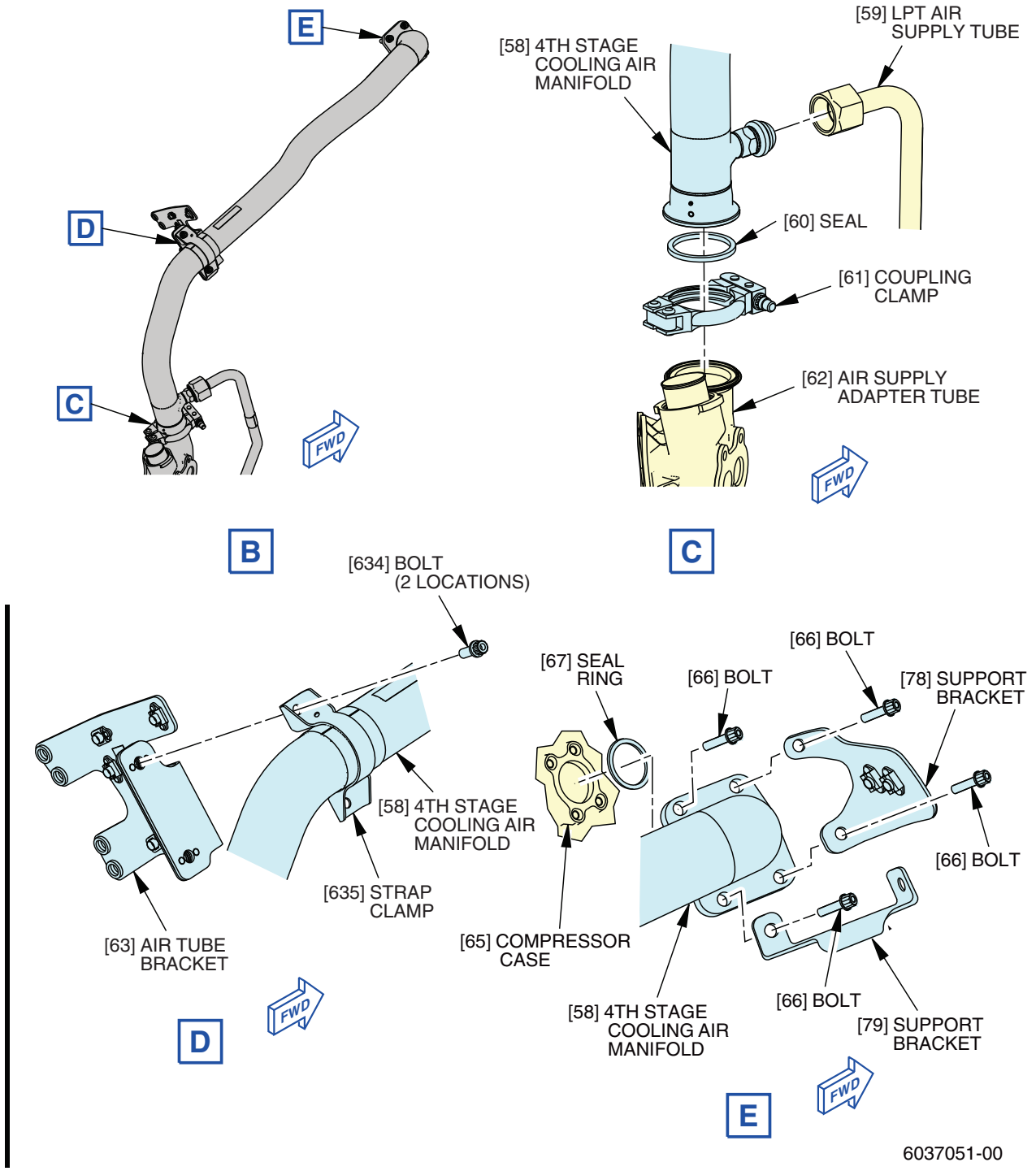
Fuel Nozzle (Position 3) Installation
Figure 404/73-11-05-990-890-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



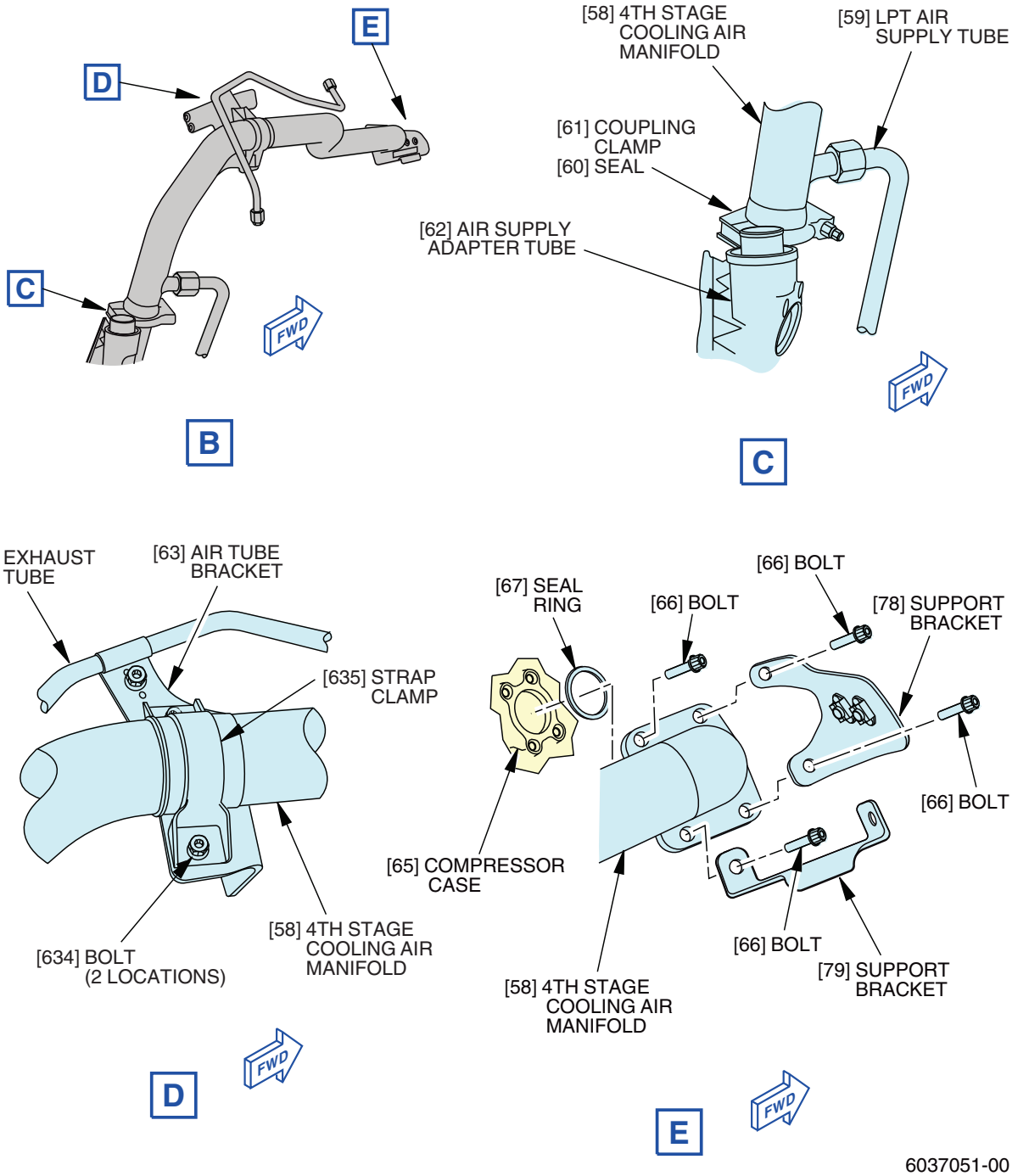
Fuel Nozzle (Position 3) Installation
Figure 404/73-11-05-990-890-G00 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



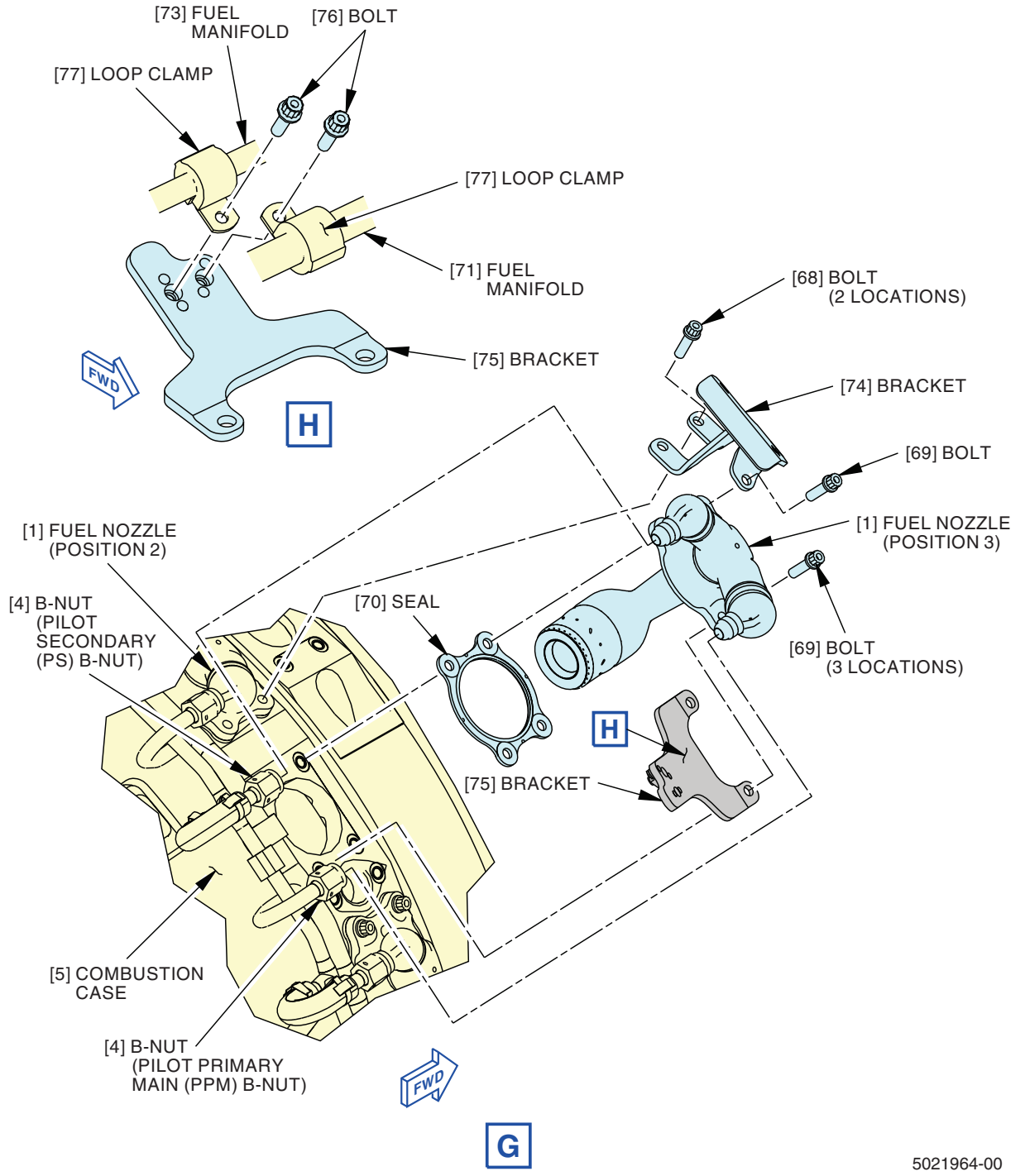
Fuel Nozzle (Position 3) Installation
Figure 404/73-11-05-990-890-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



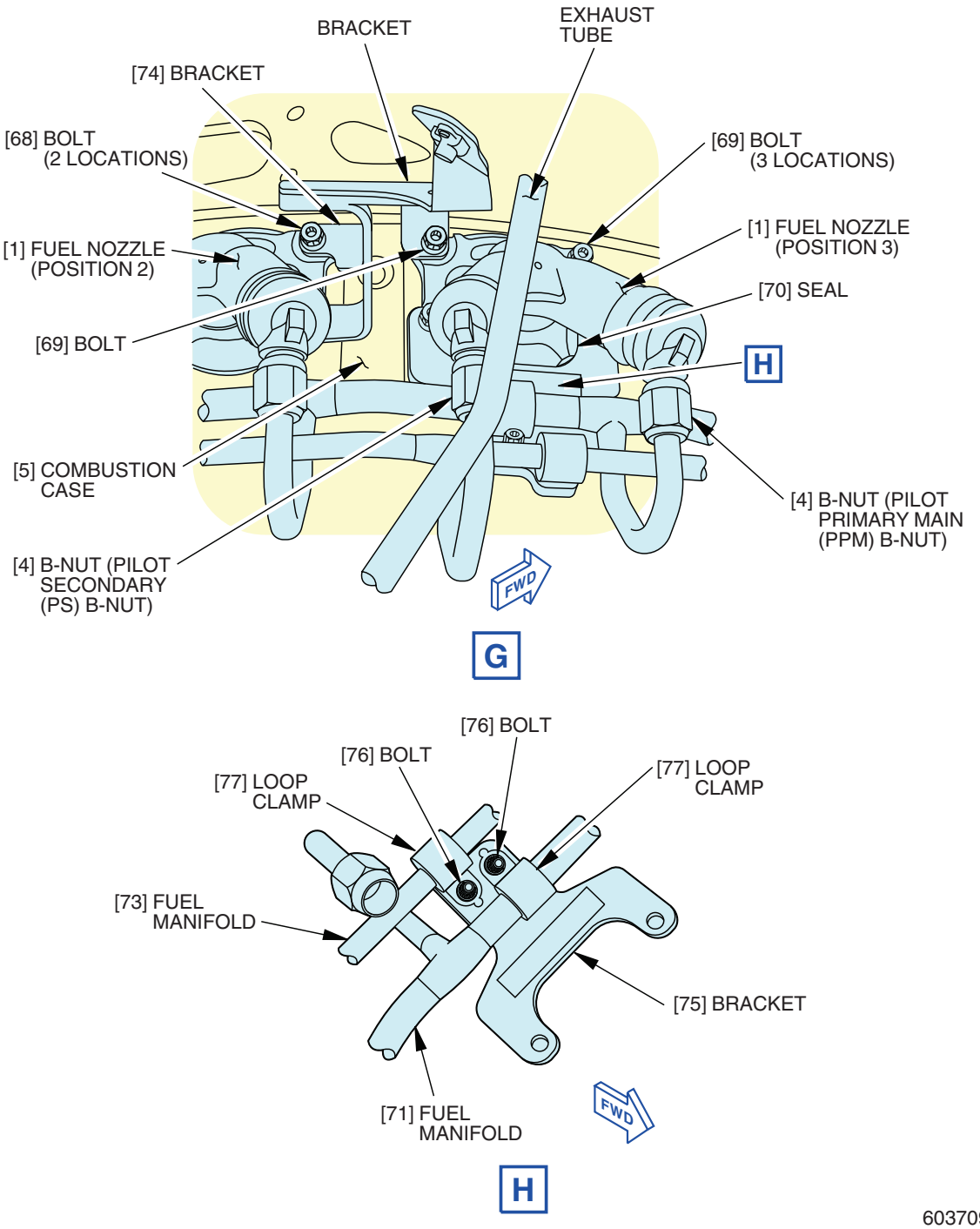
Fuel Nozzle (Position 3) Installation
Figure 404/73-11-05-990-890-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



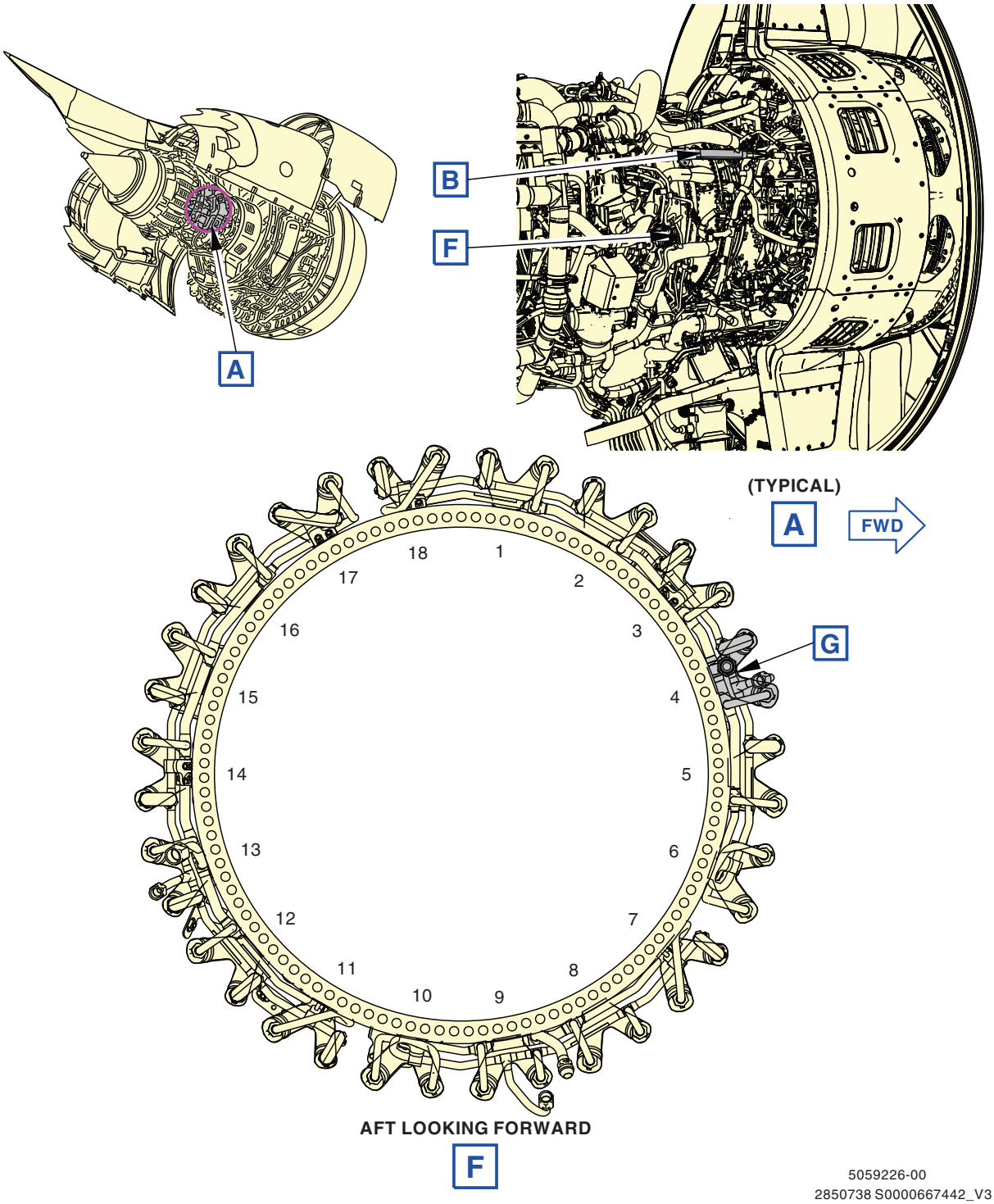
Fuel Nozzle (Position 3) Installation
Figure 404/73-11-05-990-890-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



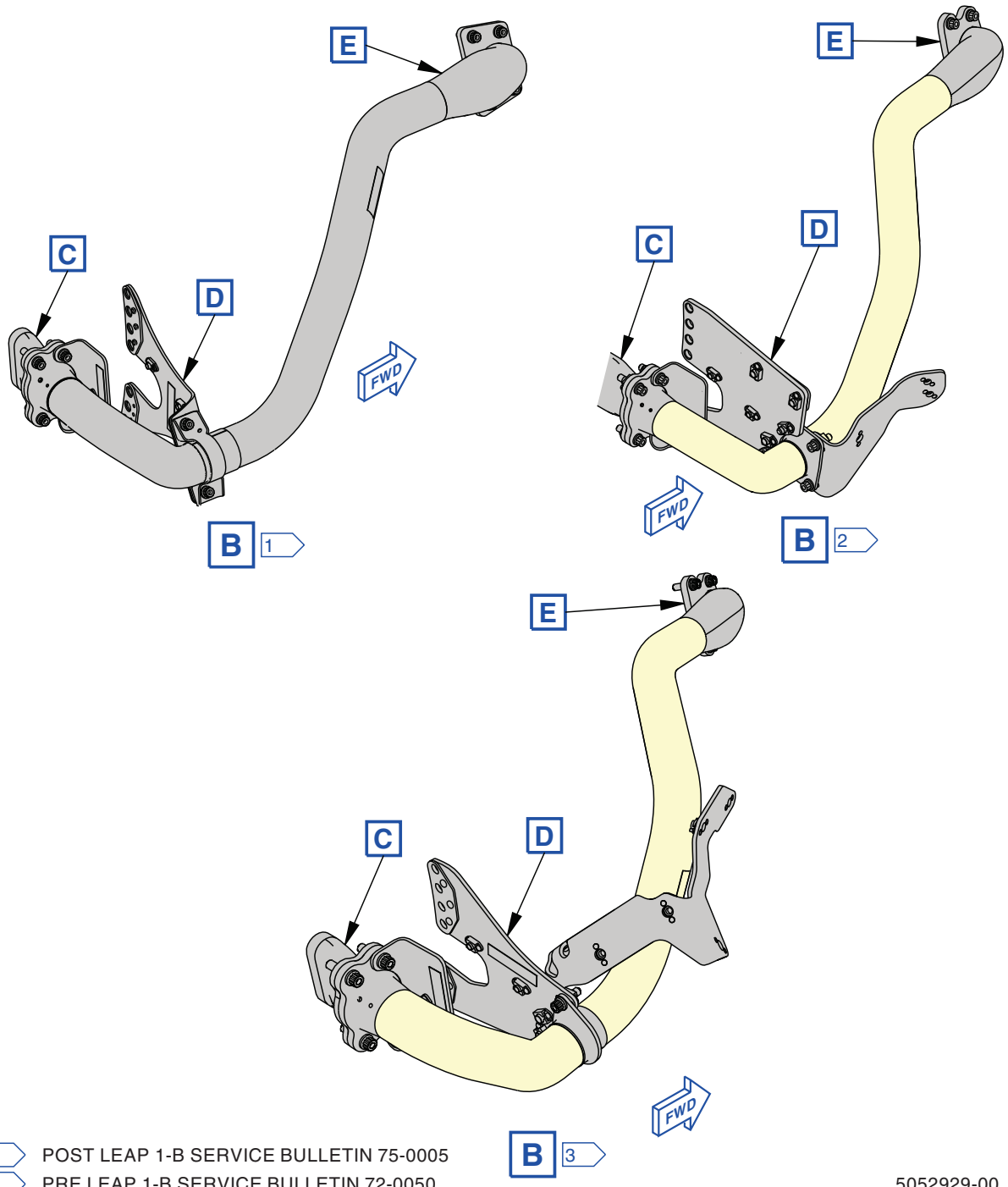
Fuel Nozzle (Position 3) Installation
Figure 404/73-11-05-990-890-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 4) Installation
Figure 405/73-11-05-990-891-G00 (Sheet 1 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



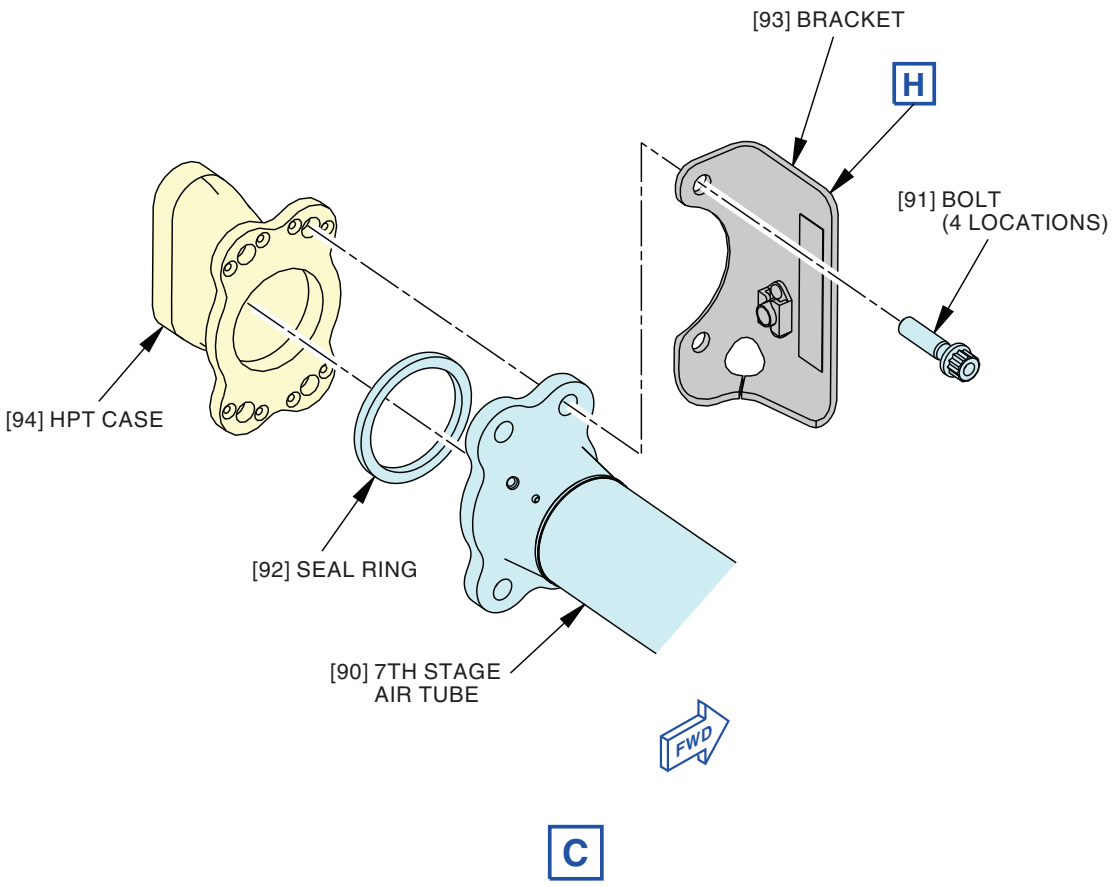
1 POST LEAP 1-B SERVICE BULLETIN 75-0005
2 PRE LEAP 1-B SERVICE BULLETIN 72-0050
3 POST LEAP 1-B SERVICE BULLETIN 72-0050

5052929-00
5058970-00

2856756 S0000669772_V2

Fuel Nozzle (Position 4) Installation
Figure 405/73-11-05-990-891-G00 (Sheet 2 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

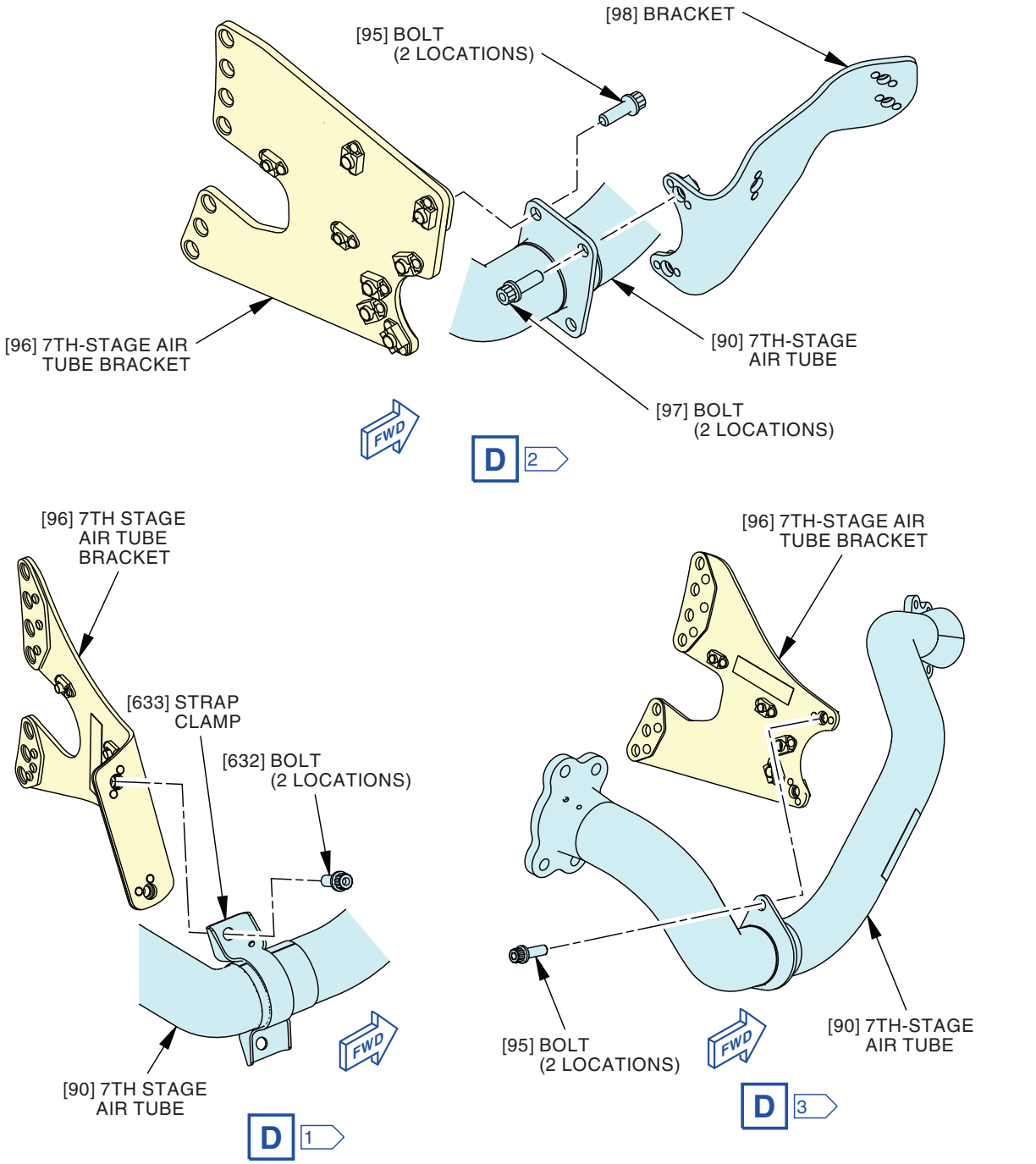


5052929-00

2441147 S0000565572_V4

Fuel Nozzle (Position 4) Installation
Figure 405/73-11-05-990-891-G00 (Sheet 3 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

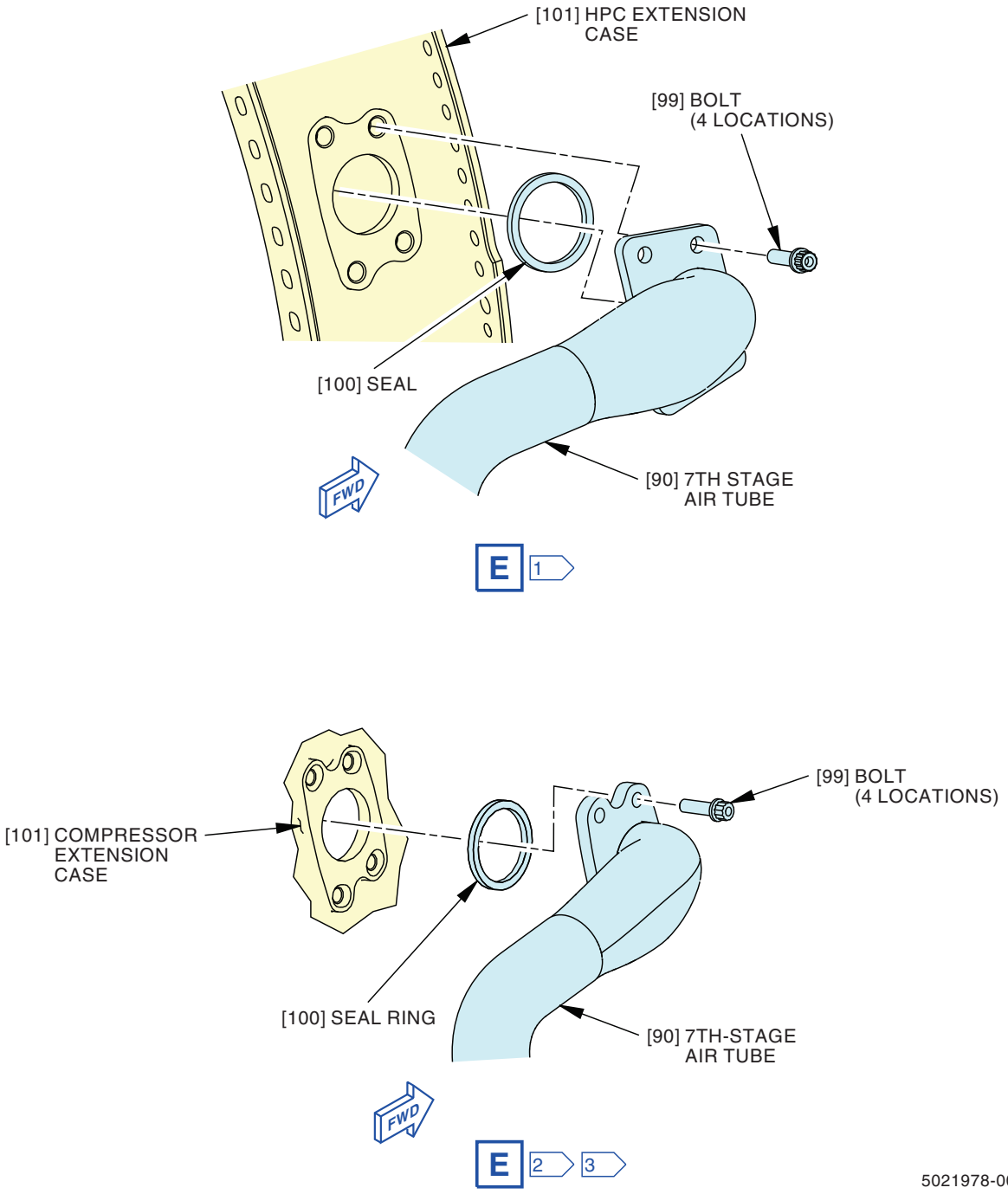


5047632-00
5021978-00

2441166 S0000565573_V6

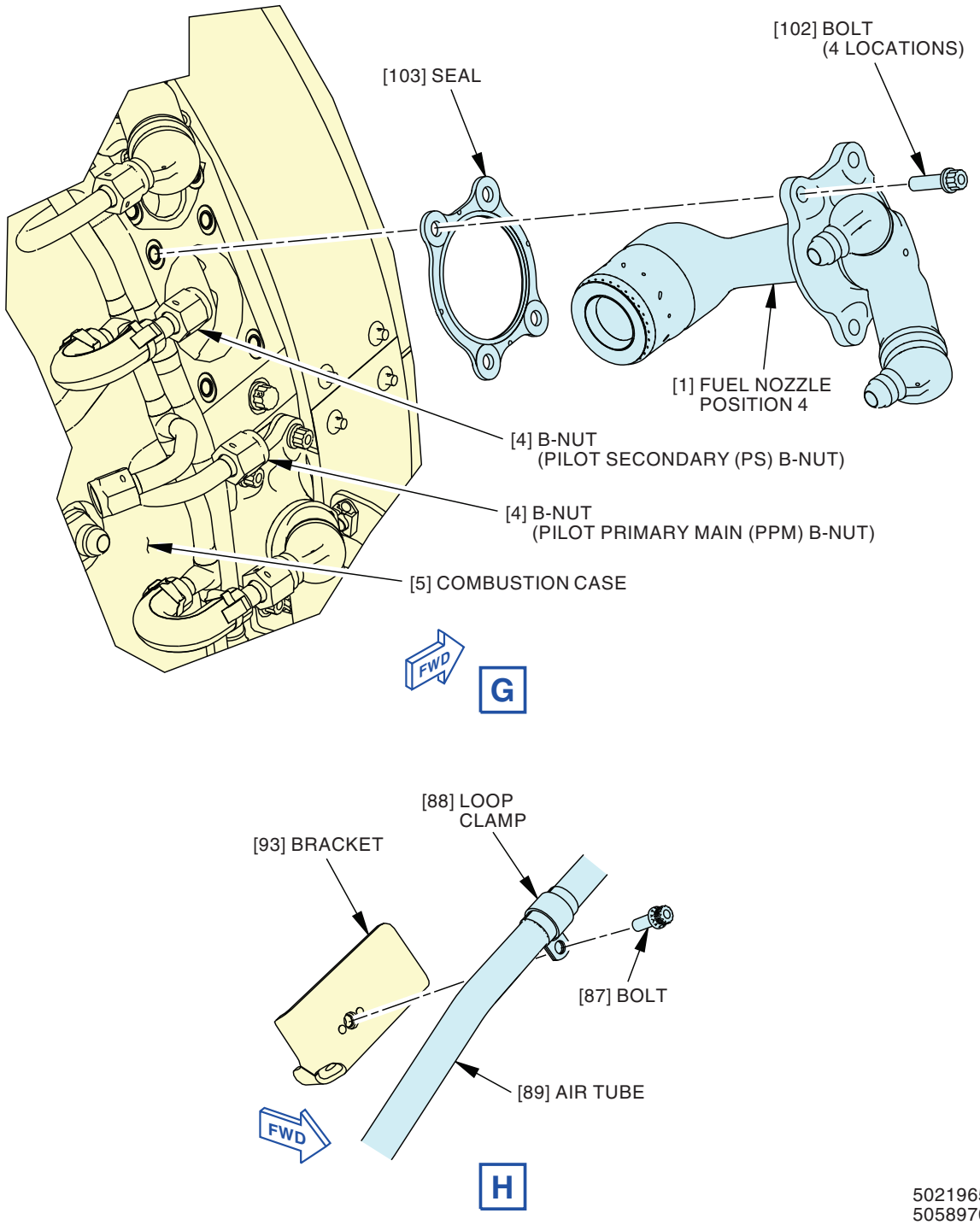
Fuel Nozzle (Position 4) Installation
Figure 405/73-11-05-990-891-G00 (Sheet 4 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 4) Installation
Figure 405/73-11-05-990-891-G00 (Sheet 5 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

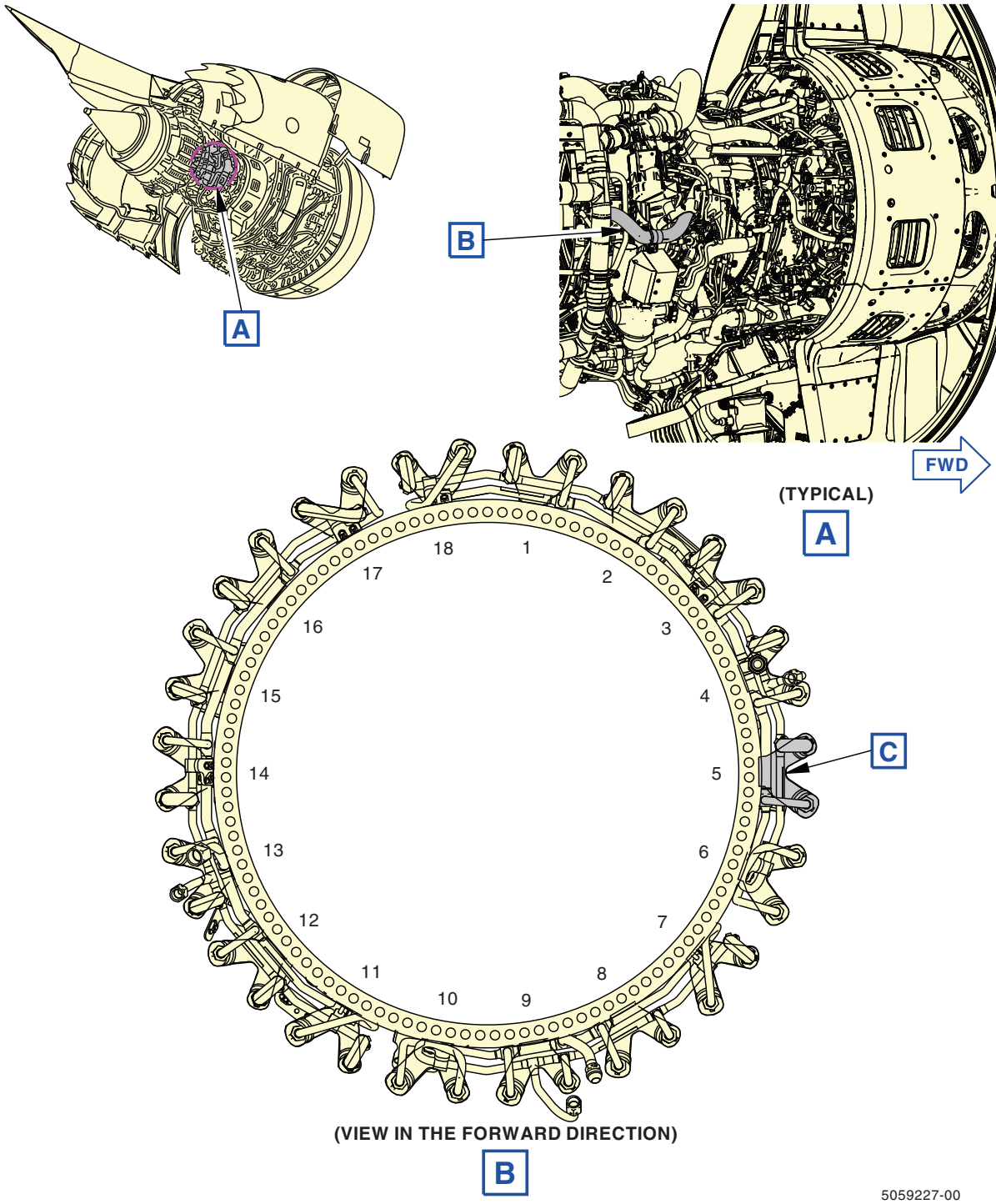


5021965-00
5058970-00

2809496 S0000646114_V4

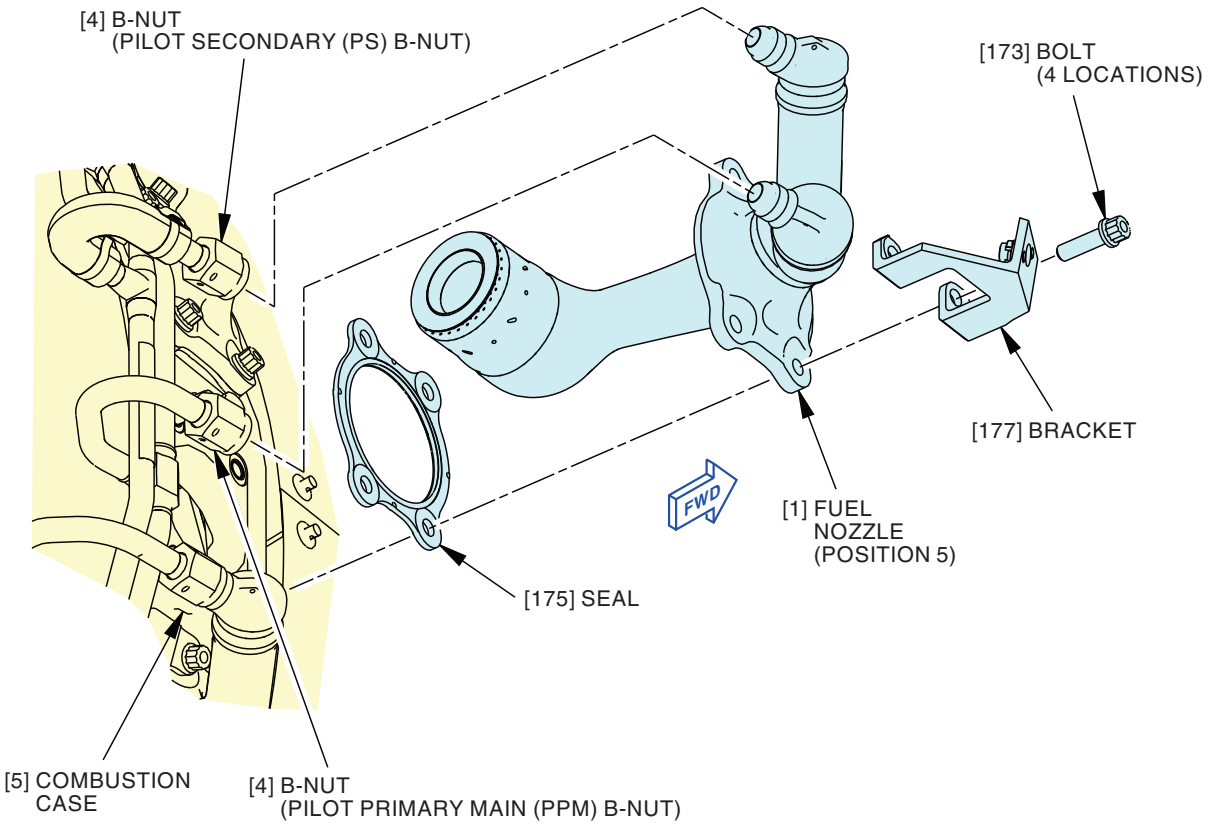
Fuel Nozzle (Position 4) Installation
Figure 405/73-11-05-990-891-G00 (Sheet 6 of 6)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 1 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

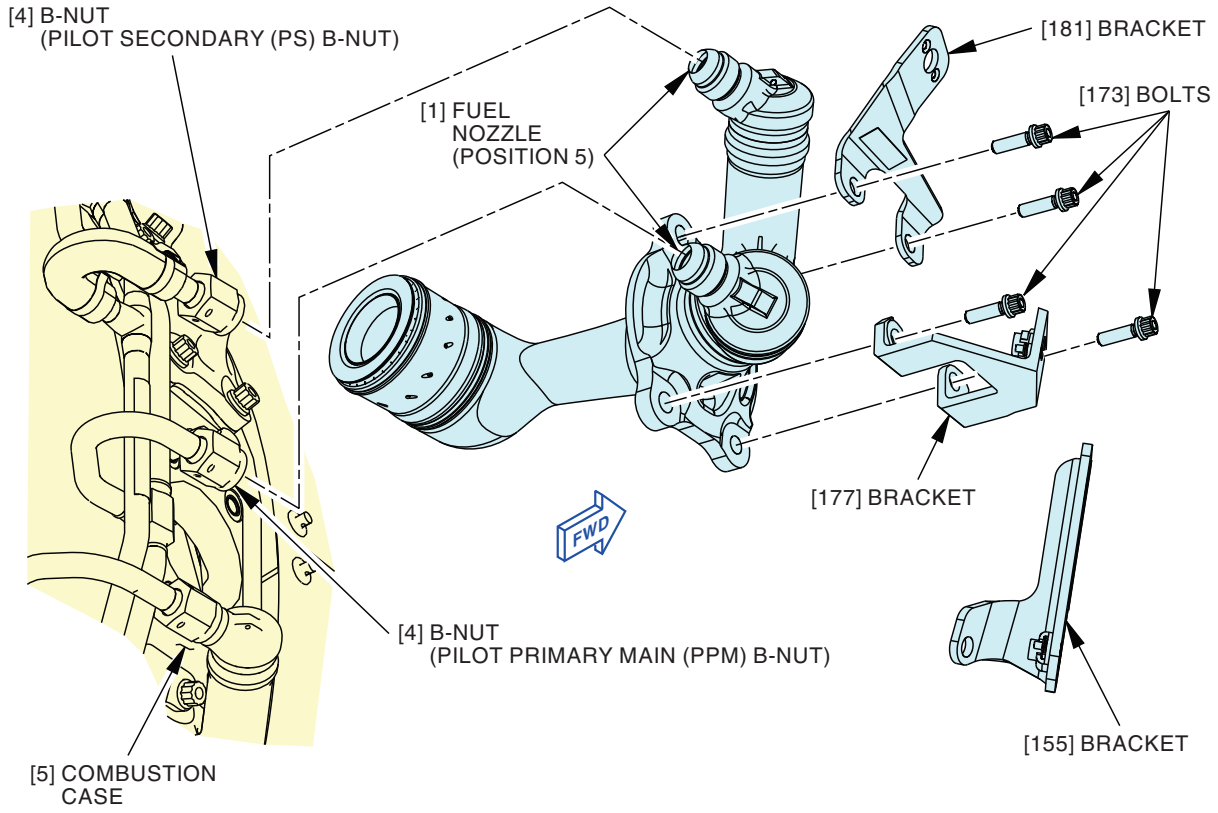


C

5022010-00
2443853 S0000566361_V3

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 2 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

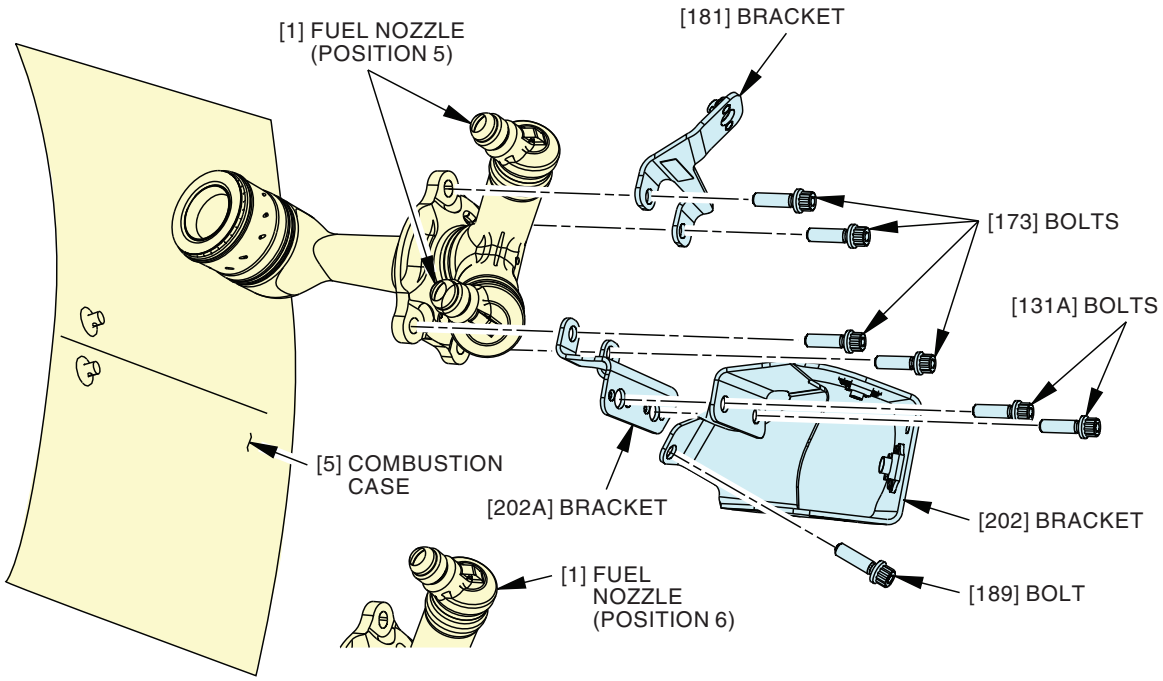


C

7100126-00
2901477 S0000693110_V2

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 3 of 38)

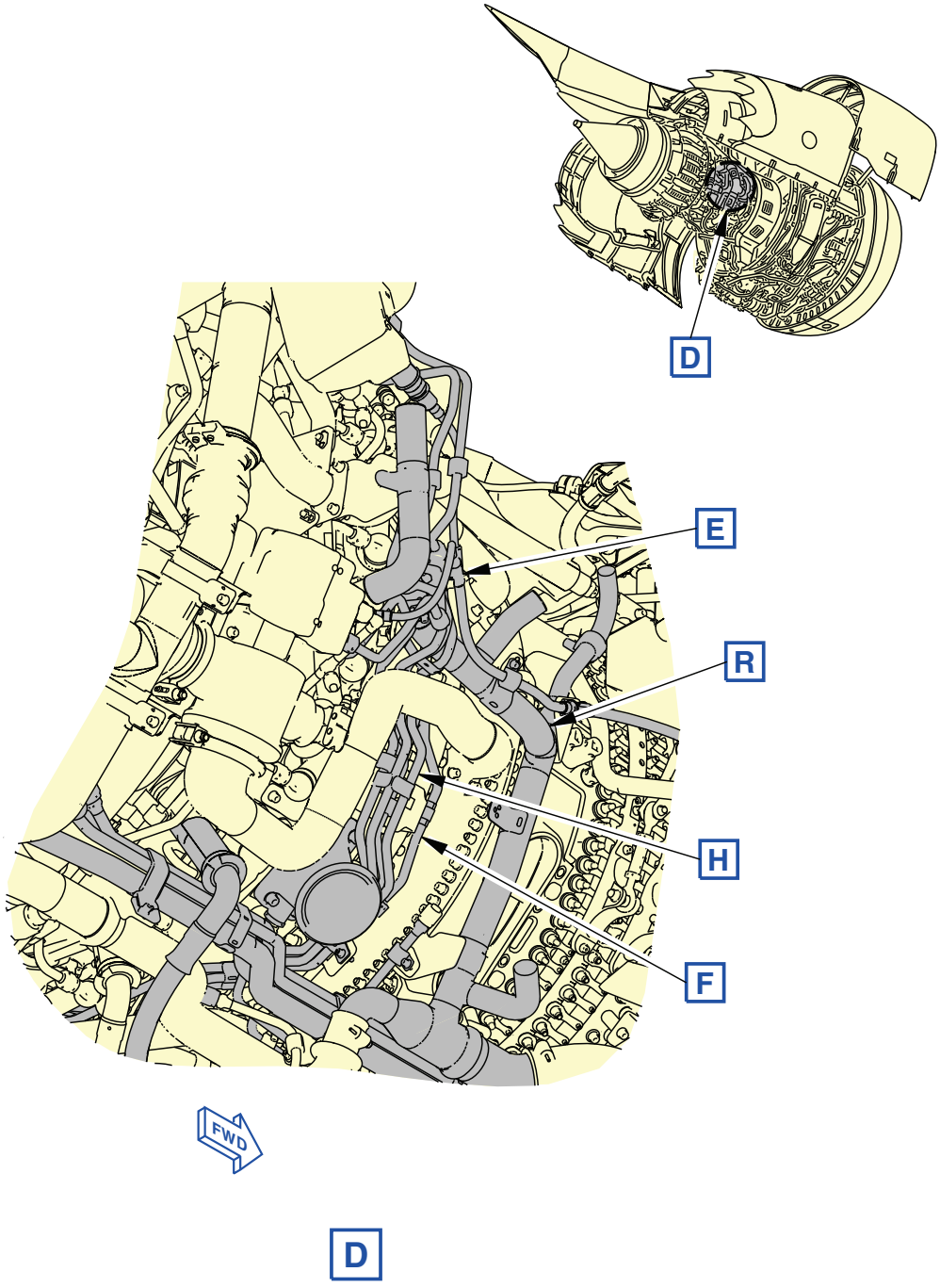
ECCN 9E991 BOEING PROPRIETARY - See title page for details



7100127-00
2901478 S0000693111_V2

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 4 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

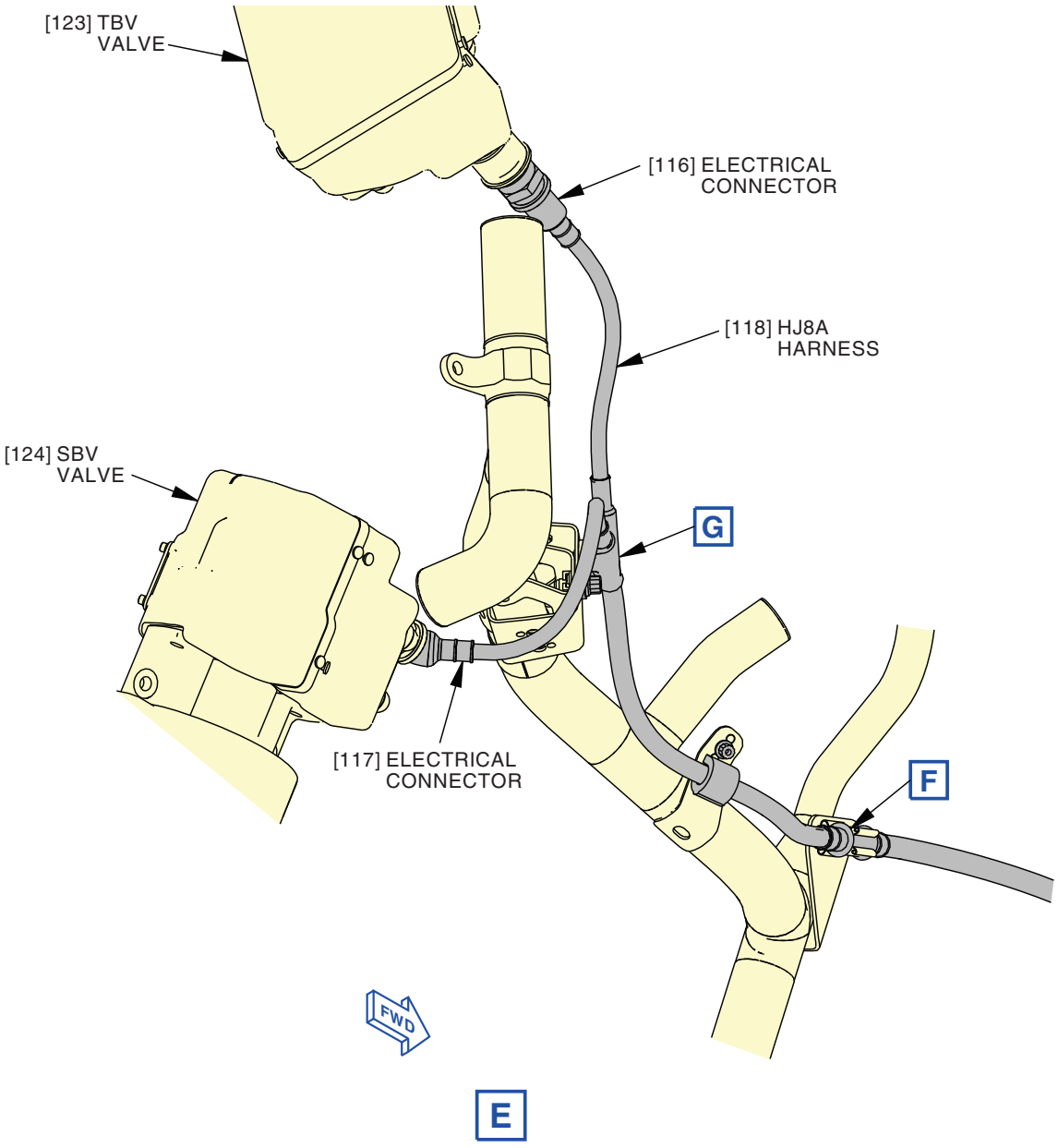


5021985-00

2443836 S0000566363_V2

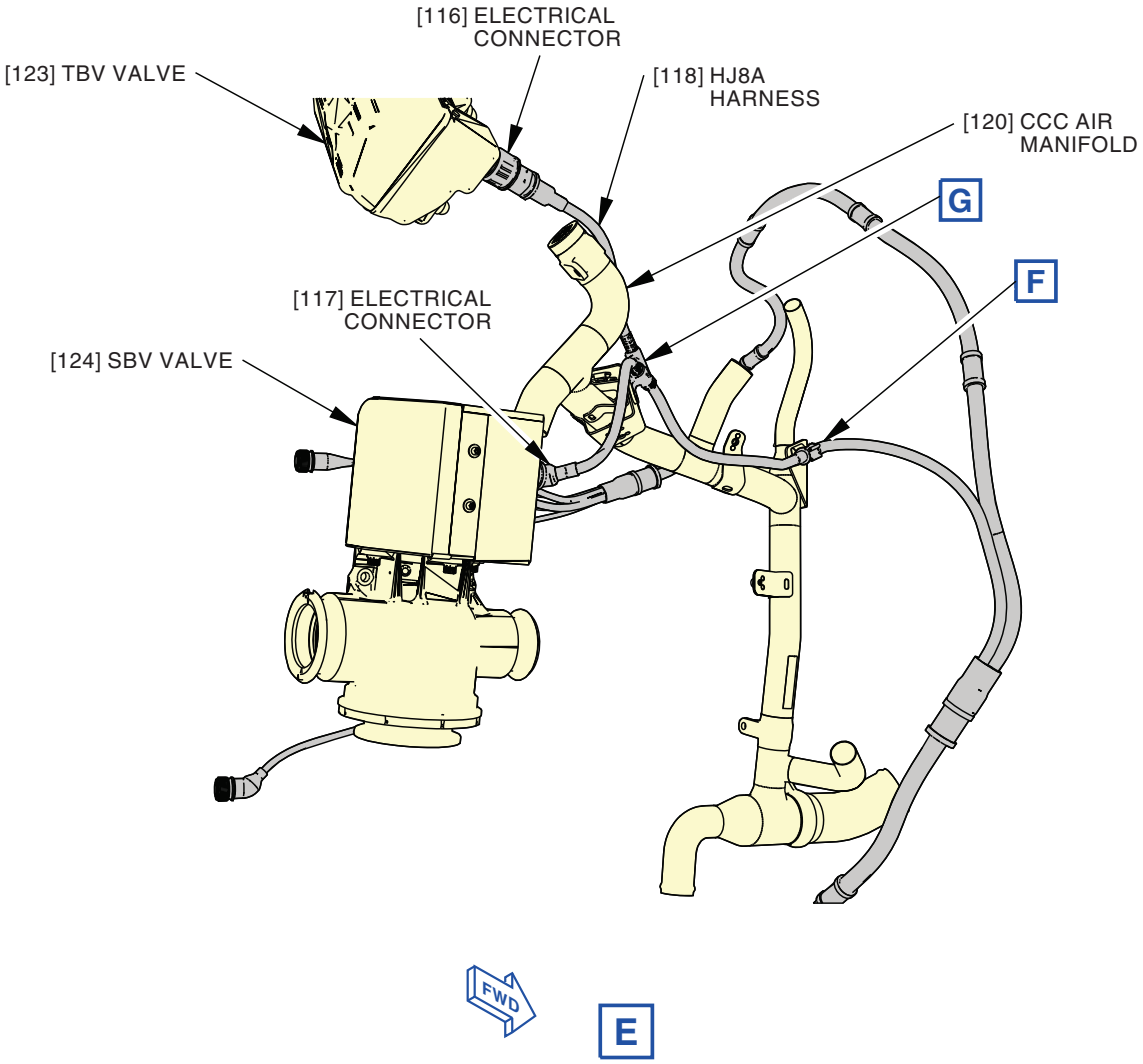
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 5 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 6 of 38)

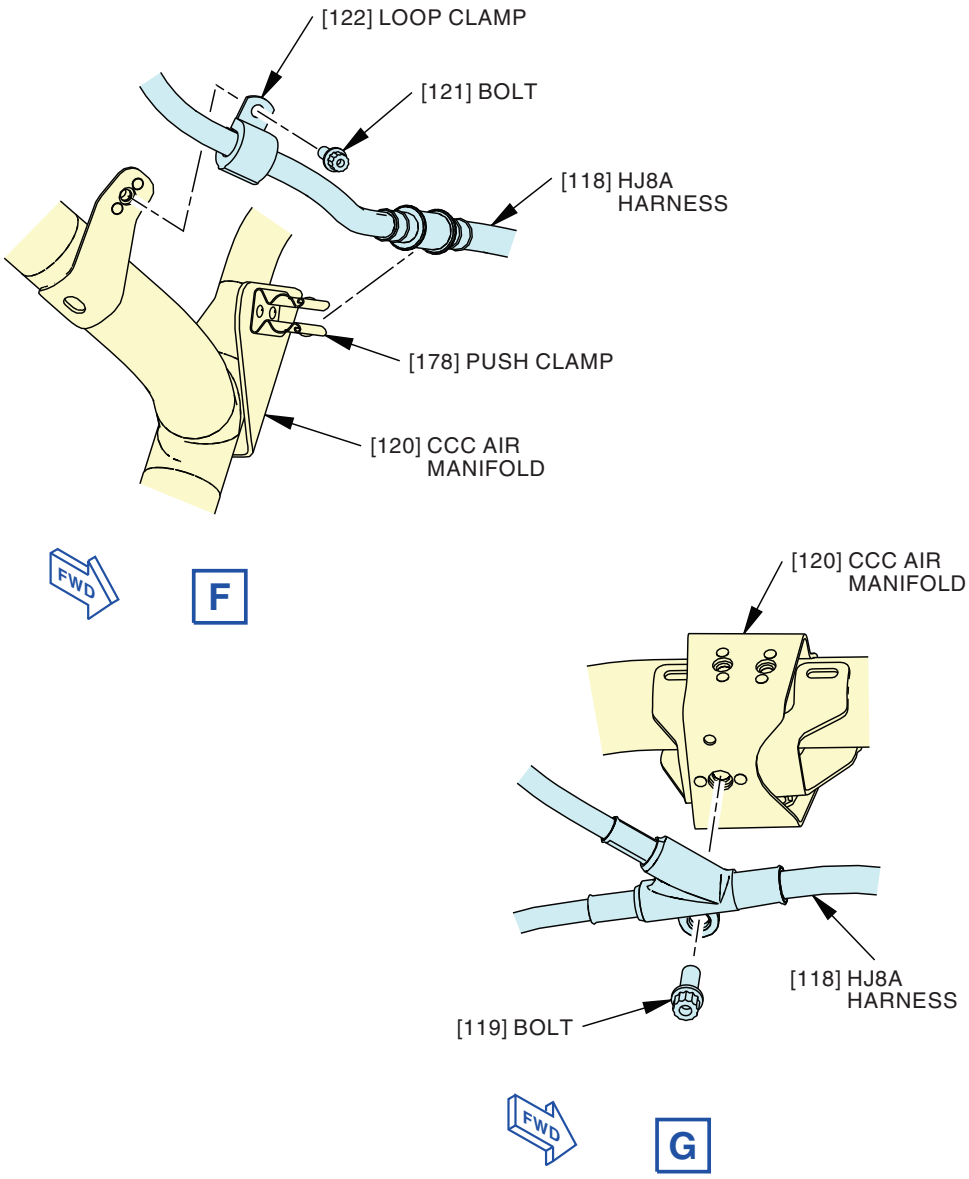
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021986-01
5021986-00
2443839 S0000566364_V4

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 7 of 38)

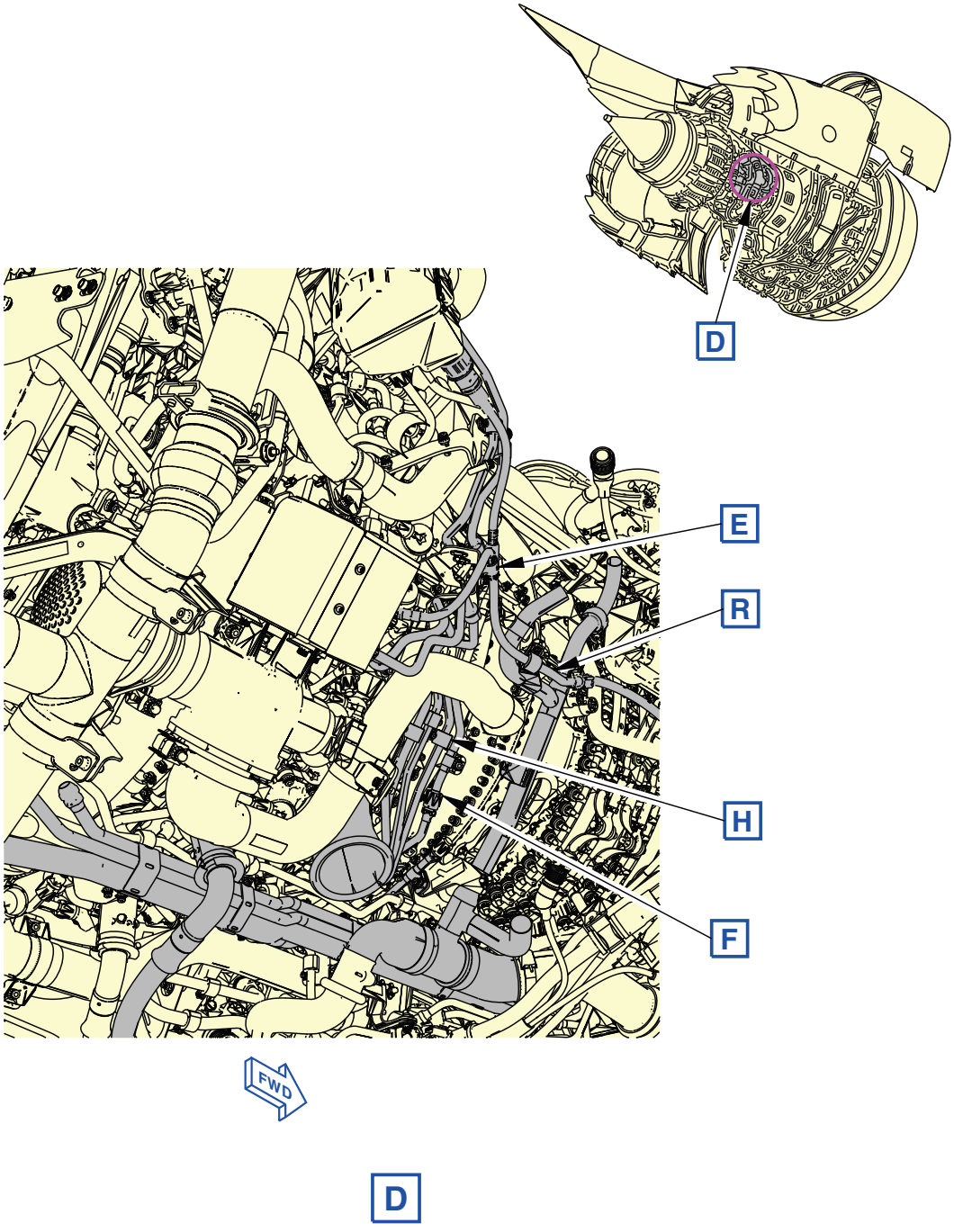
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021986-00
5021986-01
2856660 S0000670619_V3

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 8 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



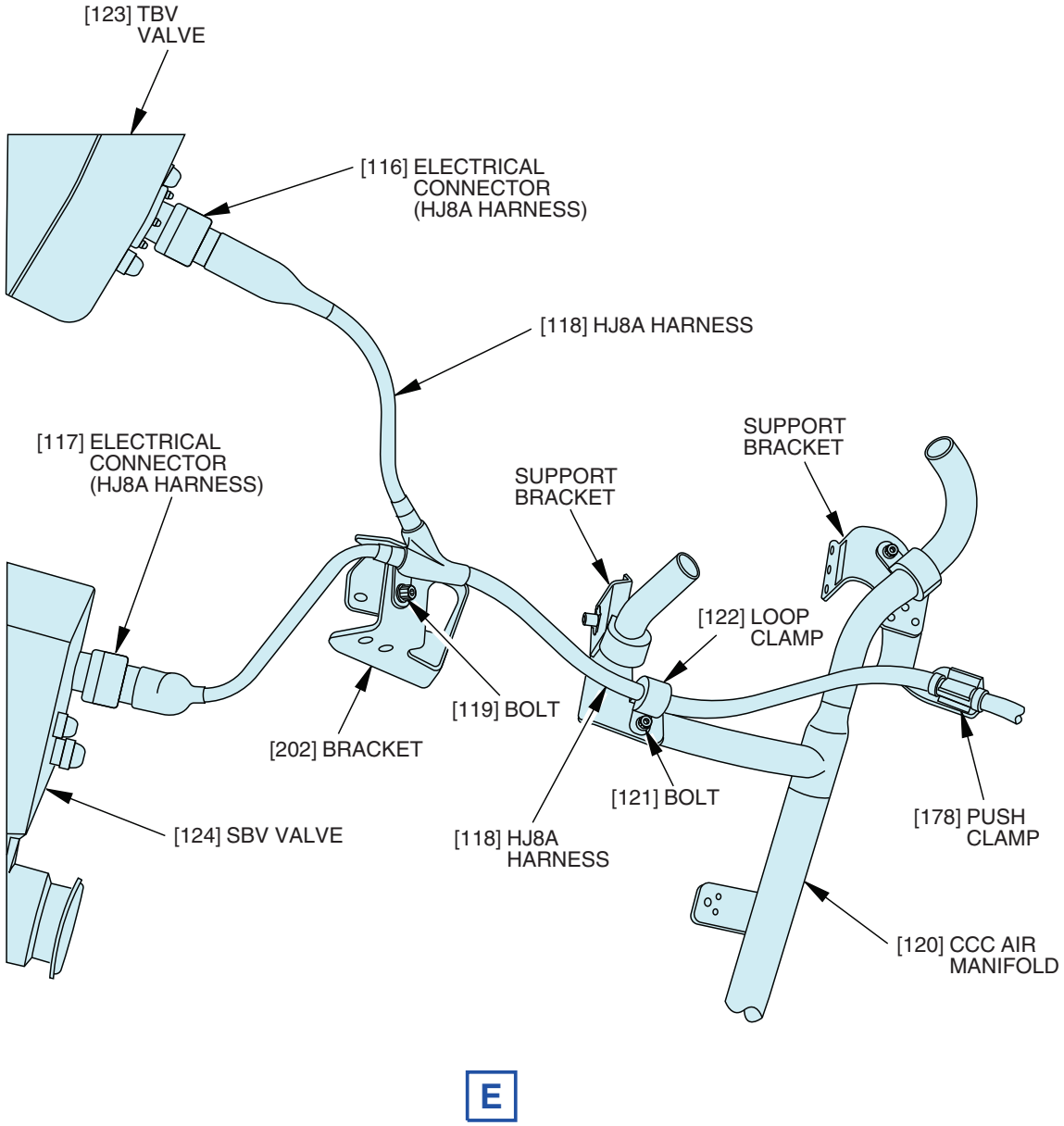
5059227-00
2850886 S0000667444_V1

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 9 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

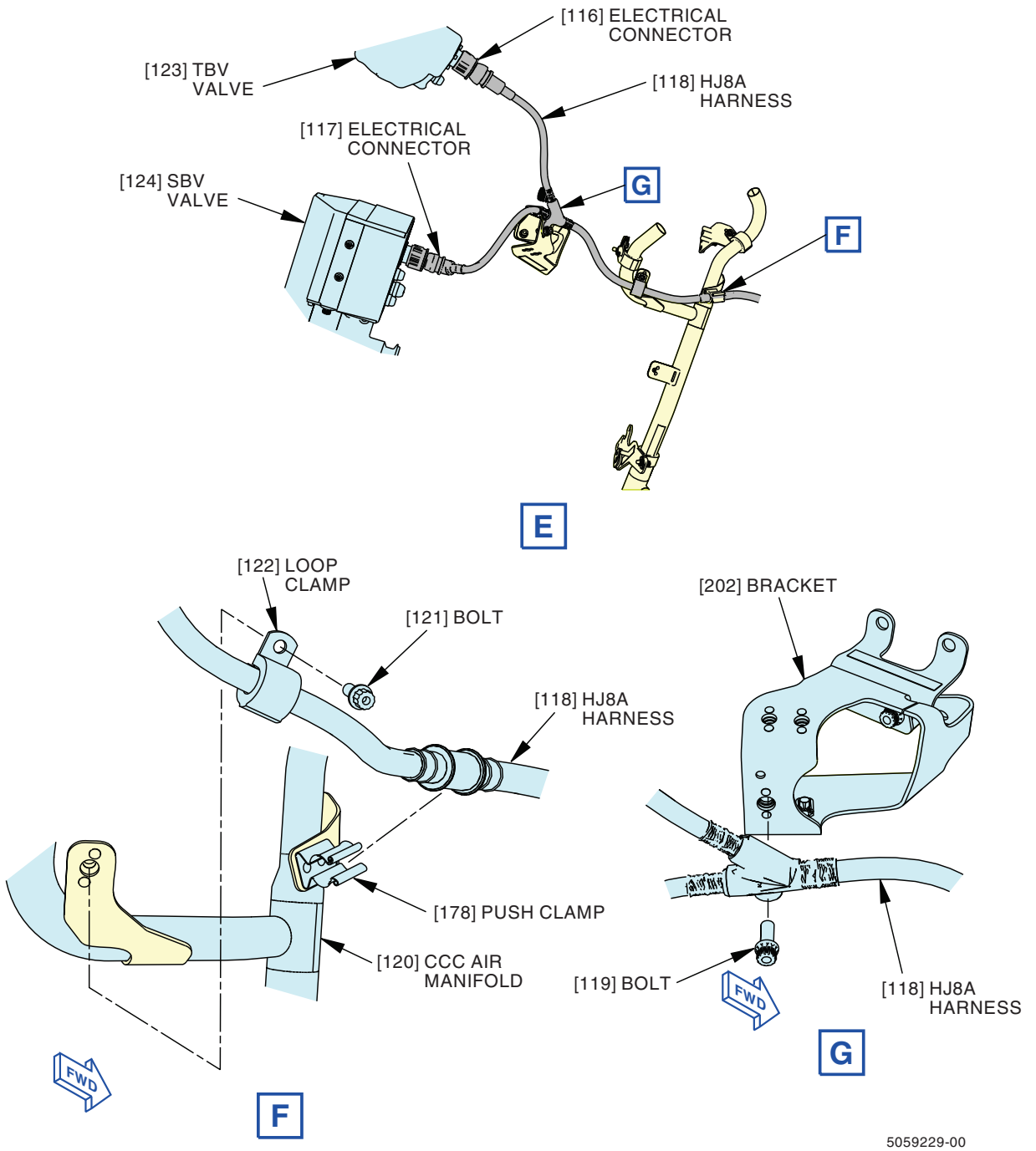
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 10 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



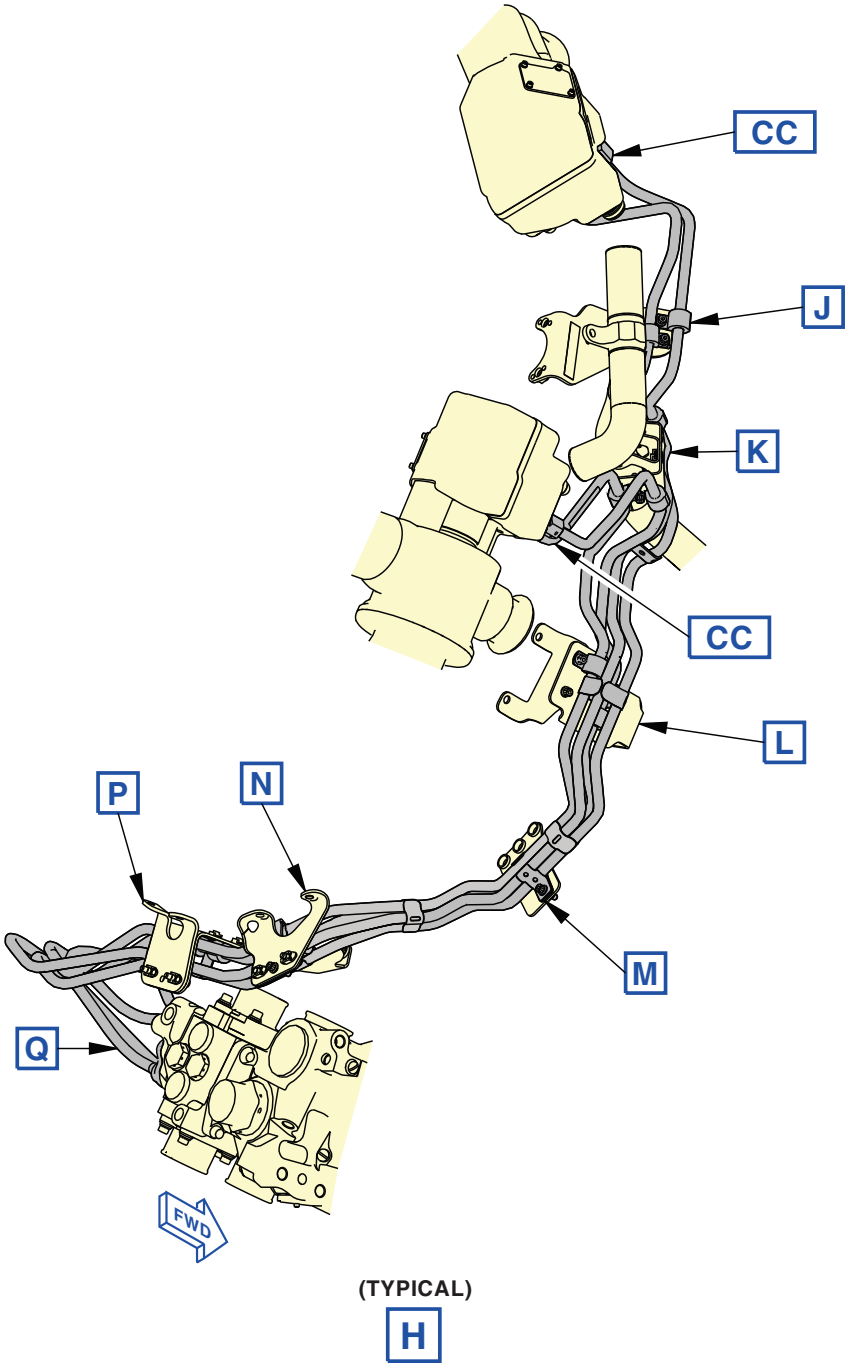
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 11 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 12 of 38)

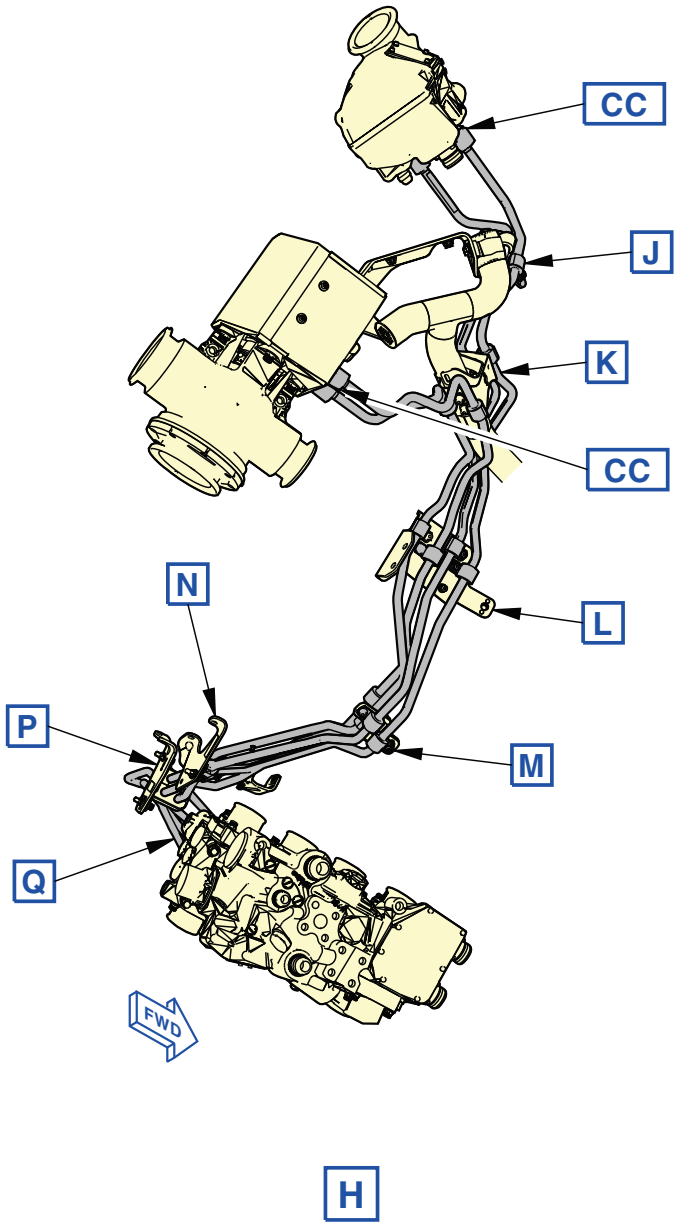
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021987-00
2443840 S0000566365_V4

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 13 of 38)

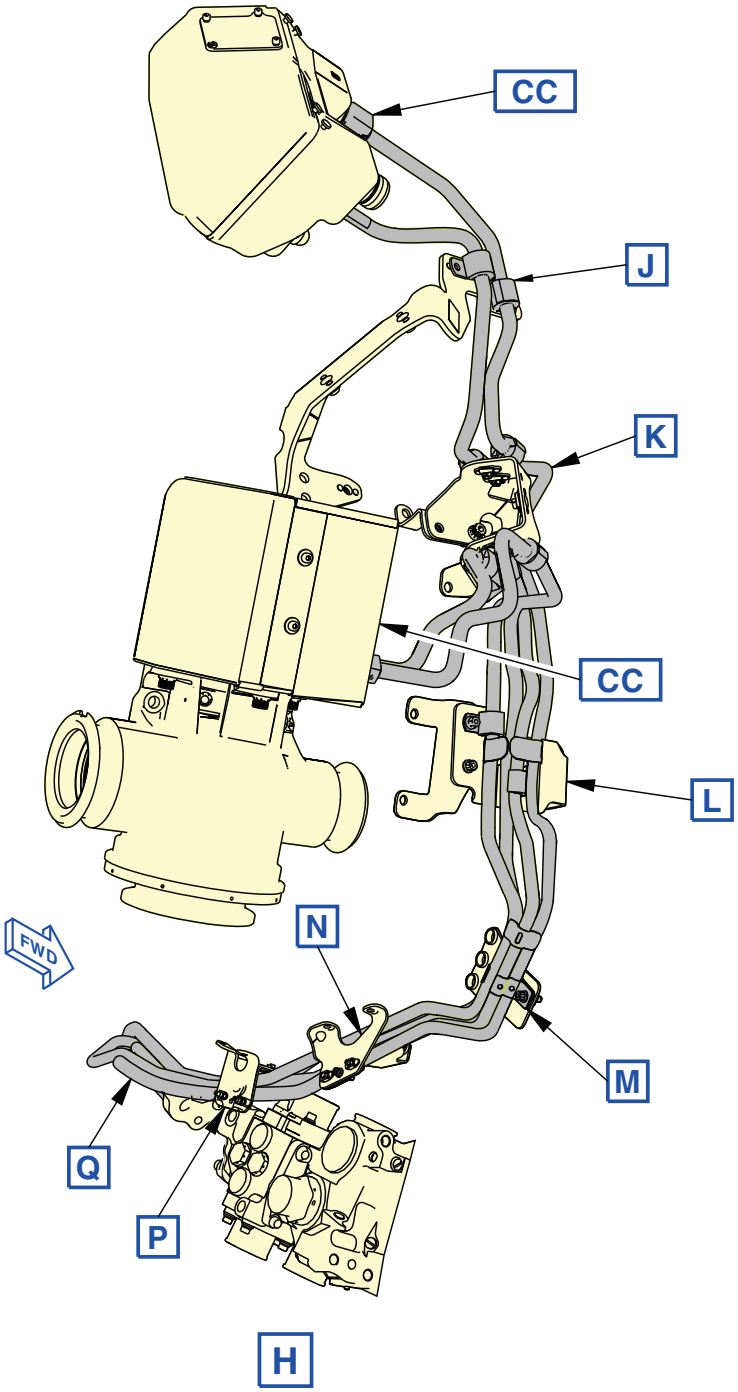
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5051821-00
2856997 S0000669774_V1

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 14 of 38)

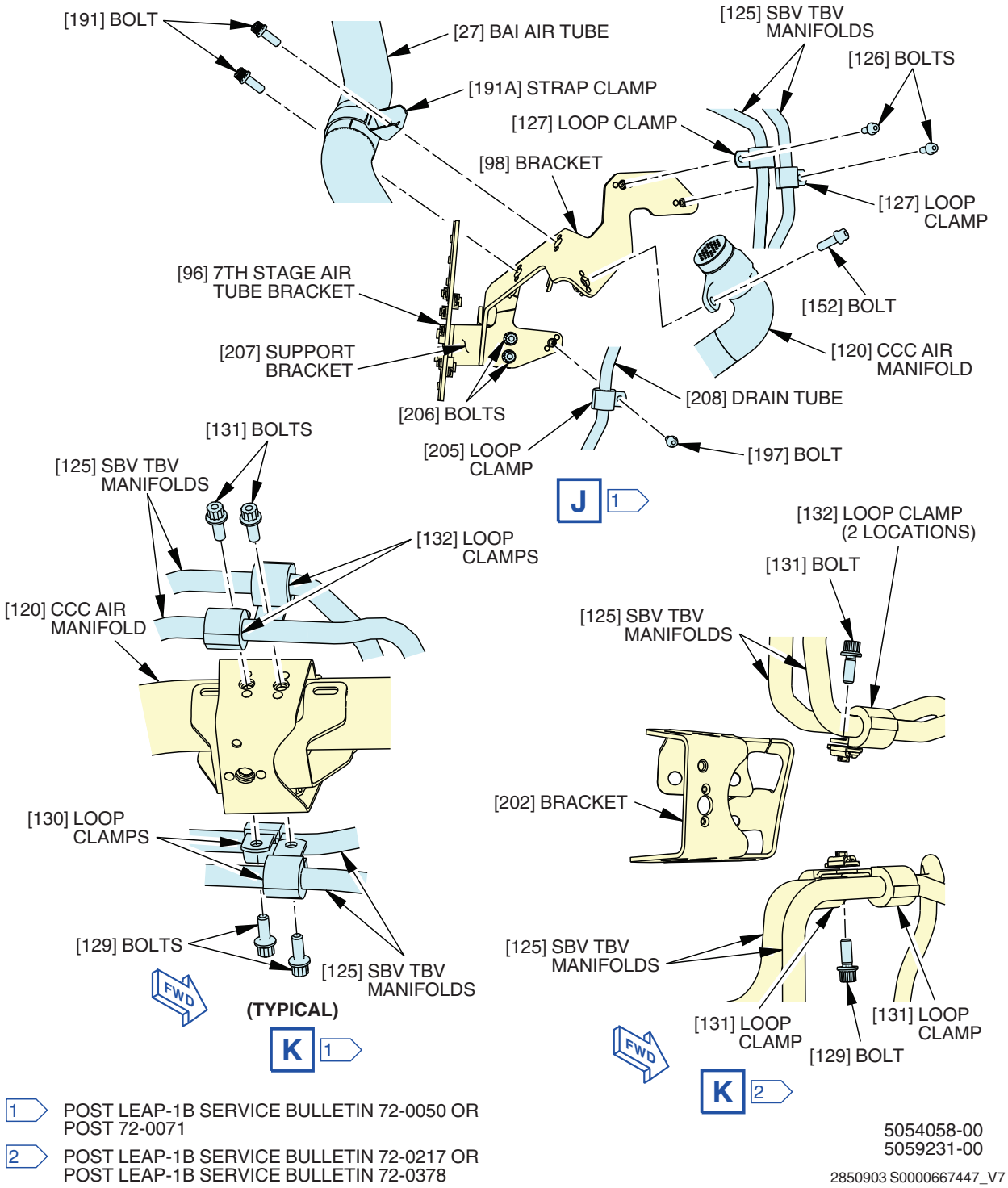
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5059230-00
2850901 S0000667446_V1

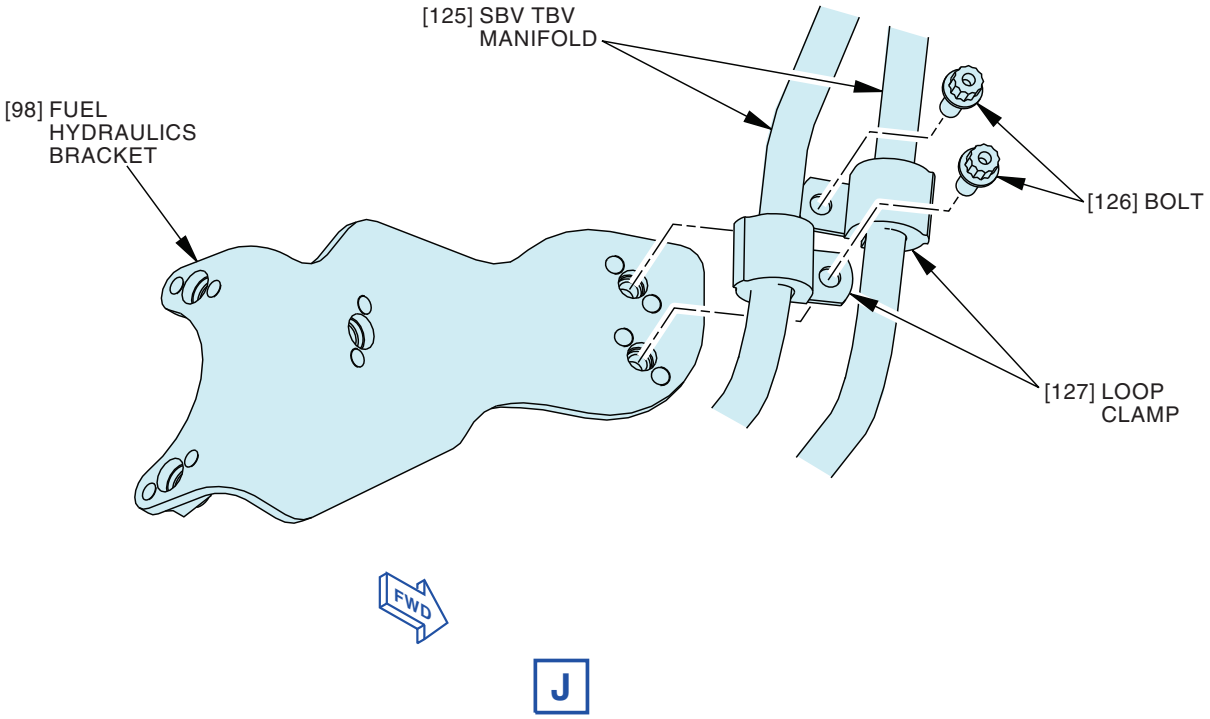
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 15 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 16 of 38)

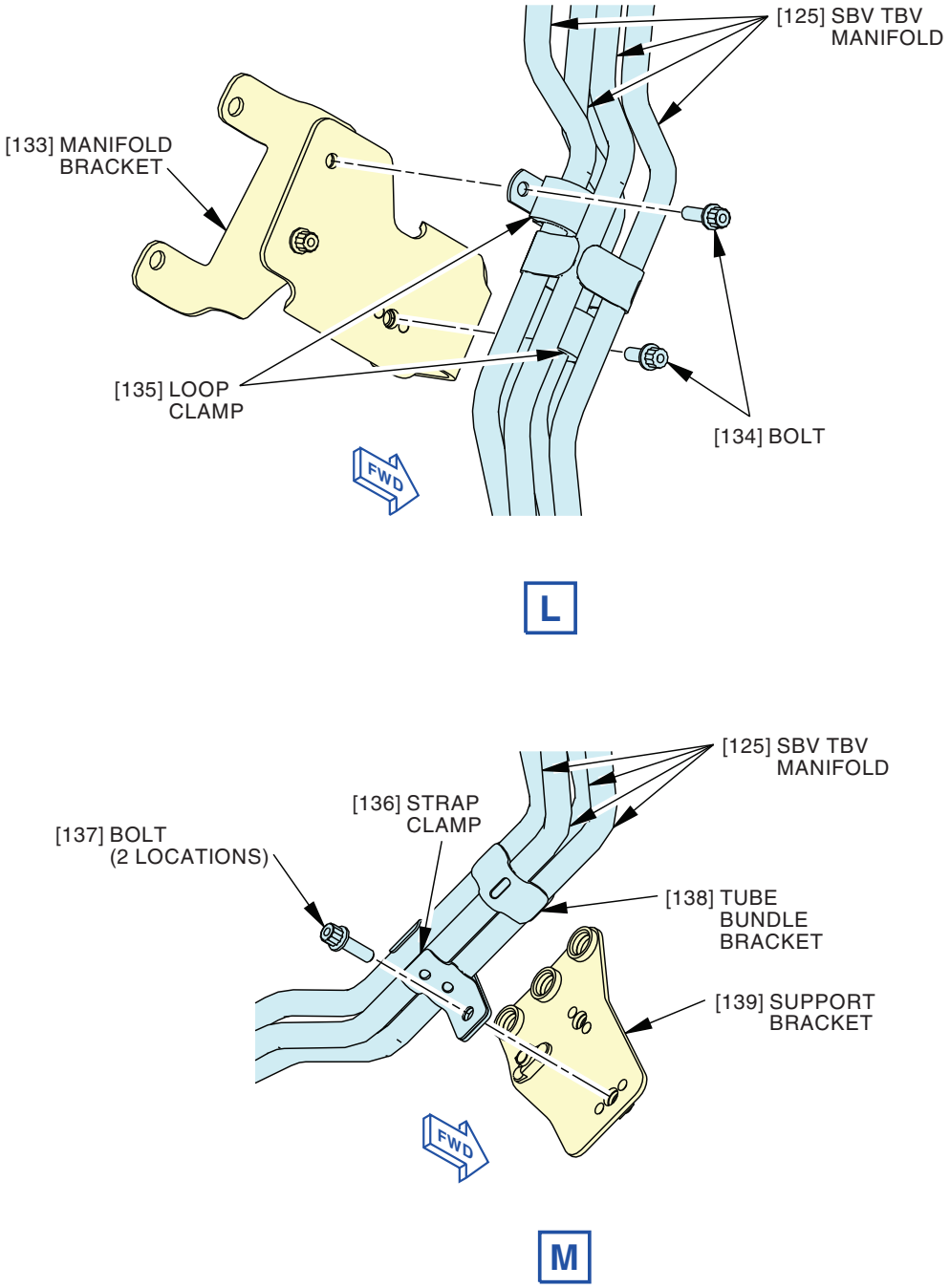
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5059231-00
2856951 S0000671365_V3

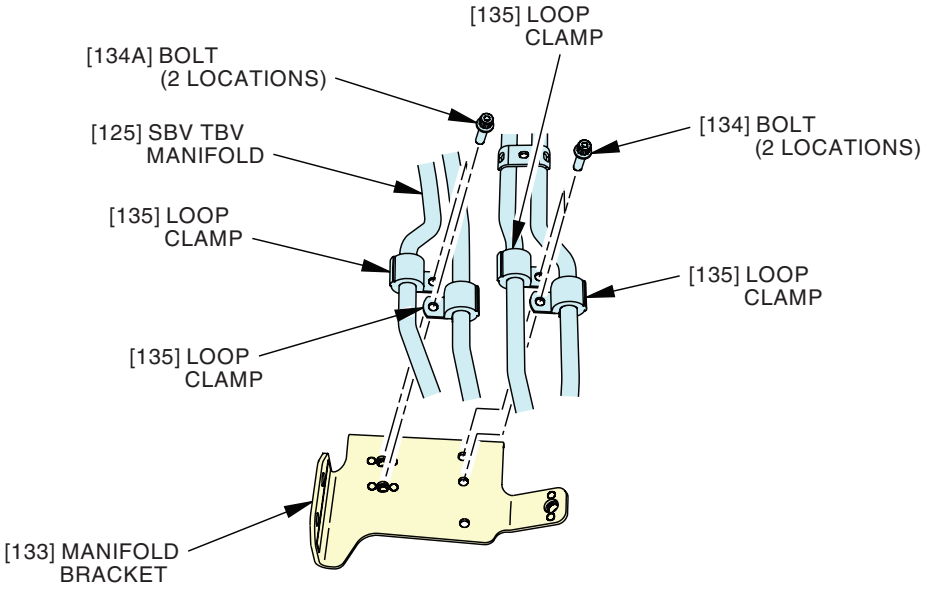
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 17 of 38)

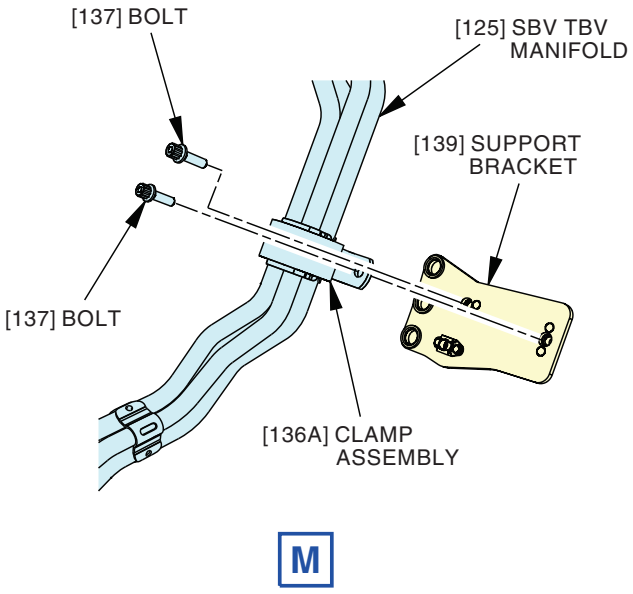
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 18 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

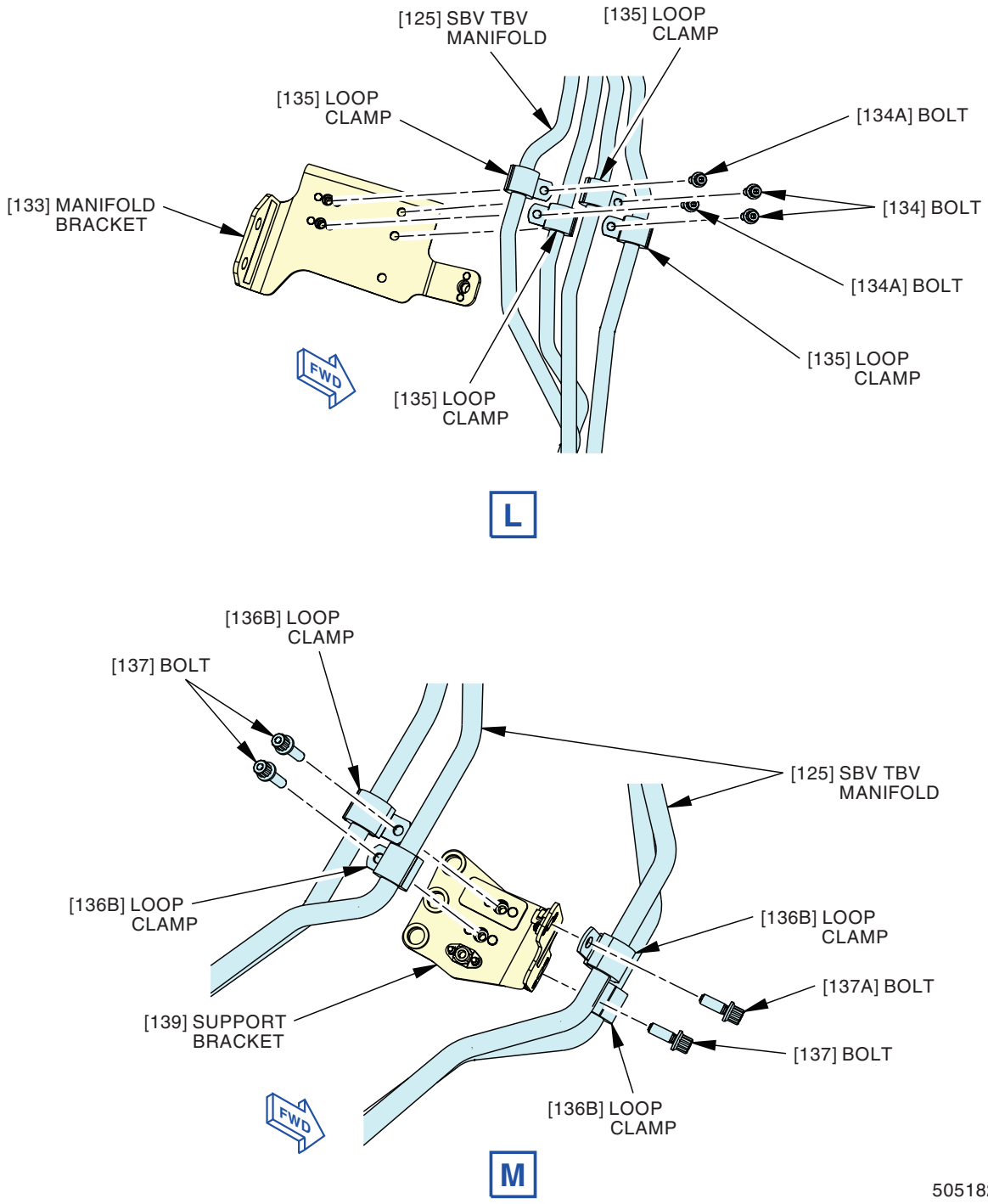




5051823-00
2857008 S0000669775_V1

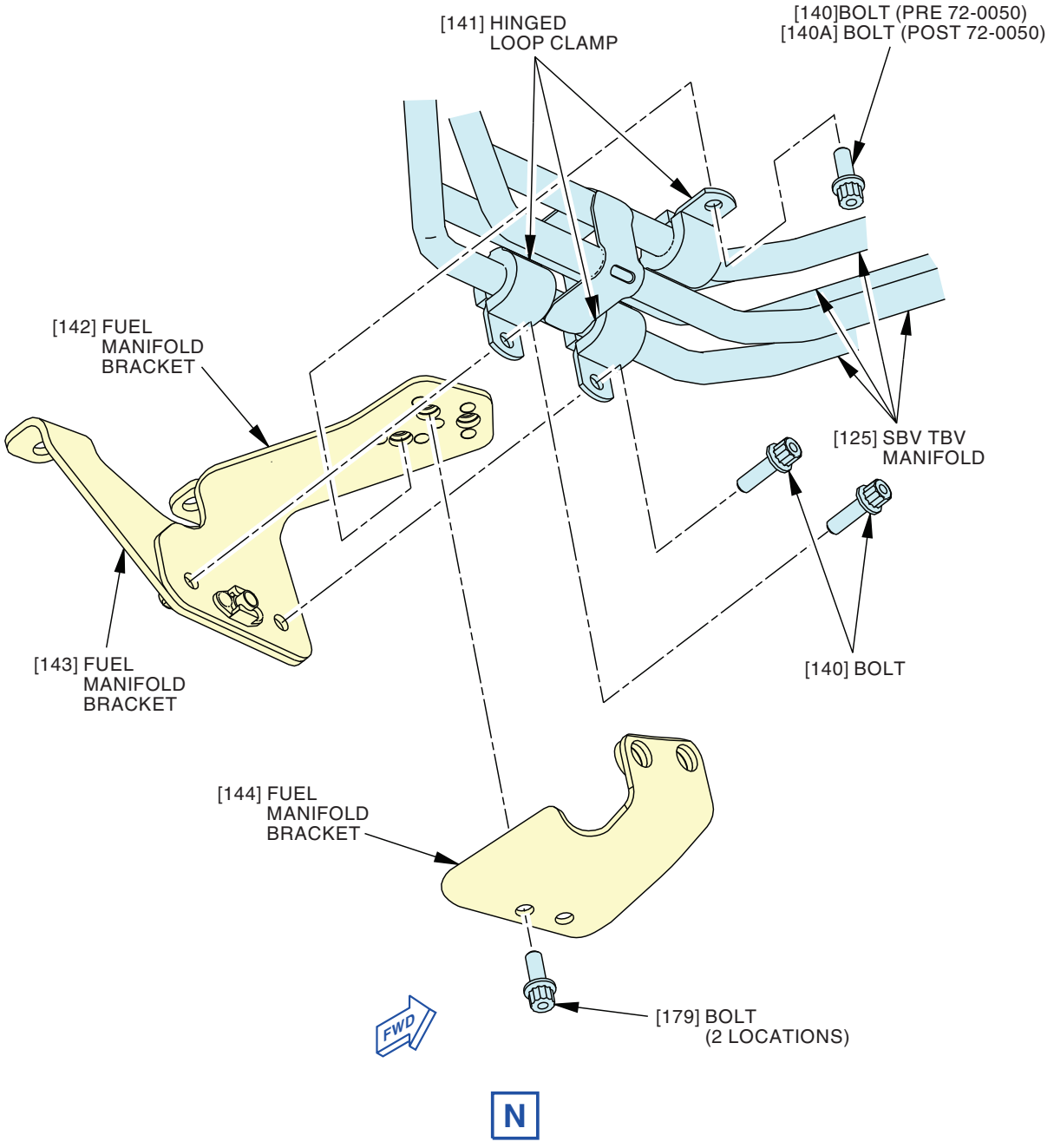
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 19 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



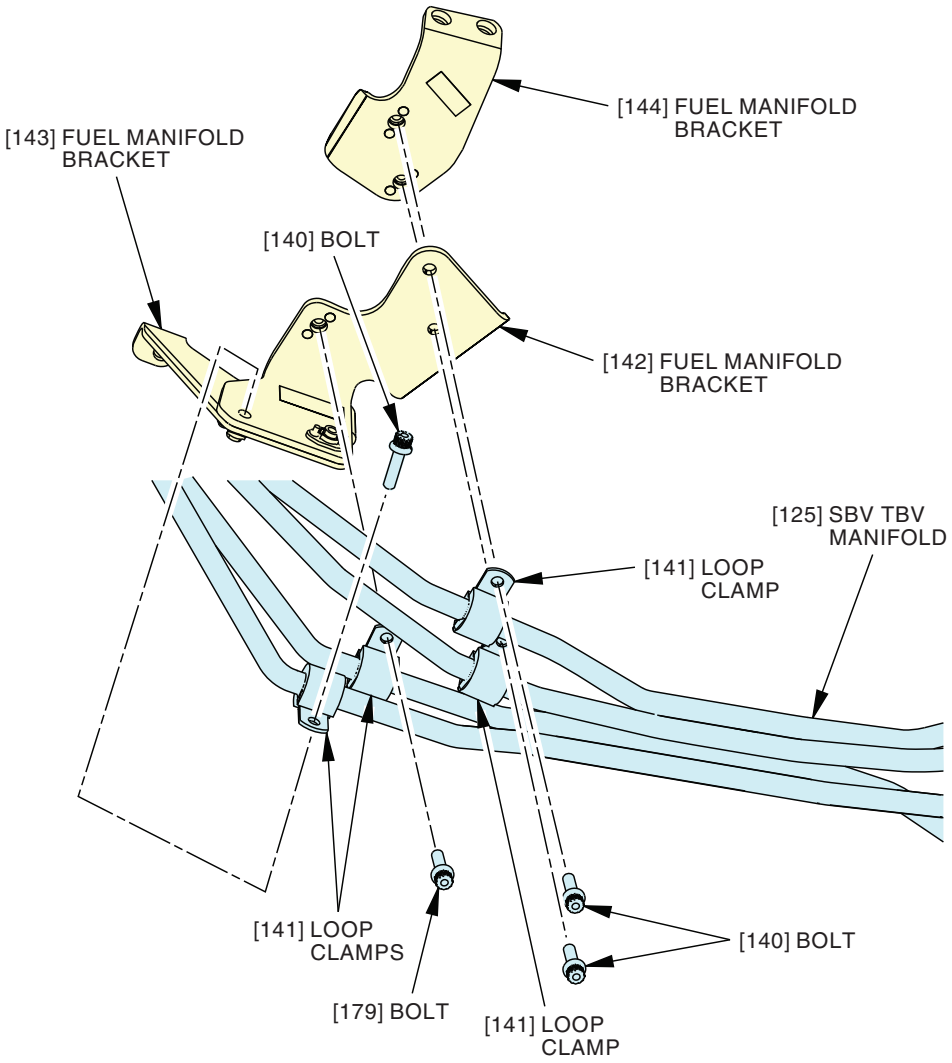
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 20 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 21 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

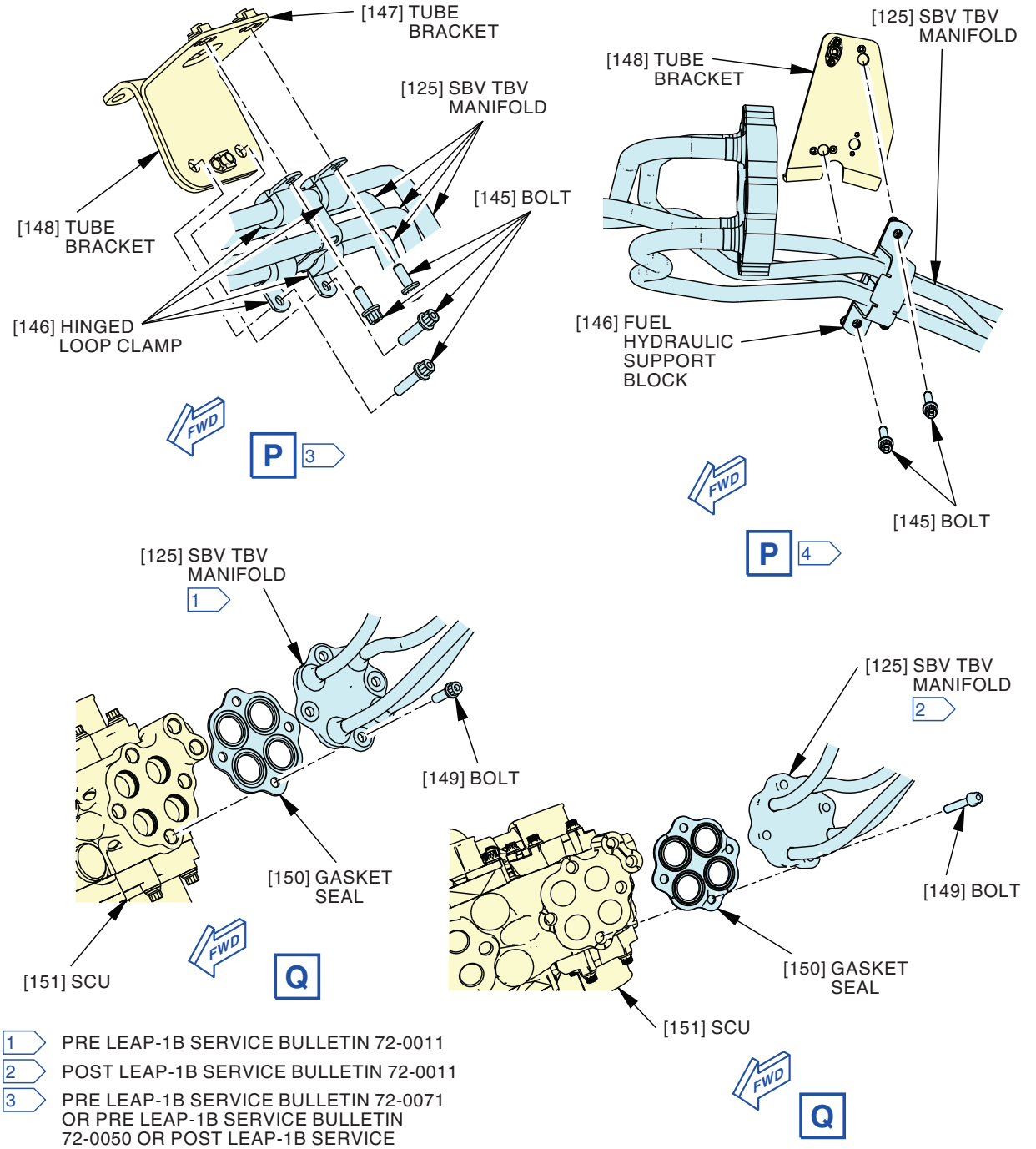


N

5051825-00
2857075 S0000670838_V2

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 22 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



1 PRE LEAP-1B SERVICE BULLETIN 72-0011

2 POST LEAP-1B SERVICE BULLETIN 72-0011

3 PRE LEAP-1B SERVICE BULLETIN 72-0071 OR PRE LEAP-1B SERVICE BULLETIN 72-0050 OR POST LEAP-1B SERVICE BULLETIN 72-0050

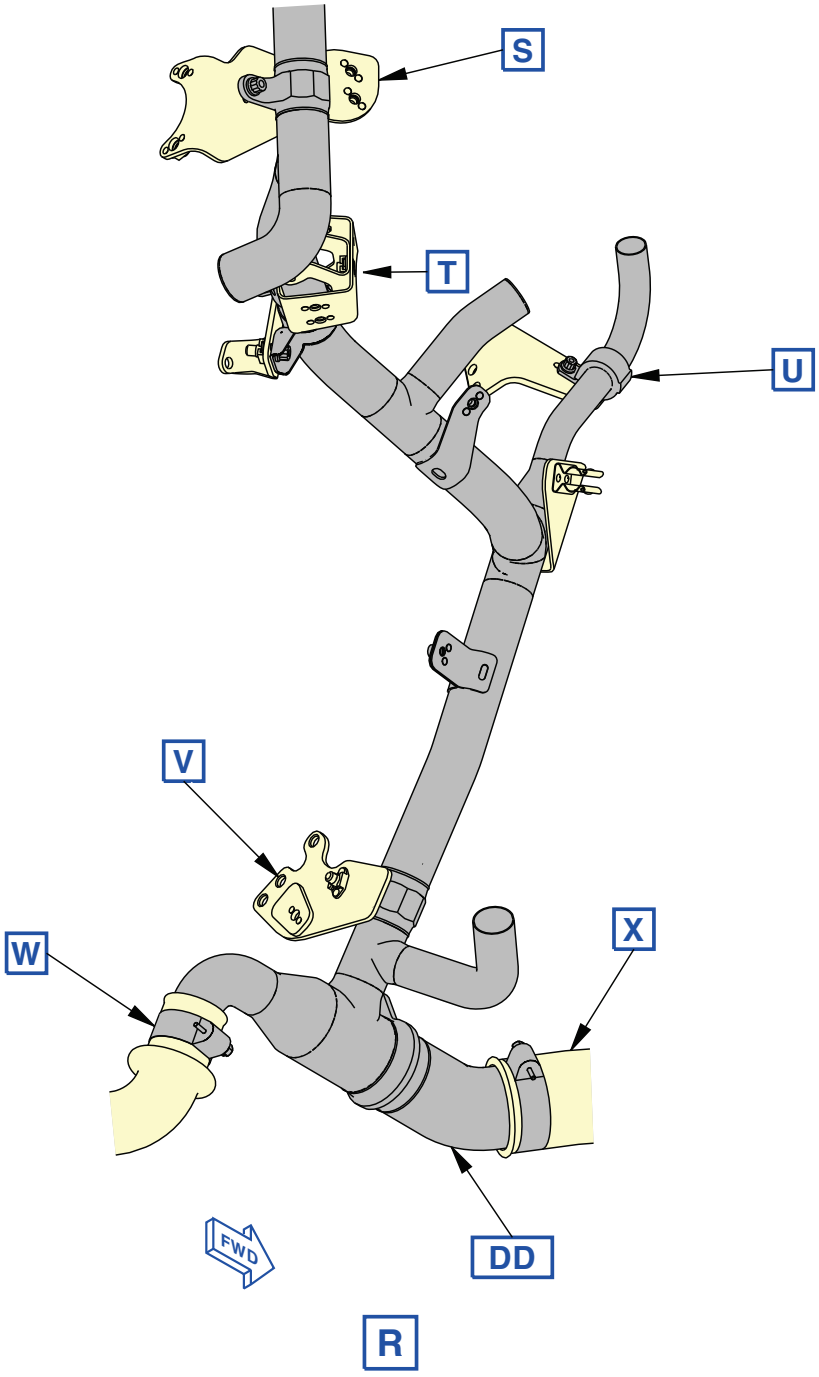
4 POST LEAP-1B SERVICE BULLETIN 72-0071

5053481-00/ 5051826-00

2443845 S0000566369_V5

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 23 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

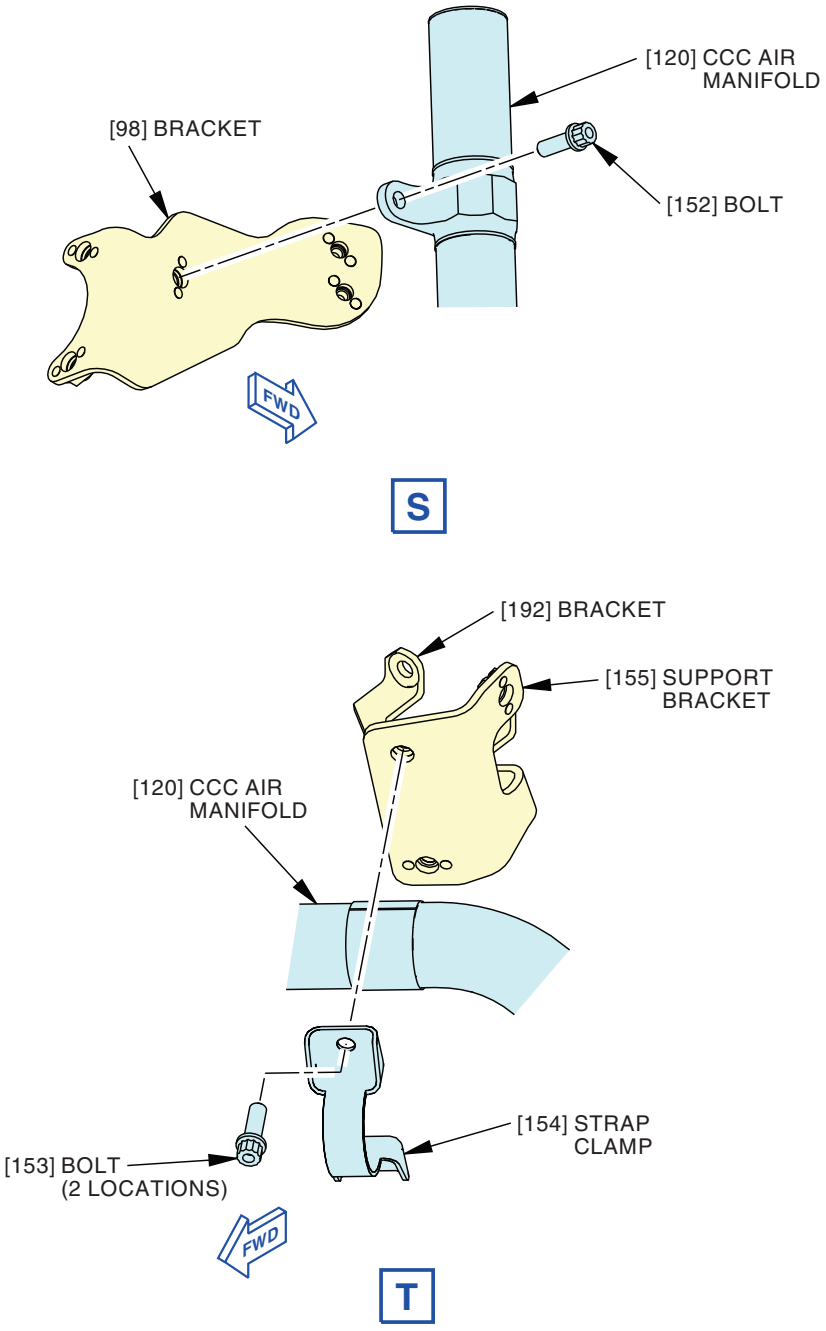


5021994-00

2443846 S0000566370_V3

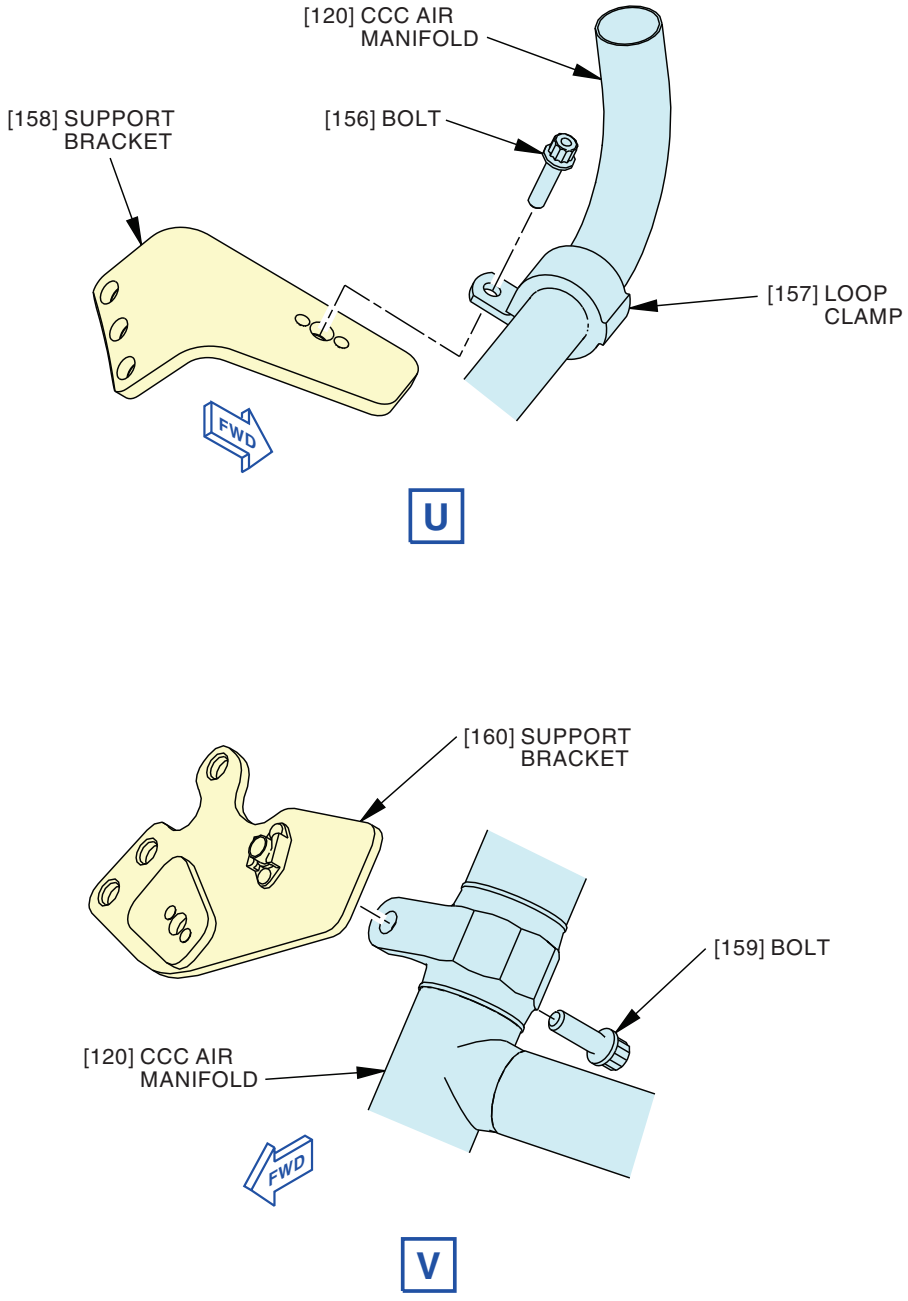
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 24 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 25 of 38)

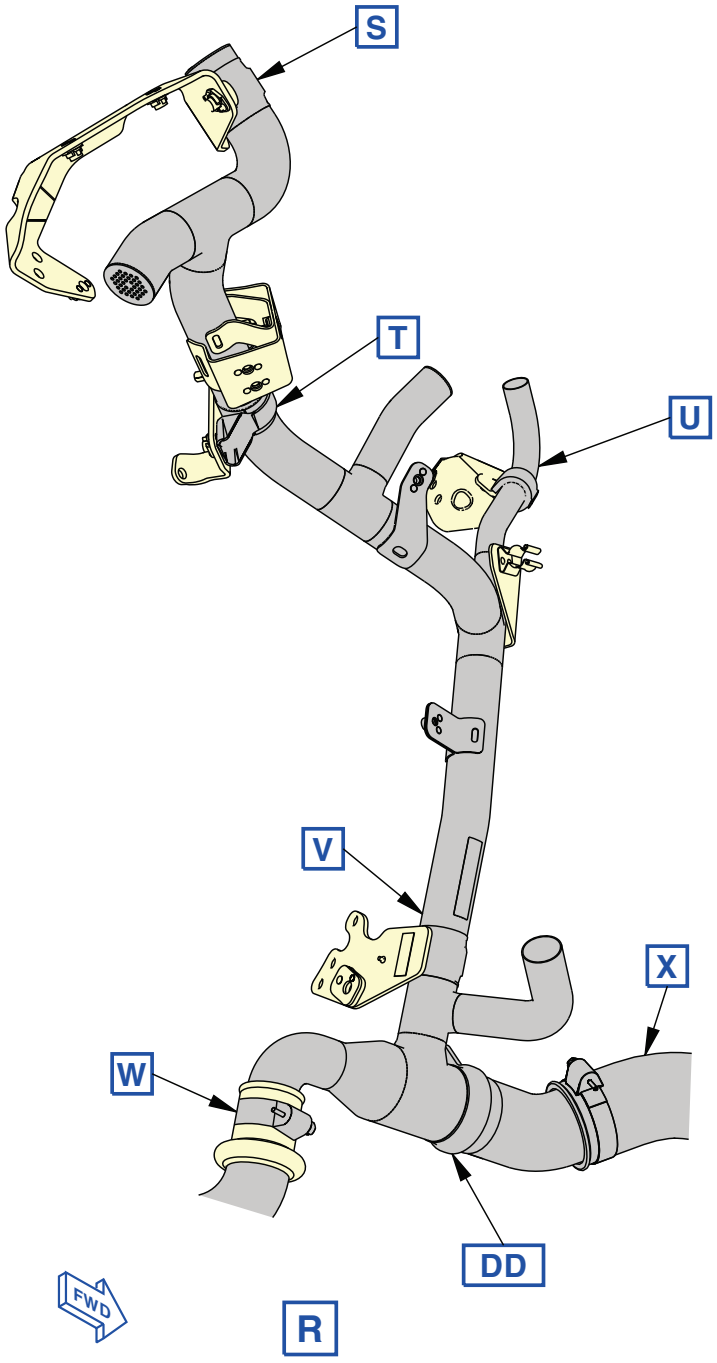
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021996-00
2443849 S0000566372_V2

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 26 of 38)

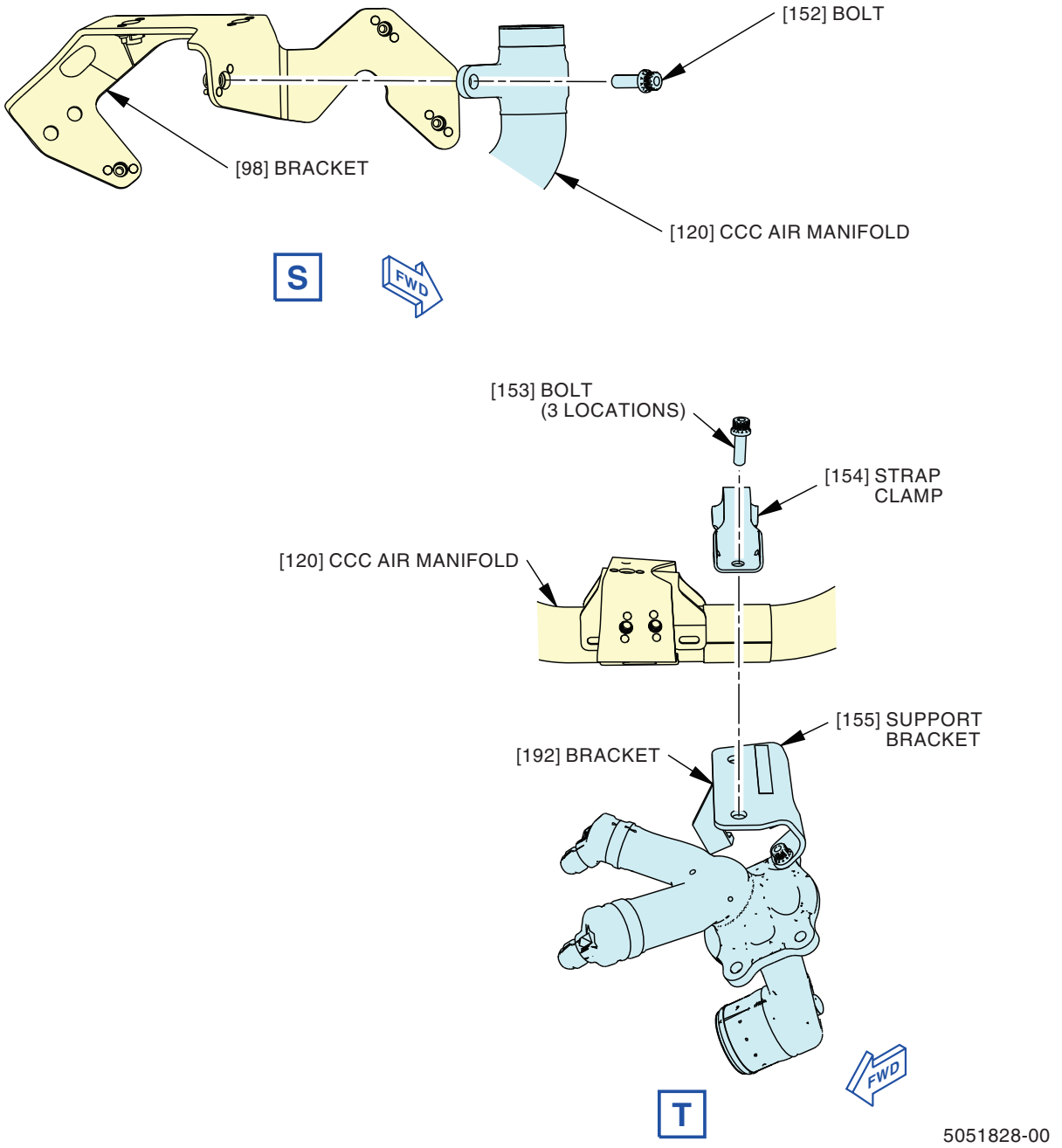
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5051827-00
2856411 S0000670621_V1

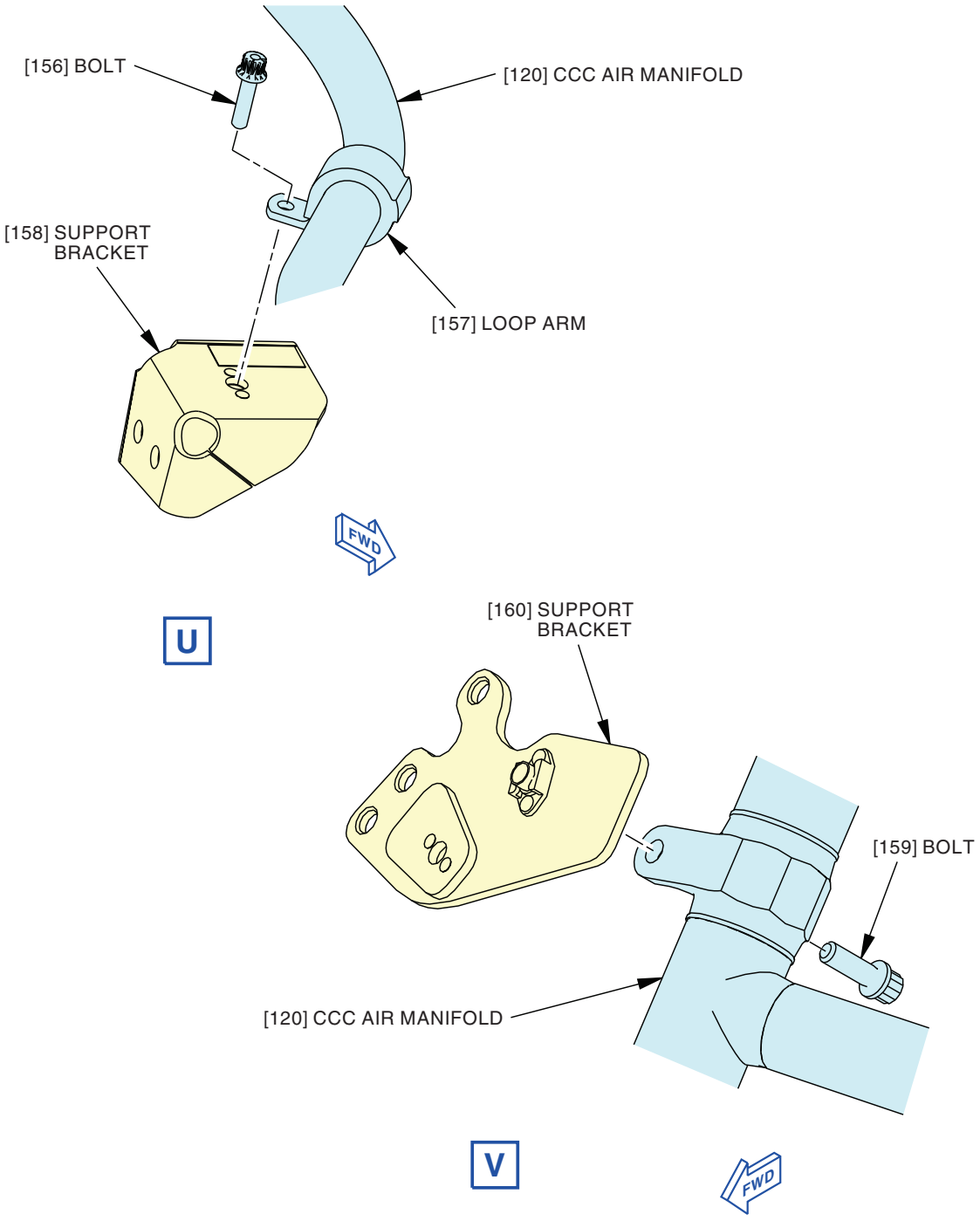
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 27 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



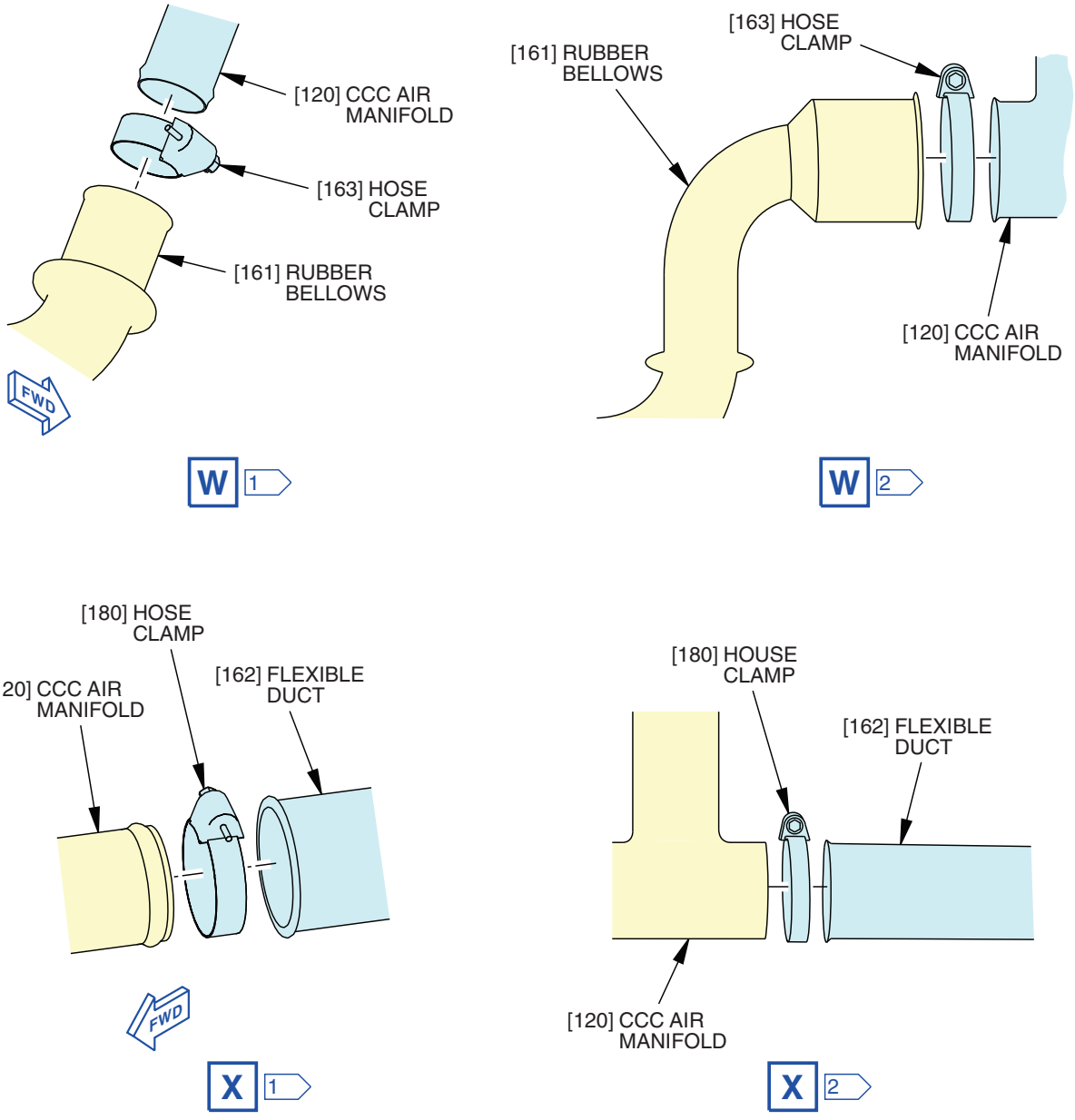
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 28 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 29 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



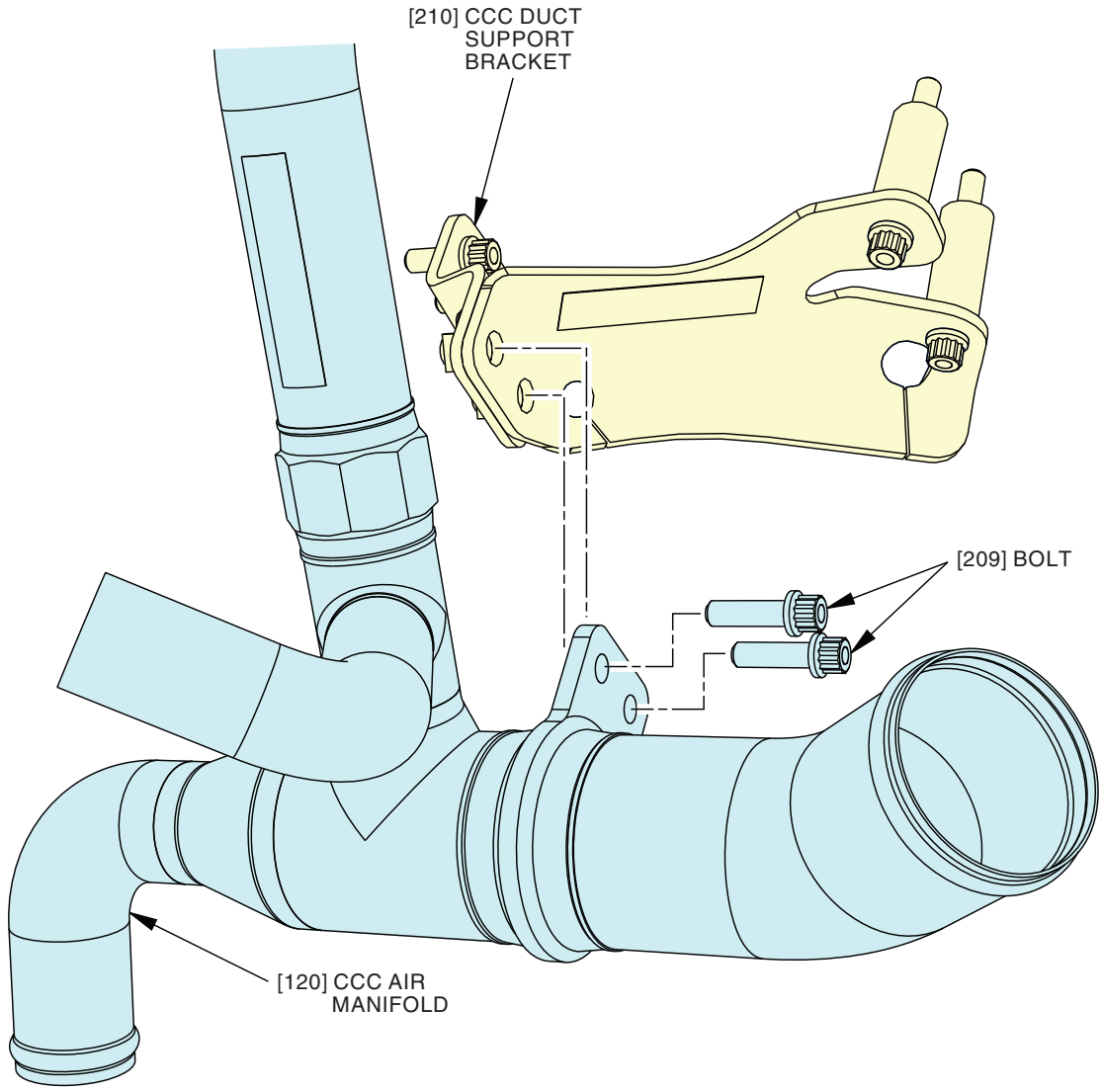
1 PRE LEAP-1B SERVICE BULLETIN 72-0050 OR
POST LEAP-1B SERVICE BULLETIN 72-0050

2 POST LEAP-1B SERVICE BULLETIN 72-0135 OR
POST LEAP-1B SERVICE BULLETIN 72-0217 OR
POST LEAP-1B SERVICE BULLETIN 72-0378

5021997-00
2443850 S0000566373_V4

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 30 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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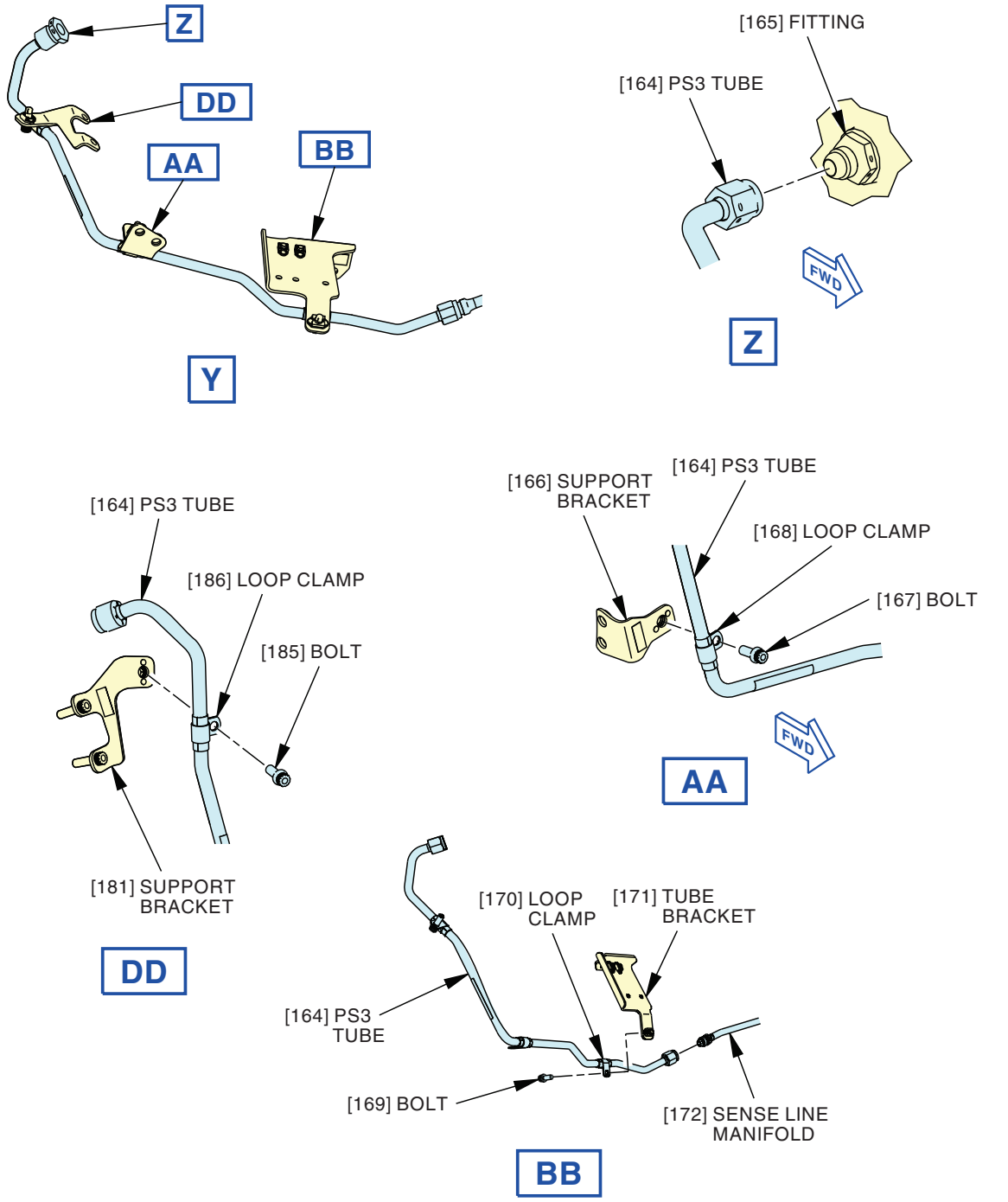
5052928-00
2856357 S0000670624_V2

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 31 of 38)

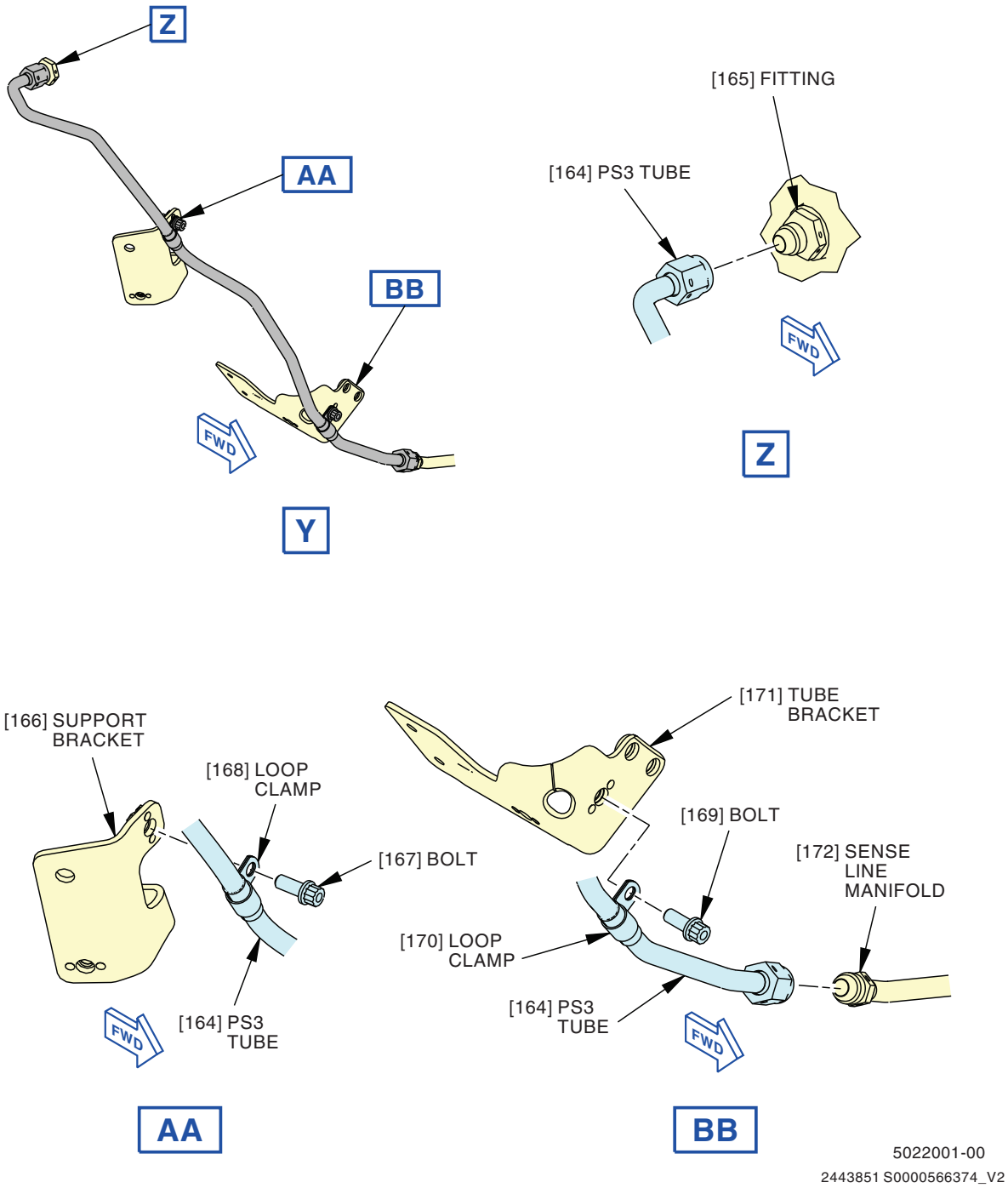
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5051836-00
2857301 S0000670839_V2

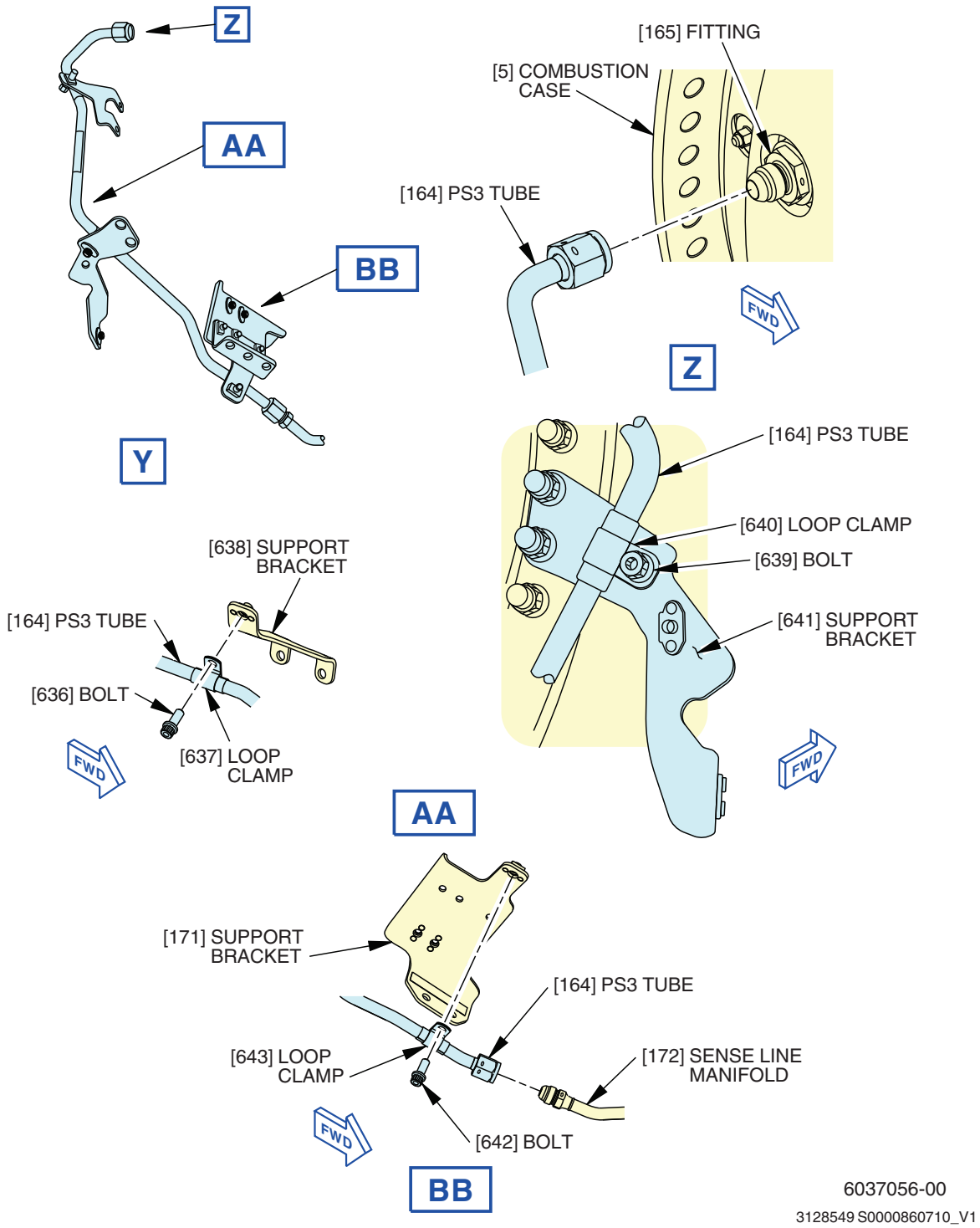
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 32 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



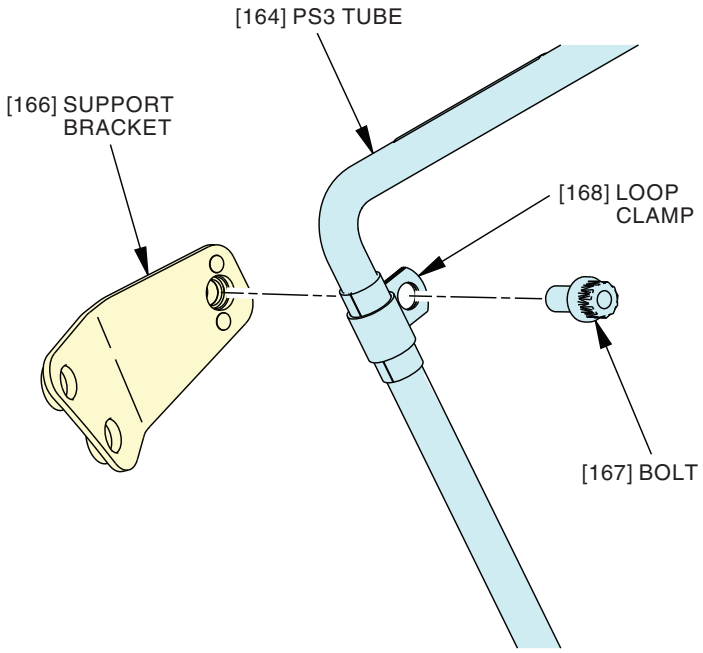
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 33 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 34 of 38)

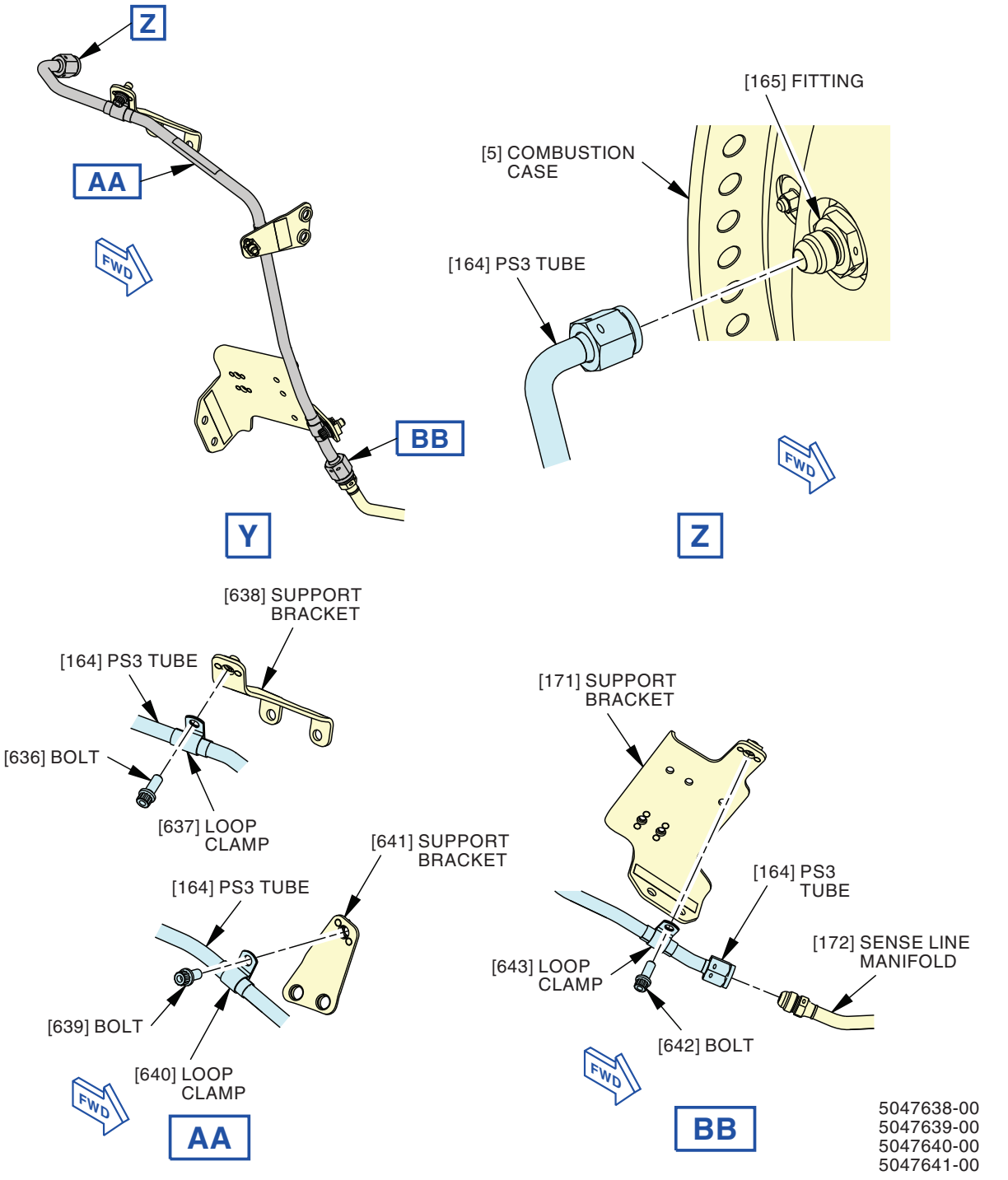
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5051837-00
2857310 S0000670840_V1

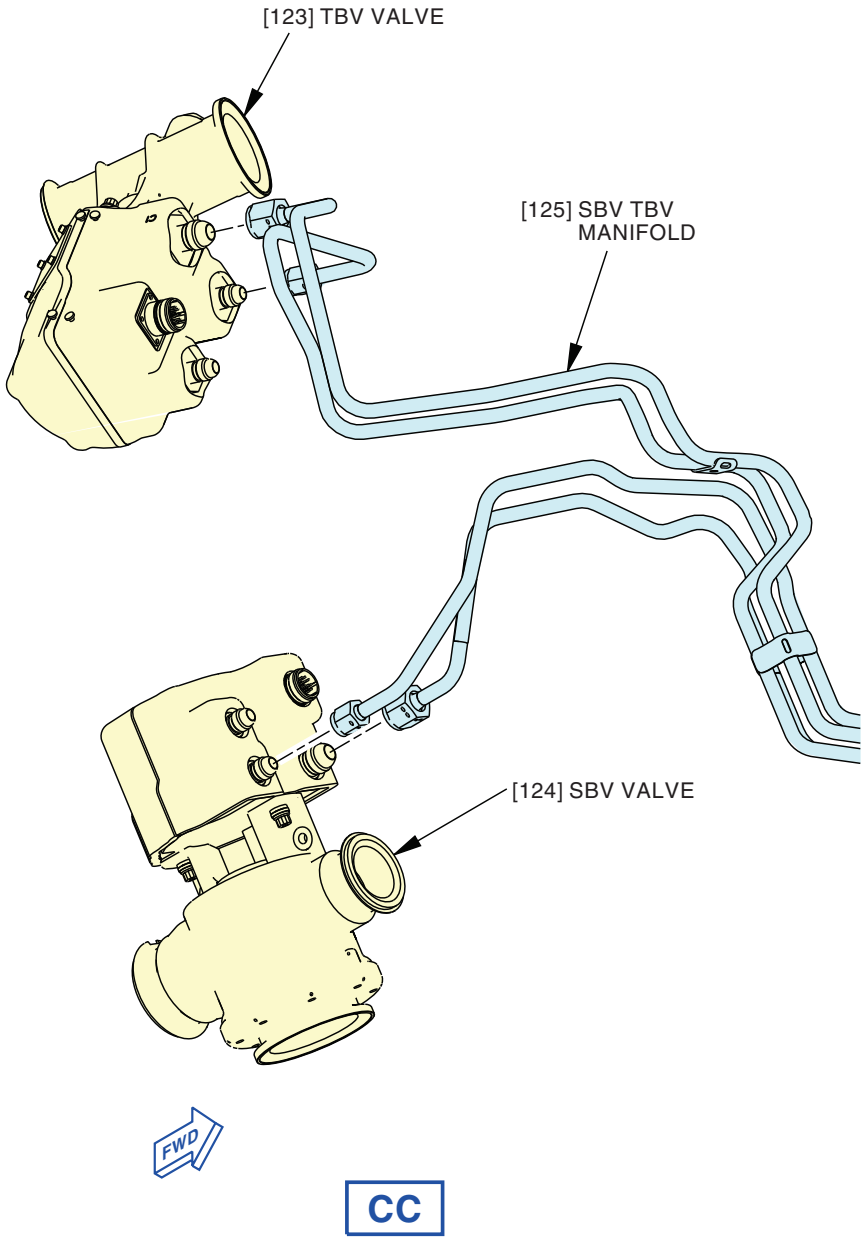
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 35 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 36 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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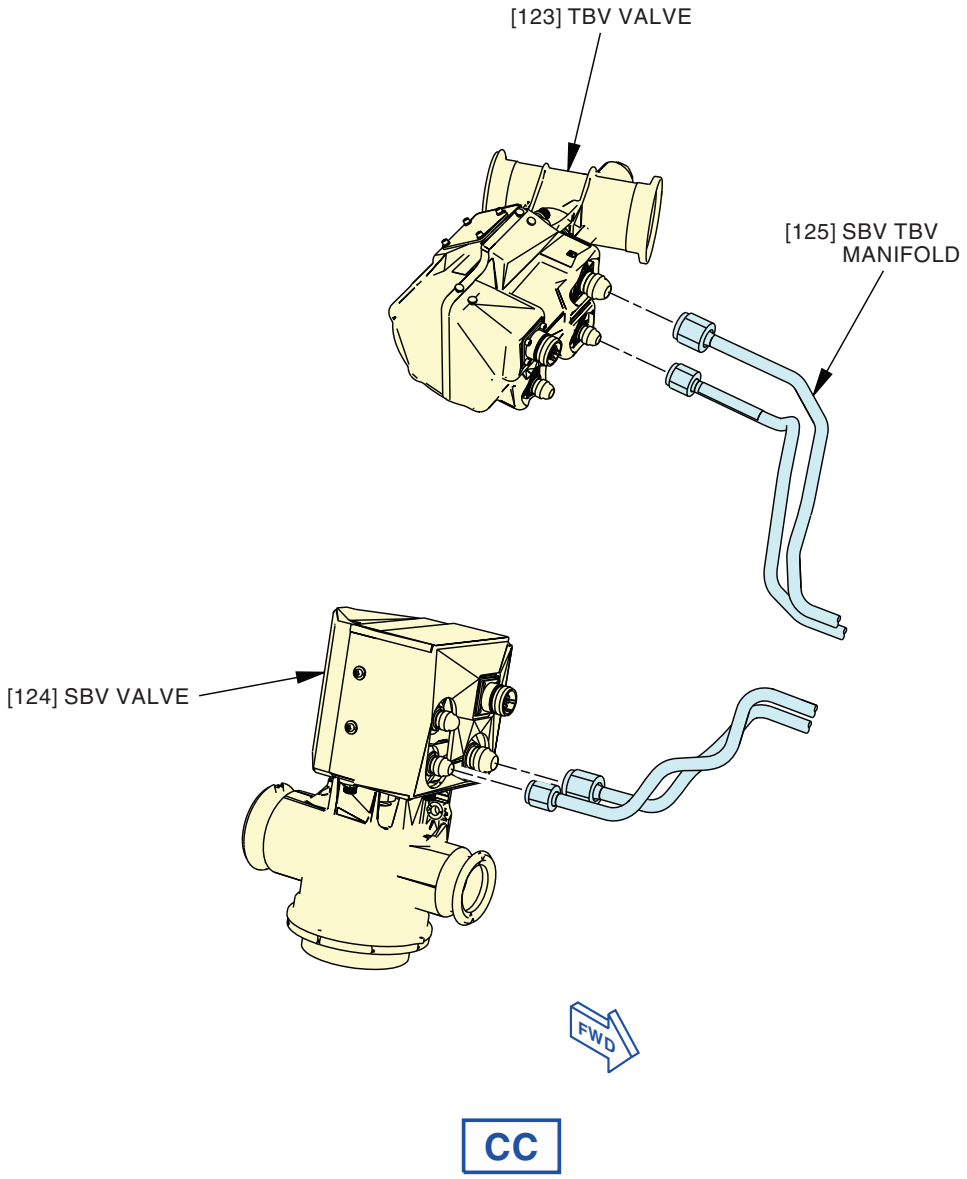
2476204 S0000580839_V1

Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 37 of 38)

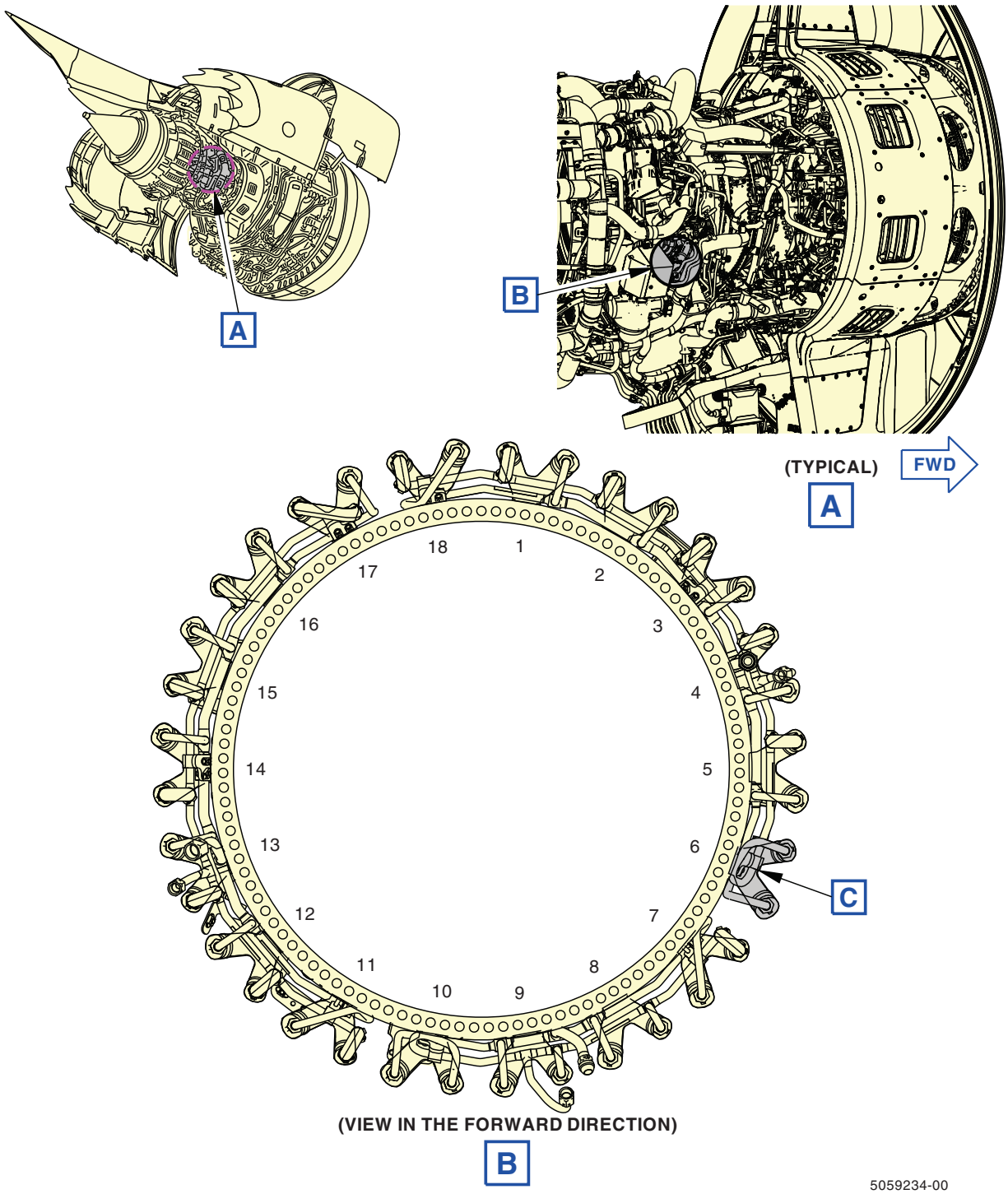
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5051839-00
2857329 S0000670841_V1

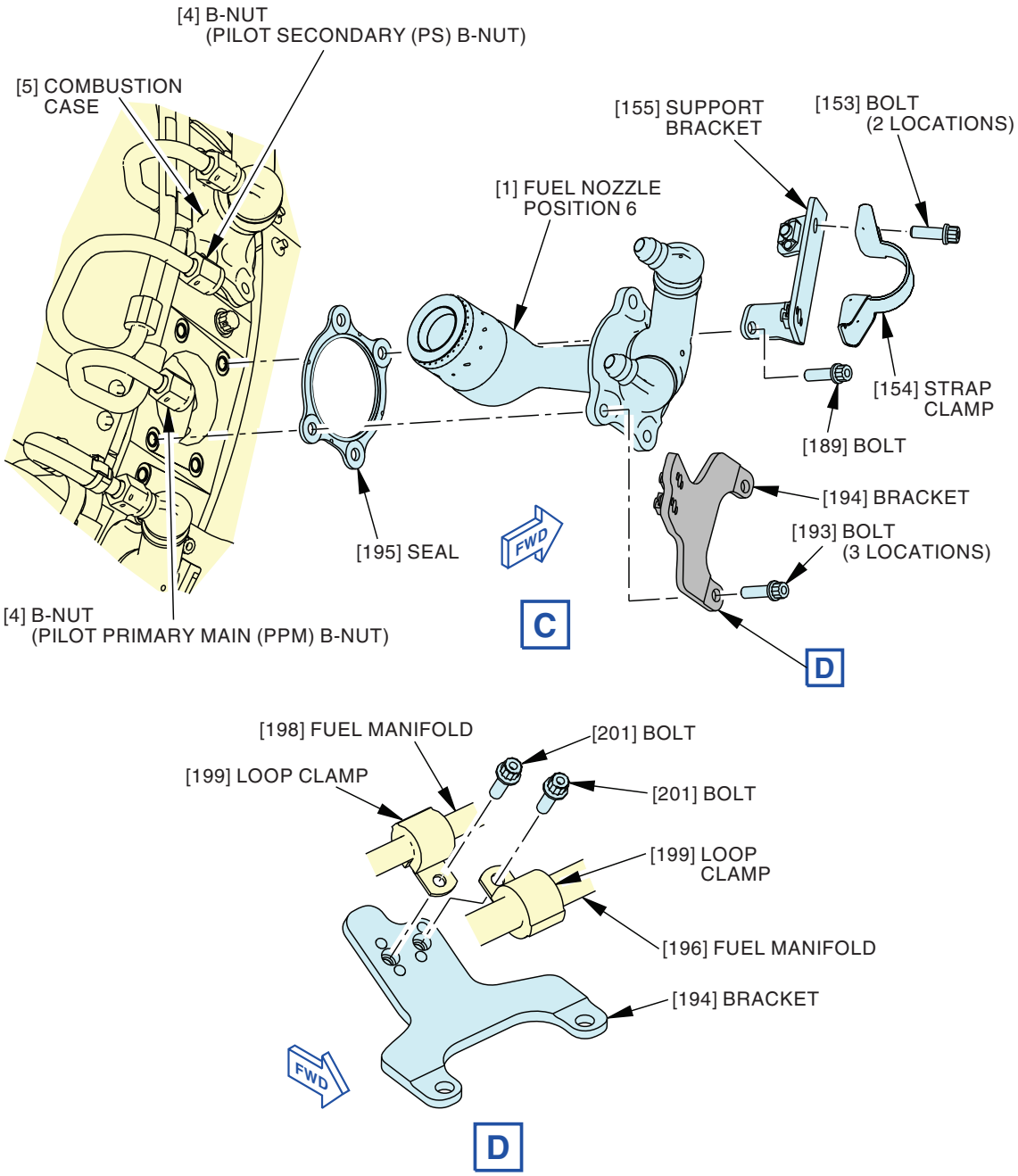
Fuel Nozzle (Position 5) Installation
Figure 406/73-11-05-990-892-G00 (Sheet 38 of 38)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



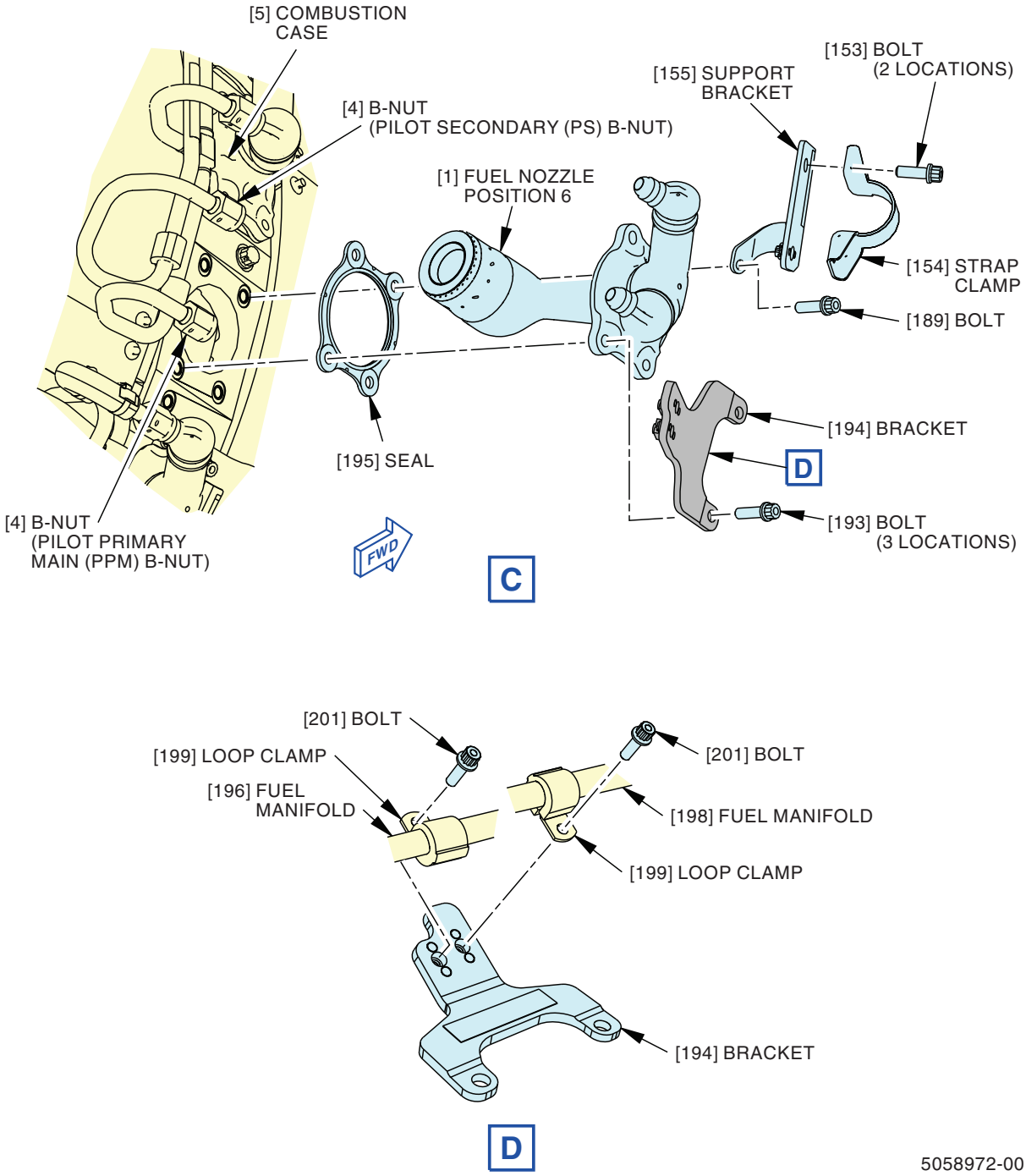
Fuel Nozzle (Position 6) Installation
Figure 407/73-11-05-990-893-G00 (Sheet 1 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



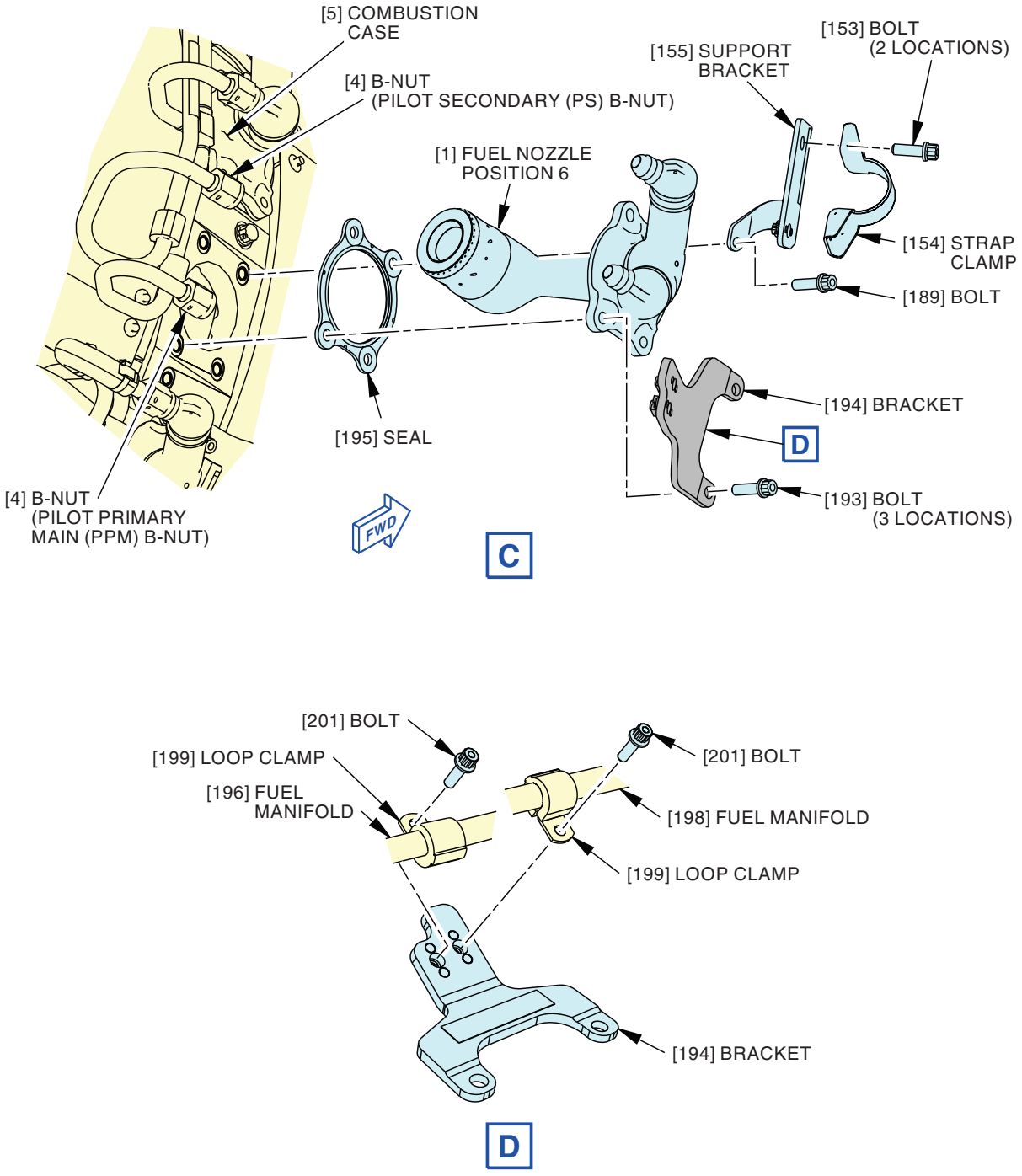
Fuel Nozzle (Position 6) Installation
Figure 407/73-11-05-990-893-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



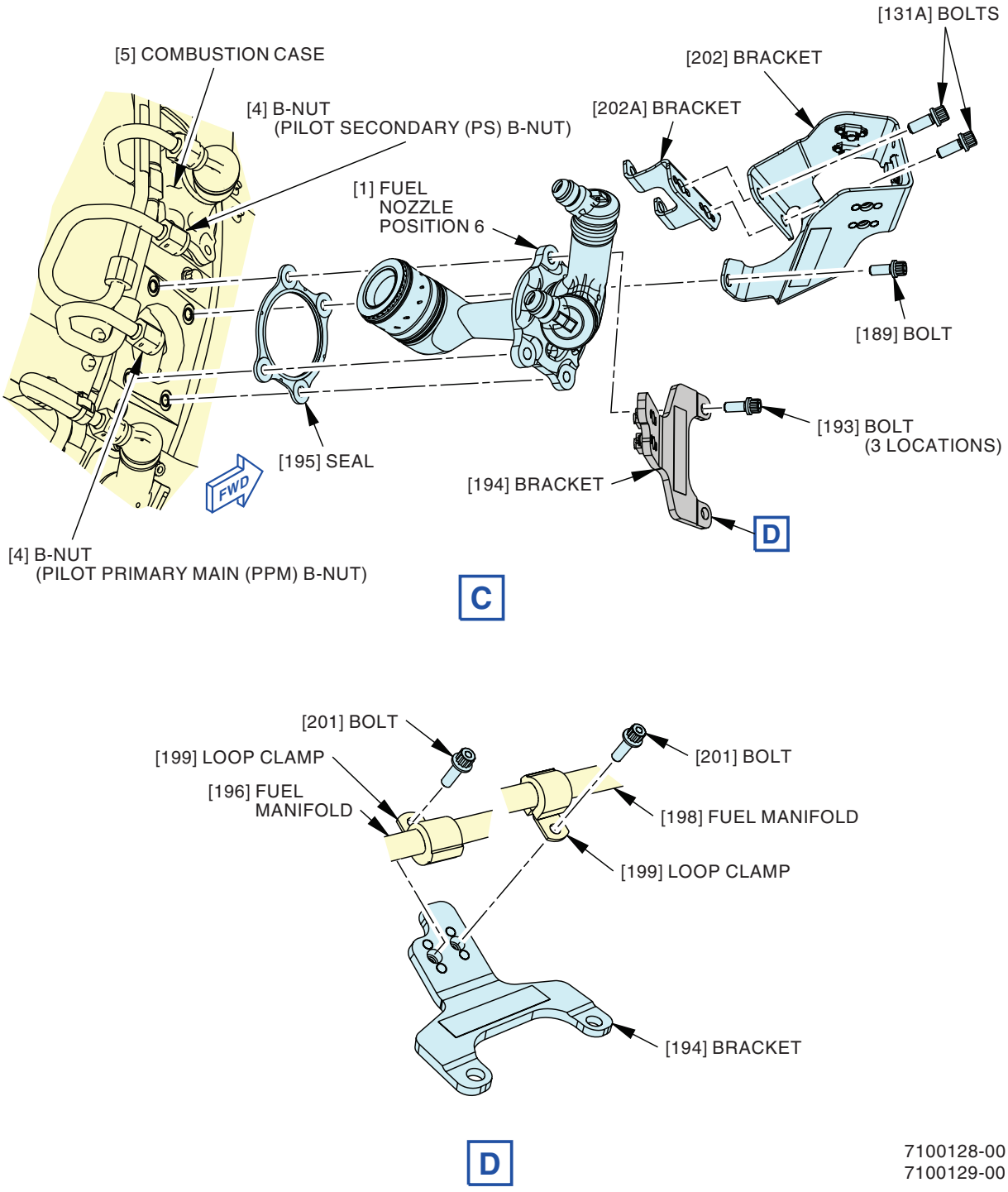
Fuel Nozzle (Position 6) Installation
Figure 407/73-11-05-990-893-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



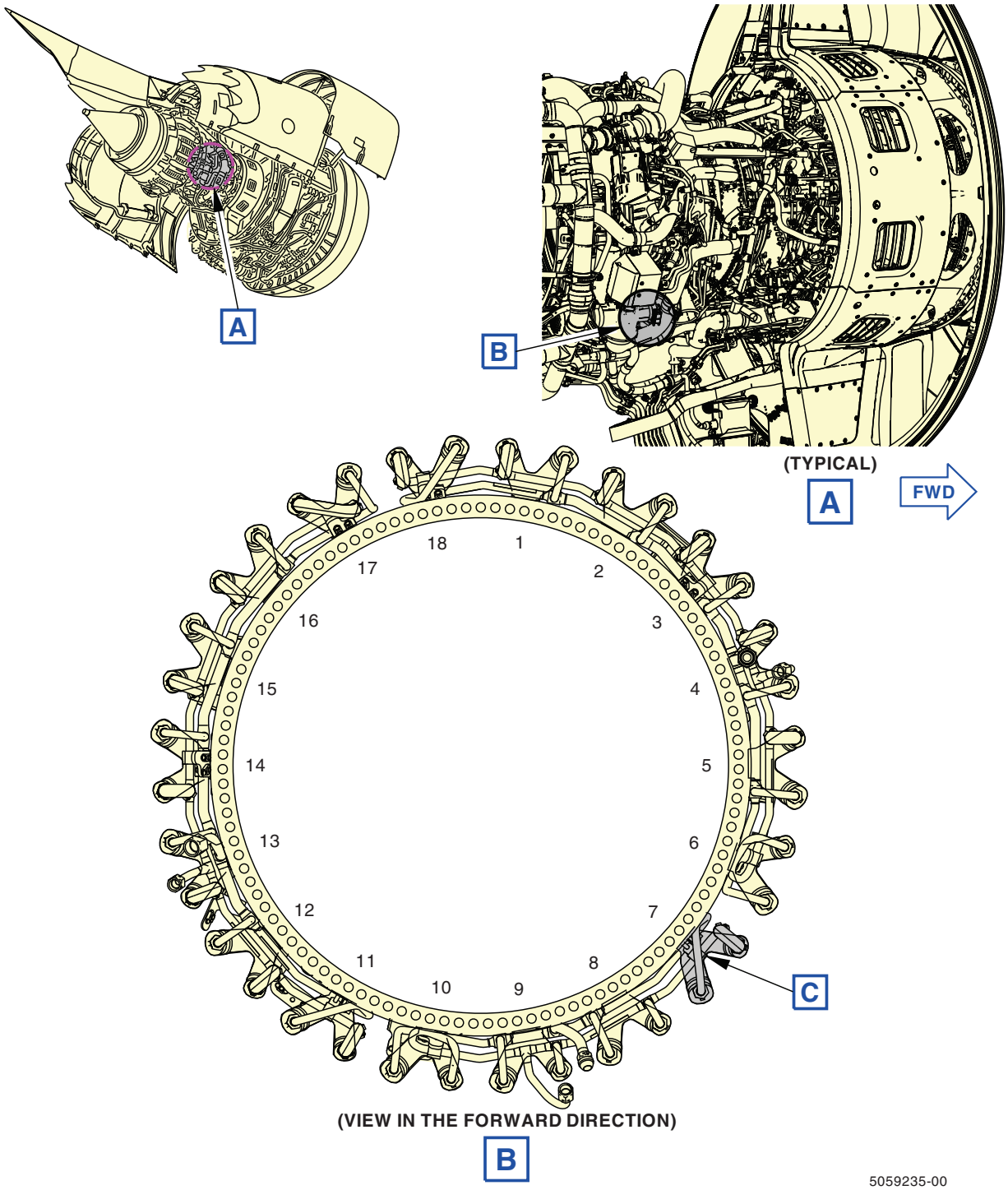
Fuel Nozzle (Position 6) Installation
Figure 407/73-11-05-990-893-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



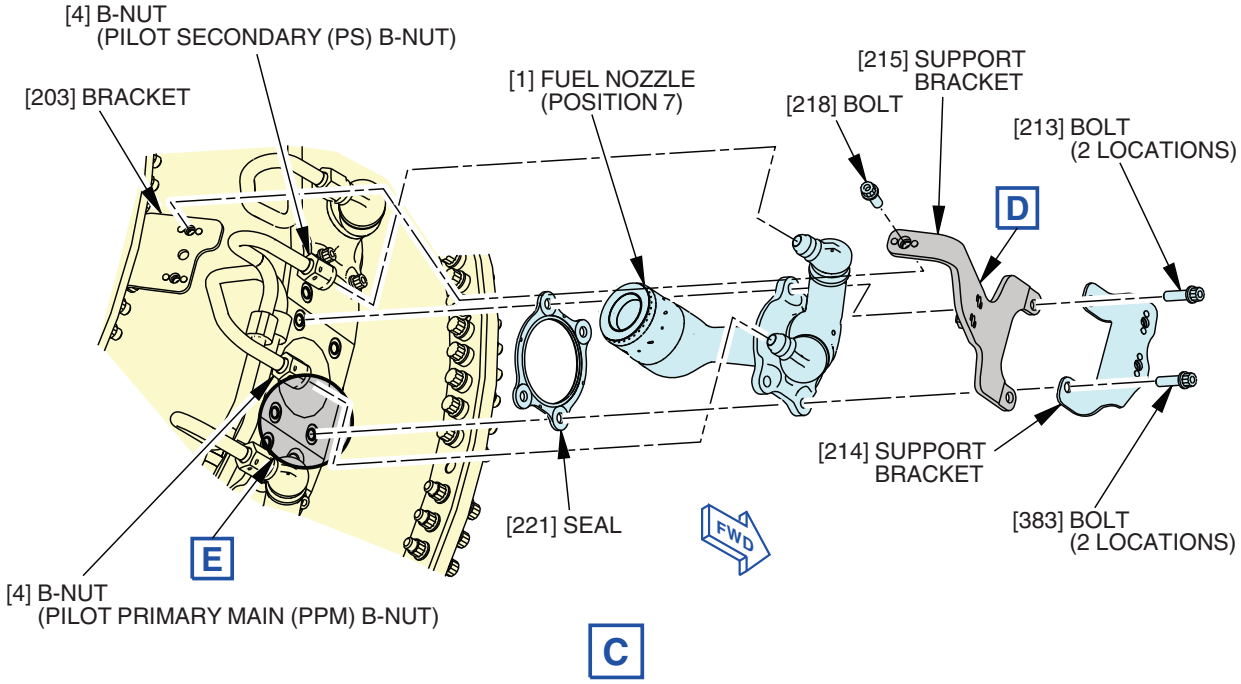
Fuel Nozzle (Position 6) Installation
Figure 407/73-11-05-990-893-G00 (Sheet 5 of 5)

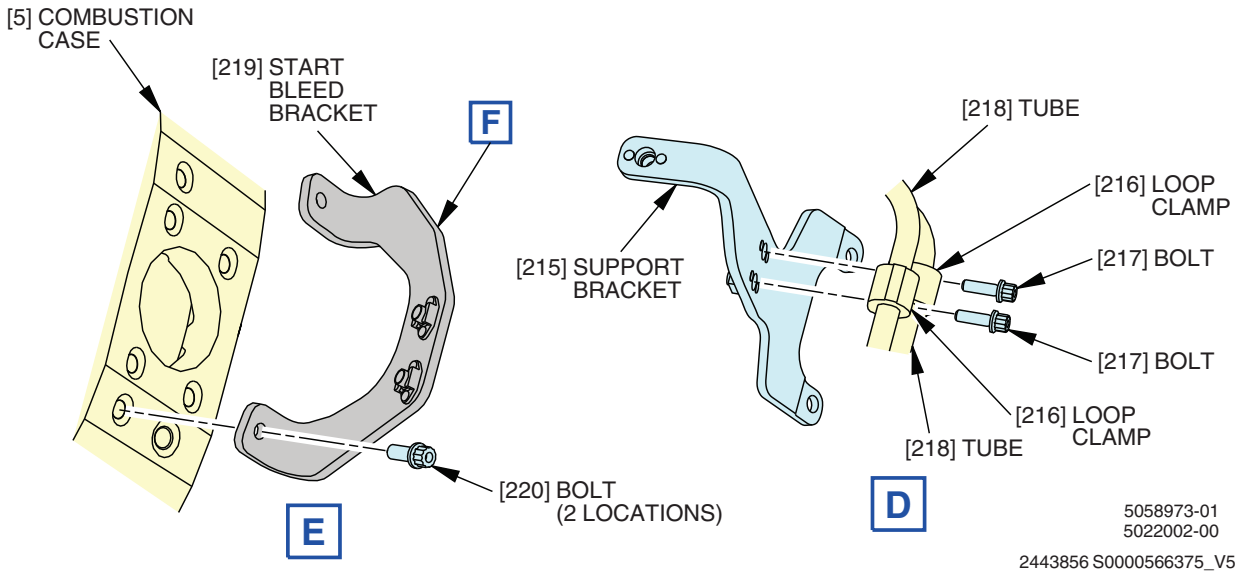
ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 7) Installation
Figure 408/73-11-05-990-894-G00 (Sheet 1 of 5)

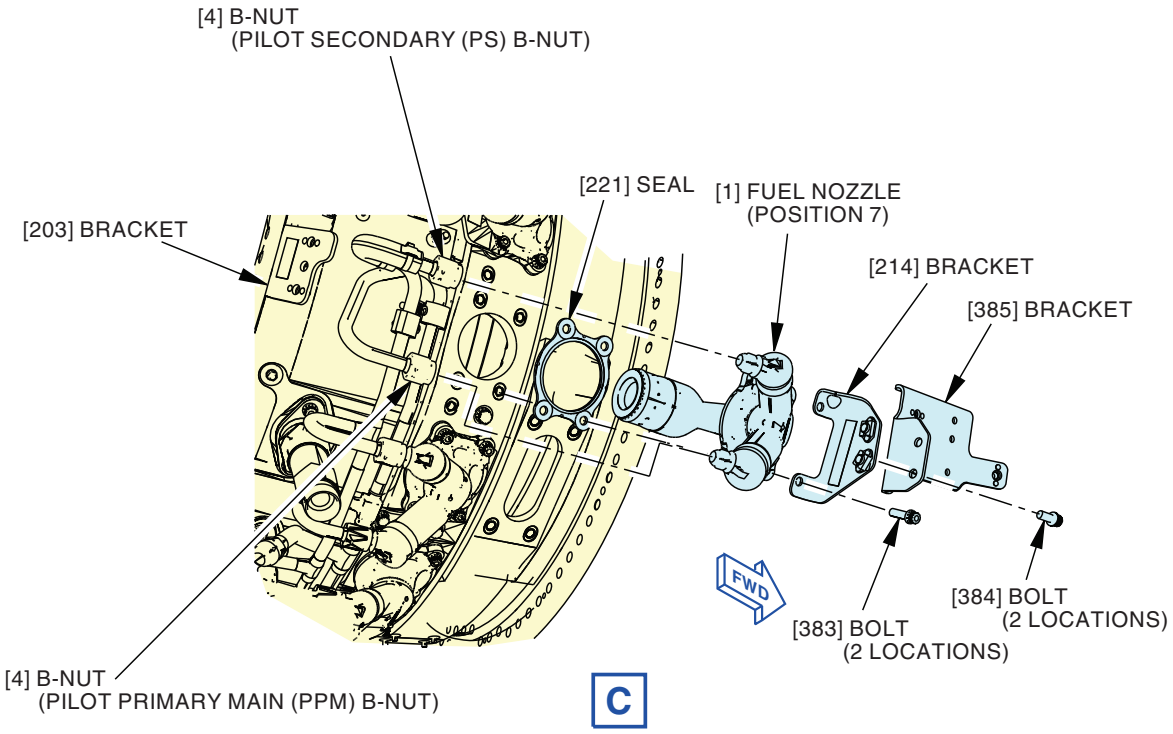
ECCN 9E991 BOEING PROPRIETARY - See title page for details





Fuel Nozzle (Position 7) Installation
Figure 408/73-11-05-990-894-G00 (Sheet 2 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

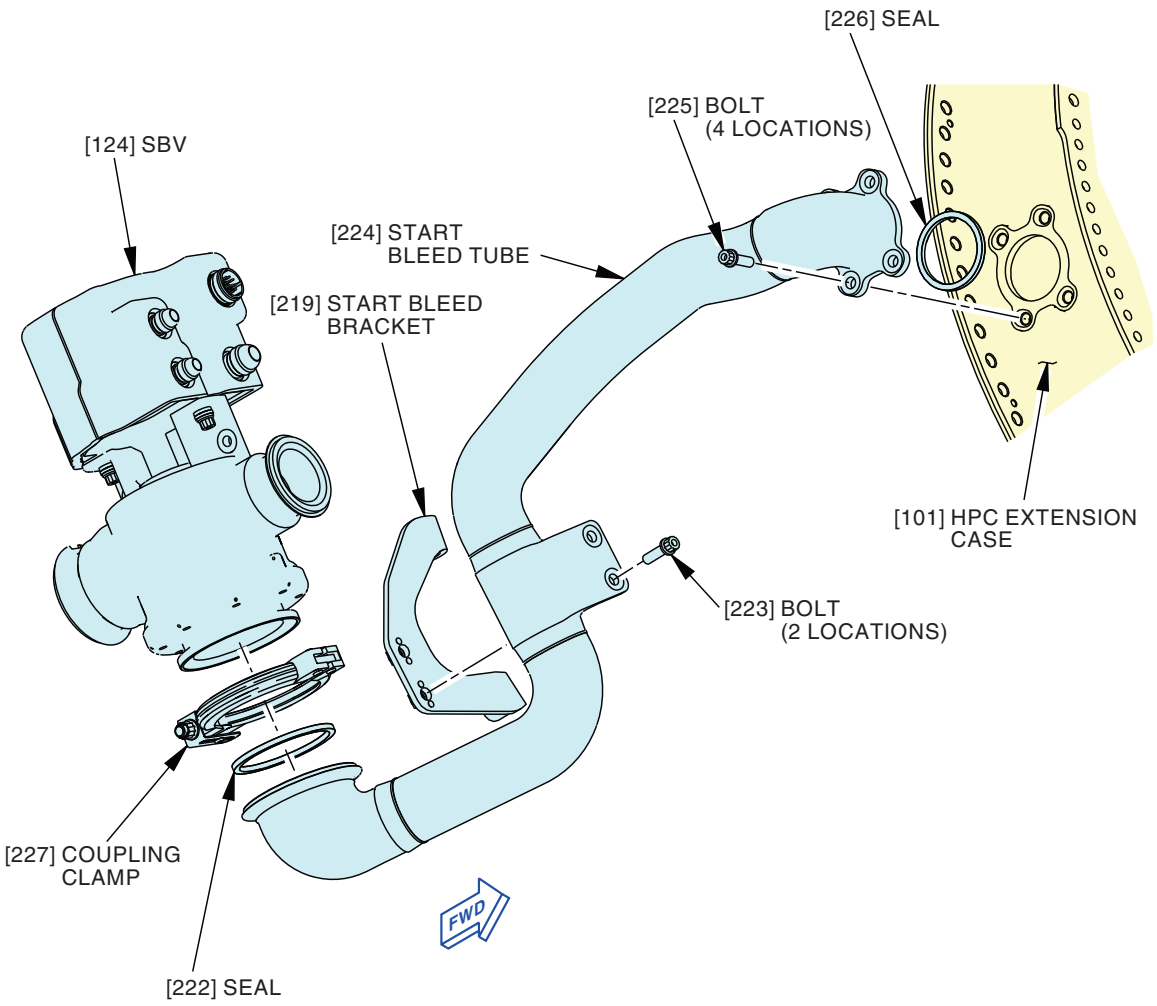


5058973-00

2855577 S0000669668_V2

Fuel Nozzle (Position 7) Installation
Figure 408/73-11-05-990-894-G00 (Sheet 3 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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5024884-00
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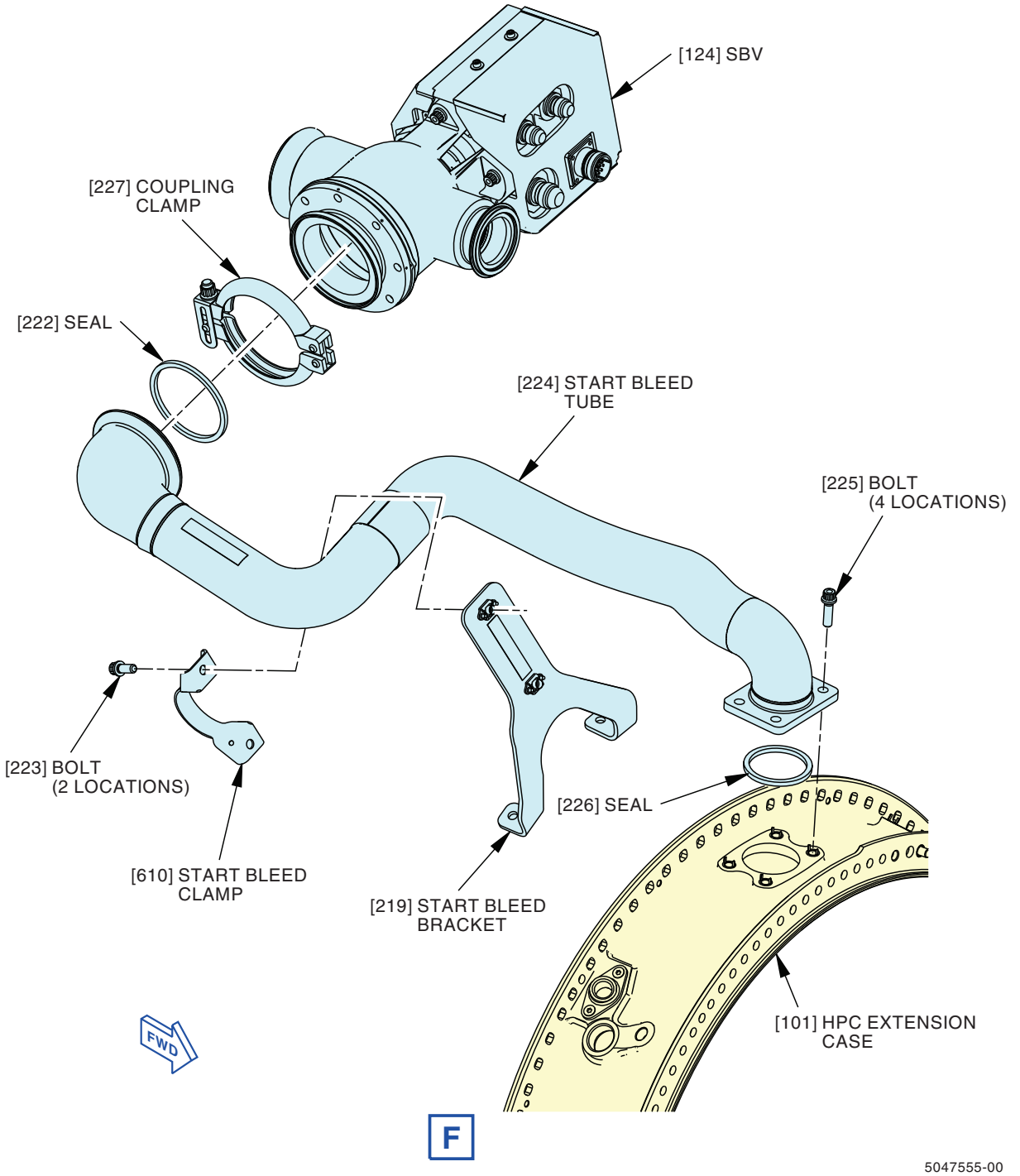
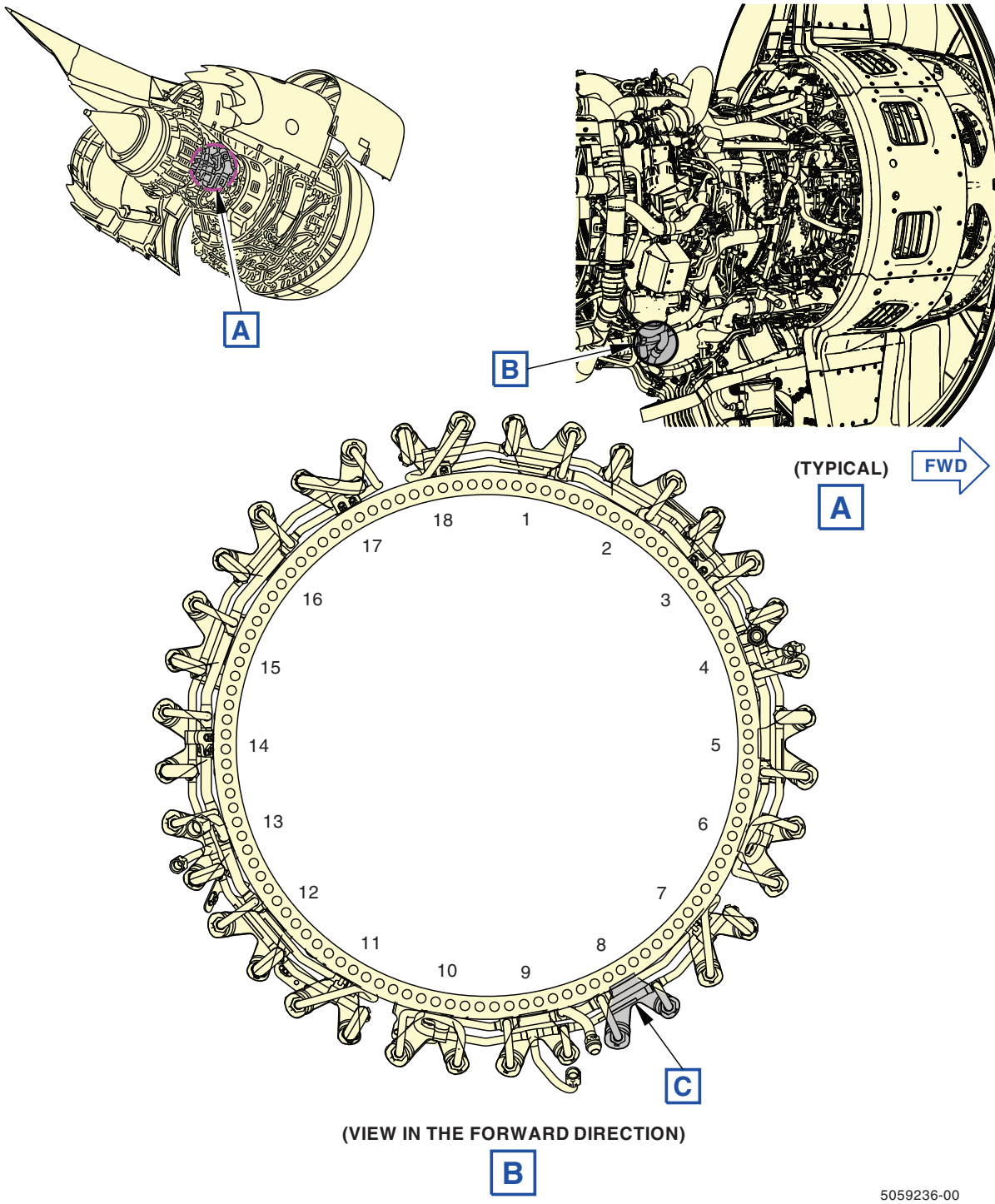
Fuel Nozzle (Position 7) Installation
Figure 408/73-11-05-990-894-G00 (Sheet 4 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



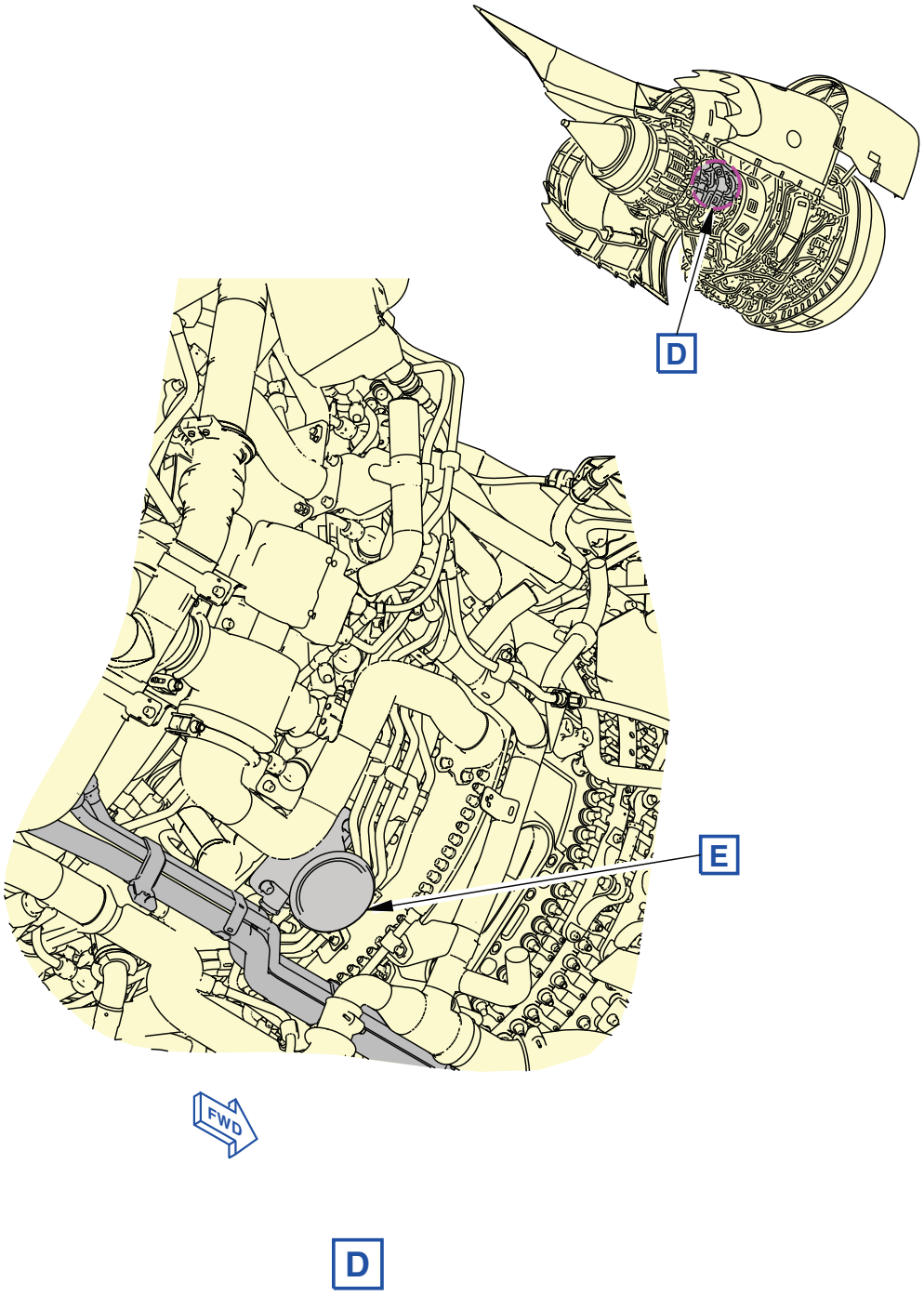
Fuel Nozzle (Position 7) Installation
Figure 408/73-11-05-990-894-G00 (Sheet 5 of 5)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 8) Installation
Figure 409/73-11-05-990-895-G00 (Sheet 1 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021985-00
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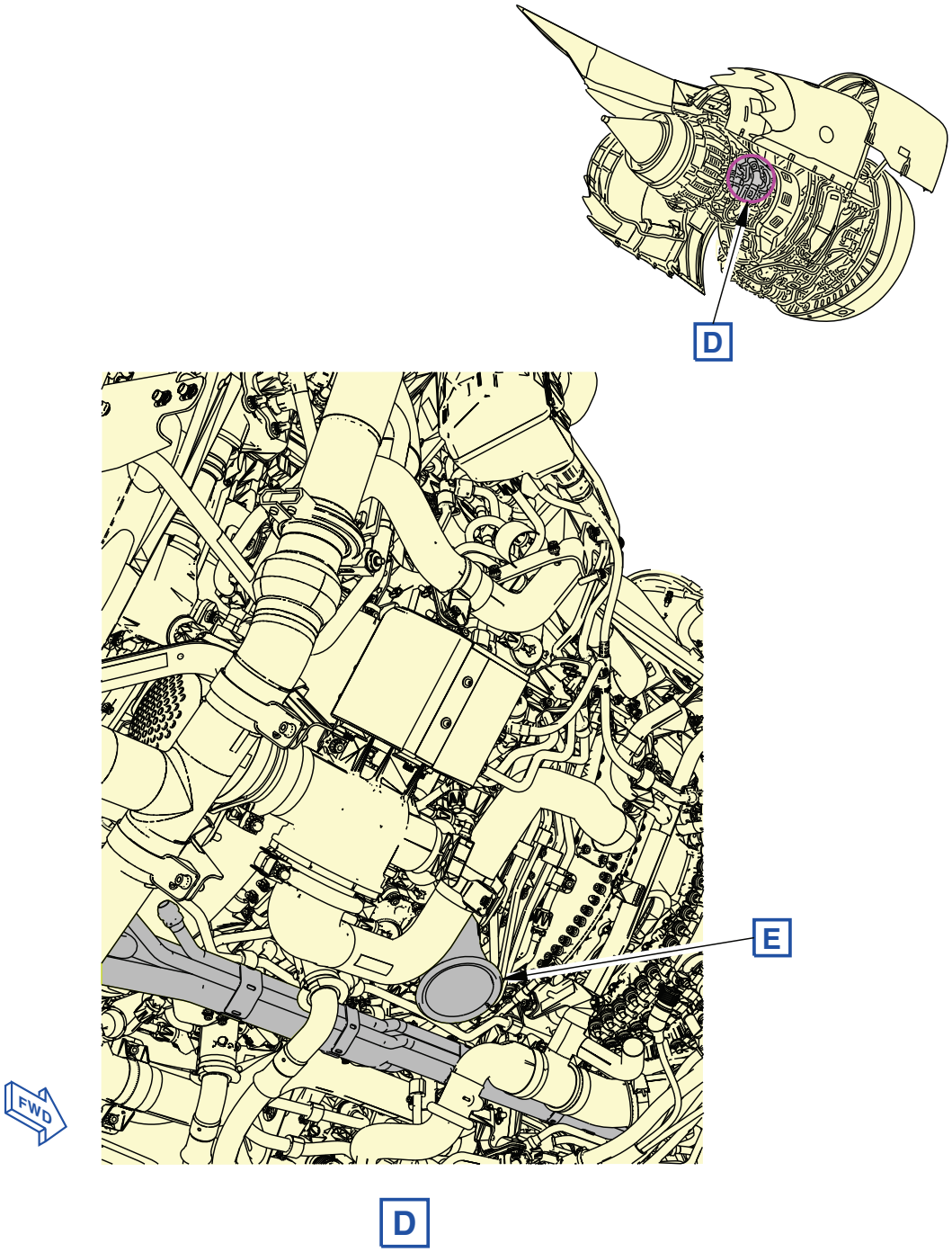
Fuel Nozzle (Position 8) Installation
Figure 409/73-11-05-990-895-G00 (Sheet 2 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5059237-00
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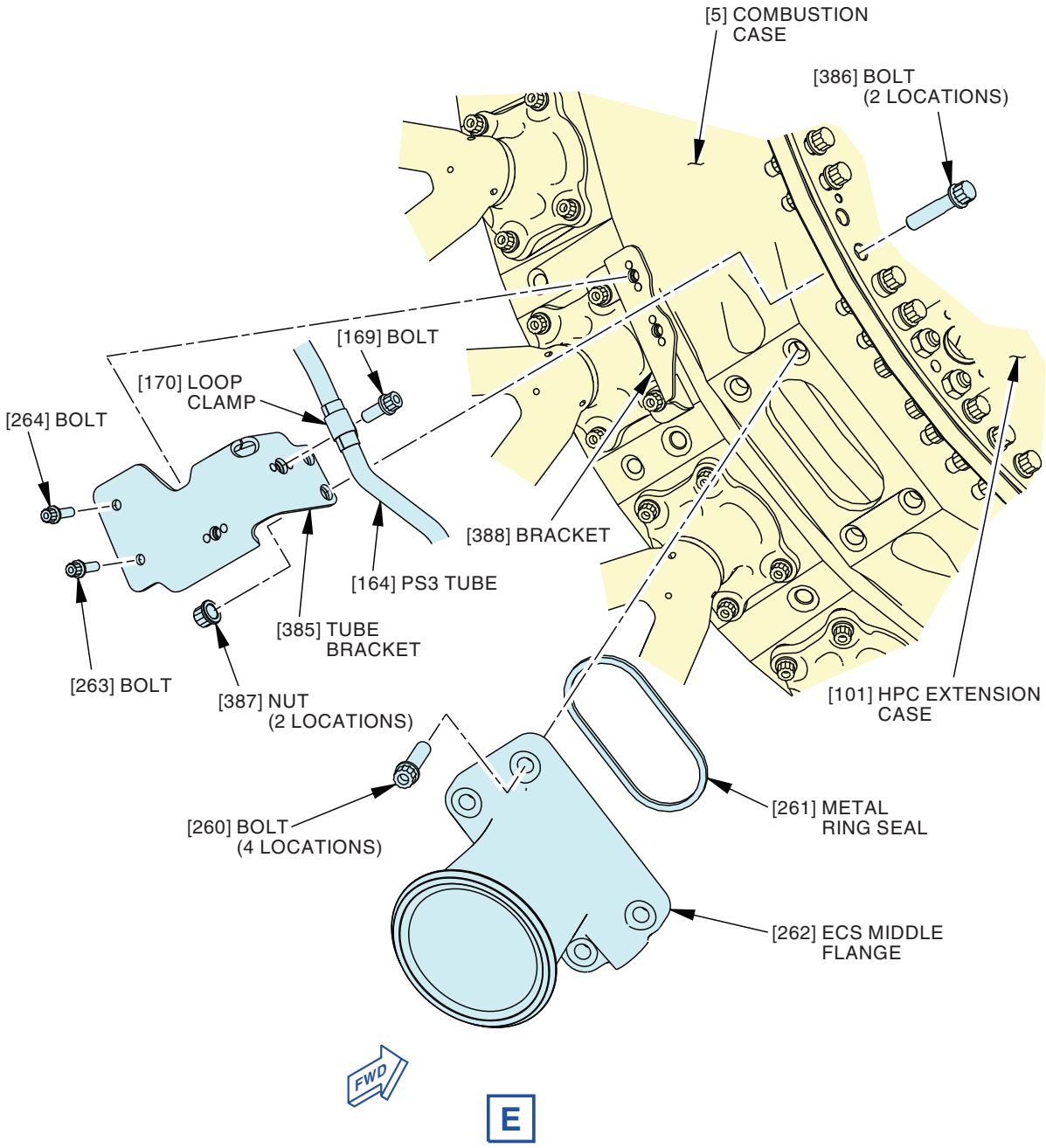
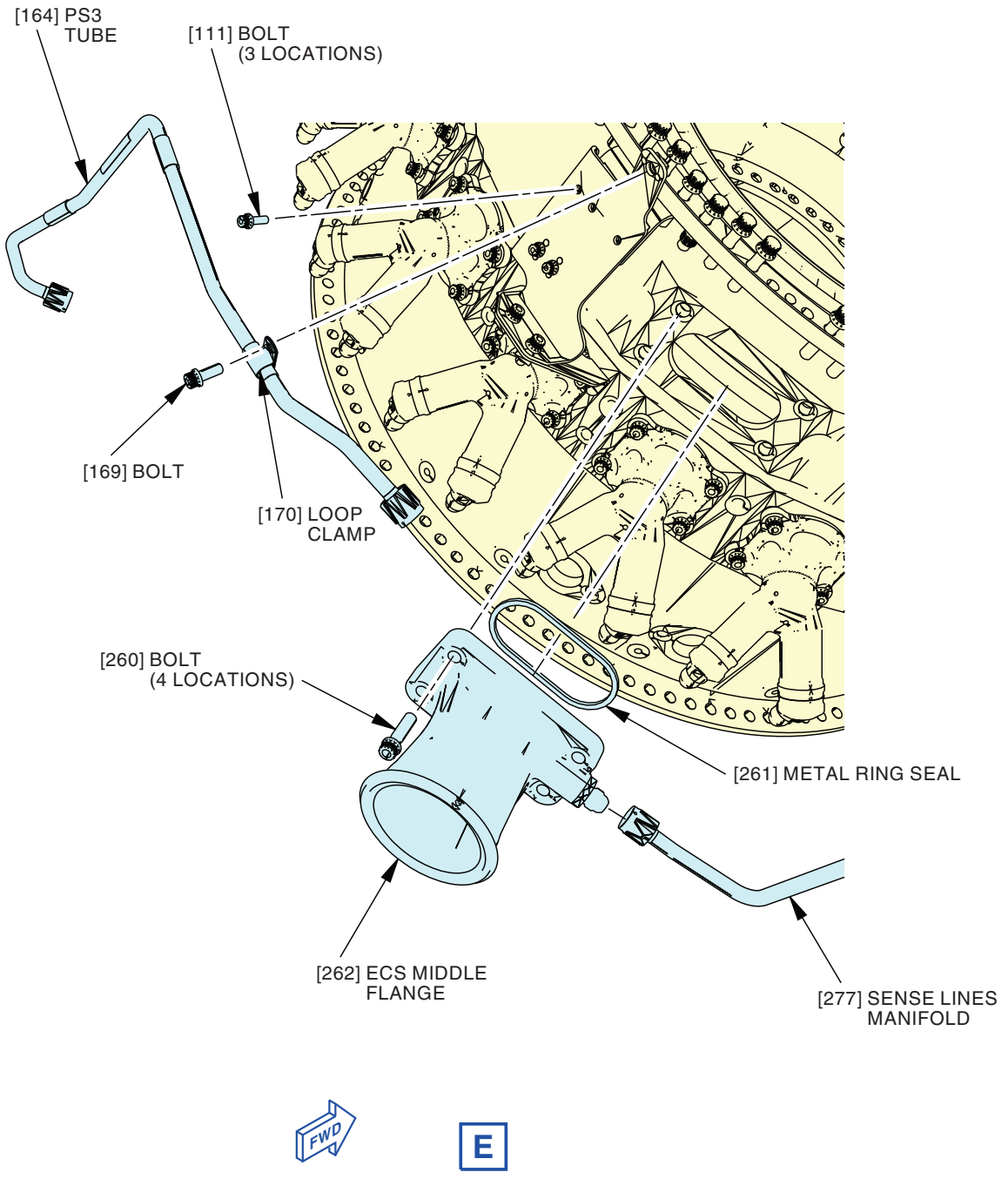
Fuel Nozzle (Position 8) Installation
Figure 409/73-11-05-990-895-G00 (Sheet 3 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



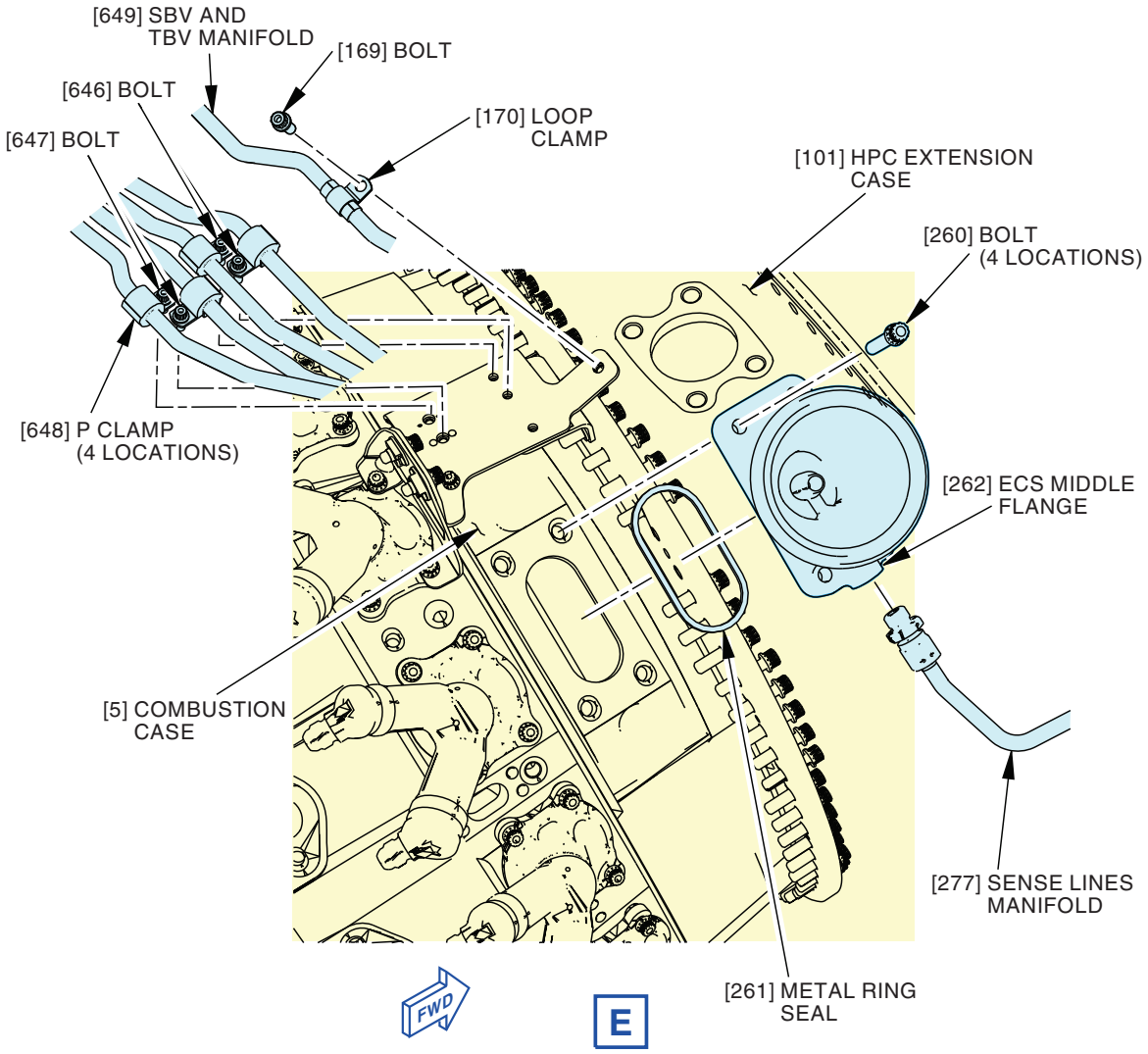
Fuel Nozzle (Position 8) Installation
Figure 409/73-11-05-990-895-G00 (Sheet 4 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 8) Installation
Figure 409/73-11-05-990-895-G00 (Sheet 5 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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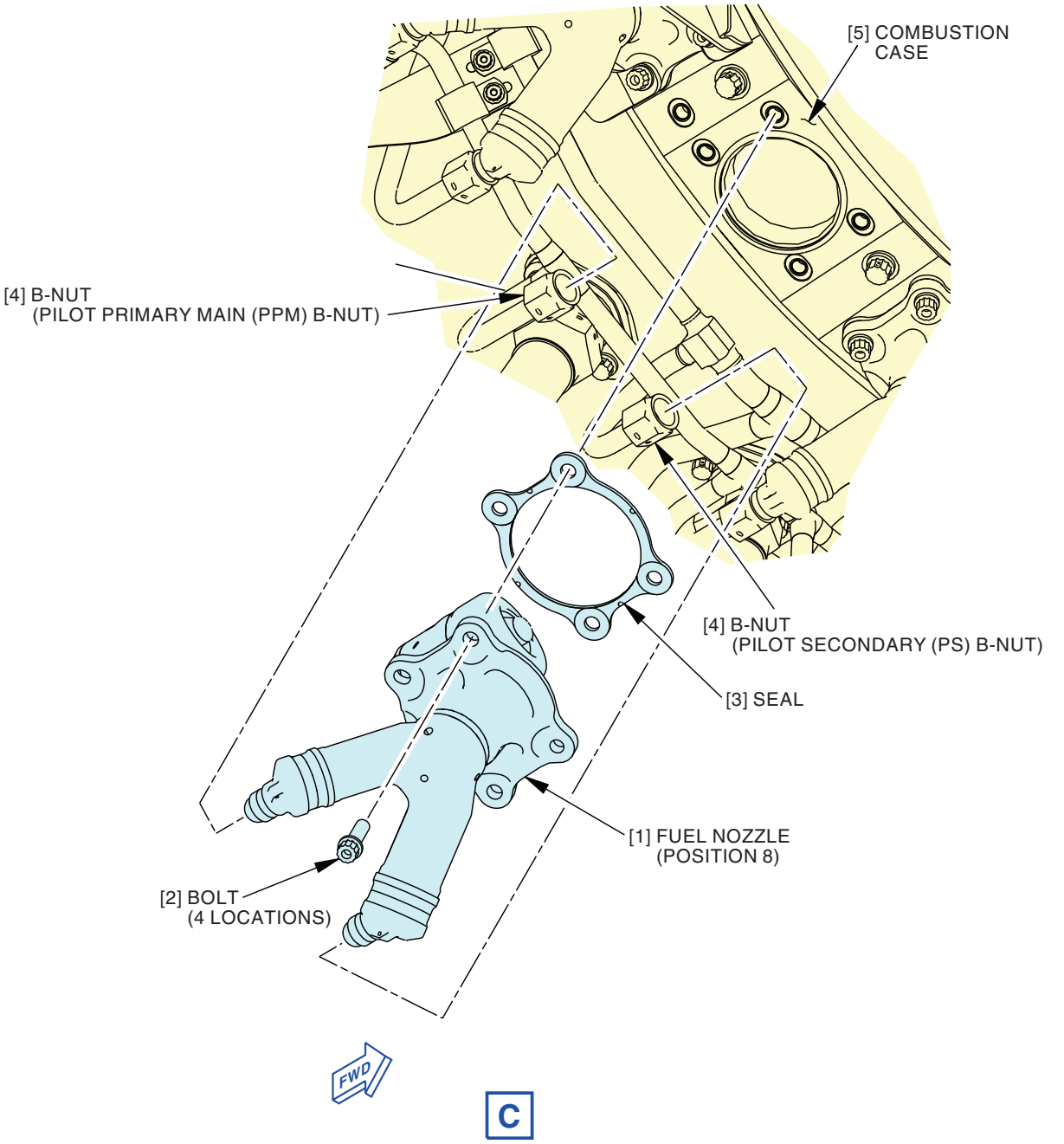
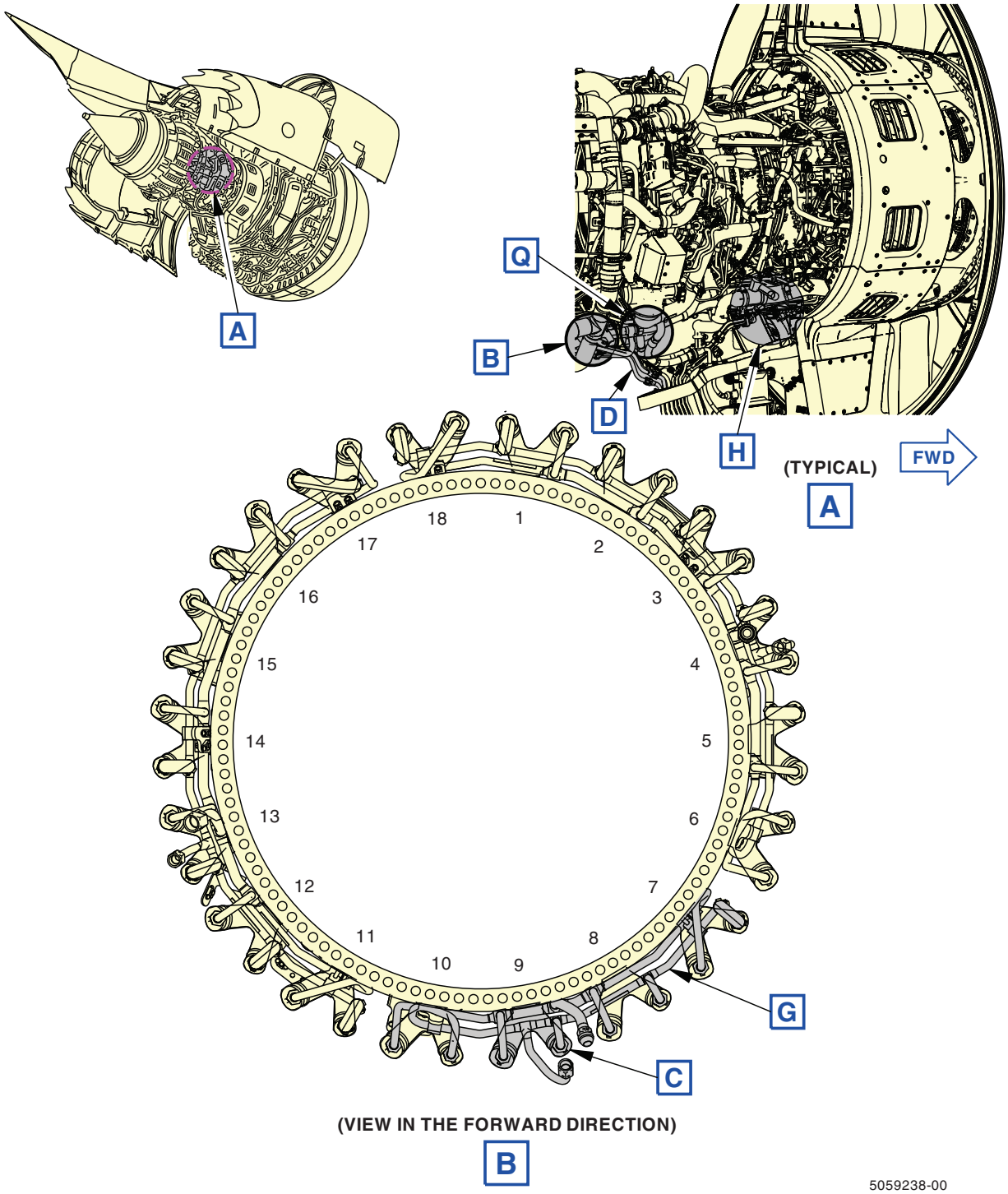
Fuel Nozzle (Position 8) Installation
Figure 409/73-11-05-990-895-G00 (Sheet 6 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 8) Installation
Figure 409/73-11-05-990-895-G00 (Sheet 7 of 7)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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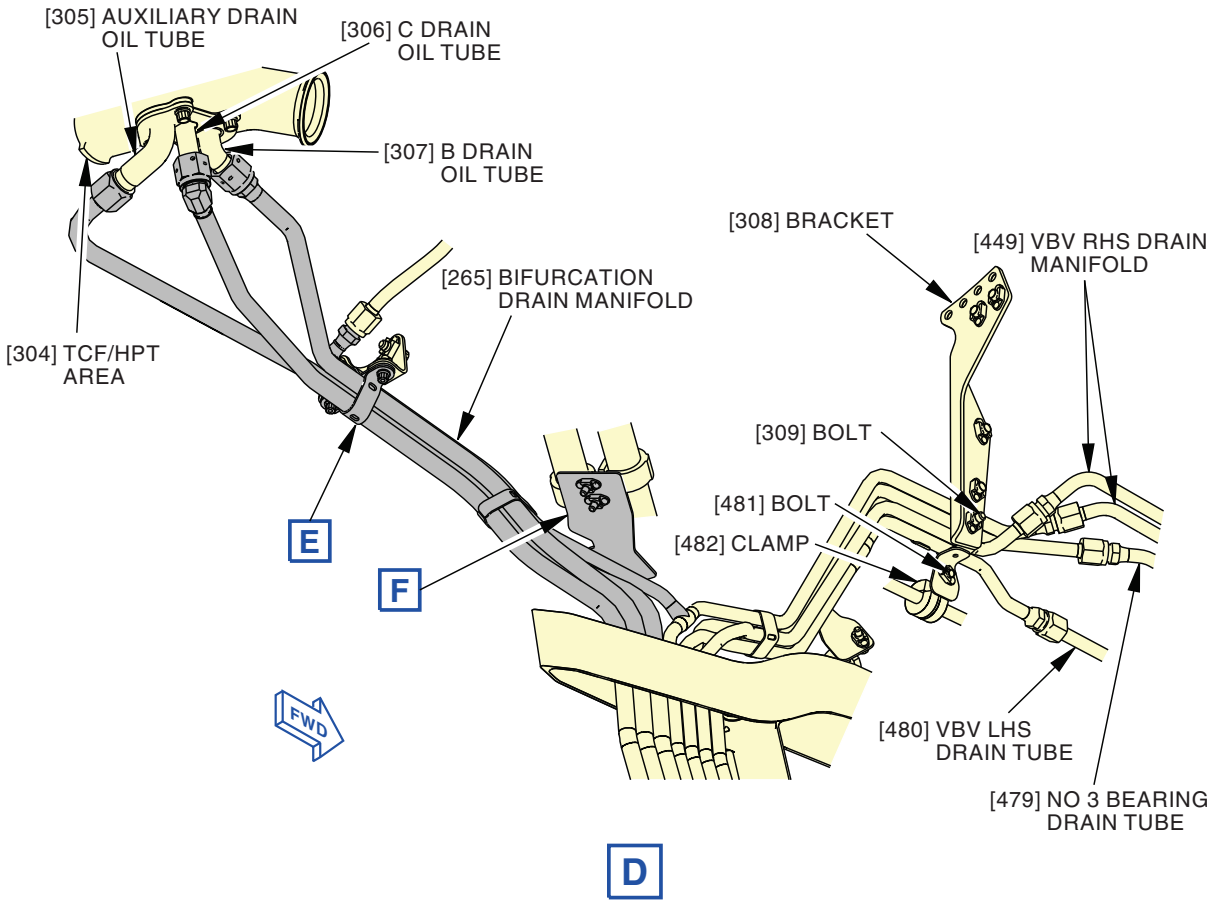
Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 1 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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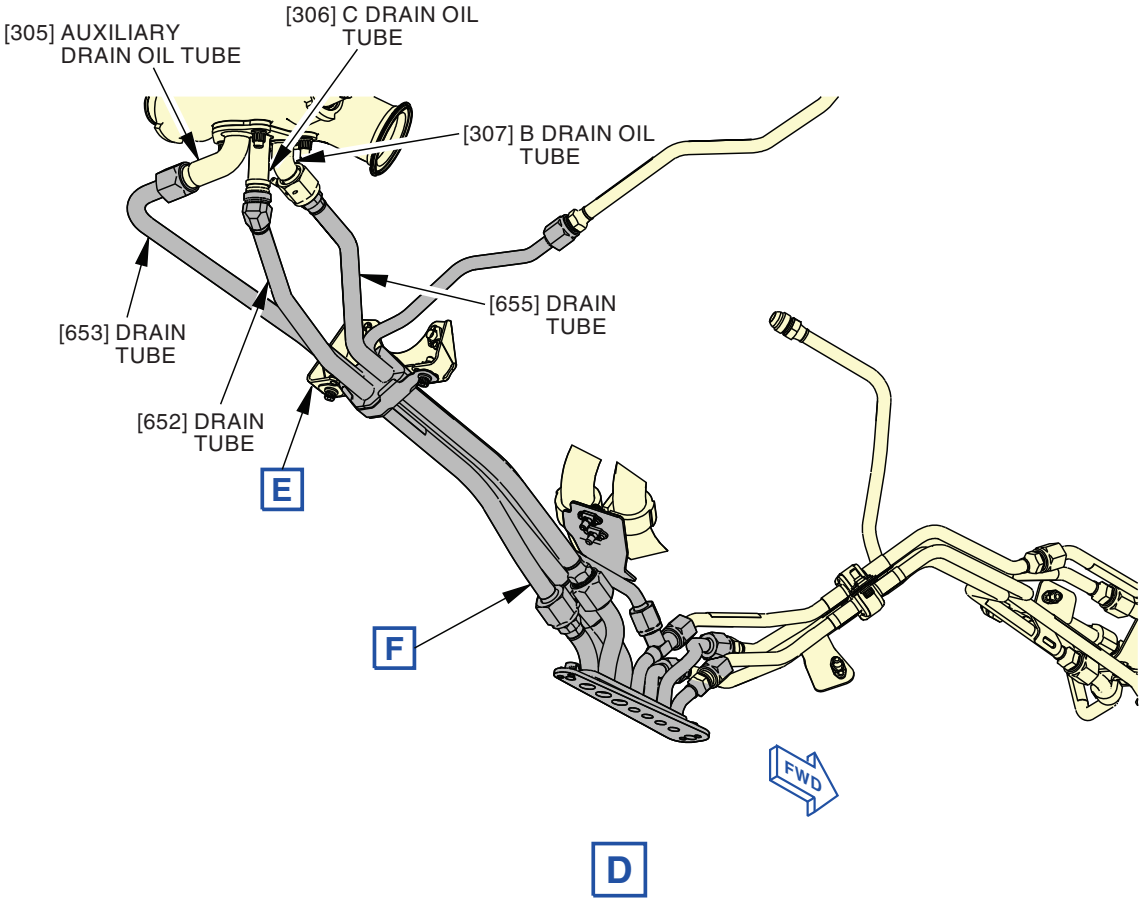
Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 2 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5058976-00
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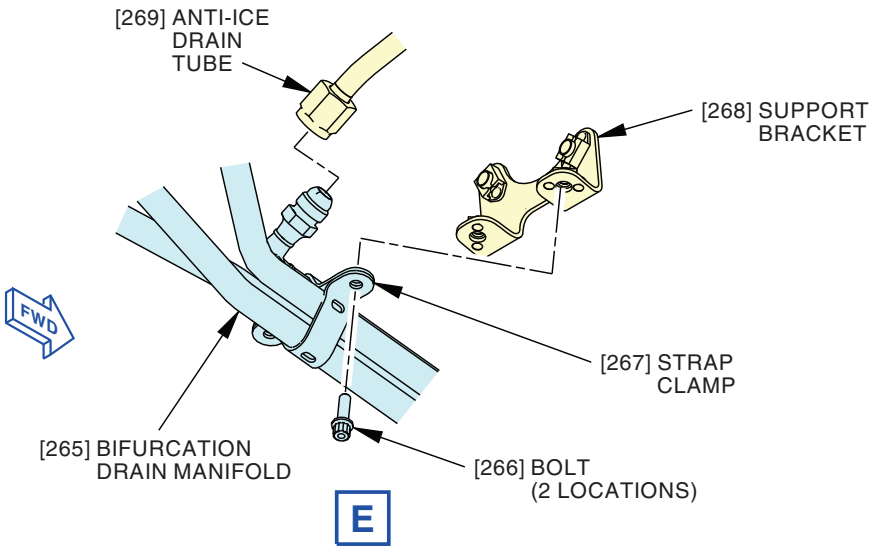
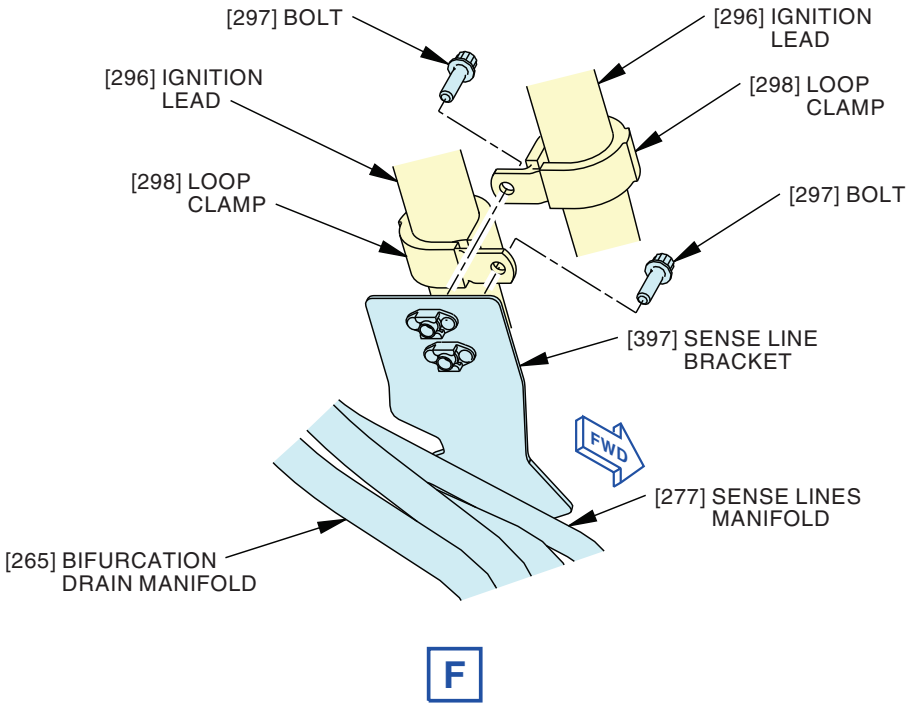
Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 3 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details





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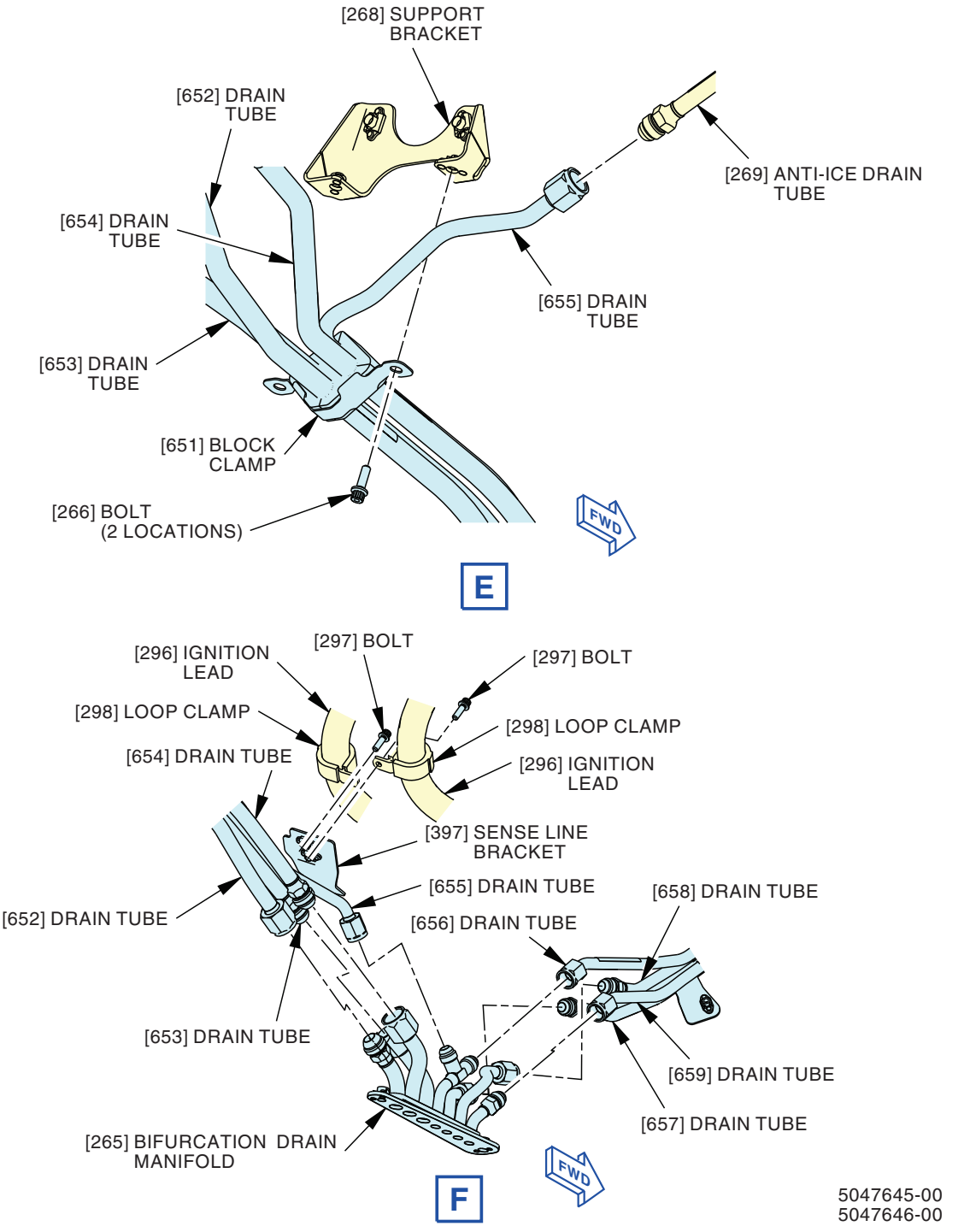
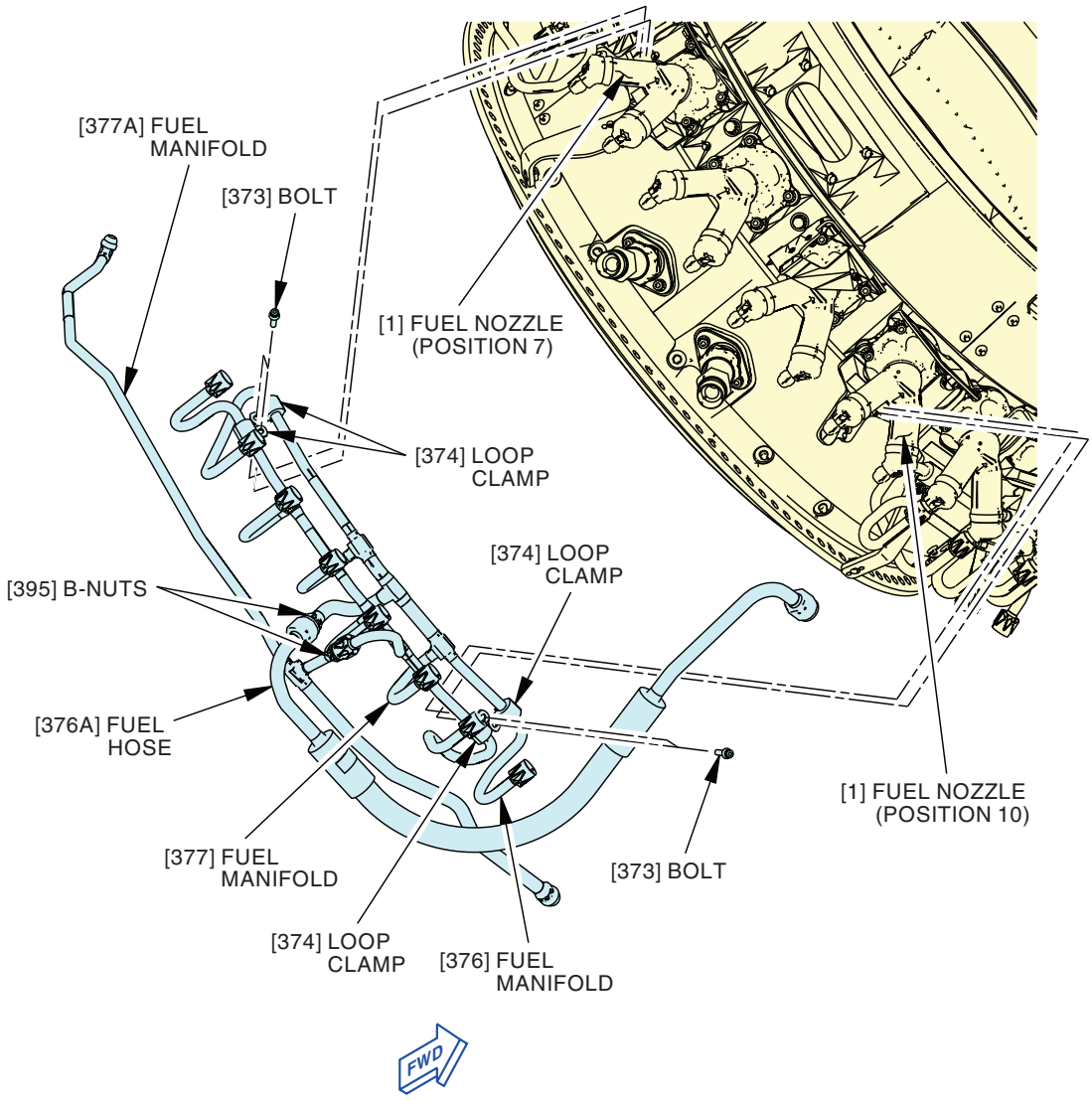
Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 4 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 5 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

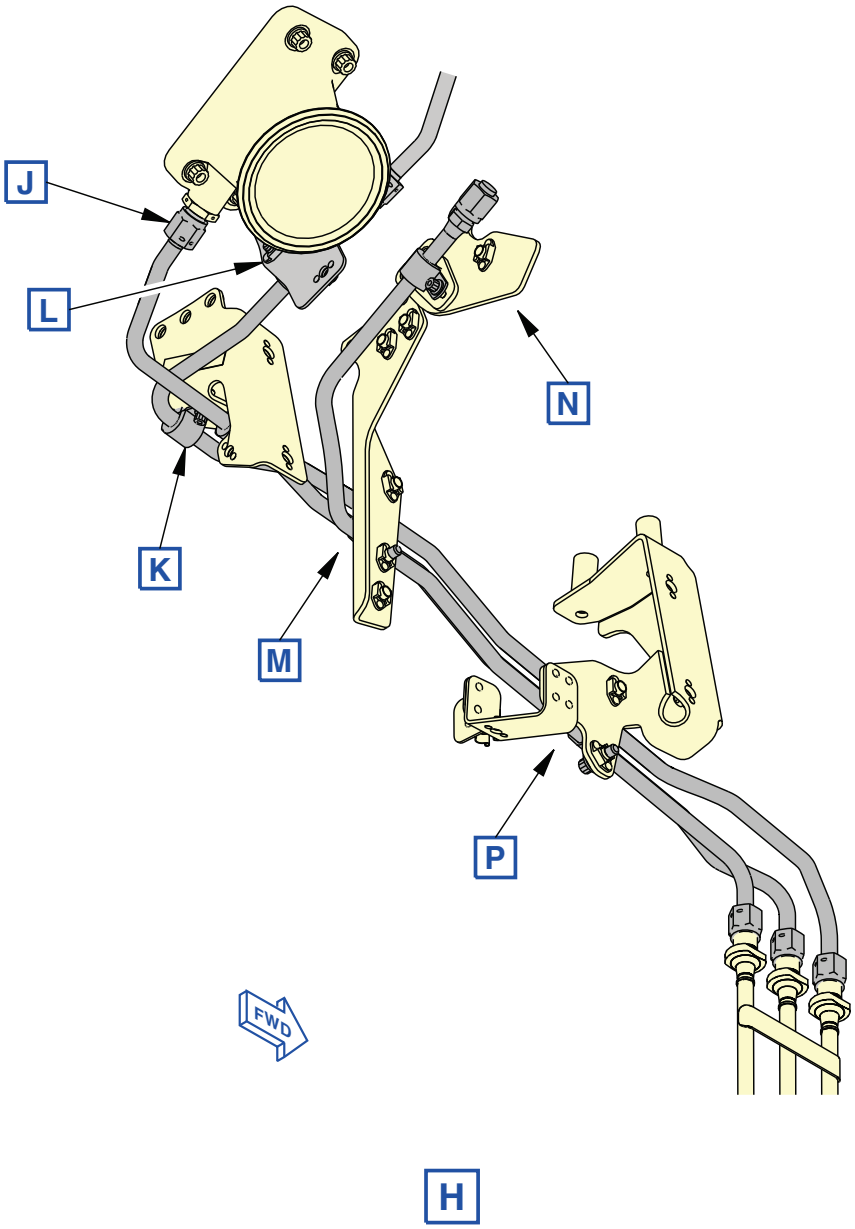


5022016-01

2440028 S0000566451_V6

Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 6 of 15)

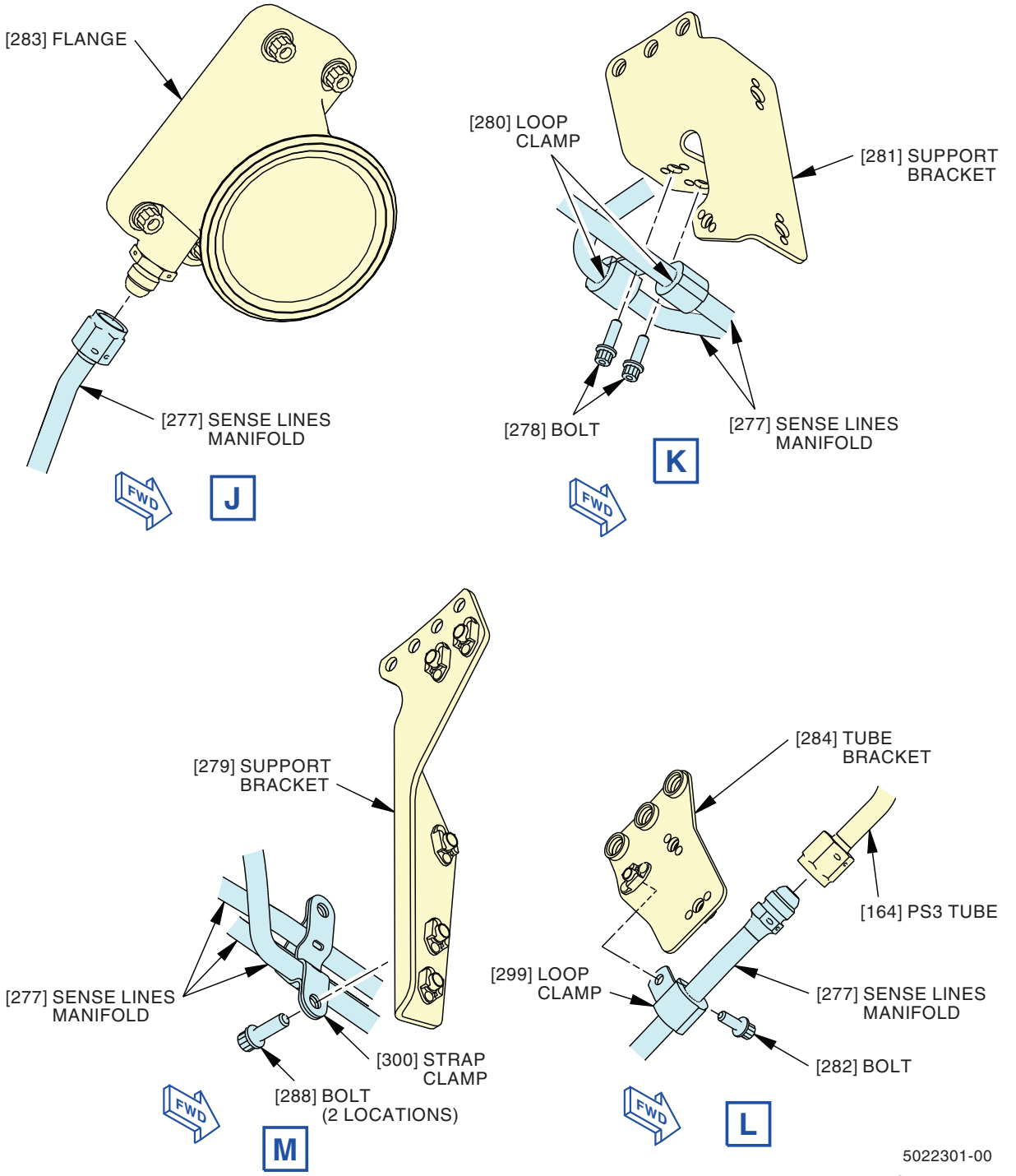
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5022018-00
2440551 S0000566453_V3

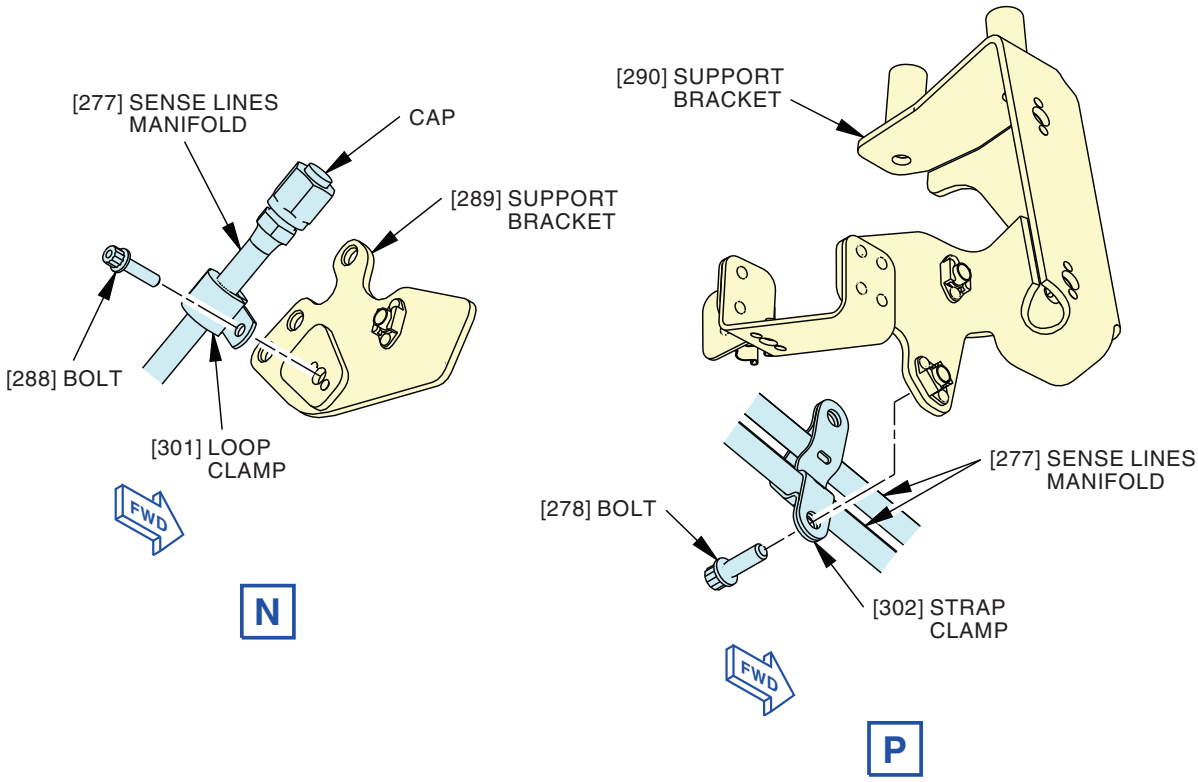
Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 7 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 8 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



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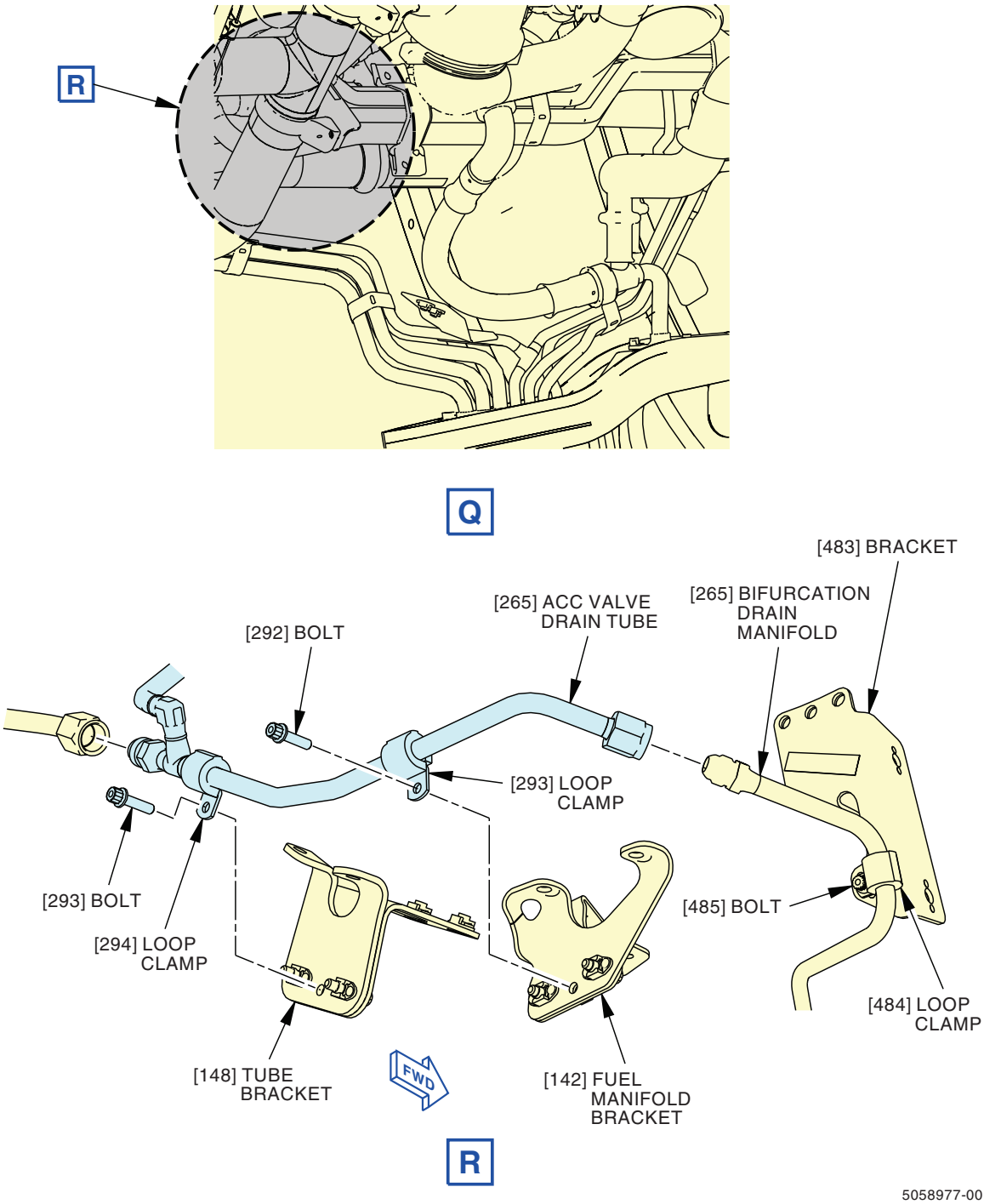
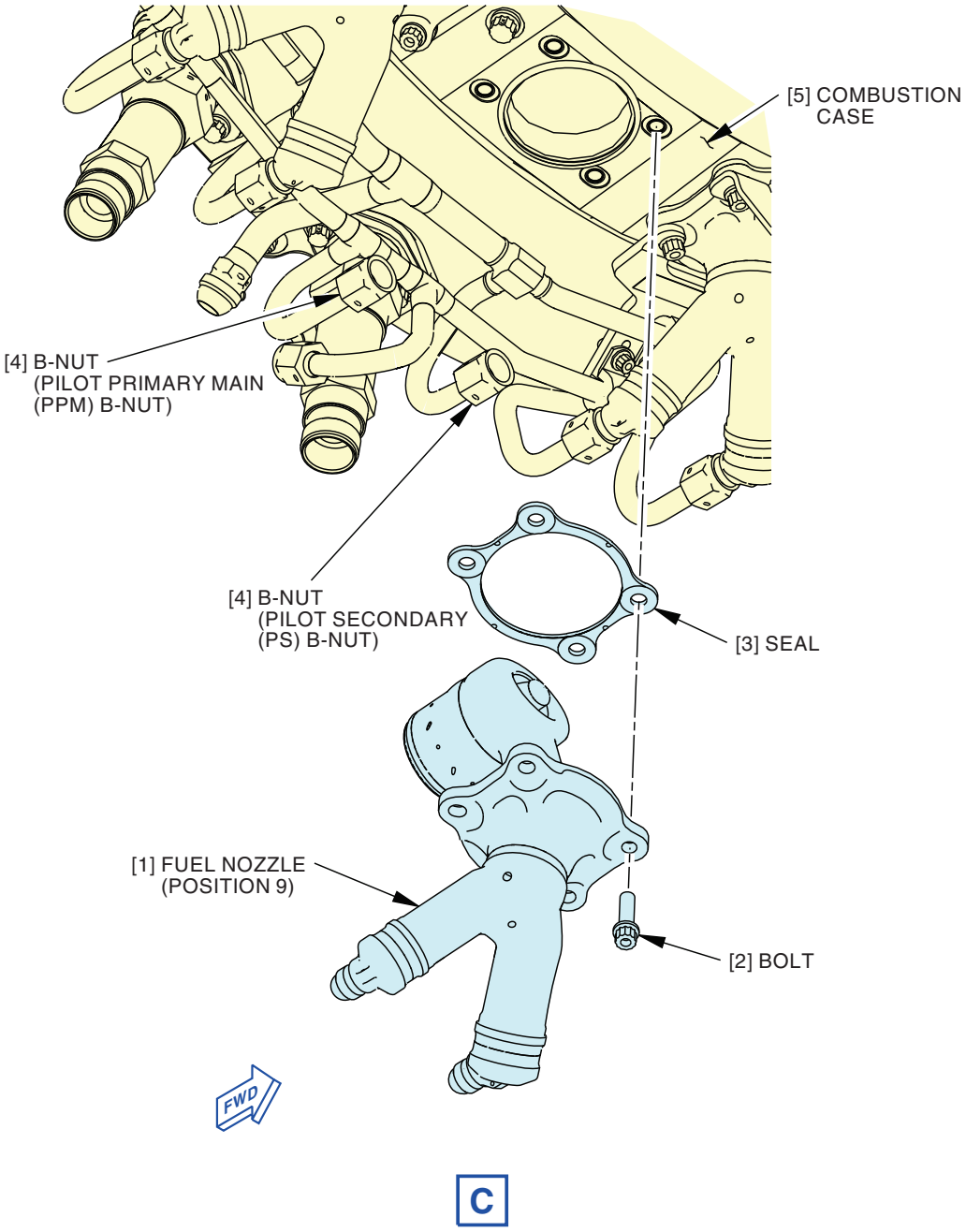
Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 9 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



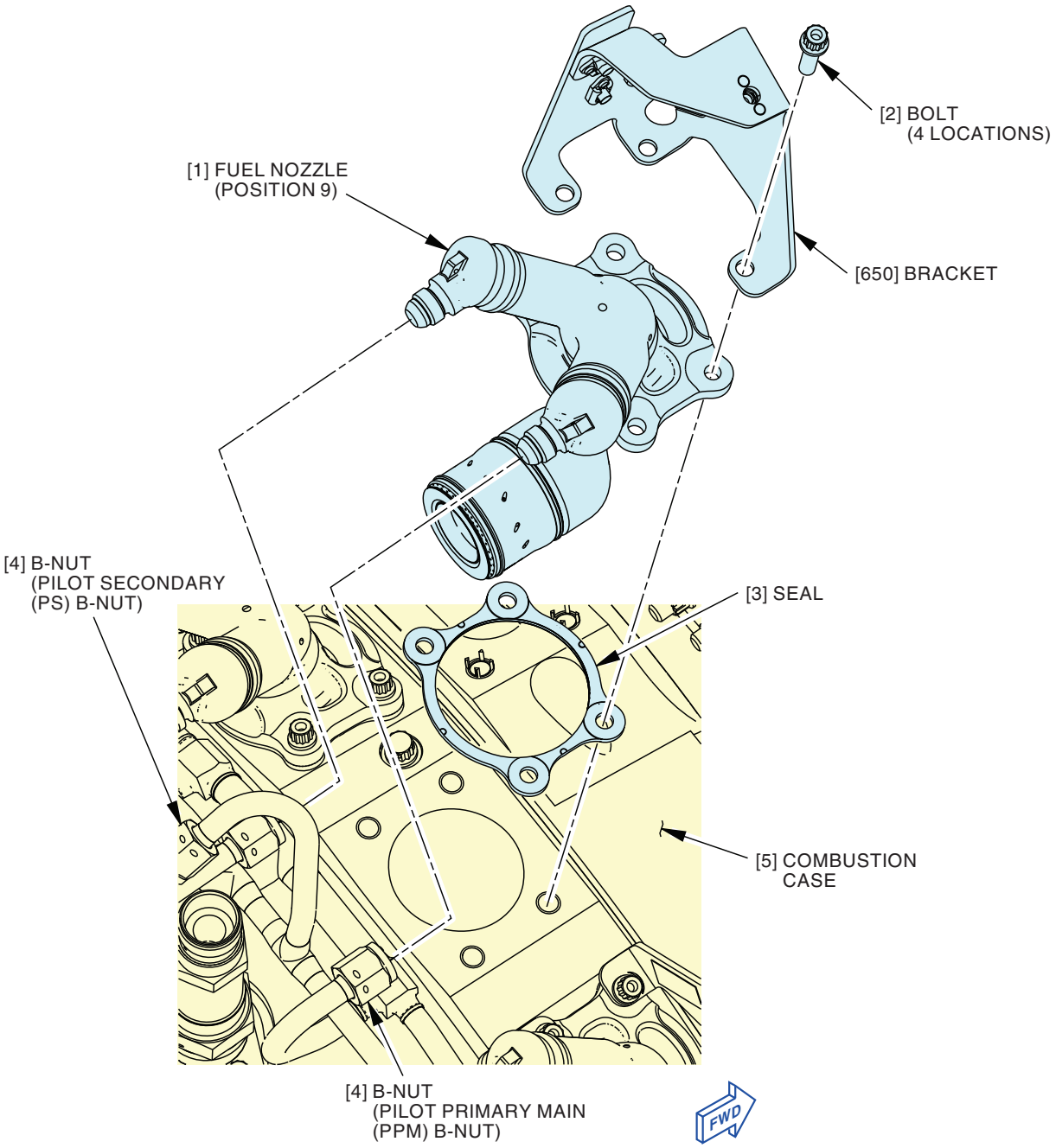
Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 10 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



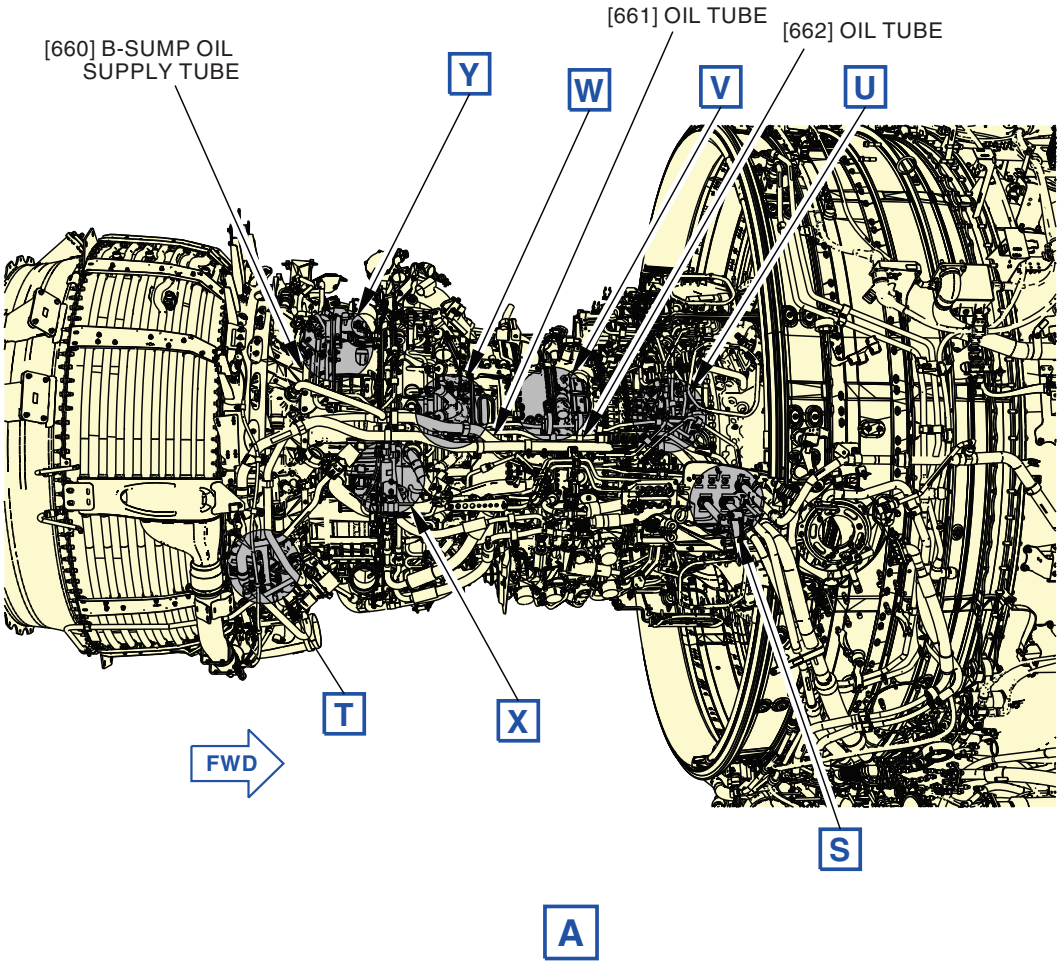
Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 11 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 12 of 15)

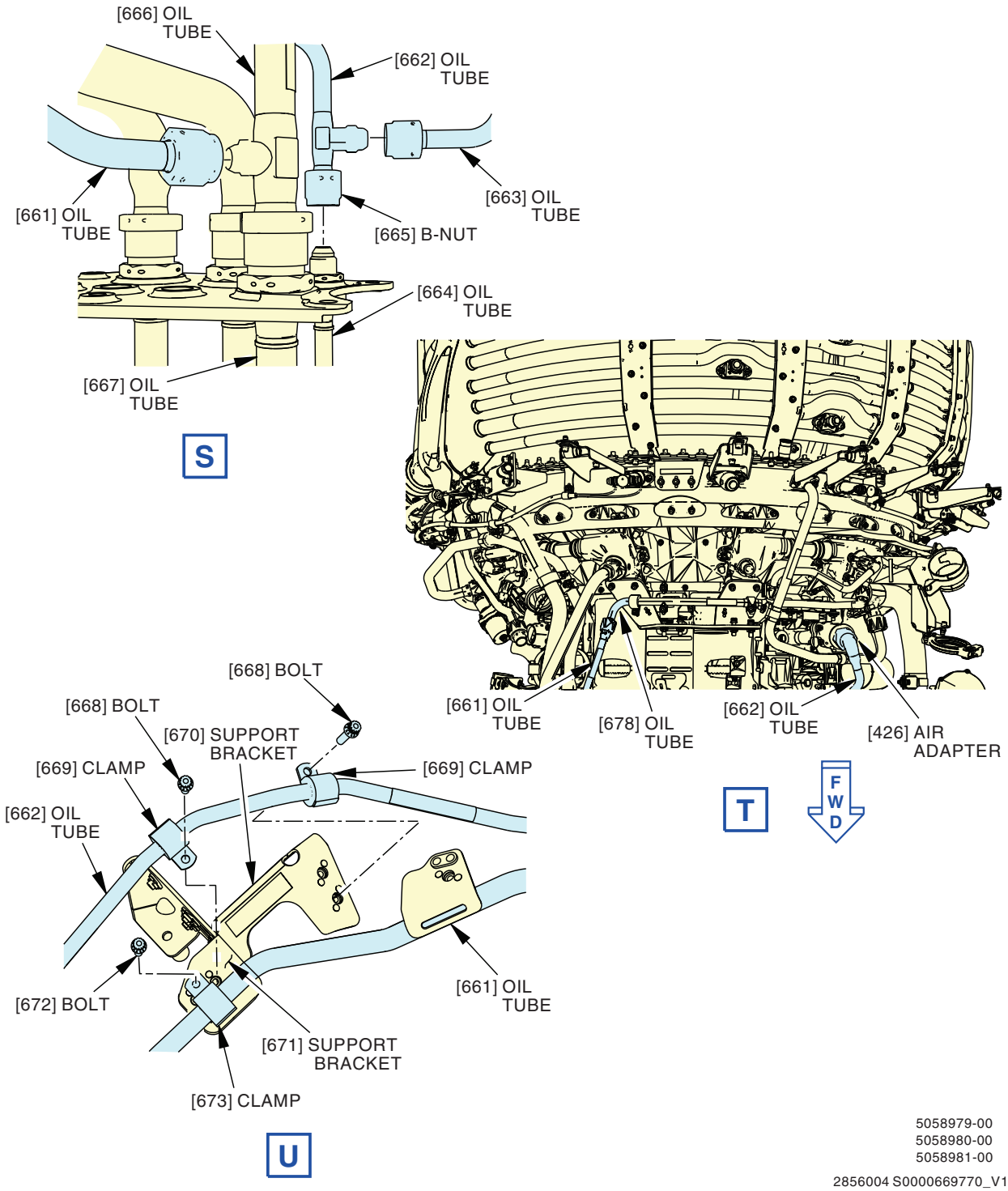
ECCN 9E991 BOEING PROPRIETARY - See title page for details



5058978-00
2855891 S0000669769_V1

Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 13 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details

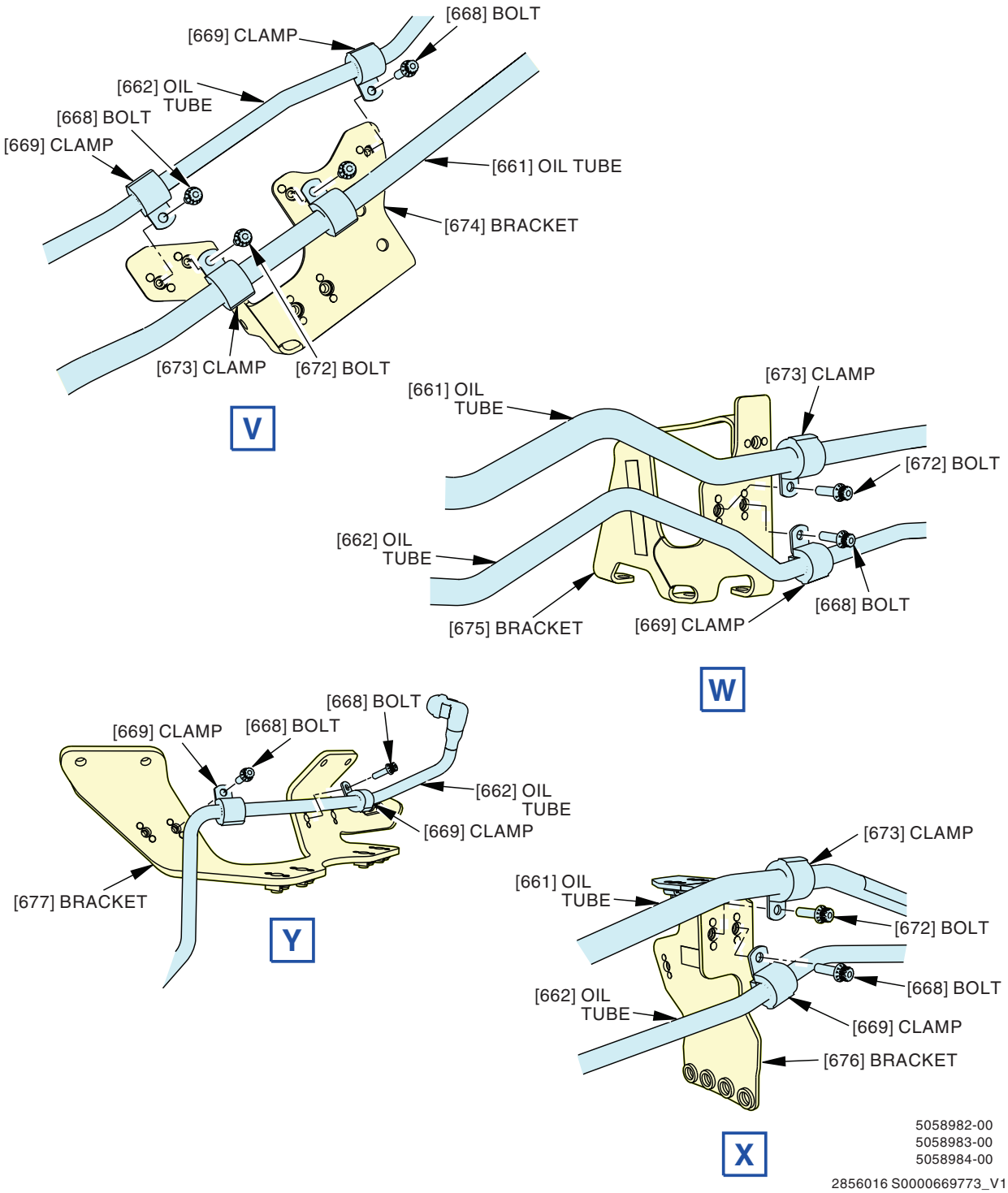


5058979-00
5058980-00
5058981-00

2856004 S0000669770_V1

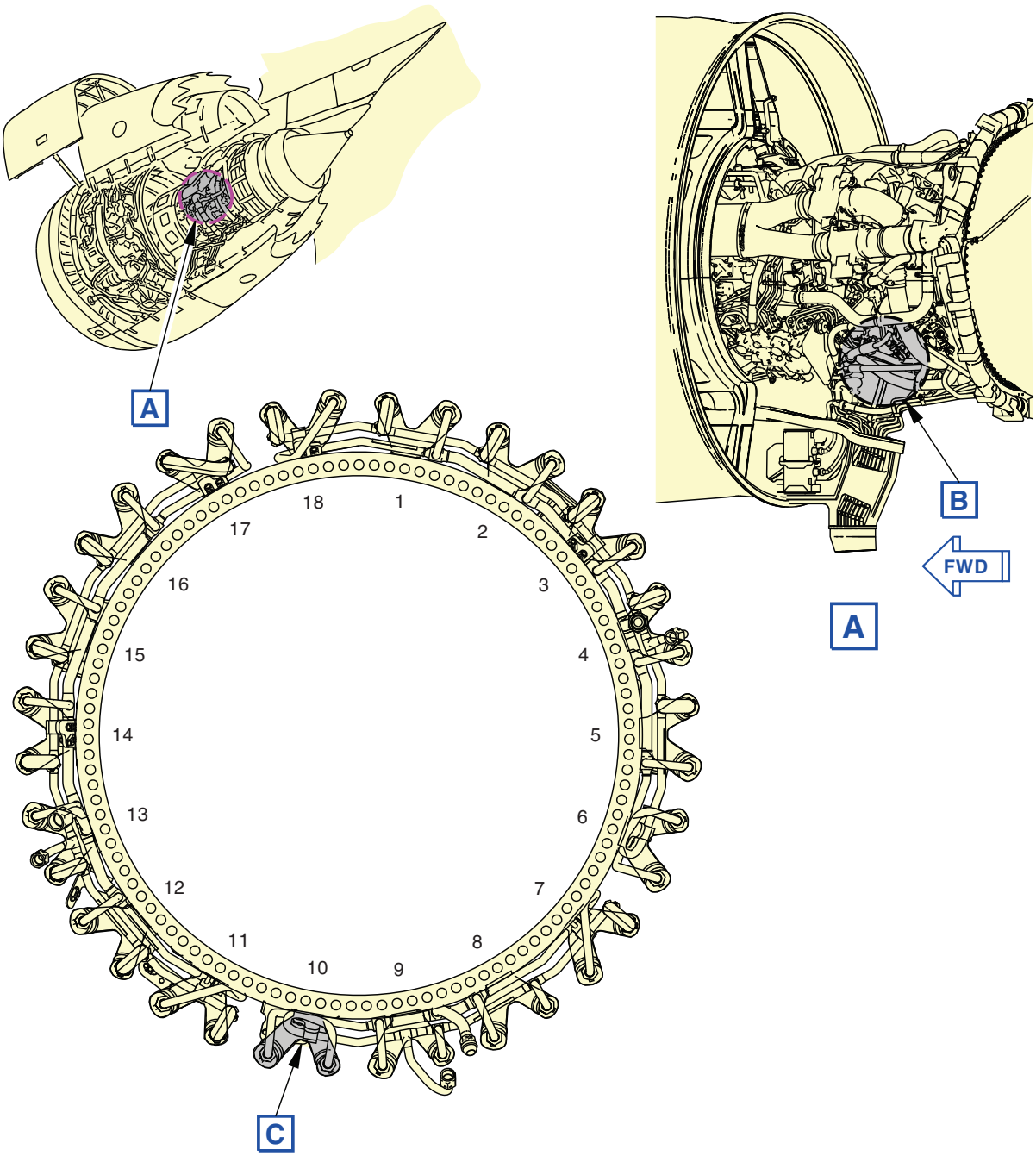
Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 14 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



Fuel Nozzle (Position 9) Installation
Figure 410/73-11-05-990-896-G00 (Sheet 15 of 15)

ECCN 9E991 BOEING PROPRIETARY - See title page for details



5021979-00
2476502 S0000580883_V1

Fuel Nozzle (Position 10) Installation
Figure 411/73-11-05-990-897-G00 (Sheet 1 of 18)

ECCN 9E991 BOEING PROPRIETARY - See title page for details